Causal Inference

Lecture notes, Chapters 1–11

Based on the course of Sanghack Lee (Graduate School of Data Science, Seoul National University)

Contents

Chapter 1. Introduction and Causal Models **12**

1.1 Why causal inference **12**
1.1.1 A brief review of machine learning . . . 12
1.1.1.1 Why machine learning? . . . 12
1.1.1.2 The classical learning paradigm . . . 12
1.1.1.3 A probabilistic view . . . 13
1.1.1.4 Reinforcement learning: intervention instead of observation . . . 13
1.1.2 Conceptual foundation: the ladder of causation . . . 13
1.1.2.1 Inference about worlds that are not realized . . . 13
1.1.2.2 Pearl's Causal Hierarchy . . . 14
1.1.2.3 Two motivations . . . 14
1.1.3 Correlation versus causation . . . 14
1.1.4 Example 1: Simpson's paradox in a 2×2 table . . . 15
1.1.4.1 Would the result be different with a different story? . . . 16
1.1.5 Example 2: a "trivial" data analysis . . . 16
1.1.6 What is going on here? . . . 16

1.2 Causal models and causal graphs **17**
1.2.1 Reviews: notation, probability, algebra . . . 17
1.2.1.1 Notation . . . 17
1.2.1.2 Probability . . . 17
1.2.1.3 Sums of products . . . 17
1.2.2 Why data alone is not enough: a prima facie example . . . 18
1.2.2.1 The story . . . 18
1.2.2.2 Current protocol . . . 18
1.2.2.3 Hypothetical protocols . . . 18
1.2.2.4 What can a learning system see? . . . 18
1.2.2.5 From statistical to causal analysis . . . 19
1.2.2.6 Modeling reality in the example . . . 19
1.2.3 Structural causal models . . . 19
1.2.3.1 Property 1: an SCM induces the observational distribution . . . 20
1.2.3.2 Property 2: an SCM induces a causal diagram . . . 20
1.2.3.3 Graph terminology . . . 21
1.2.3.4 Property 3: an SCM induces Pearl's Causal Hierarchy . . . 21
1.2.4 The first layer: what the diagram encodes about $P(\mathbf{v})$ . . . 21
1.2.5 Conditional independence and the Markov condition . . . 22
1.2.5.1 Conditional independence . . . 22
1.2.5.2 Local Markov condition . . . 22
1.2.5.3 Markovian models and the Bayesian factorization . . . 23
1.2.6 Reading independences from the graph: d-separation . . . 23
1.2.6.1 Chains . . . 23
1.2.6.2 Common causes (forks) . . . 23

1.2.6.3 Common effects (colliders) . . . 24
1.2.6.4 Triplet summary . . . 24
1.2.6.5 d-separation . . . 24
1.2.7 Implementation: graph algorithms in Python . . . 25
1.2.7.1 Topological order . . . 25
1.2.7.2 d-separation via reachability (Markovian graphs) . . . 26

Chapter 2. Identification and Adjustment 27

2.1 Identification of causal effects 27
2.1.1 Structural causal models, revisited . . . 27
2.1.1.1 Observation versus intervention . . . 27
2.1.1.2 The do-operator . . . 27
2.1.2 The second layer: causal effects . . . 28
2.1.2.1 Connecting different worlds . . . 28
2.1.2.2 Computing causal effects from observational data (Markovian case) . . . 28
2.1.3 Truncated factorization . . . 29
2.1.4 The identification problem . . . 29
2.1.4.1 Identification in Markovian models . . . 30
2.1.4.2 Adjustment by direct parents (singleton) . . . 30
2.1.4.3 Adjustment by direct parents for a set of treatments (optional) . . . 31
2.1.4.4 When a parent is latent . . . 32
2.1.5 Summary . . . 32

2.2 Confounding and the back-door criterion 33
2.2.1 Recap: identifiable and non-identifiable effects . . . 33
2.2.1.1 An identifiable effect . . . 33
2.2.1.2 A non-identifiable effect . . . 33
2.2.2 Confounding bias . . . 34
2.2.3 The back-door criterion . . . 34
2.2.3.1 Back-door sets substitute for the parents: an example . . . 35
2.2.3.2 General proof: back-door adjustment = parent adjustment . . . 35
2.2.3.3 A graphical test . . . 36
2.2.4 Evaluating back-door adjustment: inverse probability weighting . . . 36
2.2.4.1 Rewriting with the propensity score . . . 36
2.2.4.2 Sample version . . . 37
2.2.4.3 Average treatment effect . . . 37
2.2.4.4 Interpretation: pseudo-causal samples . . . 37
2.2.5 Summary . . . 37

2.3 The adjustment criterion 38
2.3.1 Back-door criterion: recap and two refinements . . . 38
2.3.1.1 Back to Simpson's paradox . . . 38
2.3.1.2 Conditional back-door . . . 38
2.3.2 Tightness of the back-door criterion . . . 39
2.3.3 The adjustment criterion . . . 39
2.3.3.1 Proper causal paths . . . 40
2.3.3.2 Disturbing a proper causal path . . . 40
2.3.3.3 The criterion . . . 40
2.3.3.4 Implementation idea . . . 41
2.3.3.5 An example . . . 41
2.3.3.6 Explicit construction of an admissible set . . . 42

2.3.3.7 Recap . . . 42
2.3.4 Listing all admissible sets (optional) . . . 42
2.3.4.1 Motivation . . . 42
2.3.4.2 Challenge: output size . . . 43
2.3.4.3 Theoretical tool: bounded separators . . . 43
2.3.4.4 Algorithm . . . 43
2.3.4.5 Analysis . . . 44
2.3.5 Connection with conditional ignorability . . . 44
2.3.6 Summary . . . 44

Chapter 3. Do-Calculus and Algorithmic Identification 46

3.1 Do-calculus 46
3.1.1 Three examples via the truncated factorization . . . 46
3.1.1.1 Back-door graph . . . 46
3.1.1.2 Bow graph . . . 47
3.1.1.3 Front-door graph . . . 47
3.1.2 Identification approaches . . . 47
3.1.3 Do-calculus . . . 47
3.1.3.1 Prelude: independence in a submodel . . . 47
3.1.3.2 Insight 1: adding/removing observations . . . 48
3.1.3.3 Insight 2: exchanging action and observation . . . 48
3.1.3.4 Insight 3: adding/removing actions . . . 48
3.1.3.5 The rules . . . 48
3.1.3.6 Understanding Rule 2 . . . 49
3.1.3.7 Understanding Rule 3 . . . 50
3.1.4 Do-calculus and identification . . . 50
3.1.4.1 Front-door by do-calculus . . . 51
3.1.4.2 Proving non-identifiability . . . 51
3.1.4.3 Catalogue of graphs . . . 52
3.1.5 Back to Simpson's paradox . . . 53
3.1.5.1 Pearson's shock . . . 53
3.1.5.2 Dissolving the paradox with the do-operator . . . 53
3.1.5.3 The paradox stems from causal interpretation . . . 53
3.1.5.4 Appendix: the Berkeley data . . . 54
3.1.6 Summary and food for thought . . . 54

3.2 Algorithmic identification 55
3.2.1 Factorizing observational distributions . . . 55
3.2.1.1 Confounded components . . . 55
3.2.1.2 Markovian case . . . 56
3.2.1.3 Semi-Markovian case . . . 56
3.2.1.4 C-factors . . . 56
3.2.1.5 C-factors are causal effects . . . 57
3.2.2 C-factor algebra . . . 57
3.2.2.1 Marginalizing variables in c-factors . . . 57
3.2.2.2 C-component factorization . . . 57
3.2.2.3 Visualization . . . 59
3.2.3 Causal queries in terms of c-factors: examples . . . 59
3.2.3.1 Back-door . . . 59
3.2.3.2 Front-door . . . 59
3.2.3.3 Napkin . . . 59

3.2.3.4 Overview of the three derivations . . . 60
3.2.4 A general identification algorithm . . . 60
3.2.4.1 Reducing the query to c-factors . . . 60
3.2.4.2 The Identify procedure . . . 61

Chapter 4. Linear Structural Causal Models 62

4.1 Linear structural causal models 62
4.1.1 Definition and conventions . . . 62
4.1.2 The identification problem . . . 63
4.1.3 Connecting observed and unobserved quantities . . . 63
4.1.4 Treks and Wright's rule . . . 63

4.2 Linear regression and causal effects 64
4.2.1 The medical researcher . . . 64
4.2.2 The implicit assumption: no confounding . . . 64
4.2.3 Regression equation versus structural equation . . . 65
4.2.4 Testing conditional independence with regression . . . 65
4.2.5 Single-door criterion: direct effects . . . 65
4.2.6 Back-door criterion: total effects . . . 66

4.3 Algorithmic identification methods 66
4.3.1 The equations of linear identification . . . 66

4.4 From instrumental variables to instrumental sets 67
4.4.1 Instrumental variables . . . 67
4.4.2 Conditional instrumental variables . . . 68
4.4.3 Instrumental sets . . . 68
4.4.4 Conclusions . . . 69

4.5 Appendix: Complexity of identification in linear SCMs 69

4.6 Appendix: Multiple regression when Y is the cause 69

4.7 Appendix: The Gröbner basis approach 69

Chapter 5. Partial Identification and Soft Interventions 71

5.1 Partial identification 71
5.1.1 The partial identification problem . . . 71
5.1.2 Natural bounds . . . 71
5.1.2.1 Tightness . . . 72
5.1.2.2 Limitations . . . 72
5.1.3 Canonical type models and LP bounds . . . 72
5.1.3.1 Experiments with non-compliance . . . 72
5.1.3.2 Optimization perspective . . . 73
5.1.3.3 U as a selector and response variables . . . 73
5.1.3.4 LP formulation . . . 74
5.1.3.5 Closed-form solution . . . 75
5.1.4 Back-door with partially observed covariates . . . 75

5.2 Soft interventions 76
5.2.1 Motivation . . . 76

5.2.2 Canonical types of regimes . . . 76
5.2.3 Reduction to known machinery . . . 77
5.2.4 σ-calculus . . . 77
5.2.4.1 Comparison with do-calculus . . . 78
5.2.4.2 Another intervention . . . 78
5.2.5 C-factors and soft interventions . . . 79
5.2.6 Summary . . . 79

Chapter 6. Potential Outcomes and Effect Estimation 80

6.1 Potential outcomes 80
6.1.1 Frameworks for causal inference . . . 80
6.1.2 Basic concepts . . . 80
6.1.2.1 The fundamental problem of causal inference . . . 80
6.1.2.2 God's eye view: individual versus average effects . . . 81
6.1.2.3 What the researcher actually sees . . . 81
6.1.3 Assumptions linking potential and observed outcomes . . . 81
6.1.3.1 SUTVA . . . 82
6.1.3.2 Consistency and well-defined interventions . . . 82
6.1.4 Causal estimands . . . 82
6.1.5 Randomization and observational studies . . . 82
6.1.5.1 Unconfounded assignment . . . 83
6.1.5.2 The role of randomization . . . 83
6.1.5.3 Observational studies . . . 83
6.1.5.4 Identification of the ATE . . . 83
6.1.5.5 Regression estimation of the ATE . . . 84
6.1.5.6 Summary . . . 84
6.1.6 Appendix: Positivity and consistency in causal diagrams . . . 84
6.1.7 Appendix: Confounding and exchangeability . . . 84

6.2 Inverse probability weighting and doubly robust estimation 85
6.2.1 The propensity score . . . 85
6.2.2 Inverse probability weighting . . . 85
6.2.2.1 The IPW identity . . . 85
6.2.2.2 Estimators . . . 86
6.2.2.3 Normalizing the weights . . . 86
6.2.2.4 The weighted population . . . 86
6.2.2.5 Pros and cons . . . 86
6.2.2.6 Trimming and truncation . . . 87
6.2.2.7 PS weighting in randomized trials . . . 87
6.2.2.8 Weighting versus balancing . . . 87
6.2.3 Doubly robust estimation . . . 87
6.2.3.1 Terminology . . . 87
6.2.3.2 Two building blocks . . . 88
6.2.3.3 Verification with the true nuisance functions . . . 88
6.2.3.4 Two equivalent forms . . . 88
6.2.3.5 What does it estimate? . . . 88
6.2.3.6 The doubly robust ATE estimator . . . 89
6.2.3.7 Concluding remarks . . . 89
6.2.3.8 Summary . . . 89

6.3 Conditional average treatment effects 89

6.3.1 Why heterogeneous treatment effects? . . . 90
6.3.1.1 The average can be misleading . . . 90
6.3.1.2 Effect modification . . . 90
6.3.1.3 Why care? . . . 90
6.3.1.4 From stratification to CATE estimation . . . 90
6.3.2 Meta-learners for CATE . . . 91
6.3.2.1 T-learner and S-learner . . . 91
6.3.2.2 Regression imputation . . . 91
6.3.2.3 X-learner . . . 91
6.3.2.4 DR-learner . . . 92

6.4 The R-learner 92
6.4.1 Motivation and identification . . . 92
6.4.2 The R-loss . . . 93
6.4.3 Algorithm . . . 93
6.4.4 Orthogonality and the quasi-oracle property . . . 93
6.4.5 Comparison and summary . . . 94

Chapter 7. Matching and Instrumental Variables 96

7.1 Matching 96
7.1.1 Matching as nonparametric imputation . . . 96
7.1.1.1 Setup . . . 96
7.1.1.2 From regression to matching . . . 96
7.1.1.3 The matching idea . . . 96
7.1.1.4 The matching estimator . . . 97
7.1.1.5 Exact matching and coarsened exact matching . . . 97
7.1.2 Nearest-neighbor matching and its theory . . . 97
7.1.2.1 Definition . . . 97
7.1.2.2 Algorithms (matching without replacement) . . . 97
7.1.2.3 The curse of dimensionality . . . 98
7.1.2.4 ATT matching: independence and consistency . . . 98
7.1.2.5 A toy example . . . 98
7.1.2.6 Bias-corrected matching (Abadie and Imbens 2011) . . . 98
7.1.2.7 Inference after matching . . . 99
7.1.2.8 Kernel matching . . . 99
7.1.3 Propensity-score matching . . . 99
7.1.4 Practical matching . . . 100
7.1.5 Model dependence and empirical evidence . . . 101
7.1.6 Appendix: ATE matching: why it is harder . . . 101
7.1.7 Appendix: Why the estimated PS can beat the true PS . . . 101
7.1.8 Appendix: The PS paradox (King and Nielsen 2019) . . . 102
7.1.9 Appendix: PS overfitting and the broader picture . . . 102
7.1.10 Appendix: Beyond binary treatment . . . 102
7.1.11 Appendix: Bias-corrected matching versus post-matching regression . . . 102
7.1.12 Appendix: Stratification . . . 102

7.2 Instrumental variables 103
7.2.1 Motivation . . . 103
7.2.1.1 Instrumental variables in action . . . 103
7.2.1.2 Why instruments? . . . 103
7.2.1.3 Conditions for a good instrument . . . 103

7.2.2 IV under linearity . . . 103
7.2.2.1 The econometric setup . . . 104
7.2.2.2 Just-identified, no covariates . . . 104
7.2.2.3 Just-identified, with covariates . . . 104
7.2.2.4 From linear to nonparametric IV . . . 105
7.2.3 Randomized experiments with non-compliance . . . 105
7.2.3.1 Setting . . . 105
7.2.3.2 Intent-to-treat effects . . . 105
7.2.3.3 Identification assumptions . . . 105
7.2.3.4 Identification . . . 106
7.2.3.5 A worked example . . . 107
7.2.3.6 CACE estimator . . . 107
7.2.3.7 The "fourth condition": homogeneity . . . 107
7.2.3.8 Summary . . . 107
7.2.4 Appendix: Candidate instruments in observational studies . . . 108
7.2.5 Appendix: IV versus other causal methods . . . 108
7.2.6 Appendix: Practical issues with classic IV studies . . . 108
7.2.7 Appendix: Weak instruments . . . 108
7.2.8 Appendix: Many instruments and robust IV . . . 109

Chapter 8. Causal Data Science 110

8.1 Motivation and big picture 110

8.2 Experimental conditions: general identifiability 111
8.2.1 The challenge . . . 111
8.2.2 Subtleties of z-ID . . . 111
8.2.3 z-ID reduced to calculus . . . 112
8.2.4 Example: drug–drug interactions . . . 112
8.2.5 Algorithm and summary . . . 112

8.3 Environmental conditions: transportability 112
8.3.1 The problem . . . 113
8.3.2 What can an experiment in LA tell us about NYC? . . . 113
8.3.3 Selection diagrams . . . 113
8.3.4 General transportability . . . 114
8.3.5 Is the gold standard golden? . . . 115

8.4 Sampling conditions: recovering from selection bias 115
8.4.1 The problem . . . 115
8.4.2 The origin of selection bias . . . 115
8.4.3 Variants . . . 116
8.4.4 Recoverability without external data . . . 116
8.4.5 Recoverability with external data . . . 116

8.5 Responding conditions: missing data 117
8.5.1 Motivation . . . 117
8.5.2 A motivating example . . . 118
8.5.3 m-graphs . . . 118
8.5.4 Three categories . . . 118
8.5.5 General results . . . 119
8.5.6 Recovering a causal effect . . . 119

8.6 Measurement: general identifiability with partial observation **119**

8.7 Conclusions **120**

Chapter 9. Regression Discontinuity, Difference-in-Differences and Synthetic Control **121**

9.1 Regression discontinuity **121**
9.1.1 Where RDD fits . . . 121
9.1.2 Concept and identification . . . 121
9.1.2.1 Examples . . . 121
9.1.2.2 The design . . . 122
9.1.2.3 Estimand and identification . . . 122
9.1.3 Validity and diagnostics . . . 123
9.1.4 Estimation . . . 123
9.1.4.1 General strategy . . . 124
9.1.4.2 Local linear regression . . . 124
9.1.4.3 Bias–variance trade-off and the MSE-optimal bandwidth . . . 124
9.1.4.4 Why the MSE-optimal bandwidth may fail for inference . . . 125
9.1.4.5 Covariates and software . . . 125
9.1.5 Fuzzy RD . . . 125
9.1.6 Case study: remedial education in Chicago . . . 126
9.1.7 Summary . . . 127
9.1.8 Appendix: Kink RD . . . 127
9.1.9 Appendix: Assumptions for the MSE-optimal bandwidth . . . 127
9.1.10 Appendix: Regularization of the bandwidth . . . 127
9.1.11 Appendix: Why local linear? Polynomial order . . . 128
9.1.12 Appendix: Global polynomials: a cautionary tale . . . 128

9.2 Difference-in-differences **128**
9.2.1 Motivation . . . 128
9.2.1.1 Case study: minimum wages (Card and Krueger 1994) . . . 128
9.2.2 The DiD design . . . 128
9.2.2.1 Setup . . . 129
9.2.2.2 The identification problem . . . 129
9.2.2.3 Three control strategies . . . 129
9.2.2.4 Parallel trends . . . 130
9.2.2.5 Estimation . . . 130
9.2.2.6 DiD and two-way fixed effects (2×2) . . . 131
9.2.3 Conditional and nonlinear DiD . . . 131
9.2.3.1 Conditional parallel trends . . . 131
9.2.3.2 Nonlinear DiD (changes-in-changes) . . . 131
9.2.3.3 Key takeaways . . . 132
9.2.4 Appendix: Pre-testing bias . . . 132
9.2.5 Appendix: Sensitivity analysis for parallel trends . . . 132
9.2.6 Appendix: Staggered treatment timing . . . 132

9.3 Staggered difference-in-differences **132**
9.3.1 Staggered adoption . . . 133
9.3.1.1 TWFE as is is not sufficient . . . 133
9.3.1.2 Terminology . . . 133
9.3.1.3 Valid and invalid comparisons . . . 133
9.3.2 Callaway–Sant'Anna group-time ATTs . . . 134

9.3.2.1 Notation . . . 134
9.3.2.2 Parallel trends . . . 134
9.3.2.3 Group-time ATT and identification . . . 134
9.3.2.4 Conditional parallel trends . . . 135
9.3.2.5 Aggregation . . . 135
9.3.2.6 Summary . . . 135

9.4 Synthetic control 135
9.4.1 From DiD to synthetic controls . . . 136
9.4.1.1 Case study: California Proposition 99 . . . 136
9.4.2 The estimator . . . 136
9.4.2.1 Setup . . . 136
9.4.2.2 Missing counterfactuals and synthetic weights . . . 136
9.4.2.3 Model-based justifications . . . 137
9.4.2.4 Convexity: no extrapolation . . . 137
9.4.2.5 Practical concerns . . . 137
9.4.3 Inference and bias . . . 138
9.4.3.1 Placebo tests . . . 138
9.4.3.2 Why imperfect pre-treatment fit causes bias . . . 138
9.4.3.3 Bias correction: augmented SCM . . . 138
9.4.4 Discussion . . . 139
9.4.5 Appendix: More treated units . . . 139
9.4.6 Appendix: Predictor importance weights V . . . 140

9.5 Synthetic difference-in-differences 140
9.5.1 DiD, SC and SDiD . . . 140
9.5.2 Formulations . . . 141
9.5.2.1 Setting and algorithm . . . 141
9.5.2.2 Unit weights . . . 141
9.5.2.3 Time weights . . . 142
9.5.3 Comparisons and takeaways . . . 142

Chapter 10. Causal Discovery 143

10.1 Causal discovery 143
10.1.1 The structure-learning problem . . . 143
10.1.1.1 DAG models . . . 143
10.1.1.2 Faithfulness . . . 143
10.1.2 Constraint-based structure learning . . . 144
10.1.2.1 What constraints a DAG encodes . . . 144
10.1.2.2 Markov equivalence . . . 144
10.1.2.3 The PC algorithm . . . 145
10.1.2.4 Extensions . . . 146
10.1.2.5 Summary . . . 146
10.1.3 Score-based approaches . . . 146
10.1.3.1 Why scores? . . . 147
10.1.3.2 Log-likelihood for DAG models . . . 147
10.1.3.3 BIC . . . 147
10.1.3.4 Search and GES . . . 147
10.1.3.5 Continuous optimization: NOTEARS and DAGMA . . . 148
10.1.3.6 Summaries . . . 149
10.1.4 Parametric approaches . . . 149

10.1.4.1 Non-linear functions . . . 150
10.1.4.2 Linear functions with non-Gaussian noise: LiNGAM . . . 150
10.1.4.3 Score matching (NoGAM) . . . 151
10.1.5 Appendix: GES operators in detail . . . 151
10.1.6 Appendix: Post-nonlinear causal model (optional) . . . 151
10.1.7 Appendix: Scores in detail . . . 152
10.1.8 Appendix: CI tests for causal discovery . . . 152

10.2 Local causal discovery 153
10.2.1 The map . . . 153
10.2.1.1 One primitive: grow–shrink . . . 153
10.2.2 Going local on CI . . . 154
10.2.2.1 The Markov blanket . . . 154
10.2.2.2 GS: grow–shrink the blanket . . . 154
10.2.2.3 HITON: target-focused, not whole-graph . . . 155
10.2.3 Bringing in the score . . . 155
10.2.3.1 MMHC: a hybrid bridge . . . 155
10.2.3.2 BOSS: best order score search . . . 156
10.2.4 Zoom out . . . 156

Chapter 11. Mediation, SWIGs and G-Methods 158

11.1 Direct and indirect effects 158
11.1.1 Preliminaries: potential responses . . . 158
11.1.2 Introduction . . . 158
11.1.2.1 Examples . . . 158
11.1.2.2 Prescriptive versus descriptive . . . 159
11.1.3 Mathematisation . . . 159
11.1.3.1 The cross-world problem . . . 160
11.1.3.2 Experimental identification . . . 160
11.1.3.3 Markovian models . . . 160
11.1.3.4 The mediation formula . . . 161
11.1.4 Natural indirect effects . . . 161
11.1.5 Path-specific effects . . . 162
11.1.6 Appendix: The linear additive relation . . . 162
11.1.7 Appendix: Counterfactual axioms . . . 163

11.2 Single-world intervention graphs 163
11.2.1 Potential outcomes and graphs . . . 163
11.2.1.1 Potential outcomes . . . 163
11.2.1.2 Graphs . . . 163
11.2.2 Node splitting . . . 164
11.2.3 SWIG construction and properties . . . 164
11.2.4 A harder inferential problem . . . 165
11.2.5 Adjustment for confounding . . . 165
11.2.6 Conclusion . . . 166

11.3 G-methods for time-varying treatments 166
11.3.1 Time-varying treatments . . . 166
11.3.1.1 Identification assumptions . . . 167
11.3.1.2 The fundamental dilemma . . . 167
11.3.2 The g-formula . . . 167
11.3.2.1 Iterated conditional expectations (ICE) . . . 168

11.3.3 IP weighting and marginal structural models 168
11.3.3.1 Marginal structural models 170
11.3.4 G-estimation of structural nested models 170
11.3.5 Doubly robust estimation 171
11.3.6 Practical considerations 171
11.3.7 Comparison of g-methods 172

Chapter 1. Introduction and Causal Models

1.1 Why causal inference

Motivation. The central question is why the tools of classical machine learning and statistics, which learn aspects of a *fixed* data-generating reality, are not sufficient to answer questions about actions and about worlds that were never realized.

1.1.1 A brief review of machine learning

Motivation. Describe what standard learning paradigms estimate, so that it becomes clear what they *cannot* estimate.

1.1.1.1 Why machine learning?

AI/ML is the design and analysis of intelligent systems. Intelligent behavior requires knowledge, i.e. a model of the environment. Writing this knowledge down explicitly is hard and often infeasible, so it is acquired by *learning from data through algorithms*.

Learning modifies an agent's decision mechanism in order to improve performance. Since environments change, an agent must adapt to unexpected situations; learning is essential when the designer is not omniscient, and is particularly useful for messy real-world data. Three fundamental questions recur:

- How can the learning process be characterized?
- How should prior knowledge be traded off against performance?
- How robust and generalizable is the resulting system?

Applications are everywhere: generative AI (LLMs, multimodal agents, digital twins), AI for science (protein folding, drug discovery, materials), medicine (personalized treatment, surgical planning), sustainability (climate, weather, energy grids), vision (scene understanding, deepfake detection), autonomous systems (humanoid robots, drone fleets) and security (fraud detection, cyber-security agents).

1.1.1.2 The classical learning paradigm

There is an unknown reality, a true model M, that generates the world. M produces examples represented as feature vectors x, and one is interested in an ideal target function $f_M(x)$ of that reality. From training data $\{x^{(i)}\}_{i=1}^N$ one selects a hypothesis $h \in \mathcal{H}$ such that $h(x) \approx f_M(x)$ for every instance. The main challenge is generalization to unseen instances.

Supervised learning maps a feature space $\mathcal{X}$ to a label space $\mathcal{Y}$ from labeled pairs $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$, looking for h that mimics an oracle $f : \mathcal{X} \to \mathcal{Y}$. Common hypothesis classes are non-parametric methods (decision trees, k-NN), probabilistic models (Bayesian networks), deep networks (CNNs, Transformers) and ensembles (XGBoost, LightGBM).

Unsupervised learning receives only $x \in \mathcal{X}$ and seeks a representation revealing intrinsic structure: clustering, dimensionality reduction (latent spaces), and generative modeling of the density $P(x)$. (A phylogenetic tree, a hierarchical clustering of species, is a typical example.)

1.1.1.3 A probabilistic view

Both paradigms estimate features of one fixed joint distribution:

- supervised learning is essentially inference of the conditional distribution $P(y \mid x)$ (e.g. the output activation of a network or a leaf of a decision tree);
- unsupervised learning infers properties of the marginal distribution $P(x)$.

In both cases *we learn to imitate aspects of an underlying static reality.* Nothing in $P(x, y)$ tells us what would happen if the mechanism producing x were changed.

1.1.1.4 Reinforcement learning: intervention instead of observation

In reinforcement learning no static labeled sample is given. The agent *intervenes* by interacting with the environment and learns from its responses (A/B testing, robotics). The key difficulty is the exploration/exploitation trade-off. RL thus touches a new kind of quantity, the effect of an action; the course asks what is still missing from this picture, in particular how to reason about actions using data that were *not* generated by our own experiments, and how to reason about events that did not happen.

1.1.2 Conceptual foundation: the ladder of causation

Motivation. Introduce the three cognitive abilities, observing, intervening, imagining, that structure the whole course.

Human cognition can be organized in three layers:

Layer 1. Observation (sensors): the ability to observe the world and infer the "next" stage.

Layer 2. Agency: the ability to intervene in the world and change it.

Layer 3. Retrospection: the ability to reflect on what we did (and did not do) and infer new behaviors.

This cognitive stratification is formalized as **Pearl's Causal Hierarchy (PCH)**.

1.1.2.1 Inference about worlds that are not realized

The classical pipeline is

$$\text{Real world / nature} \xrightarrow{\text{data}} \text{AI/ML, statistics (inference)} \longrightarrow \text{new knowledge.}$$

For an *alternative reality* that was not realized ("what if we had changed the policy?") there are no data at all. The only route is

$$\text{knowledge} \longrightarrow \text{causal model} \xrightarrow{\text{causal inference}} \text{new knowledge about the alternative reality.}$$

A causal model carries assumptions about mechanisms that let us transport information from the observed world to unobserved ones.

1.1.2.2 Pearl's Causal Hierarchy

Level (symbol)	Activity	Question	Examples / typical models
1. Association	Seeing	What is?	What does a symptom tell us about a disease? How would seeing X change my belief in Y?
$P(y \mid x)$			(Un)supervised ML: deep nets, Bayes nets, hierarchical models, decision trees.
2. Intervention	Doing	What if?	What if I take aspirin, will my headache be cured? What if I do X?
$P(y \mid \mathrm{do}(x), c)$			Reinforcement learning, causal Bayesian networks, MDPs, POMDPs.
3. Counterfactual	Imagining,	Why?	Was it the aspirin that stopped my headache? What if I had acted differently?
$P(y_x \mid x', y')$	retrospection		Structural causal models.

Causal inference = cross-layer inference. Typical tasks take data from one layer and answer a query from a higher (or different) layer:

- *Data from Layer 1, query in Layer 2.* "Will students' performance improve if we change the class time to be earlier?" using only observational records.
- *Data from Layer 2, query in Layer 2.* "Use the findings of previous experiments to predict the outcome of a new one" (transportability, generalization of experiments).
- *Data from Layers 1 and 2, query in Layer 3.* "Would I have won had I picked door #2 instead of door #1?"

Later lectures prove that, in general, lower layers *underdetermine* higher ones: without further assumptions, Layer 1 data cannot answer Layer 2 questions, and Layer 2 data cannot answer Layer 3 questions.

1.1.2.3 Two motivations

- **Scientific perspective:** understanding nature and human beings, using empirical data, logic and mathematics, and the scientific method.
- **Pragmatic perspective** (economics, medicine, engineering): building truly intelligent systems, and designing and implementing policies aimed at social, economic or political goals. Policies are interventions, so they live in Layer 2.

1.1.3 Correlation versus causation

Motivation. Show with real-world data that strong association is compatible with no causal relation at all.

Several famous spurious correlations:

- The number of active Facebook users and the yield of 10-year Greek government bonds (2005–2011) rise together (BusinessWeek, "Correlation or causation?", Dec. 2011).
- Fresh lemons imported to the USA from Mexico and total US highway fatality rate, 1996–2000, have $R^2 = 0.97$ (negative slope).
- Organic food sales and diagnosed autism prevalence, 1998–2007, have $r = 0.9971$ ($p < 0.0001$).
- The divorce rate in Maine correlates with the per-capita consumption of margarine (2000–2009).

Each pair is driven by common time trends or coincidence rather than by a mechanism linking them. The xkcd cartoon (no. 552) jokes: "I used to think correlation implied causation. Then I took a statistics class. Now I don't." "Sounds like the class helped." "Well, maybe." The real question is the converse one: *if correlation does not imply causation, then what does?* The answer developed in this course is: correlation *plus* causal assumptions, encoded in a model.

1.1.4 Example 1: Simpson's paradox in a 2×2 table

Motivation. Demonstrate that the same data support opposite conclusions depending on the causal story, so that no purely statistical rule decides which table to use.

Notation: X = took the drug, $\neg X$ = did not; Y = survived (recovered), $\neg Y$ = did not; F = female, $\neg F$ = male (Simpson 1951; Pearl 2000, Ch. 6).

All	Y	$\neg Y$	total	rate
drug (X)	20	20	40	50%
no drug ($\neg X$)	16	24	40	40%
Σ	36	44	80	

Male ($\neg F$)	Y	$\neg Y$	total	rate
drug	18	12	30	60%
no drug	7	3	10	70%
Σ	25	15	40	

Female (F)	Y	$\neg Y$	total	rate
drug	2	8	10	20%
no drug	9	21	30	30%
Σ	11	29	40	

Within each sex the drug is worse,

$$P(Y \mid F, X) = 0.2 < 0.3 = P(Y \mid F, \neg X),$$
$$P(Y \mid \neg F, X) = 0.6 < 0.7 = P(Y \mid \neg F, \neg X),$$

yet in the aggregate it is better: $P(Y \mid X) = 0.5 > 0.4 = P(Y \mid \neg X)$.

Why this is arithmetically possible. By the law of total probability,

$$P(Y \mid X) = P(Y \mid X, \neg F)P(\neg F \mid X) + P(Y \mid X, F)P(F \mid X).$$

Among drug takers $P(F \mid X) = 10/40 = 0.25$, while among non-takers $P(F \mid \neg X) = 30/40 = 0.75$. Hence

$$P(Y \mid X) = 0.75 \cdot 0.6 + 0.25 \cdot 0.2 = 0.50,$$
$$P(Y \mid \neg X) = 0.25 \cdot 0.7 + 0.75 \cdot 0.3 = 0.40.$$

The aggregate rates average the stratum rates with *different weights*; men, who recover more often regardless of treatment, are over-represented among drug takers. The reversal is not a paradox of probability, only of intuition.

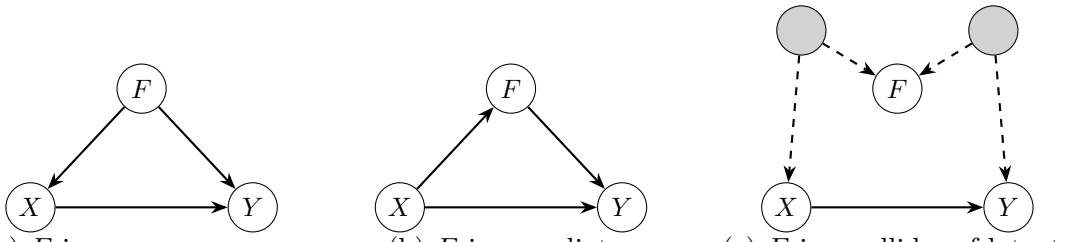
Which table should the physician consult? If sex affects both who takes the drug and recovery, the stratified table should be used, and the population-level effect is obtained by weighting the strata by the *population* distribution of sex, $P(F) = P(\neg F) = 40/80 = 1/2$:

$$\textstyle\sum_f P(Y \mid X, f)P(f) = \frac{1}{2}(0.6) + \frac{1}{2}(0.2) = 0.40,$$
$$\textstyle\sum_f P(Y \mid \neg X, f)P(f) = \frac{1}{2}(0.7) + \frac{1}{2}(0.3) = 0.50,$$

so the drug lowers recovery by 10 percentage points. (This formula is the back-door adjustment of Chapter 2.)

1.1.4.1 Would the result be different with a different story?

Keep the numbers but change the causal story behind them. Three possible graphs:



(a) F is a common cause (b) F is a mediator (c) F is a collider of latents

- **(a)** $F \to X$, $F \to Y$, $X \to Y$. F (sex) confounds the treatment–outcome relation. Consult the stratified tables; the drug is harmful.
- **(b)** $X \to F \to Y$, $X \to Y$. Suppose the third variable were not sex but something the drug changes (e.g. blood pressure). Stratifying on it would block part of the drug's effect; the aggregate table gives the total effect and the drug is beneficial.
- **(c)** $X \leftarrow U_1 \to F \leftarrow U_2 \to Y$, $X \to Y$ with U_1, U_2 unobserved. F is a collider; conditioning on it *creates* a spurious association between X and Y. The aggregate table is correct.

The data are identical in the three scenarios, yet the right answer differs. The decision therefore depends on knowledge outside the data, the causal structure, which is exactly what causal models represent.

1.1.5 Example 2: a "trivial" data analysis

Motivation. Show that a perfect fit does not reveal the direction or meaning of a dependence.

Firefighters (X)	Size of fire (Y)
10	100
40	400
60	600
200	2000

The points lie exactly on $Y = 10X$, so the regression $Y = \beta X + \beta_0$ fits perfectly with $\beta = 10$, $\beta_0 = 0$. X and Y are perfectly correlated, but does X cause Y? Read causally, $\beta = 10$ would say that each additional firefighter adds ten units of fire, which is absurd: the size of the fire determines how many firefighters are dispatched ($Y \to X$). *After all, what is the meaning of β?* A regression coefficient is a property of the joint distribution; whether it measures the effect of changing X depends on the causal mechanism, which the fit is silent about.

1.1.6 What is going on here?

Both examples ask for something that is not in the joint distribution: the effect of an action. Answering such questions requires (i) a formal language for causal assumptions, (ii) a semantics connecting assumptions with data and queries, and (iii) inference rules deciding when and how a causal query is computable from available data. Chapters 1–3 build exactly this machinery: structural causal models, causal graphs, identification, the back-door and adjustment criteria, do-calculus, and algorithmic identification.

1.2 Causal models and causal graphs

Motivation. The central question is what object can play the role of an "oracle" for causal questions, given that the observational distribution alone cannot determine the right decision, and what that object implies about the data we do observe.

1.2.1 Reviews: notation, probability, algebra

Motivation. Fix notation and the elementary manipulations used throughout the course.

1.2.1.1 Notation

- A random variable is denoted by an uppercase letter, e.g. Z, with state space $\mathfrak{X}_Z$ (rarely written explicitly). A lowercase letter denotes a value, $z \in \mathfrak{X}_Z$.
- Bold letters denote sets of variables or values, e.g. $\mathbf{V}$ and $\mathbf{v}$. Calligraphic letters denote complex objects such as graphs $\mathcal{G}$ and models $\mathcal{M}$.
- $\forall$ (for all), $\exists$ (exists); $\cup, \cap$; $\subseteq, \subset$ (note that $\subset$ is ambiguous between proper and non-proper subset); $\vee$ (or), $\wedge$ (and), $\oplus$ (xor); $\sum$, $\prod$; $\langle \cdots \rangle$ tuples, $\{\cdots\}$ sets, $\mid$ "given".
- "iff" = if and only if. $p \Rightarrow q$: if p then q; p is *sufficient* for q. Equivalently $\neg q \Rightarrow \neg p$: q is *necessary* for p.

1.2.1.2 Probability

Consider $P(A, B, C, D, E, F)$; write $P(A = a, B = b)$ as $P(a, b)$. Variables are assumed discrete without loss of generality.

Marginalization.

$$\sum_{c,e} P(a, b, c, d, e, f) = \sum_{(c,e) \in \mathfrak{X}_C \times \mathfrak{X}_E} P(a, b, c, d, e, f) = P(a, b, d, f).$$

Conditioning.

$$P(a, b, c \mid d, e) = \frac{P(a, b, c, d, e)}{P(d, e)} = \frac{\sum_f P(a, b, c, d, e, f)}{\sum_{a,b,c} \sum_f P(a, b, c, d, e, f)}.$$

Chain rule. Applying the definition of conditional probability recursively, in *any* order:

$$P(a, b, d, e) = P(a \mid b, d, e) P(b \mid d, e) P(d \mid e) P(e) = P(b \mid a, d, e) P(d \mid a, e) P(e \mid a) P(a).$$

Axioms. $0 \leq P(\cdot) \leq 1$ and $\sum_{a,b,c} P(a, b, c) = 1$. Often strictly positive distributions are assumed so that every conditional is defined.

1.2.1.3 Sums of products

Factors that do not depend on a summation index can be moved outside that sum:

$$\sum_{a,c} f(a, b) g(b, c) h(c, d) = \sum_c h(c, d) g(b, c) \Big(\sum_a f(a, b) \Big) = \sum_a f(a, b) \Big(\sum_c g(b, c) h(c, d) \Big).$$

If $f(a, b) = P(a \mid b)$, then $\sum_a P(a \mid b) = 1$ and

$$\sum_a P(a \mid b) \underbrace{\sum_c g(b, c) h(c, d)}_{\text{irrelevant to } a} = \underbrace{\Big(\sum_a P(a \mid b) \Big)}_{=1} \sum_c g(b, c) h(c, d) = \sum_c g(b, c) h(c, d).$$

This "sum out a conditional of a leaf" step is used constantly in identification proofs.

1.2.2 Why data alone is not enough: a prima facie example

Motivation. Construct a situation in which every learning algorithm, trained on observational data, gives the wrong policy recommendation.

1.2.2.1 The story

A city suffers from a contagious disease, and a drug is *believed* to help. Unknown to physicians:

- People with good living conditions (rich) always survive.
- Some people carry a gene that naturally fights the disease; they need no treatment. If treated, however, they develop an allergic reaction that is fatal under poor living conditions.

Being rich and having the gene are independent, $P(\text{rich}) = P(\text{poor}) = \frac{1}{2}$, and half the poor have the gene. The population Π splits into three groups:

$$\Pi = \text{rich} \ \cup \ \text{poor}_1 \ \cup \ \text{poor}_2,$$

$$\begin{cases} \text{rich}: & \text{alive anyway}, \\ \text{poor}_1 \ (\text{no gene}): & \text{die anyway}, \\ \text{poor}_2 \ (\text{gene}): & \text{die iff they take the drug}, \end{cases}$$

with $P(\text{poor}_1) = P(\text{poor}_2) = \frac{1}{4}$. (*Prima facie* = at first sight.)

1.2.2.2 Current protocol

Because the drug is expensive, physicians prescribe it to every rich person and only to them. Observed outcomes: all treated (rich) survive; among the untreated (poor) poor_2 survive and poor_1 die. Hence

$$P(\text{alive} \mid \text{drug}) = 1, \qquad P(\text{alive} \mid \neg\text{drug}) = 0.5.$$

Should the city give the drug to everyone for free?

1.2.2.3 Hypothetical protocols

Group (mass)	Current (drug iff rich)	Drug for all	Drug for none
rich (1/2)	alive	alive	alive
poor_1 (1/4)	dead	dead	dead
poor_2 (1/4)	alive	dead	alive
$P(\text{alive})$	3/4	1/2	3/4

Giving the drug to everyone *reduces* survival from 3/4 to 1/2; the drug never helps anyone. The apparent benefit is produced entirely by the fact that only the rich, who survive anyway, are treated.

1.2.2.4 What can a learning system see?

A data set has rows (patient, R, D, A), e.g. (1, yes, yes, yes), (2, no, no, yes), (3, no, no, no), (4, yes, yes, yes),... The best any algorithm (decision tree, SVM, DNN) can do is to learn the joint distribution

R	D	A	$P(r,d,a)$
0	0	0	1/4
0	0	1	1/4
1	1	1	1/2
others			0

What remains unseen are the exogenous factors: the probability of having the allergic gene, P(gene); the other factors that affect wealth; and how the genetic factor interacts with wealth, the drug and the reaction.

What is missing? Consider a second world in which the drug is simply inert, $A \leftarrow R \vee U_G$. Since $D = R$ in the data, this world produces exactly the same table, but there "drug for all" gives survival 3/4. Two realities with different answers generate identical data, so no algorithm using only $P(r,d,a)$ can be correct in both. The data must be supplemented with a model of how the data are generated.

1.2.2.5 From statistical to causal analysis

- **Traditional statistics/ML:** data $\rightarrow$ joint distribution $P \rightarrow$ query $Q(P)$, an aspect of P, e.g. will customers who bought A also buy B, $Q = P(B \mid A)$.
- **Causal analysis:** data $\rightarrow P \xrightarrow{\text{change}} P' \rightarrow Q(P')$, e.g. P'(sales) if we double the price, P'(cancer) if smoking is banned. How does the factual P change into the hypothetical P'? P is tied to data; P' is never observed. A new formalism must represent both.
- **SCM paradigm:** data $\leftarrow P \leftarrow \mathcal{M}$ (data-generating model) $\rightarrow Q(\mathcal{M})$. $\mathcal{M}$ is an invariant strategy (mechanism, recipe, law, protocol) by which Nature assigns values to the variables. *P is a model of data; $\mathcal{M}$ is a model of reality.*

In the big picture of Section 1.1, knowledge of the real world enters the causal model, and causal inference draws new knowledge about not-yet-realized alternative realities. (This picture should be taken with a grain of salt.)

1.2.2.6 Modeling reality in the example

Observed (endogenous) variables $\mathbf{V}$: R (1 = rich), D (1 = took the drug), A (1 = alive). Unobserved (exogenous) variables $\mathbf{U}$: U_G (1 = has the genetic factor), U_R (other factors affecting wealth). Mechanisms:

$$R \leftarrow U_R, \qquad D \leftarrow R, \qquad A \leftarrow R \vee (U_G \wedge \neg D),$$

with $P(U_G = 1) = P(U_R = 1) = \frac{1}{2}$ and $P(\mathbf{U}) = P(U_G)P(U_R)$. This is a fully specified model of reality: it implies both P and every P'. It is our new oracle, the *structural causal model.*

1.2.3 Structural causal models

Motivation. Define the object that represents reality and derive its three basic properties.

Definition (Structural causal model; Pearl 2000)

An SCM $\mathcal{M}$ is a 4-tuple $\langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$ where

- $\mathbf{V} = \{V_1, \ldots, V_n\}$ are endogenous variables;
- $\mathbf{U}$ is a set of exogenous variables;
- $\mathcal{F} = \{f_1, \ldots, f_n\}$ are functions determining $\mathbf{V}$, $v_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$, with $Pa_i \subseteq \mathbf{V} \setminus \{V_i\}$

and $\mathbf{U}_i \subseteq \mathbf{U}$;

- $P(\mathbf{U})$ is a distribution over $\mathbf{U}$.

The arrow "$\leftarrow$" stresses that f_i is an assignment (a mechanism), not a symmetric equation. Axiomatic characterizations of the resulting counterfactual logic are given by Galles and Pearl (1998) and Halpern (1998). Abstractly: the exogenous $\mathbf{U}$ carry all randomness; $\mathcal{F}$ deterministically transforms a unit $\mathbf{u}$ into its observable profile.

1.2.3.1 Property 1: an SCM induces the observational distribution

$\mathcal{F}$ is a map $\mathbf{u} \mapsto (v_1, \ldots, v_n)$. When $\mathbf{U}$ is random, so is $\mathbf{V}$. The induced $P(\mathbf{v})$ is Layer 1 of the PCH, the observational (passive) distribution. Each event, person or observation corresponds to an instantiation $\mathbf{U} = \mathbf{u}$.

Definition (Induced distribution)

$\mathcal{M}$ defines $P^{\mathcal{M}}(\mathbf{V})$ by, for each $\mathbf{Y} \subseteq \mathbf{V}$,

$$P^{\mathcal{M}}(\mathbf{y}) = \sum_{\{\mathbf{u}\,:\,\mathbf{Y}(\mathbf{u})=\mathbf{y}\}} P(\mathbf{u}),$$

where $\mathbf{Y}(\mathbf{u})$, the *potential response*, is the value of $\mathbf{Y}$ obtained by solving $\mathcal{F}$ at $\mathbf{u}$.

Example 1.1

In the drug example each citizen falls into one of four groups:

$P(\mathbf{u})$	(U_R, U_G)	$\xrightarrow{\mathcal{F}} (R, D, A)$	$P(\mathbf{v})$
1/4	(1,1)	(1,1,1)	1/2
1/4	(1,0)	(1,1,1)	
1/4	(0,1)	(0,0,1)	1/4
1/4	(0,0)	(0,0,0)	1/4

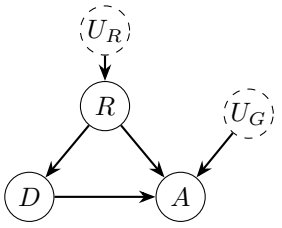
1.2.3.2 Property 2: an SCM induces a causal diagram

Definition (Causal diagram; Bareinboim et al. 2020, Def. 13)

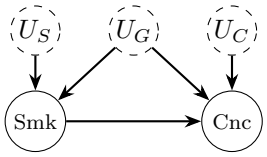
Given $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$, the causal diagram $\mathcal{G}$ is constructed by

1. adding a vertex for every $V_i \in \mathbf{V}$;
2. adding a directed edge $V_j \to V_i$ whenever V_j is an argument of f_i;
3. adding a bidirected edge $V_j \leftrightarrow V_i$ whenever $\mathbf{U}_i$ and $\mathbf{U}_j$ are correlated or f_i, f_j share some $U \in \mathbf{U}$ as an argument.

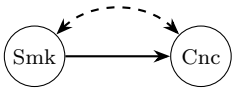
More precisely, for every set $\mathbf{W}$ of vertices with no bidirected edges among them, $\{\mathbf{U}_W\}_{W \in \mathbf{W}}$ are jointly independent. If the directed part is acyclic (assumed throughout) the diagram is a DAG with possible bidirected edges (an ADMG). For instance, $V_i \leftarrow f_i(A, B, U)$ and $V_j \leftarrow f_j(C, U)$ give $A \to V_i \leftarrow B$, $C \to V_j$, and $V_i \leftrightarrow V_j$ because of the shared U.



drug example



explicit **U**



U implicit

Example 1.2 (Smoking and cancer)

$\mathbf{V} = \{\text{Smoking}, \text{Cancer}\}$, $\mathbf{U} = \{U_S, U_C, U_G\}$, Smoking $\leftarrow f_S(U_S, U_G)$, Cancer $\leftarrow f_C(\text{Smoking}, U_C, U_G)$. The shared genetic factor U_G yields Smoking $\leftrightarrow$ Cancer.

Remark 1. The mapping goes one way only: a graph is compatible with infinitely many SCMs sharing the same function signatures and exogenous structure.

Remark 2. This is central: in practice researchers often know the *scope* of the mechanisms (which variables enter which function) but not their form. Causal inference asks what can be concluded from the graph plus data alone.

Convention. Unobserved variables are left implicit; a shared or correlated U is drawn as a dashed bidirected edge.

1.2.3.3 Graph terminology

A graph $\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$ consists of vertices (nodes) and edges. For $V \in \mathbf{V}$:

- parents $Pa(V) = \{W : W \to V \in \mathcal{G}\}$; children $Ch(V) = \{W : V \to W \in \mathcal{G}\}$;
- ancestors $An(V) = \{W : W \to \cdots \to V \in \mathcal{G}\}$; descendants $De(V) = \{W : V \to \cdots \to W \in \mathcal{G}\}$; these include V itself (zero-length paths);
- siblings (spouses through confounding) $\{W : W \leftrightarrow V \in \mathcal{G}\}$;
- a *path* is a sequence of edges joining a sequence of distinct vertices; a *directed path* from W to V has all arrowheads pointing toward V.

1.2.3.4 Property 3: an SCM induces Pearl's Causal Hierarchy

An SCM $\mathcal{M}$ answers queries of all three layers: association $P(y \mid x)$, intervention $P(y \mid \text{do}(x), c)$ and counterfactual $P(y_x \mid x', y')$. This lecture concentrates on Layer 1; Layers 2 and 3 are formalized in Chapter 2.

1.2.4 The first layer: what the diagram encodes about $P(\mathbf{v})$

Motivation. Derive the factorization of the observational distribution implied by an SCM and its diagram.

$\mathcal{M}$ induces

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{V} = \mathbf{v}, \mathbf{U} = \mathbf{u}) = \sum_{\mathbf{u}} P\big(\forall_{V_i \in \mathbf{V}}\, f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i,\ \mathbf{U} = \mathbf{u}\big),$$

where $\mathbf{pa}_i, \mathbf{u}_i$ are the observed and unobserved arguments of f_i, i.e. the parents of V_i in $\mathcal{G}$. In the examples,

$$P(r, d, a) = \sum_{u_R, u_G} P\big(\underbrace{r, d, a}_{\text{endogenous}}, \underbrace{u_R, u_G}_{\text{exogenous}}\big),$$

$$P(s,c) = \sum_{u_S,u_G,u_C} P(s,c,u_S,u_G,u_C).$$

This is the observational (passive, non-experimental) distribution.

Topological order. Because $\mathcal{G}$ is acyclic, there is an order $\prec$ on $\mathbf{V}$ in which every variable comes after its parents, $Pa_i \prec V_i$; e.g. $R \prec D \prec A$ and $S \prec C$.

Factorization. Apply the chain rule along a topological order:

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}).$$

Each V_i is fully determined by its observed and unobserved parents, and $Pa_i \cup \mathbf{U}_i \subseteq \{V_1, \ldots, V_{i-1}\} \cup \mathbf{U}$, so

$$P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P\big(f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}\big) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

Therefore

$$\boxed{P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).}$$

Examples:

$$P(r,d,a) = \sum_{u_R,u_G} P(u_R,u_G)P(r \mid u_R)P(d \mid r)P(a \mid r,d,u_G),$$
$$P(s,c) = \sum_{u_S,u_G,u_C} P(u_S,u_G,u_C)P(s \mid u_G,u_S)P(c \mid s,u_G,u_C).$$

Here $P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) \in \{0,1\}$ because the mechanisms are deterministic.

1.2.5 Conditional independence and the Markov condition

Motivation. Translate the structure of $\mathcal{G}$ into testable constraints on $P(\mathbf{v})$.

1.2.5.1 Conditional independence

X and Y are independent, $X \perp\!\!\!\perp Y$, if learning $X = x$ does not change the belief in $Y = y$:

$$X \perp\!\!\!\perp Y \iff P(y \mid x) = P(y) \;\; \forall x, y \iff P(x,y) = P(x)P(y).$$

X and Y are conditionally independent given Z if, once $Z = z$ is known, X is uninformative about Y:

$$(X \perp\!\!\!\perp Y \mid Z) \iff P(x \mid z, y) = P(x \mid z) \quad \forall x, y, z \text{ with } P(y,z) > 0.$$

These are constraints on $P(\mathbf{V})$. The guiding intuition: *lack of functional dependence leads to probabilistic independence.*

1.2.5.2 Local Markov condition

In any topological order, $P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$. What is the largest set that can precede V_i in some topological order? Any W that is not reachable by a directed path $V_i \to \cdots \to W$, i.e. all *non-descendants* Nd_i of V_i. Therefore

$$P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = P(v_i \mid \mathbf{nd}_i, \mathbf{u}),$$

which is the statement

$$V_i \perp\!\!\!\perp (Nd_i \setminus Pa_i), \; (\mathbf{U} \setminus \mathbf{U}_i) \;\mid\; Pa_i, \mathbf{U}_i.$$

1.2.5.3 Markovian models and the Bayesian factorization

If $U_1, \ldots, U_n$ are jointly independent (no bidirected edges), the model is *Markovian*. Then

$$\begin{aligned} P(\mathbf{v}) &= \sum_{\mathbf{u}} P(\mathbf{u}) \prod_i P(v_i \mid \mathbf{pa}_i, u_i) \\ &= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i) && (U_i \text{ independent}) \\ &= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i \mid \mathbf{pa}_i) && (U_i \perp\!\!\!\perp Pa_i) \\ &= \sum_{\mathbf{u}} \prod_i P(v_i, u_i \mid \mathbf{pa}_i) \\ &= \prod_i \sum_{u_i} P(v_i, u_i \mid \mathbf{pa}_i) = \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i). \end{aligned}$$

The penultimate step uses that the i-th factor depends only on u_i, so the sum over $\mathbf{u}$ distributes over the product. $U_i \perp\!\!\!\perp Pa_i$ holds because Pa_i are functions of exogenous variables of non-descendants, which are independent of U_i in a Markovian model.

Summary. Markovian SCM $\Rightarrow$ $P(\mathbf{v}) = \prod_i P(v_i \mid \mathbf{pa}_i)$ (Bayesian factorization) $\Leftrightarrow$ local Markov condition $V_i \perp\!\!\!\perp Nd_i \setminus Pa_i \mid Pa_i$.

1.2.6 Reading independences from the graph: d-separation

Motivation. Connect the unobserved mechanisms with observable independences through three elementary structures.

Each example below uses binary variables and explicit mechanisms.

1.2.6.1 Chains

$X \to Z \to Y$: good study habits $X \leftarrow U_X$, a high Suneung score $Z \leftarrow X \vee \neg U_Z$ (U_Z: difficulty), college admission $Y \leftarrow Z \wedge U_Y$ (U_Y: favorable factors). Are X and Y independent given Z? **Yes.** Knowing $Z = 1$, learning about study habits does not change the admission probability:

$$P(x, y \mid z) = \frac{P(x, y, z)}{P(z)} = \frac{P(y \mid z)P(z \mid x)P(x)}{P(z)} = \frac{P(x, z)}{P(z)} P(y \mid z) = P(x \mid z)P(y \mid z).$$

Observing Z blocks the influence from X to Y. (Marginally, X and Y are dependent.)

1.2.6.2 Common causes (forks)

$X \leftarrow Z \to Y$: rain $Z \leftarrow U_Z$ (meteorological conditions), traffic $X \leftarrow Z \vee \neg U_X$ (road design), umbrella $Y \leftarrow Z \wedge U_Y$. Are X, Y independent? **No**: seeing an umbrella raises the probability of rain, which raises the probability of bad traffic; $\exists x, y$ with $P(x, y) \neq P(x)P(y)$. Information flows from Y up to Z and down to X. Given Z? **Yes**: once we know it is raining, umbrellas tell us nothing about traffic,

$$P(x, y \mid z) = \frac{P(x \mid z)P(y \mid z)P(z)}{P(z)} = P(x \mid z)P(y \mid z).$$

1.2.6.3 Common effects (colliders)

$X \to Z \leftarrow Y$: having a cold $X \leftarrow U_X$, holiday $Y \leftarrow U_Y$, going to class $Z \leftarrow \neg Y \wedge (\neg X \wedge U_Z)$ (U_Z: motivation). Are X, Y independent? **Yes**:

$$P(x,y) = \sum_z P(x)P(y)P(z \mid x,y) = P(x)P(y)\sum_z P(z \mid x,y) = P(x)P(y).$$

Influence from X reaches Z but does not "go up" to Y. Given Z? **No!** If a student did not go to class ($Z = 0$) and it is not a holiday ($Y = 0$), a cold becomes more likely; $\exists x, y, z$ with $P(x, y \mid z) \neq P(x \mid z)P(y \mid z)$ ("explaining away"). This is the opposite of the previous two cases.

Given a descendant W of the collider (signing the attendance sheet, $W \leftarrow Z \wedge U_W$)? **No, again.** Observing $W = 0$ makes absence ($Z = 0$) more likely, which again couples X and Y. *Watch out for descendants of colliders.*

1.2.6.4 Triplet summary

Triplet	Active (dependence flows) when	Inactive when
chain $X \to Z \to Y$	Z not observed	Z observed
fork $X \leftarrow Z \to Y$	Z not observed	Z observed
collider $X \to Z \leftarrow Y$	Z or a descendant of Z observed	neither Z nor its descendants observed

1.2.6.5 d-separation

Definition (d-separation)

To decide whether $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ (with $X, Y \notin \mathbf{Z}$):

1. look at every path between X and Y;
2. a path is *active* given $\mathbf{Z}$ if every consecutive triplet on it is active;
3. if any path is active, X and Y are d-connected (not guaranteed independent); otherwise they are d-separated, written $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}}$.

Trivially, if $X = Y$ or X and Y are adjacent, they are dependent.

Soundness: if $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}}$ then $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in every distribution induced by an SCM compatible with $\mathcal{G}$ (the global Markov property, equivalent to the local one for DAGs).

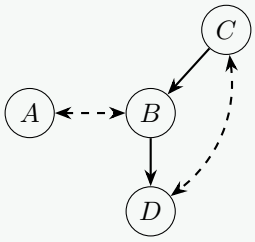
Example 1.3 (Sprinkler network)

Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet, Rain $\to$ Wet, Wet $\to$ Slippery.

1. Wet $\perp\!\!\!\perp$ Sprinkler? **No**, they are adjacent.
2. Wet $\perp\!\!\!\perp$ Season $\mid$ Sprinkler? **No**: Season $\to$ Rain $\to$ Wet is open.
3. Rain $\perp\!\!\!\perp$ Slippery $\mid$ Wet? **Yes**: the only path is the chain through Wet.
4. Season $\perp\!\!\!\perp$ Wet $\mid$ Sprinkler, Rain? **Yes**: both directed paths are blocked at observed non-colliders.
5. Sprinkler $\perp\!\!\!\perp$ Rain $\mid$ Season, Wet? **No**: Wet is an observed collider on Sprinkler $\to$ Wet $\leftarrow$ Rain.

Example 1.4 (Food for thought [Supplement])

Graph: $A \leftrightarrow B$, $C \to B$, $B \to D$, $C \leftrightarrow D$.



- $A \perp\!\!\!\perp D$? **No**: on $A \leftrightarrow B \to D$, B is an unobserved non-collider.
- $A \perp\!\!\!\perp C$? **Yes**: $A \leftrightarrow B \leftarrow C$ has a collider at B; $A \leftrightarrow B \to D \leftrightarrow C$ has a collider at D.
- $A \perp\!\!\!\perp C \mid D$? **No**: B is an ancestor of the observed D, so the collider at B is opened.
- $D \perp\!\!\!\perp C \mid B$? **No**: $D \leftrightarrow C$ are adjacent (confounded).

1.2.7 Implementation: graph algorithms in Python

Motivation. Turn the definitions into code: topological ordering and d-separation.

1.2.7.1 Topological order

Idea: a DAG has a source node (no parents); removing a node and its edges leaves a DAG. Find a source, then recursively order the rest.

```
def topological_order(G):
    for V in G.Vs:                  # for every node
        if not G.Pa(V):             # if there is no parent for V
            return [V] + topological_order(G - V)
    return []
```

This naive version is quadratic and silently returns a partial order on cyclic graphs. Tracking in-degrees (Kahn's algorithm) is efficient and detects cycles:

```
def topological_order(G):
    order = deque()
    in_degrees = {V: len(G.Pa(V)) for V in G.Vs}
    while in_degrees:
        for v, v_deg in in_degrees.items():
            if v_deg == 0:
                break
        else:
            raise ValueError('cyclic')
        order.append(v)
        del in_degrees[v]
        for c in G.Ch(v):
            in_degrees[c] -= 1
    return order
```

The `for`–`else` clause runs only if no zero-in-degree node was found, which happens iff the remaining graph contains a cycle.

1.2.7.2 d-separation via reachability (Markovian graphs)

X and Y are d-connected given $\mathbf{Z}$ iff there is a path along which every triplet $A - B - C$ satisfies:

$$\begin{aligned} A \to B \to C \quad & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \leftarrow C \quad & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \to C \quad & \text{only if } B \notin \mathbf{Z}, \\ A \to B \leftarrow C \quad & \text{only if } B \in An(\mathbf{Z}). \end{aligned}$$

So we walk from B to C only if the condition holds, using BFS/DFS over states (direction of arrival, B): arrived along an edge pointing *into* B ($\to B$) or *out of* B ($\leftarrow B$).

```
def d_separation(G, X, Y, Zs):
    q = visited_deque({('->', X), ('<-', X)})
    while q:
        direction, B = q.pop()
        if B == Y:
            return False
        if direction == '->':          # arrived with arrowhead into B
            if B in G.An(Zs):          # collider: open iff B in An(Z)
                q.extend({('<-', C) for C in G.Pa(B)})
            if B not in Zs:            # chain continues
                q.extend({('->', C) for C in G.Ch(B)})
        else:                          # arrived via tail of B (B -> previous)
            if B not in Zs:
                q.extend({('->', C) for C in G.Ch(B)})
                q.extend({('<-', C) for C in G.Pa(B)})
    return True
```

`visited_deque` is a deque that ignores states already visited, so each (direction, node) pair is processed once and the algorithm is linear in the size of the graph. The starting states allow leaving X in either direction.

Chapter 2. Identification and Adjustment

2.1 Identification of causal effects

Motivation. The central question is how an SCM defines the distribution of a population under an intervention, and when that interventional distribution can be computed from the causal diagram and the observational distribution alone.

2.1.1 Structural causal models, revisited

Motivation. Recall the framework and introduce the do(·) operator.

2.1.1.1 Observation versus intervention

Reality is an unknown black-box system. We learn about it in two ways: *observation* (observational studies, passive records) and *intervention* (clinical trials, experiments). Questions such as drug efficacy or the effect of a hypothetical public policy concern the system *under intervention*, even when only observational data are available.

An SCM $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ (unobserved variables, observed variables, mechanisms for $\mathbf{V}$, distribution of $\mathbf{U}$) is a mathematical understanding of the unknown reality. $\mathbf{V}$ are measured, $\mathbf{U}$ are unknown, and each measured variable is determined by a function of some measured and some unmeasured variables, $v \leftarrow f(\cdot)$. $\mathbf{U}$ can be viewed as what each unit actually is; $\mathbf{U}$ determines $\mathbf{V}$.

The SCM induces a causal diagram $\mathcal{G}$ and a whole family of distributions:

$$\mathcal{M} \longmapsto \mathcal{G},$$
$$\mathcal{M} \longmapsto \{P,\ P_{x_1},\ P_{x_2},\ P_{x_2,x_4}, \dots\}$$

corresponding to $\text{do}(\emptyset), \text{do}(x_1), \text{do}(x_2), \text{do}(x_2, x_4), \dots$

Layer 1 (observational). As derived in Chapter 1,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{v}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \dots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

The model also induces Layer 2 (interventional) and Layer 3 (counterfactual) distributions.

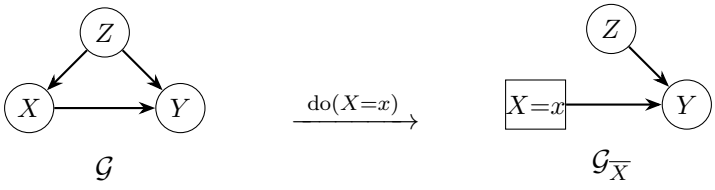
Causal diagram. Nodes for $\mathbf{V}$; a directed edge $B \to V_i$ when B is an argument of f_i (causal relationship); a bidirected edge $V_i \leftrightarrow V_j$ when $\mathbf{U}_i$ and $\mathbf{U}_j$ correlate (unobserved confounding).

2.1.1.2 The do-operator

Definition (Intervention)

Given $\mathcal{M}$, fixing $X \in \mathbf{V}$ to a constant x is written $\text{do}(X = x)$. It transforms $\mathcal{M}$ into the *submodel* $\mathcal{M}_x$, i.e. $\mathcal{M}$ with $f_X \in \mathcal{F}$ replaced by the constant x.

Graphically, the diagram of $\mathcal{M}_x$ is $\mathcal{G}_{\overline{X}}$: $\mathcal{G}$ with all edges *into* X removed.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\overline{X}}$

2.1.2 The second layer: causal effects

Motivation. Define causal effects formally and compute them in Markovian models.

2.1.2.1 Connecting different worlds

Data come from P; the query concerns P', the distribution after a change. For instance, would fewer people get cancer if smoking were banned? The query $Q = P(\text{Cancer} = \text{true} \mid \text{do}(\text{Smoking} = \text{no}))$ is *not* an aspect of P.

Observation 1. The distribution alone tells us nothing about change; it describes static conditions of a population under a specific regime.
Observation 2. We must represent change: how the population reacts when it undergoes a change in regime.

The challenge. Let Z be age/sex, X an action, W a mediator, Y an outcome, with $Z \to X \to W \to Y$ and $Z \to Y$. In the hypothetical world X is set to x, cutting $Z \to X$. The real world generates $P(z, x, w, y)$; the hypothetical world generates $P(z, w, y \mid \text{do}(x))$. The goal is to know how much Y changes with X if we vary X between two different constants, free of the influence of Z; such variations are called *causal effects*. What is the relationship between $P(z, x, w, y)$ and $P(y \mid \text{do}(x))$? Is $P(y \mid \text{do}(x)) = P(y \mid x)$?

Definition (Causal effect; Pearl 2000, Def. 3.2.1)

Given disjoint sets $\mathbf{X}, \mathbf{Y}$, the causal effect of $\mathbf{X}$ on $\mathbf{Y}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$, is a function from $\mathfrak{X}_{\mathbf{X}}$ to the space of distributions over $\mathbf{Y}$. For each $\mathbf{x}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is the probability of $\mathbf{Y} = \mathbf{y}$ induced by deleting from the model all equations of the variables in $\mathbf{X}$ and substituting $\mathbf{X} = \mathbf{x}$ in the remaining equations:

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = P_{\mathbf{x}}(\mathbf{y}), \quad \text{the probability of } \mathbf{Y} = \mathbf{y} \text{ in } \mathcal{M}_{\mathbf{x}}.$$

2.1.2.2 Computing causal effects from observational data (Markovian case)

In the example, the observational and post-interventional distributions factorize as

$$P(\mathbf{v}) = P(z)\,P(x \mid z)\,P(w \mid x)\,P(y \mid w, z),$$
$$P_x(\mathbf{v} \setminus \{X\}) = P(z) \cdot \underbrace{1}_{X=x \text{ in } \mathcal{M}_x} \cdot P(w \mid x)\,P(y \mid w, z).$$

All other mechanisms are untouched by the intervention (*modularity*), so their conditional distributions are the same in both worlds. Hence

$$P(y \mid \text{do}(x)) = \sum_{z,w} P(z)P(w \mid x)P(y \mid w, z) \;\neq\; \sum_{z,w} P(z \mid x)P(w \mid x)P(y \mid w, z) = P(y \mid x)$$

in general: conditioning weights z by $P(z \mid x)$, intervening by $P(z)$.

Example 2.1 (Sprinkler: seeing versus doing)

Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet $\leftarrow$ Rain, Wet $\to$ Slippery.

$$Q_1 = P(\text{wt} \mid \text{Sp} = \text{on}) = \frac{\sum_{\text{sn,rn}} P(\text{sn})P(\text{Sp} = \text{on} \mid \text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P(\text{Sp} = \text{on} \mid \text{sn})P(\text{sn})},$$

$$Q_2 = P(\text{wt} \mid \text{do}(\text{Sp} = \text{on})) = \frac{\sum_{\text{sn,rn}} P(\text{sn})\, P'(\text{Sp} = \text{on})\, P(\text{rn} \mid \text{sn}) P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P'(\text{Sp} = \text{on}) P(\text{sn})}$$
$$= \sum_{\text{sn,rn}} P(\text{sn}) P(\text{rn} \mid \text{sn}) P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}),$$

where $P'(\text{Sp} = \text{on}) = 1$ is the interventional probability. Seeing the sprinkler on is evidence about the season and therefore about rain; turning it on is not.

2.1.3 Truncated factorization

Motivation. State the general formula for interventional distributions in Markovian models and interpret it as reweighting.

Theorem 2.1 (Truncated factorization; manipulation theorem, g-computation)

In a Markovian model, the distribution generated by $\text{do}(\mathbf{X} = \mathbf{x})$ is

$$P(\mathbf{v} \mid \text{do}(\mathbf{x})) = \begin{cases} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) & \text{if } \mathbf{v} \text{ is consistent with } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases}$$

Derivation [Supplement]. $\mathcal{M}_{\mathbf{x}}$ is again Markovian, with the same mechanisms f_i and $P(u_i)$ for $V_i \notin \mathbf{X}$ and constant mechanisms for $\mathbf{X}$. Its Bayesian factorization (Chapter 1) is

$$\prod_{V_i \notin \mathbf{X}} P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) \prod_{X \in \mathbf{X}} \mathbb{1}[x \text{ agrees with } \mathbf{x}].$$

Since f_i and U_i are unchanged and $U_i \perp\!\!\!\perp Pa_i$ in both models,

$$P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) = \sum_{u_i} \mathbb{1}[f_i(\mathbf{pa}_i, u_i) = v_i] P(u_i) = P(v_i \mid \mathbf{pa}_i)$$

whenever the latter is defined (positivity). ■

Reweighting form. Since $P(\mathbf{v}) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) \cdot \prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)$,

$$P(\mathbf{v}' \mid \text{do}(\mathbf{x})) = \frac{P(\mathbf{v})}{\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)}, \qquad \mathbf{v}' = \mathbf{v} \setminus \mathbf{x}.$$

For a singleton X, with $\mathbf{v}'' = \mathbf{v} \setminus (\mathbf{pa}_X \cup \{x\})$,

$$P(\mathbf{v}' \mid \text{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid \mathbf{pa}_X)} = \frac{P(\mathbf{v}) P(\mathbf{pa}_X)}{P(x, \mathbf{pa}_X)} = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)\, P(\mathbf{pa}_X).$$

The passage from observational to interventional distribution is a *reweighting*: units with $X = x$ are up-weighted by $1/P(x \mid \mathbf{pa}_X)$ (the origin of inverse probability weighting, Chapter 6).

2.1.4 The identification problem

Motivation. Make precise when a causal query is determined by what we can observe and assume.

Definition (Identifiability; Pearl 2000, Def. 3.2.2)

The causal effect of $\mathbf{X}$ on $\mathbf{Y}$ is *identifiable* from $\mathcal{G}$ if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be computed uniquely from any positive distribution over the observed variables; that is, for every pair of models $\mathcal{M}_1, \mathcal{M}_2$ inducing $\mathcal{G}$,

$$P^{\mathcal{M}_1}(\mathbf{v}) = P^{\mathcal{M}_2}(\mathbf{v}) > 0 \implies P^{\mathcal{M}_1}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = P^{\mathcal{M}_2}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})).$$

Picture. The true SCM $\mathcal{M}$ is unobserved; it induces the causal graph $\mathcal{G}$ (assumed known), the observational distribution $P(\mathbf{v})$ (observed) and the experimental distribution $P(\mathbf{v} \mid \mathrm{do}(\mathbf{x}))$ (unobserved). Among all models, consider those inducing $P(\mathbf{V})$ and those inducing $\mathcal{G}$. The query is identifiable iff it is constant over the intersection of these two sets: no matter the form of $\mathcal{M}_1, \mathcal{M}_2$, all models agreeing on $\mathcal{G}$ and $P(\mathbf{v})$ agree on $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$. If two models in the intersection disagree, the query is not identifiable.

Example 2.2 (A non-identifiable effect [Supplement])

Let $\mathcal{G}$ be $X \to Y$ with $X \leftrightarrow Y$, binary variables, $U \sim \mathrm{Bern}(1/2)$ shared. $\mathcal{M}_1$: $X \leftarrow U$, $Y \leftarrow U$ (no effect). $\mathcal{M}_2$: $X \leftarrow U$, $Y \leftarrow X$. Both give $P(X = Y = 0) = P(X = Y = 1) = 1/2$, yet $P^{\mathcal{M}_1}(Y = 1 \mid \mathrm{do}(X = 1)) = 1/2$ and $P^{\mathcal{M}_2}(Y = 1 \mid \mathrm{do}(X = 1)) = 1$. (Both are compatible with $\mathcal{G}$, since a graph only permits, never requires, dependence.)

2.1.4.1 Identification in Markovian models

Theorem 2.2

Given the causal diagram $\mathcal{G}$ of any Markovian model (all relevant variables measured), $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable for all disjoint $\mathbf{X}, \mathbf{Y}$, and

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i),$$
$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v}' \setminus \mathbf{y}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i).$$

Proof [Supplement]. By the truncated factorization theorem above, $P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i)$ evaluated at $\mathbf{X} = \mathbf{x}$. Each factor is a conditional distribution of the observational $P(\mathbf{v})$; under positivity ($P(\mathbf{pa}_i) > 0$ wherever needed) it is uniquely determined by $P(\mathbf{v})$. Marginalizing over $\mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ as a function of $P(\mathbf{v})$ alone. Hence any two Markovian models with diagram $\mathcal{G}$ that agree on $P(\mathbf{v})$ agree on $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, which is the definition of identifiability. ■

The right-hand side involves only conditionals of $P(\mathbf{v})$, which are fixed by the observational distribution under positivity.

2.1.4.2 Adjustment by direct parents (singleton)

Theorem 2.3

In a Markovian model where only a subset of variables is measured, $Q = P(\mathbf{y} \mid \mathrm{do}(x))$ is identifiable whenever X, $\mathbf{Y}$ and Pa_X are measured, and

$$P(\mathbf{y} \mid \mathrm{do}(x)) = \sum_{\mathbf{pa}_X} P(\mathbf{y} \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X).$$

Proof [Supplement]. From the singleton reweighting identity, $P(\mathbf{v}' \mid \mathrm{do}(x)) = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)P(\mathbf{pa}_X)$; summing over $\mathbf{v}''\setminus\mathbf{y}$ gives the claim. Every unmeasured variable is summed out inside $P(\cdot \mid x, \mathbf{pa}_X)$, so only measured variables remain. ■

2.1.4.3 Adjustment by direct parents for a set of treatments (optional)

For $\mathbf{X} = \{X_1, \ldots, X_k\}$ adjustment by parents is not automatic: a parent of X_j may be a descendant of an earlier X_i. Notation:

$Pa_{\mathbf{X}} = \bigcup_{X\in\mathbf{X}} Pa_X$	parents of $\mathbf{X}$ (may intersect $\mathbf{X}$),
$Pa^{-}_{\mathbf{W}} = Pa_{\mathbf{W}} \setminus \mathbf{W}$	parents of $\mathbf{W}$ excluding $\mathbf{W}$,
$\mathbf{X}_{<i} = \{X_1, \ldots, X_{i-1}\}$, $\mathbf{X}_{\le i}$	predecessors in a topological order,
$Pa'_{X_i} = Pa_{X_i} \setminus (\mathbf{X}_{<i} \cup Pa^{-}_{\mathbf{X}_{<i}})$	"new" parents of X_i.

Theorem 2.4

Let $\mathbf{X}$ be topologically ordered. If no element of Pa'_{X_j} is a descendant of X_i for every $i < j$, then

$$\prod_{X\in\mathbf{X}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),$$

hence

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{pa}^{-}_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^{-}_{\mathbf{X}})P(\mathbf{pa}^{-}_{\mathbf{X}}).$$

Proof [Supplement]. By induction we show $\prod_{X\in\mathbf{X}_{\le i}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$.

Base case. $P(x_1 \mid \mathbf{pa}_{X_1}) = P(x_1 \mid \mathbf{pa}_{X_1} \setminus \{X_1\}) = P(x_1 \mid \mathbf{pa}^{-}_{X_1}) = P(\mathbf{x}_{\le 1} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le 1}})$.

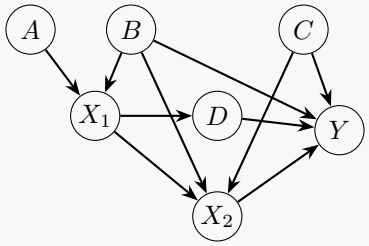
Step. Assume the claim for $i-1$. It suffices to show $P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$. Note $Pa^{-}_{\mathbf{X}_{\le i}} = Pa^{-}_{\mathbf{X}_{<i}} \cup Pa'_{X_i}$. Then

$$\begin{aligned}
&P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) && \text{Markov condition} \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})\frac{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{multiply by 1} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{chain rule, } \{X_i\}\cup\mathbf{X}_{<i} = \mathbf{X}_{\le i} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \mathbf{X}_{<i} \perp\!\!\!\perp Pa'_{X_i} \mid Pa^{-}_{\mathbf{X}_{<i}} \\
&= P(\mathbf{x}_{\le i} \mid \mathbf{pa}'_{X_i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}}).
\end{aligned}$$

The first line replaces the parents of X_i by a superset of non-descendants (local Markov: $\mathbf{X}_{<i}$ and $Pa^{-}_{\mathbf{X}_{<i}}$ are non-descendants of X_i since they precede it or are parents of predecessors). The key independence holds by d-separation: Pa'_{X_i} are non-descendants of every X_j, $j < i$, so every path from $\mathbf{X}_{<i}$ to Pa'_{X_i} either enters X_j through a parent (blocked by $Pa^{-}_{\mathbf{X}_{<i}}$) or leaves X_j along a directed edge, in which case it must meet a collider that is not an ancestor of the conditioning set. ■

Example 2.3

Graph: $A \to X_1$, $B \to X_1$, $B \to X_2$, $B \to Y$, $C \to X_2$, $C \to Y$, $X_1 \to D$, $X_1 \to X_2$, $D \to Y$, $X_2 \to Y$.



Here $Pa_{X_1} = \{A, B\}$, $Pa_{X_2} = \{X_1, B, C\}$, $Pa^-_{\mathbf{X}} = \{A, B, C\}$ and $Pa'_{X_2} = \{C\}$, which is not a descendant of X_1. First, $P(y \mid \mathrm{do}(\mathbf{x})) = \sum P(\mathbf{v})/[P(x_1 \mid a, b)P(x_2 \mid x_1, b, c)]$. Focusing on the denominator,

$$\begin{aligned} P(x_1 \mid a, b)P(x_2 \mid x_1, b, c) &= P(x_1 \mid a, b)P(x_2 \mid \underbrace{x_1}_{\mathbf{X}_{<2}}, \underbrace{a, b}_{Pa^-_{\mathbf{X}_{<2}}}, \underbrace{c}_{Pa'_{X_2}}) \\ &= P(x_1 \mid a, b)P(x_2 \mid x_1, a, b, c)\frac{P(c \mid x_1, a, b)}{P(c \mid x_1, a, b)} \\ &= \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid x_1, a, b)} = \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid a, b)} = P(x_1, x_2 \mid a, b, c) = P(\mathbf{x} \mid \mathbf{pa}^-_{\mathbf{X}}), \end{aligned}$$

so $P(y \mid \mathrm{do}(x_1, x_2)) = \sum_{a,b,c} P(y \mid x_1, x_2, a, b, c)P(a, b, c)$. D (a descendant of X_1) is correctly *not* adjusted for.

2.1.4.4 When a parent is latent

In the sprinkler model, suppose Season is unobserved. Is Q_2 still computable?

$$\begin{aligned} Q_2 &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{sn})P(\mathrm{rn} \mid \mathrm{sn})P(\mathrm{wt} \mid \mathrm{Sp} = \mathrm{on}, \mathrm{rn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{rn}, \mathrm{sn})P(\mathrm{wt} \mid \mathrm{Sp} = \mathrm{on}, \mathrm{rn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{Sp} = \mathrm{on}, \mathrm{rn})P(\mathrm{rn}). \end{aligned}$$

Yes: adjusting for Rain instead of the latent parent Season suffices. Adjustment sets other than the parents can work; characterizing them is the subject of the back-door criterion (Section 2.2).

2.1.5 Summary

- Causal effects $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ are defined through submodels (deleting and substituting equations).
- In Markovian models every causal effect is identifiable via the truncated factorization.
- Adjustment by direct parents gives a practical formula: condition on the parents of the treatment and average over their marginal distribution.
- The identification problem asks whether $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is uniquely determined by $\mathcal{G}$ and $P(\mathbf{v})$.
- With latent variables identification becomes non-trivial; this continues with the back-door criterion and beyond.

2.2 Confounding and the back-door criterion

Motivation. The central question is how a causal effect can be identified when some parents of the treatment are unobserved, and how the resulting adjustment formula can be evaluated from finite samples.

2.2.1 Recap: identifiable and non-identifiable effects

Motivation. Illustrate both sides of the identifiability definition with concrete pairs of models.

2.2.1.1 An identifiable effect

Take $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$ (no latent confounding). Any compatible model factorizes as

$$\begin{aligned} P(\mathbf{v}) &= P(z)P(x \mid z)P(y \mid x, z), \\ P(\mathbf{v}' \mid \mathrm{do}(x)) &= P(z)P(y \mid x, z), \\ P(y \mid \mathrm{do}(x)) &= \sum_z P(z)P(y \mid x, z). \end{aligned}$$

Consider two models

$$\mathcal{M}^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow f_x^{(j)}(z, u_x) \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$\mathcal{M}_x^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow x \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$j = 1, 2.$$

Whatever the functions and $P(\mathbf{u})$, if $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ agree on $\langle \mathcal{G}, P(\mathbf{v}) \rangle$ they agree on $P(z)$ and $P(y \mid x, z)$, hence on $P(\mathbf{v}' \mid \mathrm{do}(x))$.

2.2.1.2 A non-identifiable effect

Take $\mathcal{G}$: $X \to Y$ and $X \leftrightarrow Y$ (through a latent U_{xy}). Let U_y, U_{xy} be independent fair coins in both models and $X \leftarrow U_{xy}$ in the observational world. Define

$$\mathcal{M}^{(1)} : \; Y \leftarrow (X \oplus U_{xy}) \vee U_y, \qquad \mathcal{M}^{(2)} : \; Y \leftarrow U_y.$$

Observationally $X = U_{xy}$, so $X \oplus U_{xy} = 0$ and $Y = U_y$ in both models: the observational distributions coincide. Under $\mathrm{do}(x)$:

U_y	U_{xy}	X	$Y^{(1)}$	$Y^{(2)}$	$P(\mathbf{u})$
0	0	x	x	0	1/4
0	1	x	$\neg x$	0	1/4
1	0	x	1	1	1/4
1	1	x	1	1	1/4

Y	$P^{(1)}(Y \mid \mathrm{do}(x))$	$P^{(2)}(Y \mid \mathrm{do}(x))$
0	1/4	1/2
1	3/4	1/2

In $\mathcal{M}^{(1)}$, $Y = 0$ requires $U_y = 0$ and $x = U_{xy}$, probability 1/4. Both models induce $\mathcal{G}$ and the same $P(\mathbf{v})$, yet $P^{(1)}(y \mid \mathrm{do}(x)) \neq P^{(2)}(y \mid \mathrm{do}(x))$. In general $P(\mathbf{v} \mid \mathrm{do}(x)) = \sum_{u_{xy}} P(y \mid x, u_{xy})P(u_{xy})$ depends on the latent U_{xy}, which the data do not reveal.

2.2.2 Confounding bias

Motivation. Show how a common cause makes conditioning differ from intervening.

What is the causal effect of exercise on cholesterol? Age affects both (age → exercise, age → cholesterol, exercise → cholesterol). In the scatter plot, older people exercise more and also have higher cholesterol, so pooled data show cholesterol *increasing* with exercise, while within every age group more exercise means lower cholesterol. Hence

$$P(\text{cholesterol} \mid \text{exercise}) \neq P(\text{cholesterol} \mid \text{do(exercise)}).$$

This difference is **confounding bias**, one of the major obstacles to causal inference and interpretability.

Is confounding bias removable? Goal: find $Q = P(y \mid \text{do}(x))$ from measurements on $Z_1, \ldots, Z_k$ when some parents of X are unobserved. Adjustment by direct parents (Section 2.1) is then unavailable. Which subsets of measured variables can replace the parents?

2.2.3 The back-door criterion

Motivation. Give a graphical condition under which a covariate set can substitute for the parents of the treatment.

Definition (Back-door criterion; Pearl 2000, Def. 3.3.1)

A set $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

(i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$; and

(ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ that contains an arrow into X (a *back-door path*).

Theorem 2.5 (Back-door adjustment)

If $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$, the effect is identifiable and

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

Proof [Supplement]. We use two rules of do-calculus (Chapter 3), which are proved there from the truncated factorization alone, so there is no circularity. By the probability axioms,

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{z})\, P(\mathbf{z} \mid \text{do}(\mathbf{x})).$$

Step 1. $P(\mathbf{z} \mid \text{do}(\mathbf{x})) = P(\mathbf{z})$. In $\mathcal{G}_{\overline{\mathbf{X}}}$ every edge at $\mathbf{X}$ points out of $\mathbf{X}$. A path from $\mathbf{X}$ to $\mathbf{Z}$ without colliders would then be directed from $\mathbf{X}$ to $\mathbf{Z}$, making an element of $\mathbf{Z}$ a descendant of $\mathbf{X}$, which condition (i) excludes. Every other path contains a collider; since we condition on nothing, the path is blocked. So $(\mathbf{Z} \perp_d \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, and Rule 3 removes the action.

Step 2. $P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$. In $\mathcal{G}_{\underline{\mathbf{X}}}$ the edges out of $\mathbf{X}$ are deleted, so every path between $\mathbf{X}$ and $\mathbf{Y}$ begins with an arrow into $\mathbf{X}$: it is a back-door path, and condition (ii) says $\mathbf{Z}$ blocks it. Thus $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}$ and Rule 2 exchanges the action for an observation.

Step 3. Substituting both identities gives $P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$. ■

2.2.3.1 Back-door sets substitute for the parents: an example

In the sprinkler network, Rain satisfies the criterion relative to (Sprinkler, Wet):

(i) Rain is not a descendant of Sprinkler;

(ii) Rain blocks the only back-door path Sprinkler ← Season → Rain → Wet. Starting from adjustment for the direct parent Season,

$$\begin{aligned} P(\mathrm{wt} \mid \mathrm{do}(\mathrm{sp})) &= \sum_{\mathrm{sn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) \sum_{\mathrm{sn}} P(\mathrm{rn}, \mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn}). \end{aligned}$$

The third equality uses Wet ⫫ Season | Sprinkler, Rain and Rain ⫫ Sprinkler | Season, both read off by d-separation.

2.2.3.2 General proof: back-door adjustment = parent adjustment

Let $Pa_{\mathbf{X}}^{-}$ be the parents of $\mathbf{X}$ outside $\mathbf{X}$, and $\mathbf{Z}^{-} = \mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}$. The two conditions imply two independences:

$$\begin{aligned} &\text{(i)} \implies \mathbf{X} \perp\!\!\!\perp (\mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}) \mid Pa_{\mathbf{X}}^{-} && \text{(local Markov)}, \\ &\text{(ii)} \implies \mathbf{Y} \perp\!\!\!\perp (Pa_{\mathbf{X}}^{-} \setminus \mathbf{Z}) \mid \mathbf{Z}, \mathbf{X} && \text{(back-door paths blocked)}. \end{aligned}$$

The first holds because $\mathbf{Z}$ consists of non-descendants of $\mathbf{X}$; the second because every path from a parent of $\mathbf{X}$ to $\mathbf{Y}$ that avoids $\mathbf{X}$ extends to a back-door path. Then

$$\begin{aligned} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) &= \sum_{\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{parent adjustment} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}, \mathbf{z}^{-}) P(\mathbf{z}^{-} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{def. of cond. probability} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-} \mid \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{(ii) and (i)} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}) \\ &= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) \sum_{\mathbf{pa}_{\mathbf{X}}^{-} \setminus \mathbf{z}} P(\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}) && \text{by algebra} \\ &= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}). \end{aligned}$$

In the third line, $\{\mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}, \mathbf{z}^{-}\} = \{\mathbf{x}, \mathbf{z}\} \cup (\mathbf{pa}_{\mathbf{X}}^{-} \setminus \mathbf{z})$, so (ii) removes $\mathbf{pa}_{\mathbf{X}}^{-} \setminus \mathbf{z}$ from the conditioning set;

(i) removes $\mathbf{x}$ from $P(\mathbf{z}^{-} \mid \cdot)$. *Adjustment by* $\mathbf{Z}$ *is equivalent to adjustment by the direct parents whenever* $\mathbf{Z}$ *is back-door admissible.* (The parent formula holds in the Markovian model containing all variables, latent ones included; the final expression uses only measured variables.)

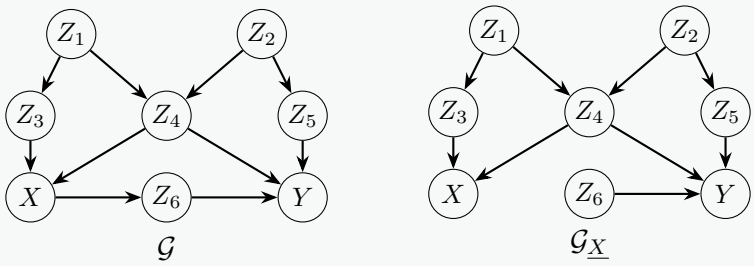
2.2.3.3 A graphical test

Condition (ii) says that $\mathbf{Z}$ d-separates $\mathbf{X}$ from $\mathbf{Y}$ once the causal paths are removed. Let $\mathcal{G}_{\underline{\mathbf{X}}}$ be $\mathcal{G}$ with all edges *out of* $\mathbf{X}$ deleted. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) \text{ is identified by } \mathbf{Z} \quad \text{if (i) and } (\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}.$$

Example 2.4 (Pearl's seven-variable model)

$Z_1 \to Z_3 \to X$, $Z_1 \to Z_4$, $Z_2 \to Z_4$, $Z_2 \to Z_5 \to Y$, $Z_4 \to X$, $Z_4 \to Y$, $X \to Z_6 \to Y$.



The back-door paths are $X \leftarrow Z_4 \to Y$, $X \leftarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \to Y$ and $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \leftarrow Z_2 \to Z_5 \to Y$. Conditioning on Z_4 blocks the first three but opens the collider Z_4 on the last, which Z_1 blocks again. Hence $\{Z_1, Z_4\}$ is admissible:

$$P(y \mid \mathrm{do}(x)) = \sum_{z_1, z_4} P(y \mid x, z_1, z_4) P(z_1, z_4).$$

Other admissible sets include $\{Z_3, Z_4\}$, $\{Z_4, Z_2\}$, $\{Z_4, Z_5\}$; $\{Z_4\}$ alone is not admissible and Z_6 never is (descendant of X).

Example 2.5 (More back-door examples [Supplement])

(a) $Z_2 \to Z_4$, $Z_2 \to Z_5$, $Z_4 \to X$, $Z_4 \to Y$, $X \leftrightarrow Z_4$, $X \to Z_6 \to Y$, $Z_5 \to Y$. Z_4 must be included to block $X \leftarrow Z_4 \to Y$; but Z_4 is a collider on $X \leftrightarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, so Z_2 or Z_5 must be added. Admissible: $\mathbf{Z} \in \{\{Z_4, Z_2\}, \{Z_4, Z_5\}, \{Z_4, Z_2, Z_5\}\}$.
(b) $X \to Z$, $X \to Y$. There is no back-door path and Z is a descendant of X, so $\mathbf{Z} = \emptyset$ and $P(y \mid \mathrm{do}(x)) = P(y \mid x)$.

2.2.4 Evaluating back-door adjustment: inverse probability weighting

Motivation. Turn the adjustment formula into an estimator whose cost does not explode with the dimension of $\mathbf{Z}$.

The criterion says when $\mathbf{Z}$ is admissible,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}),$$

$$\mathbb{E}[Y \mid \mathrm{do}(\mathbf{x})] = \mathbb{E}_{\mathbf{Z}}\big[\mathbb{E}[Y \mid \mathbf{x}, \mathbf{Z}]\big].$$

Evaluating it naively requires estimating many conditional distributions and summing over a possibly high-dimensional $\mathbf{Z}$, roughly $O(\exp(|\mathbf{Z}|))$ time, with sparse cells for each $\mathbf{z}$.

2.2.4.1 Rewriting with the propensity score

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z}) P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x, \mathbf{z})} P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})}.$$

The numerator is an entry of the joint distribution; the denominator is the assignment probability (propensity), fitted by a model $g(\mathbf{z}) \approx P(x \mid \mathbf{z})$.

2.2.4.2 Sample version

With N samples, replace $P(y, x, \mathbf{z})$ by its empirical frequency:

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &\approx \sum_{\mathbf{z}} \frac{\frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N}\sum_{i=1}^{N}\sum_{\mathbf{z}} \frac{\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N}\sum_{i=1}^{N} \frac{\mathbb{1}(y_i = y, x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

The sum over $\mathbf{z}$ collapses because only $\mathbf{z} = \mathbf{z}_i$ contributes. The cost is proportional to N, not to $|\mathfrak{X}_{\mathbf{Z}}|$. ($\mathbb{1}$(predicate) is 1 if the predicate is true and 0 otherwise.) Similarly,

$$\begin{aligned} \mathbb{E}[Y \mid \mathrm{do}(x)] &= \sum_{\mathbf{z},y} y\, \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})} \\ &\approx \frac{1}{N}\sum_{i=1}^{N}\sum_{\mathbf{z},y} \frac{y\,\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N}\sum_{i=1}^{N} \frac{y_i\,\mathbb{1}(x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

2.2.4.3 Average treatment effect

For binary X with $g(\mathbf{z}) = P(X = 1 \mid \mathbf{z})$, when the back-door criterion holds,

$$\begin{aligned} \widehat{\mathrm{ATE}} &= \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 1)] - \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 0)] \\ &= \frac{1}{N}\sum_{i=1}^{N} y_i \left(\frac{\mathbb{1}(x_i = 1)}{g(\mathbf{z}_i)} - \frac{\mathbb{1}(x_i = 0)}{1 - g(\mathbf{z}_i)} \right) \\ &= \frac{1}{N}\sum_{i=1}^{N} \left(\frac{y_i x_i}{g(\mathbf{z}_i)} - \frac{y_i(1 - x_i)}{1 - g(\mathbf{z}_i)} \right). \end{aligned}$$

Notation varies across fields: covariates are often X or L, treatment T, W or A.

2.2.4.4 Interpretation: pseudo-causal samples

IPW turns observational samples into a "pseudo-population" that mimics a randomized experiment: samples with $x_i \neq x$ are discarded and each remaining sample is replicated $1/g(\mathbf{z}_i)$ times. A unit with $P(x_1 \mid \mathbf{z}_1) = 1/3$ counts three times, one with $P(x_3 \mid \mathbf{z}_3) = 1/2$ counts twice. Units that were unlikely to receive treatment given their covariates stand in for the many similar units that were not treated, removing the dependence between X and $\mathbf{Z}$.

2.2.5 Summary

- Confounding bias arises when treatment and outcome share a common cause, making $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$.

- The back-door criterion is a graphical test of whether $\mathbf{Z}$ is admissible for adjustment.
- Back-door adjustment: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.
- It generalizes adjustment by direct parents: any back-door admissible set can substitute for the parents.
- IPW evaluates the back-door expression from samples without summing over a high-dimensional $\mathbf{Z}$.

2.3 The adjustment criterion

Motivation. The central question is exactly which covariate sets $\mathbf{Z}$ make the adjustment formula $\sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$ equal to $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, and how such sets can be tested, constructed and enumerated efficiently.

2.3.1 Back-door criterion: recap and two refinements

Motivation. Recall back-door adjustment and see how it resolves Simpson's paradox and conditional effects.

Theorem 2.6 (Back-door adjustment; Pearl 2000, Thm. 3.3.2)

If $\mathbf{Z}$ satisfies (i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$, and (ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ containing an arrow into X, then $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.

Proof [Supplement]. This is the back-door adjustment theorem of Section 2.2; the proof given there (Rule 3 for $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}))$, Rule 2 for $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$) applies verbatim. ■

2.3.1.1 Back to Simpson's paradox

Recall the drug/sex/recovery tables of Chapter 1 and the three stories:

	$F \to X,\ F \to Y,\ X \to Y$	$X \to F \to Y,\ X \to Y$	$X \leftarrow U_1 \to F \leftarrow U_2 \to Y,\ X \to Y$
$\emptyset$	not admissible	admissible	admissible
$\{F\}$	admissible	not admissible ($F \in De(X)$)	not admissible (opens collider)
Use	F-specific tables	aggregated table	aggregated table

The back-door criterion turns the physician's dilemma into a mechanical check on the graph.

2.3.1.2 Conditional back-door

Theorem 2.7 (Conditional back-door criterion)

The $\mathbf{w}$-specific effect $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ is identified whenever we can measure a set $\mathbf{Z}$ such that $\mathbf{W} \cup \mathbf{Z}$ satisfies the back-door criterion. Moreover,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathbf{w}).$$

Proof [Supplement]. By the probability axioms, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$.

Step 1. Since $\mathbf{W} \cup \mathbf{Z}$ blocks every back-door path and all paths in $\mathcal{G}_{\underline{\mathbf{X}}}$ are back-door paths, $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z}, \mathbf{W})_{\mathcal{G}_{\underline{\mathbf{X}}}}$; Rule 2 gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w})$.

Step 2. No element of $\mathbf{W}$ is a descendant of $\mathbf{X}$, so $\mathbf{X}(\mathbf{W}) = \mathbf{X}$ in Rule 3 and the relevant graph is $\mathcal{G}_{\overline{\mathbf{X}}}$. A path from $\mathbf{X}$ to $\mathbf{Z}$ there leaves $\mathbf{X}$ along an outgoing edge. If it has no collider, it is directed and $\mathbf{Z}$ would contain a descendant of $\mathbf{X}$. Otherwise its first collider C is reached by a directed path from $\mathbf{X}$, so C and all its descendants are descendants of $\mathbf{X}$, hence not in $\mathbf{W}$; the path is blocked given $\mathbf{W}$. Thus $(\mathbf{Z} \perp_d \mathbf{X} \mid \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ and Rule 3 gives $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = P(\mathbf{z} \mid \mathbf{w})$.

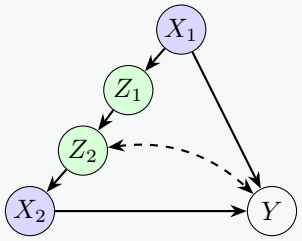
Step 3. Substituting yields the stated formula. ■

2.3.2 Tightness of the back-door criterion

Motivation. Show that the back-door criterion is sufficient but not necessary, and locate what must really be avoided.

Example 2.6

$\mathbf{X} = \{X_1, X_2\}$ with $X_1 \to Z_1 \to Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $Z_2 \leftrightarrow Y$.



Every variable other than $\mathbf{X}, Y$ descends from X_1, so no back-door admissible set exists. Yet with $\mathbf{Z} = \{Z_1, Z_2\}$,

$$P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(y \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

(Z_2 blocks $X_2 \leftarrow Z_2 \leftrightarrow Y$; conditioning on Z_2 opens the collider path $X_1 \to Z_1 \to Z_2 \leftrightarrow Y$, which Z_1 blocks.)

Condition (i) forbids descendants of $\mathbf{X}$ so as not to disturb the causal paths. But some descendants of $\mathbf{X}$ may be used, and sometimes are needed, for adjustment: in particular those that are also ancestors of $\mathbf{X}$ and do not disturb causal paths. *Which variables must really be avoided?*

2.3.3 The adjustment criterion

Motivation. Characterize all valid adjustment sets (necessary and sufficient conditions).

So far: $\mathbf{Z}$ back-door w.r.t. $\langle \mathbf{X}, \mathbf{Y} \rangle \Longrightarrow$ adjustment formula. Now we ask for a condition "???" with

$$\mathbf{Z} \text{ is ??? w.r.t. } \langle \mathbf{X}, \mathbf{Y} \rangle \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

2.3.3.1 Proper causal paths

Definition (Proper causal path; Shpitser et al. 2010)

A path from $X \in \mathbf{X}$ to $Y \in \mathbf{Y}$ is *causal* if it is directed, and *proper* if it contains a member of $\mathbf{X}$ only at its starting point.

Example 2.7

$X_1 \to W_1 \to X_2 \to W_2 \to Y$, $X_1 \to W_3 \to Y$, $X_1 \leftrightarrow Y$. Proper causal paths: $X_1 \to W_3 \to Y$ and $X_2 \to W_2 \to Y$. Non-proper: $X_1 \to W_1 \to X_2 \to W_2 \to Y$ (it passes through X_2).

2.3.3.2 Disturbing a proper causal path

A proper causal path is *disturbed* by adjusting for a variable on the path (other than $\mathbf{X}$) or for a descendant of such a variable.

Example 2.8

$X \to W \to Y$, $W \to Z$. All X–Y paths are causal, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$ (back-door with $\emptyset$). Adjusting for Z, a descendant of W on the proper causal path:

$$\sum_{z} P(y \mid x, z)P(z) = \sum_{z,w} P(y \mid x, z, w)P(w \mid x, z)P(z) = \sum_{w} P(y \mid w) \sum_{z} P(w \mid x, z)P(z),$$

$$P(y \mid \mathrm{do}(x)) = P(y \mid x) = \sum_{w,z} P(y \mid x, z, w)P(w \mid x, z)P(z \mid x) = \sum_{w} P(y \mid w) \sum_{z} P(w \mid x, z)P(z \mid x).$$

For SCMs agreeing on $P(y \mid w)$ and $P(w \mid x, z)$ but with $P(z) \neq P(z \mid x)$ (which is generic, since Z depends on X through W), the two expressions differ. The same happens if we adjust for W itself.

2.3.3.3 The criterion

Theorem 2.8 (Adjustment criterion; Shpitser, VanderWeele and Robins 2010)

$\mathbf{Z}$ satisfies the adjustment criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ iff

(i) $\mathbf{Z}$ does not disturb any proper causal path: no element of $\mathbf{Z}$ is a descendant, in $\mathcal{G}_{\overline{\mathbf{X}}}$, of any $W \notin \mathbf{X}$ lying on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$; and

(ii) all proper non-causal paths in $\mathcal{G}$ from $\mathbf{X}$ to $\mathbf{Y}$ are blocked by $\mathbf{Z}$.

$\mathbf{Z}$ satisfies the adjustment criterion *if and only if* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})P(\mathbf{z})$ for every model compatible with $\mathcal{G}$.

Proof [Supplement]. *Sufficiency when* $\mathbf{Z}$ *contains no descendant of* $\mathbf{X}$. Then condition (i) holds automatically. A non-causal path from $\mathbf{X}$ to $\mathbf{Y}$ either starts with an arrow into $\mathbf{X}$ (a back-door path) or starts with an arrow out of $\mathbf{X}$ and is not directed; in the latter case it contains a collider that is a descendant of $\mathbf{X}$, which cannot be in $\mathbf{Z}$ or have a descendant in $\mathbf{Z}$, so it is blocked anyway. Hence condition (ii) reduces to the back-door criterion, and the adjustment formula follows from the back-door theorem.

General case. When $\mathbf{Z}$ may contain descendants of $\mathbf{X}$ that are off the proper causal paths, sufficiency and necessity are proved in Shpitser, VanderWeele and Robins (2010, Theorem 3) via the proper

back-door graph; necessity constructs, for each violating $\mathbf{Z}$, two models that agree on $P(\mathbf{v})$ but disagree on the adjustment formula. The full argument is not reproduced here. ■

2.3.3.4 Implementation idea

Input: $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathcal{G}$. **Output:** whether $\mathbf{Z}$ is adjustment-admissible.

Step 1 (proper causal paths). Compute the vertices on proper causal paths,

$$\text{PCP} = (De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \setminus \mathbf{X}) \cap An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}}}},$$

descendants of $\mathbf{X}$ (after removing edges into $\mathbf{X}$) that reach $\mathbf{Y}$ without passing through $\mathbf{X}$ again. (A simpler choice is the superset $\mathbf{W} = De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \cap An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, and $\mathcal{H} = \mathcal{G}[\mathbf{W}]$ with bidirected edges removed to capture the PCP structure; computing ancestors in $\mathcal{G}_{\underline{\mathbf{X}}}$ removes vertices that reach $\mathbf{Y}$ only through another treatment.)

Step 2 (forbidden variables). $\mathcal{F} = De(\text{PCP})_{\mathcal{G}_{\overline{\mathbf{X}}}}$: variables on proper causal paths (excluding $\mathbf{X}$) and their descendants. Conditioning on any of them could block a causal path or create bias. *Check (i):* if $\mathbf{Z} \cap \mathcal{F} \neq \emptyset$, $\mathbf{Z}$ is not admissible; stop.

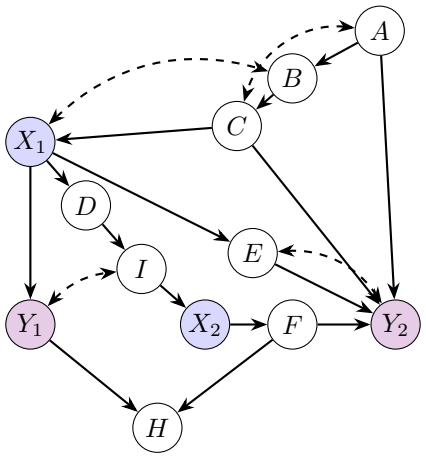
Step 3 (proper non-causal paths). Build $\mathcal{G}'$ by removing the first edge of every proper causal path,

$$\mathcal{G}' = \mathcal{G} \setminus \{X \to W : X \in \mathbf{X},\ W \in \text{PCP}\}.$$

In $\mathcal{G}'$ all causal paths are cut, so only non-causal paths remain. *Check (ii):* $\mathbf{Z}$ is admissible iff $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}'}$.

Summary: find PCPs → check $\mathbf{Z} \cap \mathcal{F} = \emptyset$ → check d-separation in $\mathcal{G}'$. Each step is a graph search, so the test is polynomial.

2.3.3.5 An example



$\mathcal{G}$: $A \to B \to C \to X_1$, $C \to Y_2$, $A \to Y_2$, $X_1 \to D \to I \to X_2$, $X_1 \to E \to Y_2$, $X_1 \to Y_1$, $X_2 \to F \to Y_2$, $Y_1 \to H \leftarrow F$, and bidirected $X_1 \leftrightarrow B$, $A \leftrightarrow C$, $Y_1 \leftrightarrow I$, $E \leftrightarrow Y_2$; $\mathbf{X} = \{X_1, X_2\}$, $\mathbf{Y} = \{Y_1, Y_2\}$.

(i) Forbidden set. Proper causal paths: $X_1 \to Y_1$, $X_1 \to E \to Y_2$, $X_2 \to F \to Y_2$. ($X_1 \to D \to I \to X_2 \to \cdots$ is not proper.) Their non-treatment vertices are $\{Y_1, E, Y_2, F\}$, whose descendants are $\{Y_1, E, Y_2, F, H\}$; apart from $\mathbf{Y}$, $\mathcal{F} = \{E, F, H\}$.

(ii) Evaluating $\mathbf{Z} = \{C, D, I\}$. $\mathbf{Z} \cap \mathcal{F} = \emptyset$, so (i) holds. Cut the first edges $X_1 \to Y_1$, $X_1 \to E$, $X_2 \to F$ to obtain $\mathcal{G}'$. Any remaining path between $\mathbf{X}$ and $\mathbf{Y}$ is non-causal and must be blocked by $\mathbf{Z}$ (with no E, F, H). But $X_1 \leftrightarrow B \leftarrow A \to Y_2$ is open: B is a collider and an ancestor of $C \in \mathbf{Z}$,

and A is an unobserved non-collider. Hence $(\mathbf{X} \not\perp_d \mathbf{Y} \mid C, D, I)_{\mathcal{G}'}$ and $\mathbf{Z}$ is *not* admissible. A path that is non-causal in $\mathcal{G}'$ corresponds to a proper non-causal path in $\mathcal{G}$.

2.3.3.6 Explicit construction of an admissible set

Lemma 2.9 (Explicit admissible set)

Let $\mathcal{F}$ be the variables forbidden by condition (i) and $An^-(\mathbf{W}) = An(\mathbf{W}) \setminus \mathbf{W}$. If any admissible set exists, then

$$\mathbf{Z}_0 = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$$

is admissible. (Ancestors should be taken in $\mathcal{G}'$, but the result is the same in $\mathcal{G}$.)

Proof [Supplement]. By the constructive back-door criterion (van der Zander, Liśkiewicz and Textor 2014), $\mathbf{Z}$ is an adjustment set iff $\mathbf{Z} \cap \mathcal{F} = \emptyset$ and $\mathbf{Z}$ d-separates $\mathbf{X}$ and $\mathbf{Y}$ in the proper back-door graph $\mathcal{G}'$ (the graph without the first edge of every proper causal path). If an admissible $\mathbf{Z}$ exists, apply the inclusion–restriction lemma in $\mathcal{G}'$ with $\mathbf{I} = \emptyset$ and $\mathbf{R} = \mathbf{V} \setminus (\mathcal{F} \cup \mathbf{X} \cup \mathbf{Y})$: the set $An(\mathbf{X} \cup \mathbf{Y}) \cap \mathbf{R} = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$ also separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}'$ and avoids $\mathcal{F}$, so it is admissible. (The equivalence of the constructive back-door criterion with the adjustment criterion is the cited theorem.) ■

In the example, $An^-(\mathbf{X} \cup \mathbf{Y}) = \{A, B, C, D, E, F, I\}$ and $\mathbf{Z}_0 = \{A, B, C, D, I\}$. Now $A \in \mathbf{Z}_0$ blocks $X_1 \leftrightarrow B \leftarrow A \rightarrow Y_2$ and $X_1 \leftrightarrow B \leftarrow A \leftrightarrow C \rightarrow Y_2$; C blocks $X_1 \leftarrow C \rightarrow Y_2$; D blocks $X_1 \rightarrow D \rightarrow I \leftrightarrow Y_1$ (the collider I is opened by conditioning); I blocks $X_2 \leftarrow I \leftrightarrow Y_1$. Hence $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z}_0)_{\mathcal{G}'}$ and $\mathbf{Z}_0$ is admissible. Note that D and I are descendants of X_1: the back-door criterion would never allow them.

Testing existence. If $\mathbf{Z}_0$ is not admissible, no admissible set exists: a *one-shot* verification of the existence of a solution.

2.3.3.7 Recap

- The adjustment criterion characterizes all valid adjustment sets (necessary and sufficient).
- Condition (i): $\mathbf{Z}$ contains no variable of $\mathcal{F}$ (descendants of variables on proper causal paths).
- Condition (ii): $\mathbf{Z}$ blocks all proper non-causal paths from $\mathbf{X}$ to $\mathbf{Y}$.
- Testing and constructing admissible sets takes polynomial time.

2.3.4 Listing all admissible sets (optional)

Motivation. Enumerate every admissible set with polynomial delay, so that sets can be chosen by cost, variance or other preferences.

2.3.4.1 Motivation

There are typically several admissible sets. Which one to use may depend on availability, measurement cost, variance, privacy or fairness. Example: $A \rightarrow X$, $A \rightarrow B \rightarrow Y$, $X \rightarrow C \rightarrow Y$. The sets $\{A, B\}$, $\{A\}$ and $\{B\}$ are all admissible. Which should we prefer? What if A is a "genetic condition" (expensive or sensitive) and B is "headache"?

2.3.4.2 Challenge: output size

Even with an oracle f returning all solutions in $O(1)$, the loop `for Zs in f(G, Xs, Ys): output(Zs)` can take exponential time, because the number of solutions can be exponential. With k parallel paths $X \to W_{i,1} \to W_{i,2} \to Y$, a set separating X from Y must contain $W_{i,1}$, $W_{i,2}$ or both for each i: 3^k separators. So total running time is the wrong yardstick.

Definition (Polynomial delay)

A listing algorithm runs with *polynomial delay* if there is a constant d such that it spends at most $O(n^d)$ time (1) between the start and either producing the first output or halting with no output, and (2) between producing one output and either producing the next or halting.

2.3.4.3 Theoretical tool: bounded separators

Lemma 2.10 (Inclusion and restriction)

Given $\mathcal{G}$ and disjoint $\mathbf{X}, \mathbf{Y}$, and $\mathbf{I} \subseteq \mathbf{R} \subseteq \mathbf{V}$: if any set $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}$, then

$$\mathbf{Z}_0 = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I}) \cap \mathbf{R}$$

also separates them.

Proof [Supplement]. We use the moralization criterion: $\mathbf{X}$ and $\mathbf{Y}$ are d-separated by $\mathbf{S}$ in $\mathcal{G}$ iff $\mathbf{S}$ separates them in the moral graph $(\mathcal{G}_A)^m$ of the ancestral set $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{S})$. For ancestral sets $A \subseteq A'$, the moral graph $(\mathcal{G}_A)^m$ is the subgraph of $(\mathcal{G}_{A'})^m$ induced by A, because every parent of a node in A lies in A.

Let $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I})$. Since $\mathbf{I} \subseteq \mathbf{R}$ and $\mathbf{I} \subseteq A$, we have $\mathbf{I} \subseteq \mathbf{Z}_0 \subseteq A$, so $An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}_0) = A$. Suppose $\mathbf{Z}_0$ does not separate: there is a path π in $(\mathcal{G}_A)^m$ from $\mathbf{X}$ to $\mathbf{Y}$ whose inner nodes avoid $\mathbf{Z}_0 = A \cap \mathbf{R}$. Its nodes lie in A, so its inner nodes lie outside $\mathbf{R}$ and hence outside $\mathbf{Z} \subseteq \mathbf{R}$. Let $A' = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}) \supseteq A$ (as $\mathbf{I} \subseteq \mathbf{Z}$). Then π is also a path in $(\mathcal{G}_{A'})^m$ avoiding $\mathbf{Z}$, so $\mathbf{Z}$ does not separate $\mathbf{X}$ and $\mathbf{Y}$, a contradiction. ■

$\mathbf{I}$ is the "include" set and $\mathbf{R}$ the "restrict to" set. Existence of a separator between the bounds is checked by testing $\mathbf{Z}_0$, in $O(n+m)$ time; call this `exist-sep`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$. With $\mathbf{I} = \emptyset$ and $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ this is the explicit construction above.

2.3.4.4 Algorithm

Divide and conquer: implement `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$, outputting all separators $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$. Let $\mathcal{F}^+ = \mathbf{X} \cup \mathbf{Y} \cup \mathcal{F}$. Initialize $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$, and use $\mathcal{G}'$ ($\mathcal{G}$ with the first edge of each proper causal path removed). Every output separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}'$ (condition (ii)) and avoids $\mathcal{F}^+$ (condition (i)), so every output satisfies the adjustment criterion.

Algorithm 1 `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$

```
if exist-sep(G, X, Y, I, R) then                          ▷ O(n + m)
    if I = R then output I
    else
        pick W ∈ R \ I
        list-seps(G, X, Y, I ∪ {W}, R)                    ▷ include W
        list-seps(G, X, Y, I, R \ {W})                    ▷ exclude W
```

Example 2.9 (Running example)

$A \to X$, $A \to B \to Y$, $D \to X$, $D \leftrightarrow Y$, $X \to C \to Y$.

1. Forbidden: variables on proper causal paths other than X and their descendants, plus X, Y: $\mathcal{F}^+ = \{C, X, Y\}$.
2. Initialize $\mathbf{I} = \emptyset$, $\mathbf{R} = \{A, B, D\}$, and remove $X \to C$.
3. Branch on D. Excluding D ($\mathbf{R} = \{A, B\}$) is pruned immediately: $X \leftarrow D \leftrightarrow Y$ cannot be blocked. Including D, branch on A, then on B. Leaves: $\{A, B, D\}$ (output), $\{A, D\}$ (output), $\{B, D\}$ (output), $\{D\}$ (pruned: $X \leftarrow A \to B \to Y$ open).

The admissible sets are $\{A, B, D\}, \{A, D\}, \{B, D\}$.

2.3.4.5 Analysis

- A node of the recursion tree is expanded only if `exist-sep` is true; otherwise no separator lies between $\mathbf{I}$ and $\mathbf{R}$ and the branch is safely ignored. Hence every expanded node has at least one output leaf below it.
- Every reached leaf produces an output $\mathbf{Z} = \mathbf{I} = \mathbf{R}$, which separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}'$ (condition ii) and, since $\mathbf{R}$ initially excludes $\mathcal{F}^+$, satisfies condition (i).
- The tree depth is at most $k = n - |\mathcal{F}^+| < n$, so at most $O(n)$ recursive calls separate two consecutive outputs.
- Each call costs $O(n + m)$ (`exist-sep`), $O(1)$ for the comparison and $O(n)$ to output. The delay is therefore $O(n(n + m))$.

The total time may still be exponential (unavoidable), but outputs arrive one after another in polynomial time, and the algorithm halts in polynomial time if there are no solutions.

2.3.5 Connection with conditional ignorability

Motivation. Relate the graphical criterion to the key assumption of the potential-outcome framework.

The central identifying assumption of the potential outcome framework is *conditional ignorability*,

$$\mathbf{Y}_{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z},$$

written with counterfactuals. It requires that, conditional on $\mathbf{Z}$, the treatment is independent of the outcomes under intervention, which gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$ and hence the adjustment formula. In the PO framework this is usually *assumed*; the adjustment criterion *justifies* it by stating under which conditions such a set exists given a model of reality.

2.3.6 Summary

- Adjustment is arguably the most popular technique across the data sciences for identifying causal effects.
- The back-door/adjustment criterion gives a principled condition for the validity of adjusting for $\mathbf{Z}$.
- Admissible sets (if any) can be found systematically and efficiently given the graph and the target effect.

- Once a set is found, estimation techniques (IPW and variants) evaluate the effect efficiently and robustly from finite samples.

Chapter 3. Do-Calculus and Algorithmic Identification

3.1 Do-calculus

Motivation. The central question is whether there is a systematic way to decide if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be expressed in terms of the observational distribution $P(\mathbf{v})$. The truncated factorization sometimes yields identifiable effects (back-door, front-door) and sometimes not (the bow graph). Do-calculus (Pearl 1995) supplies three graphical inference rules, adding/removing observations, exchanging actions and observations, adding/removing actions, which are *complete*: if an effect is identifiable, do-calculus derives it.

3.1.1 Three examples via the truncated factorization

Motivation. See how identification works, and fails, when the $\mathbf{U}$-terms of a semi-Markovian model must be eliminated by hand.

Theorem 3.1 (Generalized truncated factorization)

In a semi-Markovian model $\mathcal{M}$, the distribution generated by $\mathrm{do}(\mathbf{X} = \mathbf{x})$ is

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}),$$
$$\mathbf{V}' = \mathbf{V} \setminus \mathbf{X},$$

and for $\mathbf{Y} \subseteq \mathbf{V} \setminus \mathbf{X}$, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$.

Proof [Supplement]. Fix $\mathbf{u}$. In $\mathcal{M}_{\mathbf{x}}$ the mechanisms of $\mathbf{X}$ are replaced by constants and the others are unchanged, so $\mathbf{V}'$ is a deterministic function of $(\mathbf{x}, \mathbf{u})$ and, equivalently, $P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x}), \mathbf{u}) = \prod_{V_i \in \mathbf{V}'} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$ with the factors of the original model (each $P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$ is the indicator that $f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i$; acyclicity makes the product a valid joint law obtained by solving the equations in topological order). The intervention does not change $P(\mathbf{u})$, so averaging over $\mathbf{u}$ gives the formula. ■

Identification succeeds when the right-hand side can be rewritten without $\mathbf{U}$.

3.1.1.1 Back-door graph

$Z \to X$, $Z \to Y$, $X \to Y$ with independent U_Z, U_X, U_Y. In $\mathcal{M}_x$: $Z \leftarrow f_Z(u_Z)$, $X \leftarrow x$, $Y \leftarrow f_Y(x, z, u_Y)$.

$$\begin{aligned} P(\mathbf{v}' \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) \\ &= \Big(\sum_{u_Z} P(z \mid u_Z) P(u_Z) \Big) \Big(\sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y) \Big) \Big(\sum_{u_X} P(u_X) \Big) \\ &= P(z) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y \mid x, z) = P(z) P(y \mid x, z), \end{aligned}$$

using $U_Y \perp\!\!\!\perp \{X, Z\}$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z) P(z)$.

3.1.1.2 Bow graph

$X \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Y \leftarrow f_Y(x, u, u_Y)$.

$$P(y \mid \mathrm{do}(x)) = \sum_u P(u) \sum_{u_Y} P(y \mid x, u, u_Y) P(u_Y) \sum_{u_X} P(u_X) = \sum_u P(y \mid x, u) P(u).$$

U cannot be eliminated: $P(u)$ and $P(y \mid x, u)$ are not functions of $P(x, y)$. (Observationally, $P(\mathbf{v}) = \sum_u P(x \mid u, u_X) P(y \mid x, u, u_Y) P(\mathbf{u})$ weights u by $P(u \mid x)$ instead.)

3.1.1.3 Front-door graph

$X \to Z \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Z \leftarrow f_Z(x, u_Z)$, $Y \leftarrow f_Y(z, u, u_Y)$.

$$\begin{aligned}
P(z, y \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid x, u_Z) P(y \mid z, u, u_Y) P(\mathbf{u}) \\
&= \Big(\sum_{u_Z} P(z \mid x, u_Z) P(u_Z) \Big) \Big(\sum_{u, u_Y} P(y \mid z, u, u_Y) P(u, u_Y) \Big) \\
&= P(z \mid x) \sum_u P(y \mid z, u) \sum_{x'} P(u \mid x') P(x') \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x') P(x') && (Y \perp\!\!\!\perp X \mid Z, U) \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x', z) P(x') && (U \perp\!\!\!\perp Z \mid X) \\
&= P(z \mid x) \sum_{x'} \sum_u P(y, u \mid z, x') P(x') = P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').
\end{aligned}$$

Therefore the **front-door formula**

$$P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').$$

3.1.2 Identification approaches

Motivation. State the syntactic goal that do-calculus serves.

In both the back-door and front-door settings the goal was to reduce $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to an expression whose terms are estimable from $P(\mathbf{v})$. One way is to write the truncated factorization and derive a $\mathbf{U}$-free expression, as above. Another is to rewrite the query into a *do-free* expression without ever mentioning $\mathbf{U}$. We seek rules (axioms) that transform do-expressions into do-free ones while preserving equality with the target effect.

3.1.3 Do-calculus

Motivation. Derive the three rules from three insights and explain their graphical side conditions.

3.1.3.1 Prelude: independence in a submodel

$\mathrm{do}(\mathbf{X} = \mathbf{x})$ replaces f_X by the constant x for each $X \in \mathbf{X}$. Graphically $\mathcal{G}_{\overline{\mathbf{X}}}$ represents $\mathbf{X}$ not influenced by any other variable. Moreover $\mathbf{X}$ are now constants, which corresponds to conditioning on $\mathbf{X} = \mathbf{x}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. Hence

$$(\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}) \text{ under } \mathrm{do}(\mathbf{x}) \quad \Longleftarrow \quad (\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C}, \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}},$$

equivalently $(\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C})$ in $\mathcal{G} \setminus \mathbf{X}$ (the graph with $\mathbf{X}$ deleted) when $\mathbf{X}$ is disjoint from $\mathbf{A}, \mathbf{B}, \mathbf{C}$. For example, with $A \to X \to D$, the intervention severs A from X, and $\mathbf{X}$ fixed blocks every path through X.

3.1.3.2 Insight 1: adding/removing observations

Graph: $Z \to X \to Y$, $X \leftrightarrow Y$ (via U). Then

$$P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, u_Y, u) P(\mathbf{u}) = P(z) \sum_{u} P(y \mid x, u) P(u).$$

Marginalizing, $P(z \mid \mathrm{do}(x)) = P(z) \sum_u P(u) \sum_y P(y \mid x, u) = P(z)$, so

$$P(y \mid \mathrm{do}(x), z) = \frac{P(y, z \mid \mathrm{do}(x))}{P(z \mid \mathrm{do}(x))} = \sum_{u} P(y \mid x, u) P(u) = \sum_{z} P(y, z \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x)).$$

The observation z can be dropped, and the graphical reason is $(Z \perp_d Y \mid X)_{\mathcal{G}_{\overline{X}}}$. In the original model Z and Y are inseparable: $Z \not\perp\!\!\!\perp Y$ (through $Z \to X \to Y$) and $Z \not\perp\!\!\!\perp Y \mid X$ (X is a collider on $Z \to X \leftrightarrow Y$). In the $\mathrm{do}(X)$-world $\mathcal{M}_x$, however, the edge $Z \to X$ is cut and Y, Z are d-separated; since $P(y \mid \mathrm{do}(x), z) = P_x(y \mid z)$ is an ordinary conditional in the submodel, the observation can be removed.

3.1.3.3 Insight 2: exchanging action and observation

Graph: $Z \to X$, $Z \to Y$, $X \to Y$. After observing Z, Y reacts to X in the same way with and without intervention: given Z, Y is associated with X only through causal paths, so *seeing* $X = x$ is equivalent to *doing* $X = x$. If Z blocks all back-door paths, the remaining association is causation:

$$(Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}} \implies P(y \mid \mathrm{do}(x), z) = P(y \mid x, z).$$

Check: $P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) = P(z) P(y \mid x, z)$ and $P(z \mid \mathrm{do}(x)) = \sum_y P(z) P(y \mid x, z) = P(z)$, so $P(y \mid \mathrm{do}(x), z) = P(z) P(y \mid x, z) / P(z) = P(y \mid x, z)$.

3.1.3.4 Insight 3: adding/removing actions

Graph: $X \to Y$, $Z \leftrightarrow Y$ (latent U_{zy}), $Z \leftrightarrow X$ (latent U_{zx}). If there is no causal path from X to Z, intervening on X has no effect on Z:

$$(Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \implies P(z \mid \mathrm{do}(x)) = P(z).$$

Check: $P(z \mid \mathrm{do}(x)) = \sum_y \sum_{u_{zy}, u_{zx}} P(z \mid u_{zy}, u_{zx}) P(y \mid x, u_{zy}) P(u_{zy}, u_{zx}) = \sum_{u_{zy}, u_{zx}} P(z \mid u_{zy}, u_{zx}) P(u_{zy}, u_{zx}) = P(z)$.

3.1.3.5 The rules

Notation: $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ is $\mathcal{G}$ with edges into $\mathbf{X}$ and out of $\mathbf{Z}$ removed.

Theorem 3.2 (Rules of do-calculus; Pearl 1995)

For any do-distribution induced by a causal model $\mathcal{M}$ with diagram $\mathcal{G}$, and disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$:

Rule 1 (adding/removing observations) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Rule 2 (action/observation exchange) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$.

Rule 3 (adding/removing actions) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}}$, where $\mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Proof [Supplement]. Work in the post-intervention model $\mathcal{M}_{\mathbf{x}}$, whose diagram is $\mathcal{G}_{\overline{\mathbf{X}}}$; by the generalized truncated factorization its distribution (with the latent variables written as explicit nodes) is Markov relative to $\mathcal{G}_{\overline{\mathbf{X}}}$, so d-separation in $\mathcal{G}_{\overline{\mathbf{X}}}$ implies conditional independence in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$.

Rule 1. $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ gives $\mathbf{Y} \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{W}$ in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ (the variables $\mathbf{X}$ are constants there), which is the claim.

Augmented graph. For Rules 2 and 3 add to $\mathcal{G}_{\overline{\mathbf{X}}}$ a parent F_Z for each $Z \in \mathbf{Z}$ (an "intervention switch"): $F_Z = \text{idle}$ leaves the mechanism of Z intact, $F_Z = z'$ replaces it by the constant z'. Then $P(\cdot \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}))$ is the law given $F_{\mathbf{Z}} = \mathbf{z}$, and $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ the law given $F_{\mathbf{Z}} = \text{idle}$. The augmented model is still Markov relative to the augmented graph $\mathcal{G}^F$.

Rule 2. On the event $\mathbf{Z} = \mathbf{z}$, the two regimes $F_{\mathbf{Z}} = \mathbf{z}$ and $F_{\mathbf{Z}} = \text{idle}$ differ only in the mechanism of $\mathbf{Z}$, so it suffices that $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z to $\mathbf{Y}$ enters Z through $F_Z \to Z$. If it continues along an edge into Z, then Z is a collider on it with Z in the conditioning set, but the remainder of the path, read from Z, starts with an arrow into Z; if it continues along an edge out of Z, Z is a non-collider in the conditioning set and the path is blocked. Hence the open paths from $F_{\mathbf{Z}}$ are exactly the extensions $F_Z \to Z \leftarrow \cdots$ of paths that leave Z by an incoming edge, and these are open given $(\mathbf{X}, \mathbf{Z}, \mathbf{W})$ iff the corresponding path from Z to $\mathbf{Y}$ is open in $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ (where the outgoing edges of $\mathbf{Z}$ are removed). So $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$ implies $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W})_{\mathcal{G}^F}$, which gives Rule 2.

Rule 3. Now $\mathbf{Z}$ is not observed and we need $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z enters Z and continues either out of Z (then Z is an unconditioned non-collider) or into Z (then Z is a collider, open only if Z has a descendant in $\mathbf{W}$, i.e. $Z \in An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$). For $Z \in \mathbf{Z(W)}$ the second kind is blocked, and the first kind is open given $(\mathbf{X}, \mathbf{W})$ exactly when the corresponding path from Z is open in a graph where Z has no incoming edges, i.e. in $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}$. For $Z \in \mathbf{Z} \setminus \mathbf{Z(W)}$, the separation $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}$ blocks the paths through incoming edges as well. Hence the stated condition implies $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}^F}$, and the two regimes give the same $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$. ■

Unconditional versions ($\mathbf{W} = \emptyset$) read: Rule 1 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, Rule 2 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$, Rule 3 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}$.

Simplified form. Let $\mathcal{H} = \mathcal{G} \setminus \mathbf{X}$ and $Q = P_{\mathbf{x}}$. Then

$$\begin{aligned}
&\text{R1: } Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}},\\
&\text{R2: } Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\underline{\mathbf{Z}}}},\\
&\text{R3: } Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\overline{\mathbf{Z(W)}}}},\ \mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{H}}.
\end{aligned}$$

Every rule is a statement about one fixed submodel $\mathcal{M}_{\mathbf{x}}$; $\mathrm{do}(\mathbf{x})$ is just carried along.

3.1.3.6 Understanding Rule 2

One may add the observation $\mathbf{x}$ under $\mathrm{do}(\mathbf{x})$: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{x}, \mathbf{z}, \mathbf{w})$. So Rule 2 asks whether the model and the submodel generate the same conditional probability, e.g. (with no outer $\mathrm{do}(\mathbf{x})$) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{z}, \mathbf{w})$.

Regime indicator. Augment $\mathcal{G}$ (here $W \to Z$, $W \to Y$, $Z \to Y$) with a parent $F_Z \to Z$ selecting

the mechanism:

$$f'_Z = \begin{cases} f_Z(W, U_Z) & \text{if } F_Z = 0, \\ z & \text{if } F_Z = 1. \end{cases}$$

Then the Rule 2 equality $P(y \mid \text{do}(z), z, w) = P(y \mid z, w)$ reads $P(y \mid z, w, F_Z = 1) = P(y \mid z, w, F_Z = 0)$, i.e. $(F_Z \perp\!\!\!\perp Y \mid Z, W)$ in the augmented graph $\mathcal{G}'$. Since Z is given, every d-connecting path from F_Z must be $F_Z \to Z \leftarrow \cdots Y$ (the collider Z is opened); so we check paths that leave Z through an arrowhead, i.e. back-door paths, which is the same as checking Z versus Y in $\mathcal{G}_{\underline{Z}}$.

3.1.3.7 Understanding Rule 3

"Changing a function" subsumes "changing it to a constant", so

$$P(\mathbf{y} \mid \text{do}(\mathbf{x}), \text{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{w}) \iff P(\mathbf{y} \mid \text{do}(\mathbf{x}), F_Z = 1, \mathbf{w}) = P(\mathbf{y} \mid \text{do}(\mathbf{x}), F_Z = 0, \mathbf{w}),$$

which holds if $F_{\mathbf{Z}} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ in $\mathcal{G}'$. A path leaving F_Z has the form $F_Z \to Z \to \cdots$ (always feasible) or $F_Z \to Z \leftarrow \cdots$, feasible only if the collider Z is opened, i.e. $Z \in An(\mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. Hence edges into Z can be cut exactly for $Z \in \mathbf{Z}(\mathbf{W})$, giving the condition $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z}(\mathbf{W})}}}$.

Example 3.1 (Why Z(W) is needed)

Graph: $T \to Z \to Y$, $Z \to W$. Is $P(t \mid \text{do}(z), w) = P(t \mid w)$? In $\mathcal{G}_{\overline{Z}}$, T is isolated, so a naive check $(T \perp_d Z \mid W)_{\mathcal{G}_{\overline{Z}}}$ would say yes. But $Z \in An(W)$, so $\mathbf{Z}(\mathbf{W}) = \emptyset$ and the correct graph is $\mathcal{G}$ itself, where $T \to Z$ remains and the condition fails. Indeed, in almost every compatible model,

$$P(t \mid \text{do}(z), w) = \frac{P(t, w \mid \text{do}(z))}{P(w \mid \text{do}(z))} = \frac{\sum_y P(t)P(w \mid z)P(y \mid z)}{\sum_{y,t} P(t)P(w \mid z)P(y \mid z)} = \frac{P(t)P(w \mid z)}{P(w \mid z)} = P(t) \neq P(t \mid w),$$

since W carries information about T through Z observationally.

3.1.4 Do-calculus and identification

Motivation. State completeness, work the front-door derivation, and give tools for proving non-identifiability.

Theorem 3.3 (Completeness; Huang and Valtorta 2006, Shpitser and Pearl 2006)

$Q = P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is identifiable from $P(\mathbf{v})$ and $\mathcal{G}$ iff there is a finite sequence of applications of the rules of do-calculus and the probability axioms that reduces Q to a do-free expression.

Proof [Supplement]. *If:* each rule of do-calculus and each probability axiom preserves equality for every model compatible with $\mathcal{G}$, so a reduction to a do-free expression writes Q as a functional of $P(\mathbf{v})$, which is identifiability. *Only if:* if Q is identifiable, the ID algorithm (Chapter 3, Section 3.2) returns an expression; each of its steps is an application of the rules of do-calculus and the probability axioms (Huang and Valtorta 2006; Shpitser and Pearl 2006), and when the algorithm fails it exhibits a hedge, from which two models agreeing on $P(\mathbf{v})$ but not on Q are built. The construction of these models is not reproduced here. ■

Syntactic goal: re-express Q without do.

3.1.4.1 Front-door by do-calculus

Smoking $S \to$ Tar $T \to$ Cancer C, with $S \leftrightarrow C$.

$$
\begin{aligned}
P(c \mid \mathrm{do}(s)) &= \sum_t P(c \mid \mathrm{do}(s), t) P(t \mid \mathrm{do}(s)) && \text{probability axioms} \\
&= \sum_t P(c \mid \mathrm{do}(s), \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\overline{S}\underline{T}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 3: } (S \perp_d C \mid T)_{\mathcal{G}_{\overline{T}\,\overline{S}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid s) && \text{Rule 2: } (T \perp_d S)_{\mathcal{G}_{\underline{S}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', \mathrm{do}(t)) P(s' \mid \mathrm{do}(t)) && \text{probability axioms} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s' \mid \mathrm{do}(t)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\underline{T}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s') && \text{Rule 3: } (T \perp_d S)_{\mathcal{G}_{\overline{T}}}.
\end{aligned}
$$

Justifications: in $\mathcal{G}_{\overline{S}\underline{T}}$ the only edges are $S \leftrightarrow C$ and $S \to T$, and S is given; in $\mathcal{G}_{\overline{T}\,\overline{S}}$, S is disconnected except through $S \leftrightarrow C$, which T does not open; in $\mathcal{G}_{\underline{S}}$, $S \to T$ is removed and $S \leftrightarrow C \leftarrow T$ is blocked at the collider C; in $\mathcal{G}_{\underline{T}}$, $T \leftarrow S \leftrightarrow C$ is blocked by S; in $\mathcal{G}_{\overline{T}}$, T has no edges to S except via the collider C.

3.1.4.2 Proving non-identifiability

Example 3.2

Bow graph $X \to Y$, $X \leftrightarrow Y$, U a fair coin, $\mathcal{F}$: $X \leftarrow U$, $Y \leftarrow f_Y(X, U)$. Define two models that agree with $\mathcal{F}$ whenever $U = X$ (the only case ever observed) but differ off the diagonal:

$$
\mathcal{F}_x^{(1)}: \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 0 & \text{otherwise} \end{cases},
$$

$$
\mathcal{F}_x^{(2)}: \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 1 & \text{otherwise.} \end{cases}
$$

Under $\mathrm{do}(X = x)$, U and X need not match:

$$
P^{(i)}(Y = 1 \mid \mathrm{do}(x)) = \sum_u P^{(i)}(y \mid x, u) P(u) = P(y \mid x) P(x) + \mathbb{1}_{i=2} (1 - P(x)).
$$

Both models induce $\mathcal{G}$ and the same $P(\mathbf{v})$, but the effects differ whenever $P(x) < 1$.

Lemma 3.4 (Graph–subgraph; Tian and Pearl 2002)

If $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is not identifiable in $\mathcal{G}$, it is not identifiable in any graph obtained by adding a directed or bidirected edge to $\mathcal{G}$. Conversely, if Q is identifiable in $\mathcal{G}$, it remains identifiable after removing edges.

Proof [Supplement]. Let $\mathcal{G}^+$ be $\mathcal{G}$ with an added edge. Every model compatible with $\mathcal{G}$ is compatible with $\mathcal{G}^+$: a structural function may ignore the new parent and an added bidirected edge may correspond to a latent variable with no effect. If Q is not identifiable in $\mathcal{G}$, there are two models compatible with $\mathcal{G}$ with the same $P(\mathbf{v})$ and different Q; they are also compatible with $\mathcal{G}^+$, so Q is not identifiable in $\mathcal{G}^+$. The converse statement is the contrapositive applied to $\mathcal{G}^- \subseteq \mathcal{G}$. ■

(Adding an edge only enlarges the class of compatible models, since a graph permits but does not force dependence.)

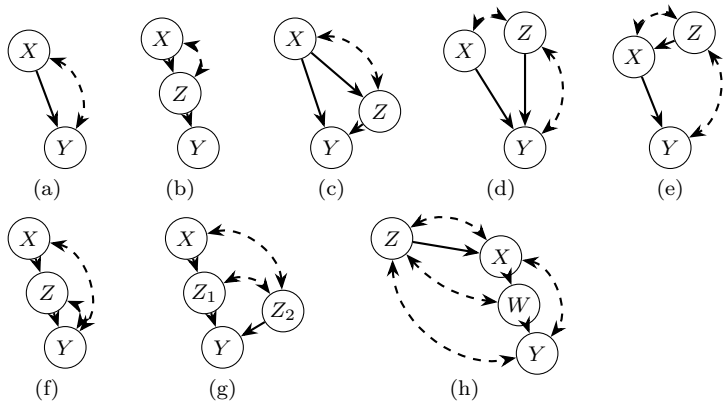
Theorem 3.5 (Non-identifiability criterion, singleton; Tian and Pearl 2002)

If there is a bidirected path connecting X to any of its children in $\mathcal{G}$, then $P(\mathbf{v}' \mid \mathrm{do}(x))$ is not identifiable from $P(\mathbf{v})$ and $\mathcal{G}$.

Proof [Supplement]. *Base case.* If $X \to Y$ and $X \leftrightarrow Y$, take binary U, X, Y. Model 1: $X = U$, $Y = X$. Model 2: $X = U$, $Y = U$. Both give $P(x, y) = P(X = Y = x) = P(U = x)$, but $P(Y = 1 \mid \mathrm{do}(X = 1))$ is 1 in model 1 and $P(U = 1)$ in model 2. *General case.* If a bidirected path $X \leftrightarrow W_1 \leftrightarrow \cdots \leftrightarrow W_k \leftrightarrow Y$ connects X to a child Y, extend the construction: let each latent variable on the path equal a common binary U and let every W_j copy U, while all other variables are independent noise. The two models above, embedded in this structure, still agree on $P(\mathbf{v})$ and disagree on $P(\mathbf{v}' \mid \mathrm{do}(x))$; by the graph–subgraph lemma, adding the remaining edges of $\mathcal{G}$ preserves non-identifiability. (This is the argument of Tian and Pearl 2002.) ■

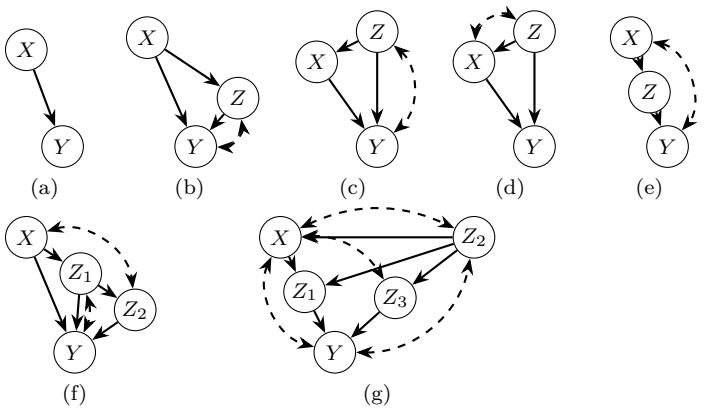
3.1.4.3 Catalogue of graphs

Graphs in which $P(y \mid \mathrm{do}(x))$ is not identifiable (Pearl 2000, Fig. 3.9):



(a) is the bow graph; (b), (c) contain the bow $X \to Z$, $X \leftrightarrow Z$ on the way to Y; in (d)–(h) X is connected to its child by a bidirected path (e.g. in (d) $X \leftrightarrow Z \leftrightarrow Y$, in (g) $X \leftrightarrow Z_2 \leftrightarrow Z_1$ with Z_1 a child of X).

Graphs in which $P(y \mid \mathrm{do}(x))$ is identifiable (Pearl 2000, Fig. 3.8):



For instance in (b) X has no incoming edges, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$;

(c) and (d) are identified by back-door adjustment for Z (in (d) the confounding between X and Z is harmless), and (e) by the front-door formula.

3.1.5 Back to Simpson's paradox

Motivation. Use the do-operator to dissolve the paradox and explain why statistics alone cannot resolve it.

3.1.5.1 Pearson's shock

Pearson, Lee and Bramley-Moore (1899) observed that mixing heterogeneous groups, each of which shows no "organic" correlation, can produce a correlation, which they called *spurious*; since no community is perfectly homogeneous, correlations are always liable to an error of unknown size, a "shock" to those who equate correlation with causation. Within each subgroup A and B may be uncorrelated, yet in the mixture a correlation appears: an artefact of heterogeneity. The same data can lead to opposite conclusions depending on whether we aggregate or stratify. This is the essence of Simpson's paradox, and it cannot be resolved by statistics alone; it requires causal reasoning.

Examples. (1) UC Berkeley's alleged sex bias in graduate admissions (Bickel et al., *Science* 1975): overall a higher admission rate among male applicants (44% vs. 30%), but within departments a slight bias in favor of women. Women tended to apply to more competitive departments, a societal pattern rather than discrimination by committees. But is conditioning on department the right thing to do? That depends on the causal model, e.g. on latent confounding between department choice and admission.

(2) "Reverse regression" (1970s–80s): in salary-discrimination cases, should one compare salaries of equally qualified men and women, or qualifications of equally paid men and women? (Pearl 2009, §4.5; Bhadane et al. 2025.)

3.1.5.2 Dissolving the paradox with the do-operator

The tables of Chapter 1 give

$$P(Y \mid F, X) < P(Y \mid F, \neg X), \qquad P(Y \mid \neg F, X) < P(Y \mid \neg F, \neg X), \qquad P(Y \mid X) > P(Y \mid \neg X).$$

If instead we have, in do-notation,

$$P(Y \mid F, \mathrm{do}(X)) < P(Y \mid F, \mathrm{do}(\neg X)), \qquad P(Y \mid \neg F, \mathrm{do}(X)) < P(Y \mid \neg F, \mathrm{do}(\neg X)),$$

and the action does not affect F (sex), then necessarily $P(Y \mid \mathrm{do}(X)) < P(Y \mid \mathrm{do}(\neg X))$. Proof:

$$P(y \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f),$$

by Rule 3 (F is not a descendant of X). Both sides are averages of the stratum-specific effects with the *same* weights $P(f)$, so the inequality is preserved.

3.1.5.3 The paradox stems from causal interpretation

Surprise surfaces only when we speak about efficacy, not about evidence for prediction. When two causal models generate the same data and one recommends the drug while the other does not, the decision must be driven by causal considerations; no statistical criterion warns us against consulting the wrong table.

Theorem 3.6 (Sure-thing principle; Savage 1972, Pearl 2000 Thm. 6.1.1)

An action X that increases the probability of an event Y in each subpopulation must also increase the probability of Y in the population as a whole, provided that the action does not change the distribution of the subpopulations.

Proof [Supplement]. Let F index the subpopulations, with $P(f \mid \mathrm{do}(x)) = P(f \mid \mathrm{do}(\neg x)) = P(f)$ by assumption. Then

$$P(Y \mid \mathrm{do}(x)) = \sum_f P(Y \mid \mathrm{do}(x), f)P(f) > \sum_f P(Y \mid \mathrm{do}(\neg x), f)P(f) = P(Y \mid \mathrm{do}(\neg x)),$$

since the inequality holds term by term and the weights $P(f)$ are the same on both sides. The reversal in Simpson's paradox is impossible here because, unlike conditional proportions, the two sides are averaged with identical weights. ■

Any statistical relationship between two variables can be reversed by including additional factors in the analysis; people think in terms of causes, not proportions. Reversal is possible in the calculus of proportions (prediction), impossible in the calculus of causation. Applications include the adjustment problem, sex and salary discrimination cases, the design of optimal treatments and fairness in machine learning.

3.1.5.4 Appendix: the Berkeley data

Department	Men applied	Men admitted	Women applied	Women admitted
A	825	62%	108	82%
B	560	56%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
Total	2691	44%	1835	30%

Women applied disproportionately to competitive departments (C–F). Which causal graph is appropriate?

- $\mathcal{G}_1$: Sex $\to$ Dept $\to$ Admit, Sex $\to$ Admit (Dept is a mediator). The total effect $P(\text{admit} \mid \text{do(sex)})$ is the aggregate rate; adjusting for Dept yields only the (controlled) direct effect. The relevant questions become: *direct effect*, is there discrimination within departments? *indirect effect*, does sex influence which department one applies to?
- $\mathcal{G}_2$: $U \to$ Dept, $U \to$ Sex, Dept $\to$ Admit, Sex $\to$ Admit, with U latent (Dept is a confounder via U). Then Dept blocks the back-door path Sex $\leftarrow U \to$ Dept $\to$ Admit, and back-door adjustment gives $P(\text{admit} \mid \text{do(sex)}) = \sum_d P(\text{admit} \mid \text{sex}, d)P(d)$, which reveals no significant discrimination within departments.

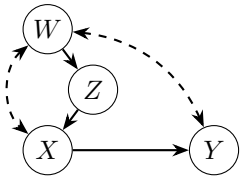
Same data, same adjustment, but different causal interpretations.

3.1.6 Summary and food for thought

- Do-calculus gives a syntactic characterization of policy evaluation for atomic interventions.
- The problem of confounding and identification is essentially solved non-parametrically.
- Simpson's paradox is mathematised and dissolved.

- Applications pervade the social and health sciences, statistics, ML and AI.

Food for thought: the napkin graph. $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$. Is $P(y \mid \mathrm{do}(x))$ identifiable?



Yes, although neither back-door nor front-door applies. Intervening on Z makes X an unconfounded treatment in the subpopulation defined by W:

$$P(y \mid \mathrm{do}(x)) = \frac{\sum_w P(x, y \mid z, w)P(w)}{\sum_w P(x \mid z, w)P(w)} \qquad \text{(for any } z\text{)}.$$

(Derivation: $P(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x), \mathrm{do}(z))$ by Rule 3; $P(x, y \mid \mathrm{do}(z)) = \sum_w P(x, y \mid z, w)P(w)$ by back-door through W; and $P(y \mid \mathrm{do}(z), \mathrm{do}(x)) = P(y \mid \mathrm{do}(z), x)$ by Rule 2, since in $\mathcal{G}_{\overline{Z}\underline{X}}$ the only paths from X to Y go through W, which is cut from X only via Z. Section 3.2 obtains this systematically.)

3.2 Algorithmic identification

Motivation. The central question is how to decide identifiability of $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ mechanically, without searching for a clever sequence of do-calculus steps.

Roadmap. (1) Define a factorization of the distributions generated by an SCM based on its causal diagram (c-components and c-factors).

(2) Establish operations that identify particular factors from a distribution.

(3) Express the target effect in terms of factors and identify each one independently.

3.2.1 Factorizing observational distributions

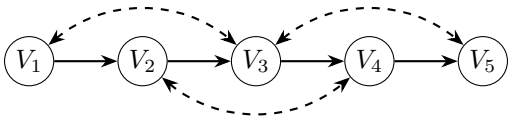
Motivation. Group variables that share latent confounders and show that the joint distribution factorizes over these groups.

3.2.1.1 Confounded components

Definition (C-component)

When V_i and V_j share a common unobservable parent U, their mechanisms are tied together through U, and V_i, V_j are in the same *confounded component* (c-component). The relation is closed under chaining: two variables are in the same c-component iff they are connected by a path of bidirected edges.

Running example: $V_1 \to V_2 \to V_3 \to V_4 \to V_5$ with U_1: $V_1 \leftrightarrow V_3$, U_2: $V_3 \leftrightarrow V_5$, U_3: $V_2 \leftrightarrow V_4$.



V_1 is in the same c-component as V_3, V_3 as V_5, hence V_1 as V_5; V_2 is with V_4. To see this easily, look at the graph induced by the bidirected edges only. The c-component relation is reflexive, symmetric and transitive, so it *partitions* the observed variables: here $\{V_1, V_3, V_5\}$ and $\{V_2, V_4\}$.

3.2.1.2 Markovian case

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

For $Z \to X$, $Z \to Y$, $X \to Y$:

$$\begin{aligned} P(z, x, y) &= \sum_{u_z} P(z \mid u_z) P(u_z) \sum_{u_x} P(x \mid z, u_x) P(u_x) \sum_{u_y} P(y \mid x, z, u_y) P(u_y) \\ &= P(z) P(x \mid z) P(y \mid x, z). \end{aligned}$$

The factors $P(v_i \mid \mathbf{pa}_i)$ are "canonical", and every one of them is computable from $P(\mathbf{v})$:

$$P(v_i \mid \mathbf{pa}_i) = \frac{\sum_{\mathbf{v} \setminus (\{v_i\} \cup \mathbf{pa}_i)} P(\mathbf{v})}{\sum_{\mathbf{v} \setminus \mathbf{pa}_i} P(\mathbf{v})},$$

$$\text{e.g. } P(z) = \sum_{x,y} P(\mathbf{v}),\ P(x \mid z) = \frac{\sum_y P(\mathbf{v})}{\sum_{x,y} P(\mathbf{v})},\ P(y \mid x, z) = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})}.$$

3.2.1.3 Semi-Markovian case

With only U_1 ($V_1 \leftrightarrow V_3$) and U_2 ($V_3 \leftrightarrow V_5$):

$$P(\mathbf{v}) = P(v_2 \mid v_1) P(v_4 \mid v_3) \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2).$$

Adding U_3 ($V_2 \leftrightarrow V_4$):

$$\begin{aligned} P(\mathbf{v}) = &\Big[\sum_{u_3} P(v_2 \mid v_1, u_3) P(v_4 \mid v_3, u_3) P(u_3) \Big] \\ &\cdot \Big[\sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2) \Big]. \end{aligned}$$

The sums separate along c-components because different c-components share no latent variable.

3.2.1.4 C-factors

These factors are long to write in terms of $P(\mathbf{v}, \mathbf{u})$, but their structure follows from the diagram. Define for $C \subseteq \mathbf{V}$

$$Q[C](\mathbf{c}, \mathbf{pa}_C) = \sum_{\mathbf{u}(C)} P(\mathbf{u}(C)) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i),$$

$$\mathbf{U}(C) = \bigcup_{V_i \in C} \mathbf{U}_i.$$

Then

$$P(\mathbf{v}) = Q[V_2, V_4](v_2, v_4, v_1, v_3) \cdot Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4),$$

abbreviated $P(\mathbf{v}) = Q[V_1, V_3, V_5]\, Q[V_2, V_4]$, without naming $\mathbf{U}$ at all. Note $Q[\mathbf{V}] = P(\mathbf{v})$, and by convention $Q[\emptyset] = 1$.

3.2.1.5 C-factors are causal effects

For $C \subseteq \mathbf{V}$, consider the effect of all other variables on C. By the truncated factorization,

$$P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = \sum_{\mathbf{u}(C)} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}(C)) = Q[C],$$

since latent variables that are not parents of C sum out. *C-factors are interventional distributions*; this is the key link between factorization and identification. For example $Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4) = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2)$ is the distribution of (V_1, V_3, V_5) when $V_2 = v_2$ and $V_4 = v_4$ are fixed by intervention.

3.2.2 C-factor algebra

Motivation. Develop the two operations, marginalizing to ancestral sets and factorizing into c-components, that drive identification.

3.2.2.1 Marginalizing variables in c-factors

C-factors behave partly like probabilities. In $Q[V_1, V_3, V_5]$ each of V_1, V_3, V_5 appears in a single term (none is a parent of another member), so summing over v_3 gives

$$\sum_{v_3} Q[V_1, V_3, V_5] = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) \Big(\sum_{v_3} P(v_3 \mid v_2, u_1, u_2) \Big) P(v_5 \mid v_4, u_2) = Q[V_1, V_5].$$

In $Q[V_1, V_2, V_3]$, by contrast, V_1 is a parent of V_2. Summing over v_1:

$$\sum_{v_1} Q[V_1, V_2, V_3] = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) \sum_{v_1} P(v_1 \mid u_1) P(v_2 \mid v_1, u_3),$$

whereas

$$Q[V_2, V_3](v_2, v_3, v_1) = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) P(v_2 \mid v_1, u_3)$$

keeps v_1 as a fixed argument. The coupling of v_1 with u_1 (shared with V_3) inside the sum prevents the reduction, so $\sum_{v_1} Q[V_1, V_2, V_3] \neq Q[V_2, V_3]$ in general: one can only sum out variables that do not feed into the remaining ones.

Lemma 3.7 (Ancestral reduction)

Let $W \subseteq C \subseteq \mathbf{V}$. If W is ancestral in $\mathcal{G}[C]$ (it contains all its ancestors within the subgraph induced by C), then

$$Q[W] = \sum_{\mathbf{c} \setminus \mathbf{w}} Q[C].$$

Proof [Supplement]. Order C topologically and sum out the variables of $C \setminus W$ one at a time from the last. When $V \in C \setminus W$ is summed, it has no child left among the remaining variables: a child in W would make V an ancestor of W in $\mathcal{G}[C]$, hence $V \in W$ since W is ancestral; children in $C \setminus W$ come later in the order and were already summed out. So v appears in $Q[C] = \sum_{\mathbf{u}} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$ only in its own factor, which sums to one; the latent variables attached only to summed variables are then integrated out as well. What remains is $Q[W]$. ■

Example: For $C = \{V_1, V_2, V_3\}$ in $\mathcal{G}[C]$ ($V_1 \to V_2 \to V_3$): $\{V_1, V_2\}$ and $\{V_1\}$ are ancestral, $\{V_2, V_3\}$ is not.

3.2.2.2 C-component factorization

Lemma 3.8 (Tian 2002)

For any $H \subseteq \mathbf{V}$, let $H_1, \dots, H_k$ be the c-components of $\mathcal{G}[H]$. Then

$$Q[H] = \prod_{j=1}^{k} Q[H_j].$$

Moreover, let $V^{(1)} < \cdots < V^{(|H|)}$ be a topological order of $\mathcal{G}[H]$ and $H^{\le i}$ the variables of H up to and including $V^{(i)}$. Then each factor is computable from $Q[H]$:

$$Q[H_j] = \prod_{V^{(i)} \in H_j} \frac{Q[H^{\le i}]}{Q[H^{\le i-1}]}, \qquad Q[H^{\le i}] = \sum_{\mathbf{h} \setminus \mathbf{h}^{\le i}} Q[H].$$

Proof [Supplement]. The latent variables split into groups, one for each c-component, since a latent variable with children in two c-components would join them by a bidirected path. In $Q[H] = \sum_{\mathbf{u}} \prod_{V_i \in H} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$, group the factors and the latent sum by c-component: the latent variables of H_j appear only in the factors of H_j, and different groups are independent under $P(\mathbf{u})$, so the sum factorizes into $\prod_j Q[H_j]$. For the second part, $H^{\le i}$ is ancestral in $\mathcal{G}[H]$ (all predecessors in a topological order), so $Q[H^{\le i}] = \sum_{h \setminus h^{\le i}} Q[H]$ by ancestral reduction; the ratio $Q[H^{\le i}]/Q[H^{\le i-1}]$ is therefore a function of $Q[H]$, and multiplying these ratios over the variables of H_j reproduces $Q[H_j]$ (each latent group of H_j is integrated exactly once). ■

($H^{\le i}$ is ancestral in $\mathcal{G}[H]$, so $Q[H^{\le i}]$ follows by ancestral reduction.)

Example 3.3 (Factorizing $P(\mathbf{v})$ by c-components)

Graph: $Z_2 \to X$, $Z_2 \to Z_1$, $Z_2 \to Z_3$, $X \to Z_1$, $Z_1 \to Y$, $Z_3 \to Y$, $Z_2 \leftrightarrow X$, $X \leftrightarrow Z_3$, $X \leftrightarrow Y$, $Z_2 \leftrightarrow Y$. The c-components are $\{Z_2, Z_3, X, Y\}$ and $\{Z_1\}$, so

$$Q[\mathbf{V}] = Q[Z_2, Z_3, X, Y]\, Q[Z_1],$$
$$P(\mathbf{v}) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, z_3, x, y)) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, x)),$$

the last step because only the parents Z_2, X of Z_1 matter (Rule 3).

Example 3.4 (Running example)

With $H = \mathbf{V}$ and the order $V_1 < \cdots < V_5$, $Q[\mathbf{V}] = Q[V_1, V_3, V_5] Q[V_2, V_4]$ and

$$\begin{aligned} Q[V_1, V_3, V_5] &= \frac{Q[V_1]}{Q[\emptyset]} \cdot \frac{Q[V_1, V_2, V_3]}{Q[V_1, V_2]} \cdot \frac{Q[V_1, \dots, V_5]}{Q[V_1, \dots, V_4]} \\ &= \frac{\sum_{v_2, \dots, v_5} P(\mathbf{v})}{\sum_{\mathbf{v}} P(\mathbf{v})} \cdot \frac{\sum_{v_4, v_5} P(\mathbf{v})}{\sum_{v_3, v_4, v_5} P(\mathbf{v})} \cdot \frac{P(\mathbf{v})}{\sum_{v_5} P(\mathbf{v})} \\ &= \frac{P(v_1)}{1} \cdot \frac{P(v_1, v_2, v_3)}{P(v_1, v_2)} \cdot \frac{P(\mathbf{v})}{P(v_1, \dots, v_4)} = P(v_1) P(v_3 \mid v_1, v_2) P(v_5 \mid v_1, \dots, v_4), \\ Q[V_2, V_4] &= \frac{Q[V_1, V_2]}{Q[V_1]} \cdot \frac{Q[V_1, \dots, V_4]}{Q[V_1, V_2, V_3]} = P(v_2 \mid v_1) P(v_4 \mid v_1, v_2, v_3). \end{aligned}$$

In $\mathcal{G}[\{V_2, V_4\}]$ there are no directed edges ($V_2 \leftrightarrow V_4$ only), so both $\{V_2\}$ and $\{V_4\}$ are ancestral there:

$$Q[V_2] = \sum_{v_4} Q[V_2, V_4] = P(v_2 \mid v_1) \sum_{v_4} P(v_4 \mid v_1, v_2, v_3) = P(v_2 \mid v_1),$$

$$Q[V_4] = \sum_{v_2} P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3).$$

3.2.2.3 Visualization

Starting from $Q[\mathbf{V}]$ one can move down a lattice of c-factors: summing out a variable that is "last" (ancestral reduction, e.g. $Q[\mathbf{V}] \to Q[V_{1234}] \to Q[V_{123}] \to Q[V_{12}] \to Q[V_1]$) or splitting along c-components (e.g. $Q[\mathbf{V}] \to Q[V_{135}] \cdot Q[V_{24}]$, $Q[V_{1234}] \to Q[V_{13}] \cdot Q[V_{24}]$, $Q[V_{123}] \to Q[V_{13}] \cdot Q[V_2]$). From $Q[V_{135}]$ every subset is reachable by summation since $\mathcal{G}[\{V_1, V_3, V_5\}]$ has no directed edges, and $Q[V_{15}] = Q[V_1]Q[V_5]$ factorizes because V_1, V_5 are not bidirected-connected inside $\{V_1, V_5\}$. Once a factor is fully factorized it cannot be factorized further; summation is commutative.

Summary of the algebra.

1. *Reduce to an ancestral set:* $Q[W] = \sum_{\mathbf{c}\setminus\mathbf{w}} Q[C]$ if W is ancestral in $\mathcal{G}[C]$.
2. *Factorize into c-components:* $Q[H] = \prod_j Q[H_j]$, with $H_1, \ldots, H_k$ the c-components of $\mathcal{G}[H]$, each computable from $Q[H]$.

These two operations yield an if-and-only-if condition for identifiability.

3.2.3 Causal queries in terms of c-factors: examples

Motivation. Rederive the back-door, front-door and napkin formulas with c-factor algebra.

3.2.3.1 Back-door

$Z \to X$, $Z \to Y$, $X \to Y$ (Markovian): $P(\mathbf{v}) = Q[X]Q[Y]Q[Z]$. In $\mathcal{G}_{\overline{X}}$, $P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$ with

$$Q[Y] = \frac{Q[X,Y,Z]}{Q[X,Z]} = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})} = P(y \mid x, z), \qquad Q[Z] = \sum_{x,y} Q[X,Y,Z] = P(z),$$

so $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid z, x)P(z)$.

3.2.3.2 Front-door

$X \to Z \to Y$, $X \leftrightarrow Y$: $P(\mathbf{v}) = Q[X,Y] \cdot Q[Z]$ with (order $X < Z < Y$)

$$Q[X,Y] = \frac{Q[X]}{Q[\emptyset]} \cdot \frac{Q[X,Z,Y]}{Q[X,Z]} = P(x)P(y \mid x, z), \qquad Q[Z] = \frac{Q[X,Z]}{Q[X]} = P(z \mid x).$$

Then

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y, z \mid \mathrm{do}(x)) = \sum_z Q[Y,Z] = \sum_z Q[Y]Q[Z],$$

and since $\{Y\}$ is ancestral in $\mathcal{G}[\{X,Y\}]$ (no directed edge between them), $Q[Y] = \sum_{x'} Q[X,Y] = \sum_{x'} P(x')P(y \mid x', z)$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(x')P(y \mid x', z)$.

3.2.3.3 Napkin

$W_1 \to W_2 \to X \to Y$, $W_1 \leftrightarrow X$, $W_1 \leftrightarrow Y$. Then $P(\mathbf{v}) = Q[W_1, X, Y]\, Q[W_2]$ and

$$P(y \mid \mathrm{do}(x)) = \sum_{w_1,w_2} Q[W_1, W_2, Y] = Q[Y],$$

because $\{Y\}$ is ancestral in $\mathcal{G}[\{W_1, W_2, Y\}]$ (in that subgraph Y has no parents). Can we get $Q[Y]$ from $Q[W_1, X, Y]$, which is computable from $Q[\mathbf{V}]$? With the order $W_1 < W_2 < X < Y$,

$$Q[W_1, X, Y] = \frac{Q[W_1]}{Q[\emptyset]} \cdot \frac{Q[W_1, W_2, X]}{Q[W_1, W_2]} \cdot \frac{Q[W_1, W_2, X, Y]}{Q[W_1, W_2, X]} = P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x).$$

$\{X, Y\}$ is ancestral in $\mathcal{G}[\{W_1, X, Y\}]$, so $Q[X, Y] = \sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)$. In $\mathcal{G}[\{X, Y\}]$, X and Y are no longer bidirected-connected (W_1 is gone), so $Q[X, Y] = Q[X]Q[Y]$, giving

$$P(y \mid \mathrm{do}(x)) = Q[Y] = \frac{Q[X, Y]}{Q[X]} = \frac{Q[X, Y]}{\sum_y Q[X, Y]} = \frac{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)}{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)}.$$

The same result by do-calculus.

$$\begin{aligned}
P(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x, w_1, w_2)) && \text{Rule 3} \\
&= P(y \mid x, \mathrm{do}(w_1, w_2)) && \text{Rule 2} \\
&= \frac{P(y, x \mid \mathrm{do}(w_1, w_2))}{P(x \mid \mathrm{do}(w_1, w_2))} = \frac{P(y, x \mid \mathrm{do}(w_2))}{P(x \mid \mathrm{do}(w_2))} && \text{Rule 3} \\
&= \frac{\sum_{w_1} P(y, x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))}{\sum_{w_1} P(x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))} && \\
&= \frac{\sum_{w_1} P(y, x \mid w_2, w_1)P(w_1)}{\sum_{w_1} P(x \mid w_2, w_1)P(w_1)} && \text{Rules 2 and 3.}
\end{aligned}$$

3.2.3.4 Overview of the three derivations

Back-door	Front-door	Napkin
$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X, Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[W_1, X, Y]\,Q[W_2]$
	$Q[X, Y] \xrightarrow{\text{sum}} Q[Y]$	$Q[W_1, X, Y] \xrightarrow{\text{sum}} Q[X, Y] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]$
$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = Q[Y]$

3.2.4 A general identification algorithm

Motivation. Combine the two operations into a sound and complete procedure.

3.2.4.1 Reducing the query to c-factors

Given $\mathcal{G}$ and $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, first use Rule 3 to enlarge the intervention: every variable that is not an ancestor of $\mathbf{Y}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$ (non-ancestors of $\mathbf{Y}$, and variables that affect $\mathbf{Y}$ only through $\mathbf{X}$) can be intervened on without changing the query. Let $C = An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c} \setminus \mathbf{y}} P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} Q[\mathbf{V} \setminus \mathbf{X}] = \sum_{\mathbf{c} \setminus \mathbf{y}} Q[C].$$

If $\mathcal{G}[C]$ has c-components $C_1, \ldots, C_k$, then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c} \setminus \mathbf{y}} \prod_{i=1}^{k} Q[C_i],$$

while the data give $P(\mathbf{v}) = \prod_j Q[T_j]$ with $T_1, \ldots, T_\ell$ the c-components of $\mathcal{G}$, each $Q[T_j]$ computable from $P(\mathbf{v})$. Each C_i lies inside a single T_j. It remains to decide whether $Q[C_i]$ is computable from $Q[T_j]$.

3.2.4.2 The Identify procedure

Algorithm 2 IDENTIFY$(C, T, Q[T], \mathcal{G})$ (decides whether $Q[C]$ is computable from $Q[T]$, $C \subseteq T$)

$A \leftarrow An(C)_{\mathcal{G}[T]}$
if $A = C$ **then** ▷ C is ancestral in T
  return $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$ ▷ marginalize
else if $A = T$ **then** ▷ nothing to marginalize, unable to factorize
  return FAIL
else
  $A_i \leftarrow$ the c-component of $\mathcal{G}[A]$ containing C
  obtain $Q[A] = \sum_{\mathbf{t}\setminus\mathbf{a}} Q[T]$, then $Q[A_i]$ by c-component factorization ▷ marginalize, then factorize
  return IDENTIFY$(C, A_i, Q[A_i], \mathcal{G})$

The three cases. (1) $A = C$: marginalize, $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$.

(2) $A = T$: C is strictly inside T, but T is its own ancestral closure and a single c-component, so T cannot be shrunk: fail.

(3) $C \subsetneq A \subsetneq T$: marginalize to A, factorize A into c-components $A_i, A_j, \ldots$, keep the one containing C (it must be contained in a single component because C is a c-component of $\mathcal{G}[C]$ and $C \subseteq A_i$), and recurse. Each step strictly shrinks T (recursive shrinking: $T \supsetneq A_i \supsetneq \cdots$), so the procedure terminates in at most $|T|$ steps.

Theorem 3.9 (Completeness; Tian 2002)

$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $\mathcal{G}$ and $P(\mathbf{v})$ if and only if each c-factor $Q[C_i]$ is identified by IDENTIFY$(C_i, T_i, Q[T_i], \mathcal{G})$, where T_i is the c-component of $\mathcal{G}$ containing C_i.

Proof [Supplement]. *If:* each call to IDENTIFY returns an expression in $Q[T_i]$, which is identified from $P(\mathbf{v})$ by the c-component factorization (the lemma above with $H = \mathbf{V}$); combining the factors by the generalized truncated factorization gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$. *Only if:* when IDENTIFY fails for some C_i, the failing configuration $A = T$ with $C_i \subsetneq T$ contains a hedge, and Shpitser and Pearl (2006) construct from it two models that agree on $P(\mathbf{v})$ but not on the query; this construction is not reproduced here. ■

See also Huang and Valtorta (2006) and Shpitser and Pearl (2006), who proved completeness of the equivalent ID algorithm and, through it, of do-calculus. The failure case $A = T$ corresponds to a graphical structure (a "hedge") from which two models agreeing on $P(\mathbf{v})$ but not on the query can be built.

Example 3.5 (Bow graph fails)

$X \to Y$, $X \leftrightarrow Y$, query $P(y \mid \mathrm{do}(x))$: $C = An(Y)_{\mathcal{G}_{\overline{X}}} = \{Y\}$, $T = \{X, Y\}$. IDENTIFY$(\{Y\}, \{X, Y\}, \ldots)$: $A = An(Y)_{\mathcal{G}[T]} = \{X, Y\} = T$, so it fails, matching Chapter 2.

Example 3.6 (Front-door succeeds)

$X \to Z \to Y$, $X \leftrightarrow Y$: $C = \{Z, Y\}$ with c-components $\{Z\}$, $\{Y\}$ in $\mathcal{G}[C]$. $Q[Z]$ is itself a c-component of $\mathcal{G}$. For $Q[Y]$ with $T = \{X, Y\}$: $A = An(Y)_{\mathcal{G}[T]} = \{Y\} = C$ (the edge $X \to Z \to Y$ is not inside T), so $Q[Y] = \sum_x Q[X, Y]$, as found above.

Chapter 4. Linear Structural Causal Models

Motivation. The central question is when, in a linear SCM with Gaussian errors, a structural coefficient (a causal effect) can be computed from the observed covariance matrix, and when ordinary regression returns it.

Outline:

(1) linear SCMs and Wright's path-tracing rules;

(2) when regression can and cannot be used to find causal effects (single-door, back-door);

(3) algorithmic identification (systems of polynomial equations, instrumental variables, conditional IVs, instrumental sets), with appendices on complexity, regression in a reversed graph, and Gröbner bases.

4.1 Linear structural causal models

Motivation. Specialize SCMs to linear mechanisms and read covariances off the graph.

4.1.1 Definition and conventions

A linear SCM is a system of linear equations representing the ground truth,

$$Y = \sum_i \lambda_{x_i y} X_i + \mathcal{E}_y,$$

where the X_i are the direct causes of Y and $\lambda_{x_i y}$ is the change in Y per unit change of X_i. All covariances among the errors $\mathcal{E}$ are explicitly specified. Without loss of generality the data are normalized ($\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$), and $\mathcal{E} \sim \mathcal{N}$, so the observed distribution is fully specified by the covariance matrix $\Sigma = (\sigma_{ij})$.

The only substantive change from the non-parametric case is that f_i becomes linear:

$$V_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i) \quad \rightsquigarrow \quad V_i \leftarrow \sum_{V_j \in Pa_i} \lambda_{ji} V_j + \mathcal{E}_i.$$

(1) λ_{ji} is the *structural coefficient*;

(2) by convention U_i is renamed $\mathcal{E}_i$;

(3) knowing all λ_{ji} yields all causal effects.

Example 4.1

Non-parametric $X \leftarrow f_X(U_X)$, $W \leftarrow f_W(U_W)$, $Y \leftarrow f_Y(X, W, U_Y)$ becomes $X \leftarrow \mathcal{E}_x$, $W \leftarrow \mathcal{E}_w$, $Y \leftarrow \lambda_{xy} X + \lambda_{wy} W + \mathcal{E}_y$. Writing the coefficients on the edges of the graph fully specifies the model.

Latent confounding. The covariance $\epsilon_{ij} = \mathbb{E}[\mathcal{E}_i \mathcal{E}_j]$ is attached to a bidirected edge $V_i \leftrightarrow V_j$. It is unobserved, but drawn on the graph like a structural coefficient. Unlike the non-parametric case, a bidirected edge here represents *error covariance*, not necessarily a shared latent variable.

Interventions. For $X \xrightarrow{\lambda} Y$,

$$\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \mathbb{E}[\lambda x + \mathcal{E}_y] = \lambda x + \mathbb{E}[\mathcal{E}_y] = \lambda x.$$

4.1.2 The identification problem

We are given (i) a hypothesized graph $\mathcal{G}$ (what causes what, and which pairs share unknown common causes), (ii) observational data $\{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$ on all observed variables, but (iii) not the structural coefficients λ (the causal effects we want) nor the error covariances ϵ.

4.1.3 Connecting observed and unobserved quantities

For $X \xrightarrow{\lambda} Y$ without confounding,

$$\sigma_{xy} = \mathbb{E}[XY] = \mathbb{E}[X(\lambda X + \mathcal{E}_y)] = \lambda \underbrace{\mathbb{E}[XX]}_{1} + \underbrace{\mathbb{E}[X\mathcal{E}_y]}_{0} = \lambda.$$

With confounding $X \leftrightarrow Y$ (ϵ_{xy}), $X = \mathcal{E}_x$ and

$$\sigma_{xy} = \lambda \mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda + \epsilon_{xy}.$$

A curious property. For the chain $X \xrightarrow{\lambda_{xz}} Z \xrightarrow{\lambda_{zy}} Y$:

$$\begin{aligned}\sigma_{xy} &= \mathbb{E}[X(\lambda_{zy}Z + \mathcal{E}_y)] = \lambda_{zy}\mathbb{E}[XZ] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{zy}\mathbb{E}[X(\lambda_{xz}X + \mathcal{E}_z)] \\ &= \lambda_{zy}\lambda_{xz}\mathbb{E}[XX] + \lambda_{zy}\mathbb{E}[X\mathcal{E}_z] = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\mathbb{E}[\mathcal{E}_x\mathcal{E}_z].\end{aligned}$$

Without confounding the last term vanishes: $\sigma_{xy} = \lambda_{zy}\lambda_{xz}$. If $X \leftrightarrow Z$ with covariance ϵ_{xz}, then $\sigma_{xy} = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\epsilon_{xz}$.

4.1.4 Treks and Wright's rule

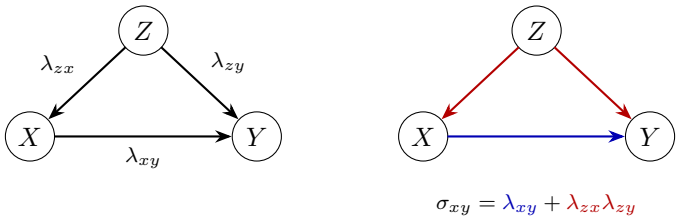


Figure. *Wright's rule for standardized variables: the covariance of X and Y is the sum, over the treks between them, of the products of path coefficients: the direct path (blue) and the trek through the common cause Z (red).*

For $Z \to X \to Y$ with $Z \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\lambda_{zy}$. The terms correspond to the paths $X \to Y$ and $X \leftarrow Z \to Y$.

Theorem 4.1 (Wright's rules, 1921)

For an acyclic linear SCM with standardized variables, σ_{xy} equals the sum, over all *open* paths between X and Y (non-self-intersecting paths without colliding arrowheads, i.e. treks; a bidirected edge counts as one step contributing ϵ), of the product of the path coefficients along the path. In particular $\sigma_{xy} = 0$ when X and Y are d-separated by $\emptyset$.

Proof [Supplement]. Write the model as $\mathbf{V} = \Lambda^{\mathsf{T}}\mathbf{V} + \boldsymbol{\epsilon}$ with Λ the matrix of path coefficients (strictly triangular in a topological order) and $\Omega = \mathrm{Cov}(\boldsymbol{\epsilon})$ (off-diagonal entries are the bidirected coefficients). Then $\mathbf{V} = (I - \Lambda^{\mathsf{T}})^{-1}\boldsymbol{\epsilon}$ and $\Sigma = (I-\Lambda)^{-\mathsf{T}}\Omega(I-\Lambda)^{-1}$. Since Λ is nilpotent, $(I-\Lambda)^{-1} = \sum_{k \geq 0} \Lambda^k$, and $(\Lambda^k)_{ij}$ is the sum over directed paths of length k from i to j of the products of coefficients. Hence

$$\sigma_{xy} = \sum_{a,b} \Big(\sum_{\text{directed } a \to x} \prod \lambda \Big) \omega_{ab} \Big(\sum_{\text{directed } b \to y} \prod \lambda \Big),$$

a sum over pairs of directed paths ending at x and y joined at their sources by either a common source $a = b$ (weight ω_{aa}, the error variance) or a bidirected edge $a \leftrightarrow b$ (weight ω_{ab}). These are exactly the treks between X and Y. With standardized variables, the error variance at the source of a trek through a common cause combines with the source's other contributions to give the variance 1 when the trek is read through the source node itself, which yields Wright's form with a factor 1 at the top of each trek. If $\emptyset$ d-separates X and Y, there are no treks and $\sigma_{xy} = 0$. ■

Consequently, if $X \xrightarrow{\lambda} Y$ is in the model, $\sigma_{xy} = \lambda +$ (other paths), and $\sigma_{xy} = \lambda$ if X and Y are d-separated in $\mathcal{G}_\lambda$ (the graph with that edge removed).

Example 4.2

$W \to X$, $W \to Y$, $Z \to W$, $Z \to X$, $X \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{wx}\lambda_{wy} + \lambda_{zx}\lambda_{wz}\lambda_{wy}$ (paths $X \to Y$, $X \leftarrow W \to Y$, $X \leftarrow Z \to W \to Y$).

Example 4.3

$W \to X$, $W \to V \to Y$, $X \to Z \to Y$, $X \leftrightarrow Z$: $\sigma_{xy} = (\lambda_{xz} + \epsilon_{xz})\lambda_{zy} + \lambda_{wx}\lambda_{wv}\lambda_{vy}$.

4.2 Linear regression and causal effects

Motivation. Clarify what regression computes and when it coincides with a structural coefficient.

4.2.1 The medical researcher

We want to know whether a new drug helps; our recommendation will be followed by doctors nationwide.

Step 1. gather data on how much drug patients took (X) and the amount of a biomarker (antibodies) in their blood (Y).

Step 2. regress $Y = \beta X + e$, giving $\beta = 0.375$; the drug seems beneficial and is authorized.

Step 3. given to everyone, the drug shows a clear *negative* relation with slope -1: it hurts. Not enough data is not the explanation. Why did regression fail (red line) and can the true causal effect (green line) be recovered?

4.2.2 The implicit assumption: no confounding

Attributing causal meaning to β presumes $X \leftarrow \mathcal{E}_x$, $Y \leftarrow \lambda_{xy}X + \mathcal{E}_y$ with $\mathcal{E}_x \perp\!\!\!\perp \mathcal{E}_y$; then $\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{xy}$.

The ground-truth model. If one cannot ascertain the absence of confounding between X and Y, the corresponding model is $X \leftarrow \mathcal{E}_x$, $Y \leftarrow \lambda_{xy}X + \mathcal{E}_y$ with $\mathcal{E}_x, \mathcal{E}_y$ *correlated* ($X \leftrightarrow Y$). Then regression of Y on X is biased:

$$\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda_{xy} + \epsilon_{xy},$$

and it is provably impossible to disentangle the effect of the drug from the confounding: λ_{xy} is not identifiable. This explains the medical researcher's failure (0.375 versus -1).

What does regression compute? For $Y = \beta X + e$, minimize $\mathbb{E}[(Y - \beta X)^2] = 1 - 2\beta\sigma_{xy} + \beta^2$. The derivative $-2\sigma_{xy} + 2\beta$ vanishes at $\beta = \sigma_{xy}$: *the regression coefficient is just the covariance* (for standardized variables).

4.2.3 Regression equation versus structural equation

- **Regression equation** $Y = \beta X + e$ with $e \perp\!\!\!\perp X$ by construction. Solved, $\beta = \sigma_{xy}$, denoted r_{yx}. It makes *no* causal claim.
- **Structural equation** $Y \leftarrow \lambda X + \mathcal{E}_y$ (for $X \to Y$). It claims $\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \lambda x$, which can be tested and falsified.

Confusing the two has been called the "confusion of the century". Causal claims via regression implicitly assume a model of the form $X_1 \xrightarrow{b_1} Y \xleftarrow{b_2} X_2$ (with the regressors unconfounded with Y), in which OLS of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ gives $\beta_1 = b_1$, $\beta_2 = b_2$.

Warning: make sure you have the right graph. If instead Y causes both, $X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$, simple regressions give $r_{yx_1} = b_1$, $r_{yx_2} = b_2$, but multiple regression gives

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}$$

(derived in Appendix 4.6).

4.2.4 Testing conditional independence with regression

For Gaussian variables, $(X \perp\!\!\!\perp Y \mid Z)$ iff $r_{yx \cdot z} = 0$, the partial regression coefficient of X in $Y = \alpha X + \beta Z$. If $\alpha \neq 0$, Z does not render X and Y independent; if $\alpha = 0$, $(X \perp\!\!\!\perp Y \mid Z)$.

4.2.5 Single-door criterion: direct effects

Model: $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$ (ϵ_{zy}). Regress $Y = \alpha X + \beta Z + e$:

$$\mathbb{E}[(Y - \alpha X - \beta Z)^2] = 1 - 2\alpha\sigma_{xy} - 2\beta\sigma_{yz} + 2\alpha\beta\sigma_{xz} + \alpha^2 + \beta^2,$$

$$\frac{\partial}{\partial \alpha} = 2\alpha - 2\sigma_{xy} + 2\beta\sigma_{xz}, \qquad \frac{\partial}{\partial \beta} = 2\beta - 2\sigma_{yz} + 2\alpha\sigma_{xz}.$$

Setting both to zero and substituting the Wright expressions ($\sigma_{xz} = \lambda_{zx}$, $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\epsilon_{zy}$, $\sigma_{yz} = \lambda_{xy}\lambda_{zx} + \epsilon_{zy}$):

$$\begin{aligned} \lambda_{xy} + \lambda_{zx}\epsilon_{zy} &= \alpha + \beta\lambda_{zx}, \\ \lambda_{xy}\lambda_{zx} + \epsilon_{zy} &= \alpha\lambda_{zx} + \beta, \end{aligned} \qquad \Longrightarrow \qquad \alpha = \lambda_{xy}, \quad \beta = \epsilon_{zy}.$$

(The linear system has a unique solution when $\lambda_{zx}^2 \neq 1$, and the stated values satisfy it.) So the coefficient of X is the causal effect, even though Z is confounded with Y.

Theorem 4.2 (Single-door criterion; Pearl 2000)

Let λ be the coefficient of $X \to Y$ and $\mathcal{G}_\lambda$ the diagram with $X \to Y$ removed. λ is identifiable if there is a set $\mathbf{Z}$ such that (i) $\mathbf{Z}$ contains no descendant of Y, and (ii) $\mathbf{Z}$ d-separates X from Y in $\mathcal{G}_\lambda$. Moreover $\lambda = r_{yx \cdot \mathbf{Z}}$.

Proof [Supplement]. Let β be the coefficient of X in the population regression of Y on X and $\mathbf{Z}$; it is the partial regression coefficient $r_{yx \cdot \mathbf{Z}}$. Write $Y = \lambda X + R$, where R collects all other parent contributions and the error of Y. Then $r_{yx \cdot \mathbf{Z}} = \lambda + r_{rx \cdot \mathbf{Z}}$, and $r_{rx \cdot \mathbf{Z}} = 0$ iff R and X are uncorrelated given $\mathbf{Z}$ (Gaussian case: conditionally independent). Every trek between X and R corresponds to a path between X and Y in $\mathcal{G}_\lambda$ (the edge $X \to Y$ removed); condition (ii) blocks all of them given $\mathbf{Z}$, and condition (i) ensures that conditioning on $\mathbf{Z}$ does not open a path through a descendant of Y (which would involve R itself). Hence $r_{rx \cdot \mathbf{Z}} = 0$ and $\lambda = r_{yx \cdot \mathbf{Z}}$. ■

Example 4.4

$A \to B$, $A \to Y$, $B \to Y$: $\lambda_{ay} = r_{ya \cdot b}$, $\lambda_{by} = r_{yb \cdot a}$.

Example 4.5

$X \to Y$, $X \to W$, $Y \to W$, $Y \to Z$, $W \to Z$, $X \leftrightarrow Z$, $Y \leftrightarrow Z$: $\lambda_{xy} = r_{yx}$ ($\emptyset$ separates X, Y once $X \to Y$ is removed, since all other paths meet a collider), and $\lambda_{wz} = r_{zw \cdot yx}$ ($\{Y, X\}$ blocks $W \leftarrow Y \to Z$, $W \leftarrow Y \leftrightarrow Z$, $W \leftarrow X \leftrightarrow Z$).

4.2.6 Back-door criterion: total effects

Theorem 4.3 (Back-door, linear; Pearl 2000)

The total effect of X on Y is identifiable if there is $\mathbf{Z}$ with no descendant of X that d-separates X from Y in $\mathcal{G}_{\underline{X}}$; it equals $r_{yx \cdot \mathbf{Z}}$.

Proof [Supplement]. In a linear Gaussian model $\mathbb{E}[Y \mid \mathrm{do}(x), \mathbf{z}]$ is linear in x with slope equal to the total effect τ (sum of products of coefficients along directed paths). By the back-door adjustment theorem (Chapter 2), $\mathbb{E}[Y \mid \mathrm{do}(x)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid x, \mathbf{z}] P(\mathbf{z})$ (an integral in the continuous case). In the Gaussian case $\mathbb{E}[Y \mid x, \mathbf{z}] = \alpha + r_{yx \cdot \mathbf{Z}} x + \boldsymbol{\beta}^{\mathsf{T}} \mathbf{z}$, so $\mathbb{E}[Y \mid \mathrm{do}(x)] = \alpha + r_{yx \cdot \mathbf{Z}} x + \boldsymbol{\beta}^{\mathsf{T}} \mathbb{E}\mathbf{Z}$, whose slope in x is $r_{yx \cdot \mathbf{Z}}$. Hence $\tau = r_{yx \cdot \mathbf{Z}}$. ■

Example 4.6

$W \to X$, $W \leftrightarrow X$, $W \to Y$, $X \to Z \to Y$, $X \to Y$, $Z \leftrightarrow Y$. The total effect is $\lambda_{xz}\lambda_{zy} + \lambda_{xy}$; by the back-door criterion with $\{W\}$ it equals $r_{yx \cdot w}$. (The coefficient λ_{xy} alone is not obtainable by regression on Z because $Z \leftrightarrow Y$.)

4.3 Algorithmic identification methods

Motivation. View identification as solving the polynomial system linking Σ to (λ, ϵ).

4.3.1 The equations of linear identification

For $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$:

$$\Sigma = \underbrace{\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}}_{\text{computable}} = \underbrace{\begin{bmatrix} 1 & \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & \lambda_{zx} \\ \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & 1 & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} \\ \lambda_{zx} & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} & 1 \end{bmatrix}}_{\text{expressions with unobservables}}.$$

Can λ_{xy} be solved in terms of σ? Here $\sigma_{xz} = \lambda_{zx}$, and

$$\sigma_{xy} = \lambda_{xy} + \sigma_{xz}\epsilon_{zy}, \qquad \sigma_{zy} = \sigma_{xz}\lambda_{xy} + \epsilon_{zy}$$

are two full-rank linear equations in two unknowns (Gaussian elimination / row reduction):

$$\lambda_{xy} = \frac{\sigma_{xy} - \sigma_{xz}\sigma_{zy}}{1 - \sigma_{xz}^2}, \qquad \epsilon_{zy} = \frac{\sigma_{zy} - \sigma_{xz}\sigma_{xy}}{1 - \sigma_{xz}^2}.$$

Non-identifiability. For $X \to Y$ with $X \leftrightarrow Y$: $\sigma_{xy} = \lambda_{xy} + \epsilon_{xy}$ is one equation in two unknowns; infinitely many $(\lambda_{xy}, \epsilon_{xy})$ give the same covariance matrix.

Finite identifiability. $X \to W$, $X \to Z$, $X \to Y$ each also confounded with X ($X \leftrightarrow W$, $X \leftrightarrow Z$, $X \leftrightarrow Y$). Wright's rules give six equations in six unknowns:

$$\begin{aligned} \sigma_{xw} &= \lambda_{xw} + \epsilon_{xw}, & \sigma_{wz} &= \lambda_{xw}\lambda_{xz} + \lambda_{xz}\epsilon_{xw} + \lambda_{xw}\epsilon_{xz}, \\ \sigma_{xz} &= \lambda_{xz} + \epsilon_{xz}, & \sigma_{wy} &= \lambda_{xw}\lambda_{xy} + \lambda_{xw}\epsilon_{xy} + \lambda_{xy}\epsilon_{xw}, \\ \sigma_{xy} &= \lambda_{xy} + \epsilon_{xy}, & \sigma_{zy} &= \lambda_{xz}\lambda_{xy} + \lambda_{xz}\epsilon_{xy} + \lambda_{xy}\epsilon_{xz}. \end{aligned}$$

Eliminating: substitute $\epsilon = \sigma - \lambda$ to get $\sigma_{wz} = \lambda_{xz}\sigma_{xw} + \lambda_{xw}\sigma_{xz} - \lambda_{xw}\lambda_{xz}$ (and similarly for σ_{wy}, σ_{zy}); solve the last two for $\lambda_{xw} = (\sigma_{wy} - \lambda_{xy}\sigma_{xw})/(\sigma_{xy} - \lambda_{xy})$ and $\lambda_{xz} = (\sigma_{zy} - \lambda_{xy}\sigma_{xz})/(\sigma_{xy} - \lambda_{xy})$; plug into the first and clear denominators. The result is a quadratic in λ_{xy}:

$$0 = (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\lambda_{xy}^2 + 2(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})\lambda_{xy} + (\sigma_{yw}\sigma_{xz}\sigma_{xy} + \sigma_{yz}\sigma_{xy}\sigma_{xw} - \sigma_{xy}^2\sigma_{wz} - \sigma_{yz}\sigma_{yw}),$$

so

$$\lambda_{xy} = \frac{-(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy}) \pm \sqrt{(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})^2 - (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\,c}}{\sigma_{xw}\sigma_{xz} - \sigma_{wz}},$$

with c the constant term. At most two values of λ_{xy} are consistent with the data: *finite identifiability*.

The three cases. *Identifiable*: a single value of λ_{xy} is consistent with the data. *Finitely identifiable*: finitely many values. *Not identifiable*: infinitely many.

Example 4.7

$W \to Z$, $W \to X$, $Z \to X$, $Z \leftrightarrow X$, $X \to Y$, $W \leftrightarrow Y$:

$$\begin{aligned} \sigma_{wz} &= \lambda_{wz}, & \sigma_{wy} &= (\lambda_{wx} + \lambda_{wz}\lambda_{zx})\lambda_{xy} + \epsilon_{wy}, \\ \sigma_{wx} &= \lambda_{wx} + \lambda_{wz}\lambda_{zx}, & \sigma_{zy} &= (\lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx})\lambda_{xy} + \lambda_{wz}\epsilon_{wy}, \\ \sigma_{zx} &= \lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx}, & \sigma_{xy} &= (\lambda_{wz}\lambda_{zx} + \lambda_{wx})\epsilon_{wy} + \lambda_{xy}. \end{aligned}$$

(Here $\lambda_{xy} = (\sigma_{zy} - \sigma_{wz}\sigma_{wy})/(\sigma_{zx} - \sigma_{wz}\sigma_{wx})$: Z is an instrument conditional on W, see below.) The goal of an ID algorithm is to find solvable subsystems efficiently; generic computer algebra is doubly exponential in the number of parameters.

4.4 From instrumental variables to instrumental sets

Motivation. Use variables that affect Y only through X to build solvable linear subsystems.

4.4.1 Instrumental variables

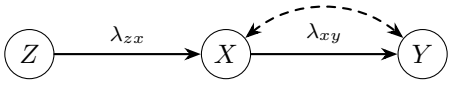


Figure. *The instrumental-variable graph. Confounding ($X \leftrightarrow Y$) makes $\sigma_{xy} \neq \lambda_{xy}$, but Z has no trek to Y except through X, so $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$ and $\sigma_{zx} = \lambda_{zx}$ give $\lambda_{xy} = \sigma_{zy}/\sigma_{zx}$.*

Suppose an independent doctor's recommendation Z, which influences how much drug patients take, is added: $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $X \leftrightarrow Y$. (Non-parametrically, $P(y \mid \mathrm{do}(x))$ is *not* identifiable here; linearity is what helps.)

Definition (Instrumental variable; Pearl 2000, p. 248)

Z is an IV for λ_{xy} if Z is d-separated from Y in $\mathcal{G}_{\lambda_{xy}}$ and Z is not d-separated from X in $\mathcal{G}_{\lambda_{xy}}$.

By Wright's rules $\sigma_{zx} = \lambda_{zx}$ and $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$, so

$$\lambda_{xy} = \frac{\sigma_{zy}}{\sigma_{zx}} = \frac{r_{zy}}{r_{zx}}.$$

Two-stage least squares. Regress $X = \alpha Z + e$, giving $\alpha = \sigma_{zx}$; set $\hat{X} = \sigma_{zx} Z$ and regress $Y = \beta \hat{X} + e$, giving $\beta = \sigma_{zy}/\sigma_{zx}$ (since $\mathrm{Cov}(Y, \hat{X}) = \sigma_{zx}\sigma_{zy}$ and $\mathrm{Var}(\hat{X}) = \sigma_{zx}^2$).

4.4.2 Conditional instrumental variables

Definition (Conditional IV; Brito and Pearl 2002)

Z is a conditional IV given $\mathbf{W}$ for λ_{xy} if (i) $\mathbf{W}$ contains only non-descendants of Y;
(ii) $\mathbf{W}$ d-separates Z from Y in $\mathcal{G}_{\lambda_{xy}}$;
(iii) $\mathbf{W}$ does not d-separate Z from X in $\mathcal{G}_{\lambda_{xy}}$.

Then $\lambda_{xy} = r_{yz \cdot \mathbf{W}} / r_{xz \cdot \mathbf{W}}$. Example: $W \to Z \to X \to Y$, $X \leftrightarrow Y$, $W \leftrightarrow Y$. Z is not an IV ($Z \leftarrow W \leftrightarrow Y$ is open) but is a conditional IV given W.

4.4.3 Instrumental sets

$Z_1 \to X_1$, $Z_1 \to X_2$, $Z_2 \to X_1$, $Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow Y$, $X_2 \leftrightarrow Y$. No single IV exists, since every path from Z_1 or Z_2 to Y passes through both X_1 and X_2. How to solve for $\lambda_{x_1 y}$? We have $\sigma_{z_i x_j} = \lambda_{z_i x_j}$ for $i, j \in \{1, 2\}$ and

$$\sigma_{z_1 y} = \sigma_{z_1 x_1}\lambda_{x_1 y} + \sigma_{z_1 x_2}\lambda_{x_2 y}, \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y},$$

a linear system built from paths from $\{Z_1, Z_2\}$ to $\{X_1, X_2\}$, solvable by Cramer's rule if its matrix is non-singular.

When does the system fail? The equations must not be collinear. If $Z_1 \to Z_2 \to \{X_1, X_2\}$, then paths from Z_1 to Y are paths from Z_2 to Y multiplied by $\lambda_{z_1 z_2}$:

$$\lambda_{z_1 z_2}\sigma_{z_2 y} = \lambda_{z_1 z_2}(\sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y}), \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y}.$$

The two equations are identical up to a constant factor: the system is degenerate.

Theorem 4.4 (Non-intersecting paths)

The system is non-degenerate (generically) iff there is a full matching between $\{Z_1, Z_2\}$ and $\{X_1, X_2\}$ by vertex-disjoint (non-intersecting) paths.

Proof [Supplement]. For the 2×2 system the instrumental equations are $\sigma_{z_i y} = \sum_j \sigma_{z_i x_j}\lambda_{x_j y}$, $i = 1, 2$, which have a unique solution iff $\det[\sigma_{z_i x_j}] \neq 0$. By Wright's rules each $\sigma_{z_i x_j}$ is a sum over treks, and the Lindström–Gessel–Viennot lemma expresses the determinant as a signed sum over pairs of vertex-disjoint path systems from $\{Z_1, Z_2\}$ to $\{X_1, X_2\}$: if no vertex-disjoint system exists, all terms cancel and the determinant is identically zero; if one exists, the determinant is a nonzero polynomial in the coefficients and hence nonzero for generic parameter values. (The general $k \times k$ statement is Brito and Pearl 2002.) ■

In the first graph, $Z_1 \to X_1$ and $Z_2 \to X_2$ are disjoint; in the degenerate graph, both paths must pass through Z_2. Such paths can be found by max-flow: treat each Z_i as a source and each X_j as a sink with unit vertex capacities; a flow of value $|\mathbf{Z}|$ certifies a full matching by non-intersecting paths.

4.4.4 Conclusions

1. Linear SCMs are the most common causal models in real-world applications.
2. Be careful with regression analysis and with what it actually computes.
3. A structural equation is not a regression equation, but regression sometimes recovers structural coefficients.
4. The causal diagram can be leveraged to solve for causal effects in many practical and general settings.

4.5 Appendix: Complexity of identification in linear SCMs

Landscape. Single-door, IV and instrumental sets are sufficient but incomplete. Other graphical criteria (half-trek criterion, auxiliary cutsets/ACID; Chen et al. 2016, Kumor et al. 2019) cover more instances, but not all identifiable ones. Is the general problem harder than these criteria?

Limits. The graphical criteria run in polynomial time but miss some identifiable instances. Brute-force computer algebra (Gröbner bases) solves all of them but takes doubly exponential time; a 4-node graph can take over 75 days (García-Puente, Spielvogel and Sullivant 2010; Bardet 2002).

Recent result (Dörfler, van der Zander, Bläser and Liśkiewicz, NeurIPS 2024). The general identification problem is provably hard, harder than NP, because it requires reasoning over real-valued unknowns rather than binary ones. Nevertheless it can be solved in polynomial space (PSPACE), hence in at most $2^{\mathrm{poly}(n)}$ time, a large improvement over the $2^{2^{\mathrm{poly}(n)}}$ of Gröbner bases. Take-away: when graphical criteria work, use them; otherwise the problem is inherently difficult, but not impossible.

4.6 Appendix: Multiple regression when Y is the cause

$X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$. Then $\sigma_{yx_1} = b_1$, $\sigma_{yx_2} = b_2$, $\sigma_{x_1x_2} = b_1b_2$. The normal equations of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ are

$$b_1 = \beta_1 + \beta_2 b_1 b_2, \qquad b_2 = \beta_2 + \beta_1 b_1 b_2.$$

Substituting $\beta_2 = b_2 - \beta_1 b_1 b_2$ into the first: $b_1 = \beta_1 + b_1 b_2^2 - \beta_1 b_1^2 b_2^2$, so $\beta_1(1 - b_1^2 b_2^2) = b_1(1 - b_2^2)$ and

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}.$$

4.7 Appendix: The Gröbner basis approach

Identification as polynomial elimination. By Wright's rules, observed covariances are polynomials in the unknowns λ, ϵ. Example: $X_1 \xrightarrow{\lambda_{12}} X_2 \xrightarrow{\lambda_{23}} X_3$ with $X_2 \leftrightarrow X_3$ (ϵ_{23}):

$$\sigma_{12} = \lambda_{12}, \qquad \sigma_{13} = \lambda_{12}\lambda_{23}, \qquad \sigma_{23} = \lambda_{23} + \epsilon_{23}.$$

From the first two, $\lambda_{23} = \sigma_{13}/\sigma_{12}$ (X_1 is an instrument); from σ_{23} alone we could not separate λ_{23} from ϵ_{23}. Gröbner bases automate this "solve and eliminate" process for any graph.

How it works (intuition).

1. *Set up:* write $\sigma_{ij} -$ (polynomial in $\lambda, \epsilon) = 0$ for every pair (i, j).
2. *Eliminate:* use elimination term orders to remove unwanted unknowns, keeping only the target (say λ_{xy}) and the σ_{ij}.
3. *Check:* a degree-1 polynomial in λ_{xy} means generically identifiable; degree d means algebraically d-identifiable (d possible values; may or may not yield unique identification); if λ_{xy} does not appear, it is not generically identifiable.

Computational results (García-Puente et al. 2010; all linear SCMs on 3 and 4 nodes):

	3 nodes ($2^6 = 64$)	4 nodes ($2^{12} = 4096$)
Generically identifiable	22	1246
Not generically identifiable	42	2844
Algebraically k-identified	0	6
Solved by single-door / IV	22/22	1093/1246
Only by Gröbner basis	0	153

On 3 nodes the graphical criteria suffice; on 4 nodes 153 cases need the algebraic method. Runtimes range from seconds to more than 75 days for a single 4-node graph.

Chapter 5. Partial Identification and Soft Interventions

5.1 Partial identification

Motivation. The central question is what can still be said about $P(y \mid \mathrm{do}(x))$ when it is not identifiable: instead of a point, we seek the tightest interval $[a(y;x), b(y;x)] \subseteq [0,1]$ consistent with the data and the graph.

5.1.1 The partial identification problem

Motivation. Extend the yes/no identification question to bounds.

Recall that $P(y \mid \mathrm{do}(x))$ is identifiable if it is uniquely determined by $P(\mathbf{v})$ and $\mathcal{G}$. When it is not, an inference engine given the query, the graph and data such as $P(x,y,z)$ answers "no". We then ask whether a non-trivial bound can be given:

$$P(y \mid \mathrm{do}(x)) \in [a(y;x),\ b(y;x)] \subseteq [0,1].$$

Definition (Partial identifiability)

The effect of X on Y is partially identifiable given $\mathcal{G}$ if $P(\mathbf{V})$ determines a bound $P(y \mid \mathrm{do}(x)) \in [a,b]$ that is a proper subset of $[0,1]$.

Identifiable $\Rightarrow$ partially identifiable (with $a = b$, ignoring the degenerate case), but not conversely.

5.1.2 Natural bounds

Motivation. Derive the simplest bounds, valid for any confounded pair, and show they cannot be improved without more structure.

Theorem 5.1 (Natural bounds; Manski 1990, Robins 1989)

For an SCM compatible with $X \to Y$, $X \leftrightarrow Y$ (latent U), given $P(x,y)$,

$$a(y;x) = P(y,x) \ \le\ P(y \mid \mathrm{do}(x)) \ \le\ P(y,x) + 1 - P(x) = b(y;x).$$

Proof [Supplement]. With $X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$, the truncated product gives

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_u P(y \mid x,u)P(u) = \sum_u P(y \mid x,u)\big(P(u,x) + P(u) - P(u,x)\big) \\ &= \sum_u P(y,x,u) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big) \\ &= P(y,x) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big). \end{aligned}$$

Since $0 \le P(y \mid x,u) \le 1$ and $P(u) - P(u,x) \ge 0$, the last sum lies between 0 and $\sum_u (P(u) - P(u,x)) = 1 - P(x)$. ∎

The width of the interval is $1 - P(x)$: the probability mass of units that did not receive x, about whom the data say nothing under $\mathrm{do}(x)$. Equivalently $b(y;x) = 1 - P(x,y')$ with y' the complement of y.

Example 5.1

$X, Y, U \in \{0,1\}$, $X \leftarrow U$, $Y \leftarrow X \oplus U$, $P(U=0) = 0.9$. Observationally $X = U$ and $Y = 0$ always: $P(X=0, Y=0) = 0.9$, $P(X=1, Y=0) = 0.1$. Under $\mathrm{do}(X=x)$, $Y = x \oplus U$:

	$P(Y=1 \mid \mathrm{do}(x))$	$a(Y=1;x)$	$b(Y=1;x)$
$x=0$	0.1	0	0.1
$x=1$	0.9	0	0.9

The true effects lie on the upper bounds.

5.1.2.1 Tightness

Theorem 5.2

For binary X, Y and any $P(x,y)$ there exist SCMs $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ with $P^{(1)}(x,y) = P^{(2)}(x,y) = P(x,y)$ and $P^{(1)}(y \mid \mathrm{do}(x)) = a(y;x)$, $P^{(2)}(y \mid \mathrm{do}(x)) = b(y;x)$.

Proof [Supplement]. Take any compatible $\mathcal{M}$ ($X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$) inducing P, and define, with the same $P(U)$,

$$\mathcal{F}^{(1)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 0 & \text{otherwise} \end{cases}, \qquad \mathcal{F}^{(2)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 1 & \text{otherwise} \end{cases},$$

both with $X \leftarrow f_X(u)$. Without intervention $X = f_X(U)$ always, so $Y = f_Y(X,U)$ in both models and $P^{(i)}(x,y) = \sum_u P(x \mid u) P(y \mid x,u) P(u) = P(y,x)$. Under $\mathrm{do}(X=x)$, U and x need not match:

$$\begin{aligned} P^{(i)}(Y=1 \mid \mathrm{do}(x)) &= P(Y=1 \mid x, f_X(U)=x) P(f_X(U)=x) \\ &\quad + P^{(i)}(Y=1 \mid x, f_X(U) \neq x) P(f_X(U) \neq x) \\ &= P(x, Y=1) + \mathbb{1}[i=2](1 - P(x)). \end{aligned}$$

Thus $\mathcal{M}^{(1)}$ attains a and $\mathcal{M}^{(2)}$ attains b (for $Y=1$; $Y=0$ is symmetric). ■

5.1.2.2 Limitations

Natural bounds are tight for the bow graph with binary variables, but not when further structure is available, e.g. an instrument Z. Can the graph be exploited for tighter bounds? Yes: canonical type models and linear programming.

5.1.3 Canonical type models and LP bounds

Motivation. Replace the unknown functional forms by a finite set of response types, turning the bounding problem into a linear program.

5.1.3.1 Experiments with non-compliance

Setting: treatment assigned Z (randomized, the instrument), treatment realized X, primary outcome Y, and exogenous U confounding X and Y: $Z \to X \to Y$, $X \leftrightarrow Y$. Task: given $P(x,y \mid z)$, bound $P(y \mid \mathrm{do}(x))$.

Theorem 5.3 (Natural bounds in the IV model)

$$a(y;x) = \max_z P(y,x \mid z), \qquad b(y;x) = \min_z \big[P(y,x \mid z) + 1 - P(x \mid z)\big].$$

Proof [Supplement]. Fix z. In the submodel $\mathcal{M}_z$, the natural bounds give $P(y,x \mid \mathrm{do}(z)) \le P(y \mid \mathrm{do}(x,z)) \le P(y,x \mid \mathrm{do}(z)) + 1 - P(x \mid \mathrm{do}(z))$. By exclusion ($Z$ affects Y only through X), $P(y \mid \mathrm{do}(x,z)) = P(y \mid \mathrm{do}(x))$; by randomization, $P(\cdot \mid \mathrm{do}(z)) = P(\cdot \mid z)$. So $P(y,x \mid z) \le P(y \mid \mathrm{do}(x)) \le P(y,x \mid z) + 1 - P(x \mid z)$ for every z, and taking the maximum of the lower and the minimum of the upper bounds over z gives the result. ■

They are not tight in IV models.

5.1.3.2 Optimization perspective

Let $\mathcal{M}$ be the set of SCMs compatible with the IV graph: $Z \leftarrow f_Z(U_Z)$, $X \leftarrow f_X(Z,U)$, $Y \leftarrow f_Y(X,U)$. The sharp bounds are

$$a(y;x) = \min_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \mathrm{do}(x)), \qquad b(y;x) = \max_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \mathrm{do}(x)).$$

This is hard to solve directly because the forms of $\mathcal{F}$ and $P(\mathbf{U})$ are unspecified.

5.1.3.3 U as a selector and response variables

For $X \leftarrow f_X(Z,U)$ and fixed $U = u$, $f_X(\cdot,u)$ is a deterministic function from Z to X: U *selects* which mapping $Z \to X$ is in effect. The domain $\mathfrak{X}_U$ can therefore be partitioned by the induced function. With binary Z, X there are $|\mathfrak{X}_X|^{|\mathfrak{X}_Z|} = 2^2 = 4$ functions:

$$\{u : f_X(0,u) = 0,\ f_X(1,u) = 0\} \quad (X = 0), \qquad \{u : f_X(0,u) = 0,\ f_X(1,u) = 1\} \quad (X = Z),$$
$$\{u : f_X(0,u) = 1,\ f_X(1,u) = 0\} \quad (X = 1 - Z), \qquad \{u : f_X(0,u) = 1,\ f_X(1,u) = 1\} \quad (X = 1).$$

(With ternary Z and binary X there would be 2^3.) A *response variable* indexes this partition:

$R_X = 0$: $X = 0$ (never-taker) $\quad R_X = 1$: $X = Z$ (complier) $\quad R_X = 2$: $X = 1 - Z$ (defier)
$R_X = 3$: $X = 1$ (always-taker)

Similarly $R_Y \in \{0,1,2,3\}$ for $Y \leftarrow f_Y(X,U)$: never-recover ($Y = 0$), helped ($Y = X$), hurt ($Y = 1 - X$), always-recover ($Y = 1$). Together (R_X, R_Y) partition $\mathfrak{X}_U$ into $4 \times 4 = 16$ cells with probabilities

$$q_{j,k} = P(R_X = j, R_Y = k), \qquad \sum_{j=0}^{3} \sum_{k=0}^{3} q_{j,k} = 1.$$

Definition (Canonical type model)

An SCM with $X, Y, Z \in \{0,1\}$ and $\mathbf{U} = \{R_X, R_Y, R_Z\}$, $R_X, R_Y \in \{0,1,2,3\}$, $R_Z \in \{0,1\}$:

$$Z \leftarrow r_Z, \qquad X \leftarrow f_X(z, r_X) = \begin{cases} 0 & r_X = 0 \\ z & r_X = 1 \\ 1 - z & r_X = 2 \\ 1 & r_X = 3 \end{cases}, \qquad Y \leftarrow f_Y(x, r_Y) = \begin{cases} 0 & r_Y = 0 \\ x & r_Y = 1 \\ 1 - x & r_Y = 2 \\ 1 & r_Y = 3 \end{cases}.$$

Theorem 5.4 (Correspondence; Balke and Pearl 1997)

For any IV model $\mathcal{M}_1$ there is a canonical type model $\mathcal{M}_2$ with $P^{\mathcal{M}_1}(x, y \mid z) = P^{\mathcal{M}_2}(x, y \mid z)$ and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x)) = P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$.

Proof [Supplement]. Partition the domain of U by the pair of response functions it induces: $u \mapsto (x(\cdot), y(\cdot))$ with $x(z) = f_X(z, u)$ and $y(x) = f_Y(x, u)$. With binary variables there are 4 functions $z \mapsto x$ and 4 functions $x \mapsto y$, so 16 cells. Define $\mathcal{M}_2$ with a discrete $R = (R_X, R_Y)$ taking value (j, k) with probability $q_{j,k}$, the $P(U)$-mass of the corresponding cell, and $X = \mathrm{response}_j(Z)$, $Y = \mathrm{response}_k(X)$. For each z, $P^{\mathcal{M}_1}(x, y \mid z)$ is the mass of the u's with $f_X(z, u) = x$ and $f_Y(x, u) = y$, which is a union of cells, so it equals $P^{\mathcal{M}_2}(x, y \mid z)$; likewise $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x))$ is the mass of $\{u : f_Y(x, u) = y\}$, a union of cells, and equals $P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$. ■

(Set $q_{j,k}$ to the $P(U)$-mass of the cell of $\mathfrak{X}_U$ inducing the pair of functions (j, k).) Hence, with $\mathcal{M}$ now the canonical models,

$$a(y; x) = \min_{M \in \mathcal{M}} P_M(y \mid \mathrm{do}(x)), \quad b(y; x) = \max_{M \in \mathcal{M}} P_M(y \mid \mathrm{do}(x)) \quad \text{s.t. } P_M(x, y \mid z) = P(x, y \mid z).$$

Summary. U is partitioned into response types by the functions it induces; for the binary IV, R_X (never-taker, complier, defier, always-taker) and R_Y; the joint $P(R_X, R_Y)$ has 16 parameters $q_{j,k}$; every IV model has an equivalent canonical model. Bounding $P(y \mid \mathrm{do}(x))$ reduces to a linear program over $\mathbf{q} = (q_{j,k}) \in \mathbb{R}^{16}$.

5.1.3.4 LP formulation

Observational constraints. Let $p_{yx.z} = P(Y = y, X = x \mid Z = z)$. Each is a linear function of $\mathbf{q}$, e.g.

$$\begin{aligned} p_{10.1} = P(Y = 1, X = 0 \mid Z = 1) &= P(\text{never-taker, hurt}) + P(\text{never-taker, always-recover}) \\ &\quad + P(\text{defier, hurt}) + P(\text{defier, always-recover}) \\ &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}. \end{aligned}$$

(With $Z = 1$, $X = 0$ requires $R_X \in \{0, 2\}$; then $Y = 1$ at $X = 0$ requires $R_Y \in \{2, 3\}$.) In full:

$$\begin{aligned} p_{00.0} &= q_{0,0} + q_{0,1} + q_{1,0} + q_{1,1}, & p_{00.1} &= q_{0,0} + q_{0,1} + q_{2,0} + q_{2,1}, \\ p_{01.0} &= q_{2,0} + q_{2,2} + q_{3,0} + q_{3,2}, & p_{01.1} &= q_{1,0} + q_{1,2} + q_{3,0} + q_{3,2}, \\ p_{10.0} &= q_{0,2} + q_{0,3} + q_{1,2} + q_{1,3}, & p_{10.1} &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}, \\ p_{11.0} &= q_{2,1} + q_{2,3} + q_{3,1} + q_{3,3}, & p_{11.1} &= q_{1,1} + q_{1,3} + q_{3,1} + q_{3,3}. \end{aligned}$$

Causal effect. Y under $\mathrm{do}(X = 0)$ equals 1 iff $R_Y \in \{\text{hurt, always-recover}\}$:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) = P(R_Y = 2) + P(R_Y = 3) = \textstyle\sum_j (q_{j,2} + q_{j,3}),$$

$$P(Y = 1 \mid \mathrm{do}(X = 1)) = \textstyle\sum_j (q_{j,1} + q_{j,3}).$$

Optimization. The bound $P(Y = 1 \mid \mathrm{do}(X = 0)) \in [a_{10}, b_{10}]$ solves

$$a_{10} = \min_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3}), \qquad b_{10} = \max_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3})$$

subject to $\sum_{j,k} q_{j,k} = 1$, $q_{j,k} \geq 0$, and the eight observational equalities above.

5.1.3.5 Closed-form solution

The LP has a closed-form solution: each bound is a linear combination of the $p_{yx.z}$ whose gap from the objective is provably non-negative.

$$P(Y=1 \mid \mathrm{do}(X=0)) \;\geq\; \max \begin{cases} p_{10.0} \\ p_{10.1} \\ p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} \\ p_{01.0} + p_{10.0} - p_{00.1} - p_{01.1} \end{cases}$$

$$P(Y=1 \mid \mathrm{do}(X=0)) \;\leq\; \min \begin{cases} 1 - p_{00.0} \\ 1 - p_{00.1} \\ p_{10.0} + p_{11.0} + p_{01.1} + p_{10.1} \\ p_{01.0} + p_{10.0} + p_{10.1} + p_{11.1} \end{cases}$$

The first two lower bounds and upper bounds are the natural bounds; the extra terms make the LP bounds tighter.

Why is $p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}$ a valid lower bound? Substituting,

$$p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} = q_{0,2} + q_{0,3} + q_{1,2} + q_{2,3} - q_{0,0} - q_{0,1} - q_{2,0} - q_{1,1}.$$

Subtracting from the objective,

$$\sum_j (q_{j,2} + q_{j,3}) - (p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}) = q_{2,2} + q_{3,2} + q_{1,3} + q_{3,3} + q_{0,0} + q_{0,1} + q_{2,0} + q_{1,1} \;\geq 0.$$

Every closed-form bound is certified the same way (this is LP duality: the coefficients are a feasible dual solution).

5.1.4 Back-door with partially observed covariates

Motivation. Bound an effect when the admissible set is only partly measured but the marginal of the missing part is known.

Suppose $\{W, U\}$ is back-door admissible ($W \to X$, $W \to Y$, $U \to X$, $U \to Y$, W–U related, $X \to Y$), W observed, U not:

$$P(y \mid \mathrm{do}(x)) = \sum_{w,u} P(y \mid x, w, u) P(w, u) = \sum_{w,u} \frac{P(x, y, w, u) P(w, u)}{P(x, w, u)}.$$

Given $P(X, Y, W)$ and $P(U)$ but not $P(X, Y, W, U)$, the naive (natural) bound is $P(x, y) \leq P(y \mid \mathrm{do}(x)) \leq 1 - P(x, y')$, where $y' = 1 - y$, $x' = 1 - x$ denote complements. Li and Pearl (2022) tighten it:

$$a(y; x) = \min_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}}, \qquad b(y; x) = \max_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}},$$

where $a_{w,u}$, $b_{w,u}$, $c_{w,u}$ stand for the unknown $P(x, y, w, u)$, $P(w, u)$, $P(x, w, u)$, subject to, for all $w \in \mathfrak{X}_W$ and $u \in \mathfrak{X}_U$,

$$\begin{gathered} \textstyle\sum_u a_{w,u} = P(x, y, w), \qquad \sum_u b_{w,u} = P(w), \qquad \sum_u c_{w,u} = P(x, w), \\ b_{w,u} \geq c_{w,u} \geq a_{w,u} \quad (\text{since } P(x, y, w, u) \leq P(x, w, u) \leq P(w, u)), \\ \max\{0, P(x, y, w) + P(u) - 1\} \leq a_{w,u} \leq \min\{P(x, y, w), P(u)\}, \\ \max\{0, P(w) + P(u) - 1\} \leq b_{w,u} \leq \min\{P(w), P(u)\}, \\ \max\{0, P(x, w) + P(u) - 1\} \leq c_{w,u} \leq \min\{P(x, w), P(u)\}. \end{gathered}$$

The last three lines are Fréchet bounds for a joint probability given its two marginals. (In fact $a_{w,u}$ alone is a sufficient parametrization.)

5.2 Soft interventions

Motivation. The central question is how to evaluate realistic policies, which change a variable's mechanism to a conditional or stochastic rule rather than fixing it to a constant, and how the do-calculus generalizes to such soft interventions.

5.2.1 Motivation

Motivation. Show why atomic $\text{do}(x)$ interventions are often not the policies we can implement.

A tutoring program. For each student we observe the previous GPA W, motivation Z (low/high), tutoring X, and the final GPA Y. Motivation depends on previous GPA (and unobserved factors); students get tutoring depending on motivation; the final GPA depends on previous GPA, motivation and tutoring: $W \to Z \to X \to Y$, $W \to Y$, $Z \to Y$.

With large data and ML, $P(y \mid w, z, x)$ can be predicted with small error. But this distribution reflects the current, *natural regime* (regime: the mode a system operates in). Decisions intervene on the regime, so we must predict GPA under a hypothetical, unrealised regime.

What we know so far. From $P(W, Z, X, Y)$ and $\mathcal{G}$ we can identify

$$\begin{aligned} P(y \mid \text{do}(x)) &= \sum_z P(y \mid \text{do}(x), z) P(z \mid \text{do}(x)) && \text{condition on } Z \\ &= \sum_z P(y \mid \text{do}(x), z) P(z) && \text{Rule 3: } (Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \\ &= \sum_z P(y \mid x, z) P(z) && \text{Rule 2: } (Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}}. \end{aligned}$$

Suppose $P(Y = 1 \mid \text{do}(X = 1)) > P(Y = 1)$: we predict improvement if $\text{do}(X = 1)$ is implemented. But what does $\text{do}(X = 1)$ mean in the real world? Mandatory tutoring for every single student. Is this realizable?

Implementing do-interventions. Even when the effect is identifiable, resources may be insufficient (not enough teachers and hours), and effectiveness cannot be guaranteed (non-compliance). For practicality we care about realizable interventions:

Do-like intervention	Realistic intervention
make sure no one smokes	reduce tobacco consumption to 20% of current
provide treatment to all patients	treat iff a patient is in critical condition
move a robotic arm to coordinate (x, y, z)	move it to (x, y, z) with a small error
mark all applicants as males	mark all applicants as males on paper

5.2.2 Canonical types of regimes

Motivation. Classify interventions by the new mechanism they install.

- **Idle/natural:** $X = f_X(\mathbf{pa}_X, u_X)$ (no intervention).
- **Hard/atomic:** $X = x$, $\text{do}(X = x)$, the type considered so far. Every student gets tutoring.
- **Conditional:** $X = g(\mathbf{w})$ sets X to the output of a function of observable variables $\mathbf{W}$. Students get tutoring iff they have a low GPA.
- **Stochastic:** $X \sim P^*(x \mid \mathbf{w})$ makes X follow a given distribution conditional on $\mathbf{W}$. Students with low GPA enter a raffle for 80% of the spots, other interested students for the remaining 20%.

We write σ_X for such an intervention and $P(\cdot;\sigma_X)$ for the induced distribution.

Changes in regime. The unperturbed world, where data are generated, is the natural regime with graph $\mathcal{G}$. An intervention creates a new regime differing in the mechanisms of the intervened variables. Its graph $\mathcal{G}_{\sigma_X}$ is drawn with a regime indicator (a small square σ_X) pointing into the intervened variables, and edges into X replaced according to the new mechanism.

5.2.3 Reduction to known machinery

Motivation. See when soft-intervention effects follow from hard-intervention results.

Can identification of soft (conditional, stochastic) interventions be reduced to that of hard interventions? **A reducible example.** Suppose $P(X=1 \mid Z=1)=0.3$: only 30% of motivated students get tutoring. What if this were doubled, σ_X: $P^*(X=1 \mid Z=1)=0.6$? In $\mathcal{G}_{\sigma_X}$, X's parents remain $\{Z\}$ but its mechanism changes. Then

$$\begin{aligned} P(y;\sigma_X) &= \sum_z P(y \mid z;\sigma_X)P(z;\sigma_X) = \sum_z P(y \mid z;\sigma_X)P(z) \\ &= \sum_{z,x} P(y \mid z,x;\sigma_X)P(x \mid z;\sigma_X)P(z) = \sum_{z,x} P(y \mid z,x;\sigma_X)P^*(x \mid z)P(z) \\ &= \sum_{z,x} P(y \mid z,x)P^*(x \mid z)P(z). \end{aligned}$$

Z is not affected by changing X's mechanism; the new assignment is $P^*(x \mid z)$; and given Z and X, Y responds as in the natural regime. Is a do-calculus-like derivation always sufficient?

An intentionally wrong derivation. A patient receives treatment X_1; a second physician observes $Z=z$ and decides a second treatment X_2; finally the patient survives or not (Y). Graph: $X_1 \to Z$, $X_1 \to X_2$, $X_1 \to Y$, $Z \to X_2$, $X_2 \to Y$, $X_1 \leftrightarrow Z$, $Z \leftrightarrow Y$. Consider fixing $X_1=x_1$ and $X_2=g(x_1,z)$, i.e. $\mathrm{do}(x_1), \mathrm{do}(X_2=g(x_1,z))$, or $\sigma_{\mathbf{X}}=\{X_1=x_1, X_2=g(x_1,z)\}$. One might write

$$P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2=g(x_1,z))) = P(y \mid x_1, \mathrm{do}(X_2=g(x_1,z))),$$

citing Rule 2 with $(Y \perp_d X_1 \mid X_2)$ in $\mathcal{G}_{\overline{X_2}\underline{X_1}}$. This is wrong: under $\mathrm{do}(X_2=g(x_1,z))$ the edges into X_2 remain active, so $\mathcal{G}_{\overline{X_2}\underline{X_1}}$ is not the right graph. The check must use $\mathcal{G}_{\sigma_{X_2}\underline{X_1}}$, where separation fails because, given the intervention on X_2, the path $X_1 \leftrightarrow Z \to X_2 \to Y$ is open.

Where do-calculus intuition breaks. Under do-interventions the intervened variables lose all dependence on their parents; soft interventions may keep some or all dependencies on parents, change their distribution, or even add new dependencies.

5.2.4 σ-calculus

Motivation. State inference rules for soft interventions that consult both pre- and post-interventional graphs.

Theorem 5.5 (σ-calculus, simplified; Correa and Bareinboim 2020)

For disjoint $\mathbf{X},\mathbf{Y},\mathbf{Z} \subseteq \mathbf{V}$ and disjoint $\mathbf{T},\mathbf{W} \subseteq \mathbf{V} \setminus (\mathbf{Z} \cup \mathbf{Y})$:

Rule 1 (insertion/deletion of observations) $P(\mathbf{y} \mid \mathbf{w},\mathbf{t};\sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w};\sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$.

Rule 2 (change of regimes under observation) $P(\mathbf{y} \mid \mathbf{w},\mathbf{x};\sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w},\mathbf{x})$ if $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\underline{\mathbf{X}}}$ and in $\mathcal{G}_{\underline{\mathbf{X}}}$.

Rule 3 (change of regimes without observation) $P(\mathbf{y} \mid \mathbf{w};\sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w})$ if $(\mathbf{Y} \perp_d$

$\mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\overline{\mathbf{X}(\mathbf{W})}}$ and in $\mathcal{G}_{\overline{\mathbf{X}(\mathbf{W})}}$.

Proof [Supplement]. The proof parallels that of do-calculus (Chapter 3), with the regime indicator now selecting between the natural mechanism and the stochastic policy $\sigma_{\mathbf{X}}$. **Rule 1** is d-separation in $\mathcal{G}_{\sigma_{\mathbf{X}}}$, to which the post-regime distribution is Markov. For **Rules 2 and 3**, add a regime switch F_X with $F_X \to X$; the two sides correspond to $F_X = \sigma$ and $F_X =$ idle. Exactly as in the proof of Rules 2 and 3 of do-calculus, $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ (Rule 2) or $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}$ (Rule 3) holds if the corresponding separation holds with the outgoing (respectively incoming) edges of $\mathbf{X}$ removed. Because a soft regime may add new parents of $\mathbf{X}$, the separation must hold in the graph of each regime, which is why both $\mathcal{G}_{\sigma_{\mathbf{X}}\ldots}$ and $\mathcal{G}_{\ldots}$ appear in the conditions. ■

Full version. With an outer regime $\sigma_{\mathbf{X}}$ and a regime change on $\mathbf{Z}$ from $\sigma_{\mathbf{Z}}$ to $\sigma'_{\mathbf{Z}}$:

- Rule 1: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$;
- Rule 2: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\underline{\mathbf{Z}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\underline{\mathbf{Z}}}$;
- Rule 3: $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\overline{\mathbf{Z}(\mathbf{W})}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\overline{\mathbf{Z}(\mathbf{W})}}$.

5.2.4.1 Comparison with do-calculus

Remark. Under $\mathrm{do}(x)$, X is deterministic, so $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x), w, x)$. Under a non-atomic intervention this is not necessarily so: for $X \to Y$, $P(y; \sigma_X) \neq P(y \mid x; \sigma_X)$ in general.

	do-calculus	σ-calculus
Rule 1	$P(y \mid \mathrm{do}(x), w, t) = P(y \mid \mathrm{do}(x), w)$ if $(Y \perp_d T \mid W, X)_{\mathcal{G}_{\overline{X}}}$	$P(y \mid w, t; \sigma_X) = P(y \mid w; \sigma_X)$ if $(Y \perp_d T \mid W)_{\mathcal{G}_{\sigma_X}}$
Rule 2	$P(y \mid \mathrm{do}(x), w) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\underline{X}}}$	$P(y \mid x, w; \sigma_X) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \underline{X}}$ and $\mathcal{G}_{\underline{X}}$
Rule 3	$P(y \mid \mathrm{do}(x), w) = P(y \mid w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\overline{X(W)}}}$	$P(y \mid w; \sigma_X) = P(y \mid w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \overline{X(W)}}$ and $\mathcal{G}_{\overline{X(W)}}$

In Rule 1 the relevant graph depends on the specification of the intervention; in Rules 2 and 3 the independence must hold in both the pre- and post-interventional graphs.

Revisiting Rule 2. For the sequential-treatment example,

$$\begin{aligned} P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) &= P(y \mid x_1, \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) \\ &= P(y \mid x_1; \sigma_{X_1, X_2}) \overset{?}{=} P(y \mid x_1; \sigma_{X_2}). \end{aligned}$$

To invoke Rule 2 we must check the absence of a back-door path from X_1 to Y in two graphs: the one where X_2 is set by g ignoring Z, and the one where X_2 still listens to X_1 and Z. It holds only in the first, so the rule is not applicable.

5.2.4.2 Another intervention

Resources are limited, so tutoring is focused on those who need it most: students with low GPA must get tutoring, and only they can, i.e. $P^*(X = 1 \mid W = 0) = 1$, $P^*(X = 1 \mid W = 1) = 0$, a conditional intervention ($W = 0$ denoting low GPA). In $\mathcal{G}_{\sigma_X}$, X now depends on W (not on Z):

$$\begin{aligned} P(y; \sigma_X) &= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w, z; \sigma_X) P(w, z; \sigma_X) \\ &= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w; \sigma_X) P(w, z; \sigma_X) \qquad \text{Rule 1} \end{aligned}$$

$$= \sum_{w,z,x} P(y \mid x, w, z) P(x \mid w; \sigma_X) P(w, z; \sigma_X) \qquad \text{Rule 2}$$

$$= \sum_{w,z,x} P(y \mid x, w, z) P^*(x \mid w) P(w, z) \qquad \text{Rule 3.}$$

5.2.5 C-factors and soft interventions

Motivation. Identify soft-intervention effects systematically with c-factors.

For $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$,

$$P(z, x, y) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u_X} P(x \mid z, u_X) P(u_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y) = Q[Z]\, Q[X]\, Q[Y].$$

Under $\mathcal{M}' = \mathcal{M}_{\sigma_X}$ with diagram $\mathcal{G}' = \mathcal{G}_{\sigma_X}$, only X's mechanism changes:

$$P(z, x, y; \sigma_X) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u'_X} P'(x \mid z, u'_X) P(u'_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y)$$

$$= Q[Z]\, Q[X; \sigma_X]\, Q[Y].$$

In general, $P(\mathbf{y}; \sigma_{\mathbf{X}}) = \sum_{\mathbf{d} \setminus \mathbf{y}} \prod_{\mathbf{D}_i \in \mathcal{C}(\mathbf{D})} Q[\mathbf{D}_i]$, the product over the c-components of $\mathcal{G}_{\sigma_{\mathbf{X}}}$ restricted to the relevant ancestors; $Q[\mathbf{X}; \sigma_{\mathbf{X}}]$ is given by the definition of $\sigma_{\mathbf{X}}$, and every other factor must be identified from $P(\mathbf{V})$, e.g. with IDENTIFY (Chapter 3).

Example 5.2 (Systematic identification 1)

$\mathcal{G}$: $W \to Z$, $X \to Z \to Y$, $X \leftrightarrow Y$; intervention $\sigma_X = P^*(X \mid W)$, so $\mathcal{G}_{\sigma_X}$ has $W \to X$ and loses $X \leftrightarrow Y$. C-components: in $\mathcal{G}$, $\mathbf{C}_1 = \{X, Y\}$, $\mathbf{C}_2 = \{W\}$, $\mathbf{C}_3 = \{Z\}$; in $\mathcal{G}_{\sigma_X}$, $\mathbf{D}_1 = \{Y\}$, $\mathbf{D}_2 = \{W\}$, $\mathbf{D}_3 = \{Z\}$, $\mathbf{D}_4 = \{X\}$. Subgoal: identify each $Q[\mathbf{D}_i]$; $Q[\mathbf{D}_4] = Q[X; \sigma_X]$ is given by the intervention. IDENTIFY$(\{Y\}, \{X, Y\}, Q[T], \mathcal{G})$: $A = An(\{Y\})_{\mathcal{G}[\{X,Y\}]} = \{Y\} = C$, so $Q[Y] = \sum_x Q[X, Y]$. IDENTIFY$(\{W\}, \{W\}, \cdot)$ and IDENTIFY$(\{Z\}, \{Z\}, \cdot)$ return immediately. Hence

$$P(y; \sigma_X) = \sum_{w,x,z} Q[X; \sigma_X]\, Q[Z]\, Q[W]\, Q[Y] = \sum_{w,x,z} Q[X; \sigma_X]\, Q[Z]\, Q[W] \sum_{x'} Q[X, Y]$$

$$= \sum_{w,x,z} P^*(x \mid w)\, P(z \mid x, w)\, P(w) \sum_{x'} P(x')\, P(y \mid x', z, w).$$

Example 5.3 (Systematic identification 2: failure)

$\mathcal{G}$: $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$; intervention $\sigma_X = P^*(X \mid Z)$. C-components: in $\mathcal{G}$, $\mathbf{C}_1 = \{W, X, Y\}$, $\mathbf{C}_2 = \{Z\}$; in $\mathcal{G}_{\sigma_X}$, $\mathbf{D}_1 = \{Y, W\}$, $\mathbf{D}_2 = \{X\}$, $\mathbf{D}_3 = \{Z\}$. IDENTIFY$(\{W, Y\}, \{W, X, Y\}, Q[T], \mathcal{G})$: $A = An(\{W, Y\})_{\mathcal{G}[\{W,X,Y\}]} = \{W, X, Y\} \neq C$ and $A = T$, so FAIL. (Recall $Q[\mathbf{C}_j] = \prod_{V_i \in \mathbf{C}_j} Q[\mathbf{V}^{\leq i}]/Q[\mathbf{V}^{\leq i-1}]$ and $Q[\mathbf{V}^{\leq i}] = \sum_{\mathbf{v}^{>i}} Q[\mathbf{V}]$.)

5.2.6 Summary

- In decision-making settings, where interventions must be implemented, non-atomic (conditional and stochastic) interventions are needed.
- In the spirit of do-calculus, analogous rules consulting both pre- and post-interventional graphs identify the effects of soft interventions.
- Soft-intervention effects can also be identified by systematic procedures based on c-factors.

Chapter 6. Potential Outcomes and Effect Estimation

6.1 Potential outcomes

Motivation. The central question is how causal effects are defined and identified when causation is formulated as a comparison of potential outcomes, a missing-data problem, rather than through structural equations and graphs.

6.1.1 Frameworks for causal inference

Motivation. Place the potential outcome (PO) framework next to the causal-diagram framework.

A framework for causal inference must be rich enough to formalize basic intuitions about cause and effect. Two are in common use:

- the **potential outcome framework**, also called the counterfactual framework or Neyman–Rubin causal model (Neyman 1923; Rubin 1974);
- the **causal diagram framework** (Pearl 2009).

Mathematically the two are connected (Richardson and Robins 2013; see the SWIG lecture).

The PO framework is arguably the most widely used across medicine, health care, policy and the social sciences. Causal inference is viewed as a *missing-data problem*, with explicit modeling of the *assignment mechanism* as the process that reveals the observed data. It is rooted in Fisher's (1918, 1925) and Neyman's (1923) work on randomized experiments, extended by Rubin (1974, 1978) and many others to non-randomized studies.

6.1.2 Basic concepts

Motivation. Define units, treatments, potential outcomes and causal effects, and state the fundamental problem.

- **Unit:** the person, place or thing on which a treatment operates, at a particular time. (The same person at two different times constitutes two units.)
- **Treatment:** an intervention whose effect on some measurement of the units the investigator wishes to assess relative to no intervention (control).
- **Potential outcomes:** the values of a unit's measurement after application of the active treatment, $Y(1)$, and after non-application (control), $Y(0)$. Synonymously, *counterfactuals*.
- **Causal effect:** for each unit, a comparison of $Y(1)$ and $Y(0)$, e.g. $Y_i(1) - Y_i(0)$.

6.1.2.1 The fundamental problem of causal inference

At most one potential outcome is observed for each unit; the other is missing, i.e. counterfactual (Holland 1986).

Unit	$Y(x{=}0)$	$Y(x{=}1)$	X	Y
1	?	1	1	1
2	0	?	0	0
3	?	0	1	0
4	?	1	1	1
5	1	?	0	1

Neyman's urn model. Picture each unit as a ticket carrying both Y_1 and Y_0, placed in an urn. Assignment draws tickets and, for each, reveals only one half: Y_1 for treated units, Y_0 for controls.

6.1.2.2 God's eye view: individual versus average effects

Outcome Y: death (1) or survival (0). With both potential outcomes visible (adapted from Hernán and Robins, *What If*, Table 1.1):

	$Y(0)$	$Y(1)$	effect		$Y(0)$	$Y(1)$	effect
Rheia	1	0	-1	Hestia	1	0	-1
Zeus	0	0	0	Poseidon	1	0	-1
Kronos	0	1	$+1$	Hera	1	0	-1
Demeter	0	0	0	Artemis	1	1	0
Hades	0	0	0	Apollo	0	1	$+1$
Risk: $\mathbb{E}[Y(0)] = 0.5$, $\mathbb{E}[Y(1)] = 0.3$							

The average causal effect is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = 0.3 - 0.5 = -0.2$: treatment reduces risk. But individual effects vary: 4 helped (-1), 2 harmed ($+1$), 4 unaffected. We can never observe both columns for the same person.

6.1.2.3 What the researcher actually sees

(Adapted from *What If*, Table 1.2.)

	$Y(0)$	$Y(1)$	X	Y^{obs}		$Y(0)$	$Y(1)$	X	Y^{obs}
Rheia	?	0	1	0	Hestia	?	0	1	0
Zeus	0	?	0	0	Poseidon	1	?	0	1
Kronos	?	1	1	1	Hera	?	0	1	0
Demeter	0	?	0	0	Artemis	1	?	0	1
Hades	0	?	0	0	Apollo	?	1	1	1

The "?" entries are counterfactual and forever missing. By consistency $Y^{\text{obs}} = Y(X)$. The naive comparison gives $\bar{Y} = 2/5 = 0.4$ among the treated and $2/5 = 0.4$ among controls, a difference of $0 \neq -0.2$. Why? Treatment was not randomly assigned: sicker patients (those with $Y(0) = 1$) chose treatment more often (confounding). This is why we need exchangeability.

6.1.3 Assumptions linking potential and observed outcomes

Motivation. State SUTVA and consistency, and illustrate their failures.

6.1.3.1 SUTVA

Assumption 1 (Stable Unit Treatment Value Assumption; Rubin 1980). SUTVA comprises (1) *no interference* and (2) *no different versions* of a treatment (e.g. the same surgery performed by different surgeons). It connects the observed treatment X with the intervention of interest x:

$$X_i = 1 \Rightarrow Y_i = Y_i(1), \qquad X_i = 0 \Rightarrow Y_i = Y_i(0),$$

equivalently

$$Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

Violation by interference: $Y_i(x)$ depends on the treatments received by other units, e.g. vaccination, advertising, infectious diseases, social networks, agricultural plots. Then other assumptions are needed (Rosenbaum 2007; Hudgens and Halloran 2008), and spillover effects are often themselves of interest.

On the history of SUTVA (VanderWeele and Hernán, "Causal inference under multiple versions of treatment"). Rubin (1980) pointed out that the notation $Y_j(a)$ presupposes (i) that j's outcome under a does not depend on the treatments of other individuals and (ii) that there are no multiple versions of a leading to different outcomes. The first is "no interference" (attributed to Cox 1958), the second "no versions of treatment" (attributed to Neyman 1935). Also included in SUTVA is *consistency* (Robins 1986): $Y_j(a) = Y_j$ when $A_j = a$, tying counterfactual data to observed data. These assumptions matter in both experimental and observational studies; although usually stated only for observational research, they also govern the interpretation of randomized trials.

6.1.3.2 Consistency and well-defined interventions

Consistency: if $X_i = x$ then $Y_i = Y_i(x)$. It sounds obvious, but requires that "$X = x$" means the same thing for everyone. Example: the effect of "losing 20 kg" on heart-attack risk. Losing 20 kg through diet and exercise ($Y(x)$ = survived) and losing it through starvation or illness ($Y(x)$ = died) are the same nominal treatment with different outcomes, so $Y(x)$ is not well-defined. Fix: specify the intervention precisely ("20 kg loss via a supervised diet and exercise program"). Vague treatments give vague causal questions and meaningless answers (*What If*, Fine Point 3.4).

6.1.4 Causal estimands

Motivation. List the quantities typically targeted.

Differences.

$$\begin{aligned} \text{ATE:} \quad & \tau = \mathbb{E}[Y(1) - Y(0)], \\ \text{CATE:} \quad & \tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z], \\ \text{ATT:} \quad & \text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid X = 1]. \end{aligned}$$

Ratios. Individual ratio $r_i = Y_i(1)/Y_i(0)$; population ratio $r = \mathbb{E}[Y(1)]/\mathbb{E}[Y(0)]$; relative effect (lift) $l = \mathbb{E}[Y(1) - Y(0)]/\mathbb{E}[Y(0)]$. Note that the population ratio is not an average of individual ratios unless the effect is homogeneous. Continuous outcomes (income, test scores) dominate in the social sciences; binary outcomes (mortality, readmission) are more common in medicine and health.

6.1.5 Randomization and observational studies

Motivation. Show why randomization identifies causal effects and which assumptions replace it in observational data.

6.1.5.1 Unconfounded assignment

An assignment mechanism is *unconfounded* if it does not depend on the potential outcomes, only on pre-treatment covariates:

$$p(X = 1 \mid Z, Y(0), Y(1)) = p(X = 1 \mid Z).$$

$p(X = 1 \mid Z)$ is the *propensity score* (Rosenbaum and Rubin 1983). In randomized experiments the mechanism is unconfounded, and moreover known and controlled by the investigator.

6.1.5.2 The role of randomization

Randomization balances observed covariates ($X \perp\!\!\!\perp Z$), unobserved covariates ($X \perp\!\!\!\perp U$) and potential outcomes, i.e. it guarantees ignorability

$$X \perp\!\!\!\perp (Y(1), Y(0)).$$

("Covariates", "pre-treatment variables" and "baseline characteristics" are used interchangeably for Z.) Under randomization causal effects are identified:

$$P(Y(x)) = P(Y(x) \mid X = x) = P(Y \mid X = x), \qquad x \in \{0, 1\},$$

the first equality by ignorability and the second by consistency. Association then implies causation (within the PO framework and its assumptions):

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid X = 1] - \mathbb{E}[Y(0) \mid X = 0] = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0].$$

As graphs. In a randomized experiment nothing points into X (e.g. $X \to Y \leftarrow Z$). A single-world intervention graph (SWIG) splits the treatment node into the observed X and the intervened value x, making potential outcomes explicit: $X \mid x \to Y(x) \leftarrow Z$. Since X and $Y(x)$ are d-separated, $Y(x) \perp\!\!\!\perp X$.

6.1.5.3 Observational studies

The most common assumptions:

1. **No unmeasured confounding:** $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$. Also called unconfoundedness, ignorability, selection on observables, no omitted variables, exogeneity, conditional exchangeability. Conditional on covariates, X_i is effectively randomly assigned.
2. **Positivity (overlap):** $0 < P(X_i = 1 \mid Z_i) < 1$; both treatment and control are possible at every value of Z_i.

We take Z as given for now; choosing it is exactly the adjustment-criterion problem. These assumptions can be wrong. Graphically, $Z \to X$, $Z \to Y$, $X \to Y$; in the SWIG, $Z \to X$, $Z \to Y(x)$, $x \to Y(x)$, and $Y(x) \perp\!\!\!\perp X \mid Z$ is read off by d-separation.

6.1.5.4 Identification of the ATE

Under positivity and no unmeasured confounding,

$$\begin{aligned}
\tau &= \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_Z\{\mathbb{E}[Y(1) - Y(0) \mid Z]\} = \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid Z] - \mathbb{E}[Y(0) \mid Z]\} \\
&= \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid X = 1, Z] - \mathbb{E}[Y(0) \mid X = 0, Z]\} && \text{(ignorability)} \\
&= \mathbb{E}_Z\{\mathbb{E}[Y \mid X = 1, Z] - \mathbb{E}[Y \mid X = 0, Z]\} && \text{(consistency).}
\end{aligned}$$

Positivity ensures that the conditional expectations in the last line are defined for every z. Define the treated and control conditional expectation functions (CEFs)

$$\mu_1(z) = \mathbb{E}[Y(1) \mid Z = z], \qquad \mu_0(z) = \mathbb{E}[Y(0) \mid Z = z],$$

describing how the means of the potential outcomes vary with covariates. The key step of the proof is

$$\underbrace{\mu_1(z)}_{\text{counterfactual}} = \underbrace{\mathbb{E}[Y \mid X = 1, Z = z]}_{\text{observational}}, \qquad \mu_0(z) = \mathbb{E}[Y \mid X = 0, Z = z].$$

6.1.5.5 Regression estimation of the ATE

Fit regressions $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$ (linear or non-linear); the safest practice is to estimate separate regressions in each treatment group. Then

$$\hat{\tau}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)).$$

Procedure: predict for all units with $X_i = 1$; predict for all units with $X_i = 0$; average the differences.

6.1.5.6 Summary

Potential outcomes	$Y(1), Y(0)$
Conditional ignorability (back-door)	$Y(1), Y(0) \perp\!\!\!\perp X \mid Z$

Identification: $\mathbb{E}[Y(1)] = \mathbb{E}[\mathbb{E}[Y(1) \mid Z]] = \mathbb{E}[\mathbb{E}[Y(1) \mid X = 1, Z]] = \mathbb{E}[\mathbb{E}[Y \mid X = 1, Z]]$.

6.1.6 Appendix: Positivity and consistency in causal diagrams

Positivity concerns arrows into the treatment node. If X is a deterministic function of a pre-treatment Z (a bold/deterministic edge $Z \to X$), positivity is violated by design. In causal diagrams positivity is implicit unless otherwise specified.

Consistency concerns arrows out of the treatment node. $X \to Y$ must correspond to a well-defined intervention. If $X = x$ can be achieved in several ways with different effects on Y (weight loss via diet, exercise or surgery), the effect is ill-defined. Treatment nodes thus have a special status: they must correspond to sufficiently well-defined interventions (*What If*, §6.4).

6.1.7 Appendix: Confounding and exchangeability

- **Marginal exchangeability** $Y(x) \perp\!\!\!\perp X$ holds when X and Y have no common causes (no open back-door paths), as in a marginally randomized experiment ($X \to Y$ only).
- **Conditional exchangeability** $Y(x) \perp\!\!\!\perp X \mid Z$ holds when all back-door paths are blocked by Z, as in a conditionally randomized experiment ($Z \to X \to Y$, $Z \to Y$).
- If Z satisfies the back-door criterion, conditional exchangeability holds. The converse requires the more general adjustment criterion (Perković et al. 2018).
- Confounding (graphical, structural) = lack of exchangeability (counterfactual) (*What If*, §7.2 and Technical Point 7.1).

6.2 Inverse probability weighting and doubly robust estimation

Motivation. The central question is how to estimate $\mathbb{E}[Y(x)]$ and the ATE from observational data once unconfoundedness and positivity are granted, and how to protect the estimate against misspecification of the models we must fit.

Recall the key assumptions of the PO framework:

(1) unconfoundedness $\{Y(1), Y(0)\} \perp\!\!\!\perp X \mid Z$;

(2) positivity $0 < P(X = 1 \mid Z) < 1$;

(3) hence identification by adjustment, $\mathbb{E}[Y(x)] = \mathbb{E}_Z[\mathbb{E}[Y \mid X = x, Z]]$.

6.2.1 The propensity score

Motivation. Reduce the covariate vector to a scalar that suffices for confounding control.

Definition (Propensity score; Rosenbaum and Rubin 1983)

$e(Z) = P(X = 1 \mid Z) = \mathbb{E}[X \mid Z]$, where $Z = (Z_1, \ldots, Z_p)$ collects the p pre-treatment covariates.

Proposition 6.1 (Property 1: balancing)

$X \perp\!\!\!\perp Z \mid e(Z)$. More generally a *balancing score* is any $b(Z)$ with $X \perp\!\!\!\perp Z \mid b(Z)$; $e(Z)$ is one (Z itself trivially is another), and Rosenbaum and Rubin (1983) show that $e(Z)$ is the *coarsest* balancing score.

Proof [Supplement]. It suffices to show $P(X = 1 \mid Z, e(Z)) = P(X = 1 \mid e(Z))$. The left side is $P(X = 1 \mid Z) = e(Z)$ since $e(Z)$ is a function of Z. The right side is $\mathbb{E}[\mathbb{E}[X \mid Z] \mid e(Z)] = \mathbb{E}[e(Z) \mid e(Z)] = e(Z)$. ■

Proposition 6.2 (Property 2: unconfoundedness given the PS)

If $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$, then $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid e(Z_i)$.

Proof [Supplement]. $P(X = 1 \mid Y(1), Y(0), e(Z)) = \mathbb{E}[P(X = 1 \mid Y(1), Y(0), Z) \mid Y(1), Y(0), e(Z)] = \mathbb{E}[e(Z) \mid Y(1), Y(0), e(Z)] = e(Z)$, which does not depend on the potential outcomes. ■

Given covariates that ensure unconfoundedness, adjusting for differences in the propensity score removes all biases associated with differences in the covariates; $e(Z)$ is a summary score of the observed covariates.

6.2.2 Inverse probability weighting

Motivation. Estimate potential-outcome means by reweighting observed outcomes.

6.2.2.1 The IPW identity

$$\text{ATE} = \mathbb{E}\left[\frac{XY}{e(Z)} - \frac{(1 - X)Y}{1 - e(Z)}\right].$$

Using consistency ($XY = XY(1)$) and unconfoundedness ($X \perp\!\!\!\perp Y(1) \mid Z$),

$$\mathbb{E}\left[\frac{XY}{e(Z)}\right] = \mathbb{E}\left[\frac{\mathbb{E}[XY(1) \mid Z]}{e(Z)}\right] \overset{\text{uncf.}}{=} \mathbb{E}\left[\frac{\mathbb{E}[X \mid Z]\,\mathbb{E}[Y(1) \mid Z]}{e(Z)}\right] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid Z]\big] = \mathbb{E}[Y(1)],$$

and symmetrically for the control term. This is the Horvitz–Thompson (HT) estimator of survey sampling, also called inverse probability of treatment weighting (IPTW).

6.2.2.2 Estimators

With an estimated PS $\hat{e}$, define the weights $\hat{w}_1(Z_i) = 1/\hat{e}(Z_i)$ for $X_i = 1$ and $\hat{w}_0(Z_i) = 1/(1-\hat{e}(Z_i))$ for $X_i = 0$. The difference of weighted means

$$\hat{\tau}_{\text{ipw},1} = \frac{1}{N}\left(\sum_{i=1}^{N} \frac{Y_i X_i}{\hat{e}(Z_i)} - \sum_{i=1}^{N} \frac{Y_i(1-X_i)}{1-\hat{e}(Z_i)}\right) = \frac{1}{N}\sum_{i=1}^{N} \{Y_i X_i \hat{w}_1(Z_i) - Y_i(1-X_i)\hat{w}_0(Z_i)\}$$

is unbiased when $\hat{e} = e$.

6.2.2.3 Normalizing the weights

Since $\mathbb{E}\big[\mathbb{E}[X \mid Z]/e(Z)\big] = 1$, the weights average to one in expectation, but $\frac{1}{N}\sum_i X_i/\hat{e}(Z_i)$ need not equal one in a sample. Good practice for any weighting method is to normalize so that the weights within each group sum to one: divide $\hat{w}_i$ by $\sum_{j:X_j=x} \hat{w}_j$. This gives the Hájek estimator ("normalized" or "self-normalized" weights):

$$\hat{\tau}_{\text{ipw},2} = \frac{\sum_i Y_i X_i \hat{w}_1(Z_i)}{\sum_i X_i \hat{w}_1(Z_i)} - \frac{\sum_i Y_i(1-X_i)\hat{w}_0(Z_i)}{\sum_i (1-X_i)\hat{w}_0(Z_i)}.$$

It typically reduces variance, $V(\hat{\tau}_{\text{ipw},2}) \leq V(\hat{\tau}_{\text{ipw},1})$, and gives more stable estimates (Hirano, Imbens and Ridder 2003).

6.2.2.4 The weighted population

IPW creates a weighted (target) population that the sample represents. In it, covariate distributions are balanced between groups:

$$\mathbb{E}\left[\frac{X}{e(Z)}Z\right] = \mathbb{E}\left[\frac{1-X}{1-e(Z)}Z\right] \ (= \mathbb{E}[Z]),$$

since $\mathbb{E}[XZ/e(Z)] = \mathbb{E}[\mathbb{E}[X \mid Z]Z/e(Z)] = \mathbb{E}[Z]$. Intuition: the less likely a subject was to be treated, the larger the population it should represent.

6.2.2.5 Pros and cons

Advantages: simple, with a theoretical foundation (Horvitz–Thompson); model-free for the outcome; explicit target population (ATE, ATT, ...) via the choice of weights; extends to complex settings (time-varying treatments, marginal structural models).
Disadvantages: more sensitive to misspecification of the PS; propensity scores near 0 or 1 yield extreme weights, handled by many ad hoc decisions.

6.2.2.6 Trimming and truncation

- **Symmetric trimming** (Crump et al. 2009): exclude units whose estimated PS lies outside $[\alpha, 1-\alpha]$; rule of thumb $\alpha = 0.1$.
- **Asymmetric trimming** (Stürmer et al. 2010): exclude units outside the common PS range of treated and controls; then further exclude units whose PS is below the q quantile of the treated, and units whose PS is above the $(1-q)$ quantile of the controls.
- **Truncation:** set estimated PS below α to α and above $1-\alpha$ to $1-\alpha$.

Picture: the PS histograms of untreated (left-heavy) and treated (right-heavy) units have non-overlapping tails; range restriction removes the lowest percentiles of the treated and the highest percentiles of the untreated.

6.2.2.7 PS weighting in randomized trials

Although developed for observational studies, PS weighting can be used for covariate adjustment in RCTs: chance imbalances are common, and adjustment improves precision. The procedure is the same as in observational studies (with an estimated, rather than the known, PS).

6.2.2.8 Weighting versus balancing

IPW balances covariates *in expectation* but not in finite samples. A recent line of work finds weights w_i directly so that weighted covariate means are (approximately) balanced, bypassing PS estimation:

$$\frac{1}{N_1}\sum_{i=1}^{N} w_i X_i Z_i \approx \frac{1}{N_0}\sum_{i=1}^{N} w_i (1-X_i) Z_i,$$

typically minimizing a measure of weight dispersion (e.g. entropy) subject to balance constraints, by numerical optimization. Examples: entropy balancing (Hainmueller 2011), covariate balancing propensity score (Imai and Ratkovic 2014), stabilized balancing weights (Zubizarreta 2015). These balance only the functions (moments) of Z the analyst specifies. If the true outcome model is $Y \sim Z_1 + Z_2 + Z_1 Z_2$ and we balance $Z_1, Z_2, Z_1^2, Z_2^2, \ldots$ but not $Z_1 Z_2$, the interaction may be severely imbalanced and the result biased.

6.2.3 Doubly robust estimation

Motivation. Combine outcome regression and IPW so that consistency requires only one of the two models to be correct.

6.2.3.1 Terminology

- **Correctly specified model:** the true function belongs to the assumed class (e.g. $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$ fitted with a linear model with interaction).
- **Misspecified model:** it does not (e.g. the truth is non-linear in Z but a linear model is fitted). The fit converges to the best approximation within the class, which is generally biased.
- **Consistent estimator:** $\hat{\theta}_N \xrightarrow{p} \theta$ as $N \to \infty$. Correct specification gives consistency (under regularity conditions); misspecification generally does not.

6.2.3.2 Two building blocks

Since $\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$ is symmetric under $X \leftrightarrow 1-X$, $e \leftrightarrow 1-e$, focus on $\mu_1 = \mathbb{E}[Y(1)]$. Let $m_x(Z) = \mathbb{E}[Y(x) \mid Z]$ (written $\mu_x(z)$ in Section 6.1).

- **Outcome regression:** $\hat{\mu}_{1,\text{adj}} = N^{-1}\sum_i \hat{m}_1(Z_i)$; consistent if m_1 is correctly specified.
- **IPW:** $\hat{\mu}_{1,\text{ipw}} = N^{-1}\sum_i Y_i X_i/\hat{e}(Z_i)$; consistent if the PS model (e.g. logistic) is correct, but can be inefficient.

Can the virtues be combined? Yes:

$$\hat{\mu}_{1,\text{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right].$$

6.2.3.3 Verification with the true nuisance functions

$$\mathbb{E}\left[\frac{XY}{e(Z)} - \frac{X-e(Z)}{e(Z)}m_1(Z)\right] = \mathbb{E}\left[\frac{XY - Xm_1(Z) + e(Z)m_1(Z)}{e(Z)}\right] = \mathbb{E}\left[m_1(Z) + \frac{X}{e(Z)}\{Y - m_1(Z)\}\right] = \mu_1,$$

since by iterated expectations and $\mathbb{E}[X \mid Z] = e(Z)$ (as in the IPW derivation) $\mathbb{E}[X\{Y(1) - m_1(Z)\}/e(Z)] = \mathbb{E}[\mathbb{E}[Y(1) - m_1(Z) \mid Z]] = 0$.

6.2.3.4 Two equivalent forms

$$\begin{aligned}\hat{\mu}_{1,\text{dr}} &= \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right] \qquad \text{(IPW augmented by OR)}\\ &= \frac{1}{N}\sum_{i=1}^{N}\left[\hat{m}_1(Z_i) + \frac{X_i\{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}(Z_i)}\right] \qquad \text{(OR augmented by IPW residual correction).}\end{aligned}$$

The forms are algebraically identical but suggest different statistical interpretations.

6.2.3.5 What does it estimate?

With arbitrary limits $\hat{e}$, $\hat{m}_1$, using $XY = XY(1)$:

$$\begin{aligned}\mathbb{E}\left[\frac{XY}{\hat{e}(Z)} - \frac{X-\hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right] &= \mathbb{E}\left[Y(1) + \frac{X-\hat{e}(Z)}{\hat{e}(Z)}Y(1) - \frac{X-\hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right]\\ &= \mathbb{E}[Y(1)] + \underbrace{\mathbb{E}\left[\frac{X-\hat{e}(Z)}{\hat{e}(Z)}\{Y(1) - \hat{m}_1(Z)\}\right]}_{R}.\end{aligned}$$

$\hat{\mu}_{1,\text{dr}}$ estimates $\mathbb{E}[Y(1)]$ iff $R = 0$: a regression estimator augmented by weighted residuals.

Scenario 1: PS wrong, OR right ($\hat{m}_1 = m_1$). Conditioning on Z and using unconfoundedness,

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right], \qquad \mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z] = m_1(Z) - \hat{m}_1(Z) = 0.$$

Scenario 2: PS right, OR wrong ($\hat{e} = e$).

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right] = 0 \quad \text{since } e(Z) - \hat{e}(Z) = 0.$$

In both cases $R = 0$ and $\hat{\mu}_{1,\mathrm{dr}}$ is consistent. By the same argument with $X \to 1 - X$, $e \to 1 - e$, $m_1 \to m_0$,

$$\hat{\mu}_{0,\mathrm{dr}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(1 - X_i) Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)} \hat{m}_0(Z_i) \right]$$

is consistent for $\mathbb{E}[Y(0)]$, and if both models are correct it is of course consistent too.

6.2.3.6 The doubly robust ATE estimator

$$\hat{\tau}_{\mathrm{dr}} = \underbrace{\frac{1}{N} \sum_{i=1}^{N} \left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)} \hat{m}_1(Z_i) \right]}_{\hat{\mu}_{1,\mathrm{dr}}} - \underbrace{\frac{1}{N} \sum_{i=1}^{N} \left[\frac{(1 - X_i) Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)} \hat{m}_0(Z_i) \right]}_{\hat{\mu}_{0,\mathrm{dr}}}.$$

6.2.3.7 Concluding remarks

- **Double robustness:** $\hat{\tau}_{\mathrm{dr}}$ is consistent if either the PS model or the outcome model (not necessarily both) is correctly specified: two chances to get it right. DR is a large-sample property.
- If both e and m_x are correct, $\hat{\tau}_{\mathrm{dr}}$ has smaller asymptotic variance than IPW. If the outcome model is correct, it has larger asymptotic variance than the direct regression estimator.
- Empirically the outcome model matters more: misspecifying $\hat{m}$ causes much larger bias than misspecifying $\hat{e}$, and even with a correct $\hat{e}$ a misspecified $\hat{m}$ usually gives notable bias; the extra protection from PS augmentation is somewhat limited.
- Kang and Schafer (2007, *Stat. Sci.*) give an example where moderate misspecification of both models (a likely scenario) leads to large bias and variance of DR.

6.2.3.8 Summary

1. The propensity score $e(Z) = P(X = 1 \mid Z)$ is a scalar summary of covariates that balances treatment groups.
2. IPW reweights by $1/e(Z)$ to remove confounding. Pros: simple, outcome-model-free. Cons: extreme weights.
3. Outcome regression models $\mathbb{E}[Y \mid X, Z]$ directly. Pros: efficient when correct. Cons: sensitive to misspecification.
4. Doubly robust estimation combines both and is consistent if either model is correct.

6.3 Conditional average treatment effects

Motivation. The central question is how treatment effects vary with covariates, why that matters, and how flexible machine-learning "meta-learners" estimate the conditional average treatment effect $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$.

6.3.1 Why heterogeneous treatment effects?

Motivation. Show that a single average can hide opposite effects in subgroups.

6.3.1.1 The average can be misleading

Consider 20 individuals, half women ($Z = 1$) and half men ($Z = 0$), with a null average effect:

$$P(Y(1) = 1) - P(Y(0) = 1) = 0.5 - 0.5 = 0.$$

Yet among women the causal risk difference is $0.6 - 0.4 = +0.2$ (treatment increases the risk of death) and among men $0.4 - 0.6 = -0.2$ (it decreases it). Indeed $\frac{1}{2}(+0.2) + \frac{1}{2}(-0.2) = 0$. *There is generally no such thing as "the causal effect of treatment"; it depends on the characteristics of the population studied* (Hernán and Robins, *What If*, §4.1).

6.3.1.2 Effect modification

Definition

Z is a *modifier* of the effect of X on Y when the average causal effect of X on Y varies across levels of Z.

- **Additive** effect modification: $\mathbb{E}[Y(1) - Y(0) \mid Z = 1] \neq \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$.
- **Multiplicative** effect modification: $\dfrac{\mathbb{E}[Y(1) \mid Z = 1]}{\mathbb{E}[Y(0) \mid Z = 1]} \neq \dfrac{\mathbb{E}[Y(1) \mid Z = 0]}{\mathbb{E}[Y(0) \mid Z = 0]}$.

Modification may appear on one scale but not on the other (e.g. risks $0.2 \to 0.4$ and $0.4 \to 0.8$ have equal ratios but different differences). *Qualitative* effect modification (effects in opposite directions across strata) implies modification on both scales.

6.3.1.3 Why care?

1. **Policy targeting:** identify who benefits. If treatment harms women and helps men (qualitative modification), a physician would, absent other information, treat only men.
2. **Transportability:** if the distribution of modifiers (sex, age) differs between the study and a target population, so does the ATE.
3. **Mechanisms:** modification may reveal biological, social or other mechanisms.

Conditioning on the full covariate vector gives the CATE, $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$ (*What If*, §4.3).

6.3.1.4 From stratification to CATE estimation

The traditional approach computes the effect within each stratum of Z and compares stratum-specific measures. Its limitations: continuous modifiers must be discretized; the number of strata grows exponentially with the number of covariates (curse of dimensionality); strata may contain too few observations. Modern CATE estimators (meta-learners, causal forests) use flexible ML models to estimate $\tau(z)$ without explicit stratification (*What If*, §4.2).

6.3.2 Meta-learners for CATE

Motivation. Build CATE estimators from arbitrary regression algorithms (Künzel et al. 2019, PNAS).

Under unconfoundedness and positivity, $\mu_x(z) := \mathbb{E}[Y(x) \mid Z = z] = \mathbb{E}[Y \mid X = x, Z = z]$, so any regression method can be used.

6.3.2.1 T-learner and S-learner

T-learner (two models): fit $\hat{\mu}_1(z)$ on the treated and $\hat{\mu}_0(z)$ on the controls separately;

$$\hat{\tau}_T(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z), \qquad \widehat{\mathrm{ATE}}_T = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\right).$$

S-learner (single model): fit one model $\hat{\mu}(z, x) \approx \mathbb{E}[Y \mid Z = z, X = x]$ with X as an ordinary feature;

$$\hat{\tau}_S(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0), \qquad \widehat{\mathrm{ATE}}_S = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\mu}(Z_i, 1) - \hat{\mu}(Z_i, 0)\right).$$

The T-learner can be noisy when one arm is small; the S-learner may shrink the effect of X towards zero if the learner regularizes it away.

6.3.2.2 Regression imputation

Plug in $\hat{\mu}$ only for the *unobserved* counterfactual and use Y_i for the observed one:

$$\hat{\tau}_{RI} = \frac{1}{n}\sum_{i=1}^{n}\Big[\big(X_i Y_i + (1 - X_i)\hat{\mu}_1(Z_i)\big) - \big((1 - X_i)Y_i + X_i\hat{\mu}_0(Z_i)\big)\Big].$$

For treated units ($X_i = 1$): observed Y_i for $Y(1)$, imputed $\hat{\mu}_0(Z_i)$ for $Y(0)$; for controls: imputed $\hat{\mu}_1(Z_i)$ for $Y(1)$, observed Y_i for $Y(0)$.

6.3.2.3 X-learner

1. Estimate the response functions $\hat{\mu}_0(z) \approx \mathbb{E}[Y(0) \mid Z = z]$ and $\hat{\mu}_1(z) \approx \mathbb{E}[Y(1) \mid Z = z]$.
2. Impute individual treatment effects: for the treated $\tilde{D}_i^1 := Y_i - \hat{\mu}_0(z_i)$; for the controls $\tilde{D}_i^0 := \hat{\mu}_1(z_i) - Y_i$. Regress to obtain $\hat{\tau}_1(z) \approx \mathbb{E}[\tilde{D}_i^1 \mid Z = z]$ (from the treated) and $\hat{\tau}_0(z) \approx \mathbb{E}[\tilde{D}_i^0 \mid Z = z]$ (from the controls).
3. Combine with a weight function $g(z) \in [0, 1]$, e.g. $g(z) = \hat{e}(z)$:

$$\hat{\tau}(z) = g(z)\,\hat{\tau}_0(z) + (1 - g(z))\,\hat{\tau}_1(z).$$

Illustration. In a simulation (one covariate $x \in [-1, 1]$) the true effect is $\tau = 1$ and treated units are rare ($g = \hat{e} \approx 0.05$). Step 1 fits $\hat{\mu}_0$ as a piecewise-linear function (many control data) and $\hat{\mu}_1$ as a linear function (few treated data). $\hat{\mu}_0$, fitted on the large control group, is flexible and contains jumps; $\hat{\mu}_1$, fitted on few treated points, is nearly linear. The T-learner $\hat{\tau}_T = \hat{\mu}_1 - \hat{\mu}_0$ inherits the jumps of $\hat{\mu}_0$, while the X-learner $\hat{\tau}_X \approx (1 - g)\hat{\tau}_1$ is nearly linear and close to the true τ: $\hat{\tau}_1$ uses the imputed effects of the treated, computed against the accurate $\hat{\mu}_0$, and is smooth because the effect itself is simple.

6.3.2.4 DR-learner

Apply the doubly robust score of Section 6.2 to learn $\tau(z)$:

1. Estimate the nuisances $\hat{\mu}_x(z)$ and $\hat{e}(z)$.
2. Construct the pseudo-outcome
$$\tilde{Y}_i = \frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1-X_i)(Y_i - \hat{\mu}_0(Z_i))}{1-\hat{e}(Z_i)} + \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i).$$
3. Regress $\tilde{Y}_i$ on Z_i with any flexible method to obtain $\hat{\tau}(z)$.

Why the DR-learner?

- $\mathbb{E}[\tilde{Y}_i \mid Z_i = z] = \tau(z)$ when the nuisances are correct (by the same computation as for the DR estimator, conditionally on Z); in fact one correct nuisance suffices.
- *Neyman orthogonality:* errors in $\hat{\mu}, \hat{e}$ affect $\hat{\tau}$ only at second order (through products of the two errors).
- It adapts to the complexity of $\tau(z)$, even when the nuisance functions are more complex (Kennedy 2023).
- In practice use K-fold cross-fitting: estimate the nuisances on folds $\neq k$ and construct $\tilde{Y}_i$ on fold k, to avoid overfitting bias.

6.4 The R-learner

Motivation. The central question is how to estimate the conditional average treatment effect with a loss function whose minimizer is the CATE itself, so that errors in the nuisance regressions affect the estimate only at second order.

6.4.1 Motivation and identification

Motivation. Set up the target and the residualised representation.

We observe i.i.d. $O_i = (X_i, Z_i, Y_i)$ with $X_i \in \{0,1\}$ and want $\tau(z) = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = z]$. S-, T- and X-learners are easy to explain but can pass nuisance-regression bias into $\hat{\tau}$. DR-learners build a pseudo-outcome from a Neyman-orthogonal DR score and regress it on Z. The R-learner instead builds a *loss function whose target is $\tau(\cdot)$ itself* (Nie and Wager 2021).

Assumptions: consistency $Y_i = Y_i(X_i)$, unconfoundedness $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i \mid Z_i$, overlap $0 < e(Z_i) < 1$. **Nuisances:**

$$e(z) = \Pr(X = 1 \mid Z = z), \qquad m(z) = \mathbb{E}[Y \mid Z = z],$$

where m is the Z-only outcome mean, not an arm-specific mean μ_x. Residuals: $\tilde{Y}_i = Y_i - m(Z_i)$, $\tilde{X}_i = X_i - e(Z_i)$. The R-learner uses

$$\tilde{Y}_i = \tilde{X}_i \, \tau(Z_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$

Robinson's partially linear decomposition with a varying coefficient.

Derivation. Fix $Z = z$ and let $\mu_x(z) = \mathbb{E}[Y(x) \mid Z = z]$, $\tau(z) = \mu_1(z) - \mu_0(z)$. By consistency and unconfoundedness, $\mathbb{E}[Y \mid X = x, Z = z] = \mu_0(z) + x\,\tau(z)$. Since m averages over the observed treatment mix,

$$m(z) = \mathbb{E}[Y \mid Z = z] = e(z)\mu_1(z) + \{1 - e(z)\}\mu_0(z) = \mu_0(z) + e(z)\tau(z),$$

so $\mathbb{E}[Y - m(Z) \mid X, Z = z] = \{X - e(z)\}\tau(z)$, i.e. $\mathbb{E}[\tilde{Y} \mid X, Z = z] = \tilde{X}\tau(z)$; the remainder is ε.

6.4.2 The R-loss

Motivation. Turn the representation into a loss and interpret its minimizer.

Oracle R-loss. Since $\mathbb{E}[\tilde{Y} \mid X, Z] = \tilde{X}\tau(Z)$, an oracle knowing m and e could estimate τ by minimizing

$$L_R(\tau) = \mathbb{E}\big[\{\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\}^2\big], \qquad \tilde{\tau} = \arg\min_{\tau} \Big\{ \frac{1}{n}\sum_{i=1}^{n} \big(\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\} \Big\},$$

with a penalty/regularizer Λ_n (Robinson 1988; Nie and Wager 2021, §2).

A residual regression. Conditionally on $Z = z$ the oracle loss is $Q_z(t) = \mathbb{E}[(\tilde{Y} - \tilde{X}t)^2 \mid Z = z]$, a least-squares regression of $\tilde{Y}$ on $\tilde{X}$ through the origin. Its slope solves $0 = \mathbb{E}[\tilde{X}(\tilde{Y} - \tilde{X}t_z) \mid Z = z]$, so

$$t_z = \frac{\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z]}{\mathbb{E}[\tilde{X}^2 \mid Z = z]}.$$

Why this slope is the CATE. By iterated expectations, $\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z] = \mathbb{E}[\tilde{X}\,\mathbb{E}[\tilde{Y} \mid X, Z = z] \mid Z = z] = \mathbb{E}[\tilde{X}^2 \mid Z = z]\tau(z)$, hence $t_z = \tau(z)$. *The CATE is the population slope of outcome residuals on treatment residuals.*

6.4.3 Algorithm

Motivation. Implement the R-learner with cross-fitting.

1. Split the data into K folds (cross-fitting: each unit's residual uses nuisance models trained on other folds).
2. For each fold k, fit nuisances on the other folds: $\hat{m}^{(-k)}(z) \approx \mathbb{E}[Y \mid Z = z]$, $\hat{e}^{(-k)}(z) \approx \Pr(X = 1 \mid Z = z)$.
3. Out-of-fold residuals: $Y_i^r = Y_i - \hat{m}^{(-k_i)}(Z_i)$, $X_i^r = X_i - \hat{e}^{(-k_i)}(Z_i)$.
4. Estimate $\hat{\tau} = \arg\min_\tau \Big\{ \frac{1}{n}\sum_{i=1}^{n} \big(Y_i^r - X_i^r\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\} \Big\}$.

Workflow: data $(X, Z, Y) \to$ cross-fit $\hat{m}, \hat{e} \to$ residualise $Y - \hat{m}(Z)$, $X - \hat{e}(Z) \to$ minimize the R-loss $\to \hat{\tau}(z)$. First-stage prediction creates residualised signals; the second stage learns τ.

The R-loss weights residual treatment variation.

$$\sum_i \big(Y_i^r - X_i^r\tau(Z_i)\big)^2 = \sum_i (X_i^r)^2 \Big(\frac{Y_i^r}{X_i^r} - \tau(Z_i)\Big)^2.$$

Units with large $|X_i^r|$ (treatment poorly predicted by Z_i) carry information about $\tau(Z_i)$; units with small $|X_i^r|$ (treatment nearly determined by Z_i) put little pressure on the local $\tau(Z_i)$. R-learners learn the CATE where residual treatment variation remains.

What can optimize the R-loss? The second stage is empirical risk minimization, $\min_{\tau\in\mathcal{F}} \frac{1}{n}\sum_i (Y_i^r - X_i^r\tau(Z_i))^2 + \Lambda_n\{\tau\}$: simple choices (linear or sparse linear $\tau(z)$, splines, kernels) or flexible ones (boosting, random forests, neural networks, any learner with a squared-loss interface, using weights $(X_i^r)^2$ and targets Y_i^r/X_i^r). The learner for τ is tuned by *validation R-loss*, not by outcome prediction loss.

6.4.4 Orthogonality and the quasi-oracle property

Motivation. Show why nuisance errors matter only at second order.

A population moment is an expectation equation that is zero in the target population (before finite-sample fluctuation). For a candidate τ, use

$$\psi(O;\tau,m,e) = \{Y - m(Z) - (X - e(Z))\tau(Z)\}\{X - e(Z)\}, \qquad \mathbb{E}[\psi(O;\tau^*,m,e) \mid Z = z] = 0 \text{ at the truth.}$$

Proposition 6.3 (Neyman orthogonality)

Let $\hat{m} = m + \delta_m$, $\hat{e} = e + \delta_e$ and hold $\tau = \tau^*$. Then

$$\mathbb{E}[\psi(O;\tau^*,\hat{m},\hat{e}) \mid Z = z] = \underbrace{\delta_m(z)\delta_e(z)}_{\text{product}} - \underbrace{\tau^*(z)\delta_e(z)^2}_{\text{square}}.$$

Proof [Supplement]. With $\tilde{Y} = Y - m(Z)$, $\tilde{X} = X - e(Z)$, the moment equals $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^* - \delta_m + \delta_e\tau^*)(\tilde{X} - \delta_e) \mid Z]$. The term $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^*)\tilde{X} \mid Z] = \mathbb{E}[\varepsilon\tilde{X} \mid Z] = 0$; the term $-\delta_e\mathbb{E}[\tilde{Y} - \tilde{X}\tau^* \mid Z] = -\delta_e\mathbb{E}[\varepsilon \mid Z] = 0$; and $(-\delta_m + \delta_e\tau^*)\mathbb{E}[\tilde{X} - \delta_e \mid Z] = (-\delta_m + \delta_e\tau^*)(-\delta_e)$ since $\mathbb{E}[\tilde{X} \mid Z] = 0$. ■

No first-order δ_m or δ_e term remains: small nuisance errors create only second-order bias, so flexible nuisance fits are usable (Nie and Wager 2021; Chernozhukov et al. 2018).

The quasi-oracle property. Measure excess risk with the oracle population R-risk $R(\tau) = \mathbb{E}[\{Y - m(Z) - (X - e(Z))\tau(Z)\}^2]$, via $R(\tau) - R(\tau^*)$. Let $\hat{\tau}$ fit residuals from $\hat{m}, \hat{e}$ and $\tilde{\tau}$ the oracle version with true m, e. Nie and Wager show

$$R(\hat{\tau}) - R(\tau^*) \approx R(\tilde{\tau}) - R(\tau^*),$$

provided the nuisances are estimated well enough (RMSE $o(n^{-1/4})$, so their product is $o(n^{-1/2})$). The rate depends mainly on the complexity of τ: after residualisation, the second stage learns τ, not the full complexity of m and e.

Why cross-fitting matters. If observation i helps train $\hat{m}, \hat{e}$, then $\hat{m}(Z_i), \hat{e}(Z_i)$ can memorize noise in (Y_i, X_i): residuals become too small, the noise fitted by the nuisances leaks into the R-loss, and the second stage may mistake first-stage overfitting artefacts for heterogeneity in $\tau(Z)$. With cross-fitting, residuals are out-of-fold predictions, unit i's noise was not used to train its nuisances, and orthogonality can remove first-order nuisance bias. Orthogonality handles nuisance error; cross-fitting prevents reusing the same noise twice.

6.4.5 Comparison and summary

Motivation. Contrast the R-learner with the DR-learner.

	DR-learner	R-learner
Second stage	regress $\hat{\phi}_i^{DR}$ on Z_i	minimize a residualised weighted loss
Orthogonality	via the EIF / DR score	via Robinson residualisation
Main nuisances	μ_0, μ_1, e	m, e
Model selection	validation MSE for $\hat{\phi}^{DR}$	validation R-loss

DR pseudo-outcome:

$$\hat{\phi}_i^{DR} = \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{X_i\{Y_i - \hat{\mu}_1(Z_i)\}}{\hat{e}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{\mu}_0(Z_i)\}}{1 - \hat{e}(Z_i)}.$$

Validation losses on a held-out set V: DR, $\sum_{i\in V}\{\hat{\phi}_i^{DR} - \hat{\tau}(Z_i)\}^2$; R, $\sum_{i\in V}\{Y_i^r - X_i^r\hat{\tau}(Z_i)\}^2$.

What the R-learner is for. Residualised CATE estimation (estimate $\tau(Z)$ after removing prognosis $m(z)$ and assignment $e(z)$); model selection (tune and compare candidate τ models by validation

R-loss); regularized heterogeneity (put shrinkage or structure directly on τ when the effect signal is weak).

Summary.

- R-learning starts from Robinson's decomposition $Y - m(Z) = \{X - e(Z)\}\tau(Z) + \varepsilon$.
- Estimate m and e by cross-fitting, then minimize the residualised R-loss.
- The R-loss is orthogonal to nuisance error and can be optimized with many ML tools.
- The quasi-oracle result says the difficulty lies mainly in the complexity of τ, not of m and e.

Chapter 7. Matching and Instrumental Variables

7.1 Matching

Motivation. The central question is how to estimate the effect of treatment on the treated by imputing each treated unit's missing control outcome from similar control units, and how to design such comparisons so that the result does not hinge on modeling choices.

7.1.1 Matching as nonparametric imputation

Motivation. Motivate matching as a model-free way to impute counterfactual control outcomes.

7.1.1.1 Setup

N units, treatment $X_i \in \{0,1\}$, covariates Z_i, potential outcomes $Y_i(0), Y_i(1)$; $N_1 = |\{i : X_i = 1\}|$ treated, $N_0 = N - N_1$ controls. Assumptions: ignorability $(Y(0), Y(1)) \perp\!\!\!\perp X \mid Z$ and positivity $0 < P(X = 1 \mid Z) < 1$. Focus: the ATT.

7.1.1.2 From regression to matching

Under ignorability, with $\mu_t(Z) = \mathbb{E}[Y \mid X = t, Z]$,

$$\text{ATT} = \mathbb{E}_{Z|X=1}\big[\underbrace{\mu_1(Z)}_{\text{observed}} - \underbrace{\mu_0(Z)}_{\text{impute}}\big].$$

Regression performs model-based imputation of $\mu_0(Z_i)$:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{\mu}_0(Z_i)).$$

When the groups differ in Z, the estimate depends heavily on the functional form of $\hat{\mu}_0$ (model dependence, Section 7.1.5). Matching performs *nonparametric* imputation instead.

7.1.1.3 The matching idea

Often there is a large reservoir of potential controls, so each treated unit can be matched to one or more distinct controls with similar covariates.

Matching as study design. Matching is a design tool, not only an analysis technique. In many observational studies outcomes are not yet measured when matching is done: identify treated units, select which controls to recruit/follow, then measure outcomes for both. This separates the design stage (whom to compare) from the analysis stage (how to estimate); design choices are made without seeing outcomes, removing the temptation to tune the analysis toward a desired result (Stuart 2010, §1).

Bias–variance trade-off in M. Each treated unit may be matched to M controls. $M = 1$: lower bias, higher variance. $M = 3$: farther matches increase bias, while averaging three outcomes reduces variance (Abadie and Imbens 2006).

7.1.1.4 The matching estimator

Let $\mathcal{M}_i$ be the matched set of treated unit i (units with covariates similar to i's). Still assuming ignorability,

$$\hat{\tau}_{\text{match}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \underbrace{\frac{1}{|\mathcal{M}_i|}\sum_{j\in\mathcal{M}_i} Y_j}_{\approx Y_i(0)}\Big).$$

Matching can be one-to-one or one-to-many, with or without replacement, etc.

7.1.1.5 Exact matching and coarsened exact matching

Exact matching: for each treated i find a control j with $Z_j = Z_i$. It ensures perfect balance and no model dependence but is usually infeasible with many or continuous covariates.
Coarsened exact matching (CEM; Iacus et al. 2012): discretize covariates (categories may have substantive meaning), then match exactly on the coarsened values. Some treated units may still have no matches (lack of overlap).

7.1.2 Nearest-neighbor matching and its theory

Motivation. Define NN matching, compare matching algorithms, and state its large-sample behavior.

7.1.2.1 Definition

Let $\mathcal{M}_i$ be the indices of the M closest units of the opposite group in a distance $\|\cdot\|$. The imputed potential outcomes are

$$\hat{Y}_i(t) = \begin{cases} Y_i, & X_i = t, \\ \dfrac{1}{M}\sum_{j\in\mathcal{M}_i} Y_j, & X_i = 1-t, \end{cases} \qquad t \in \{0,1\},$$

and the ATT estimator is $\hat{\tau}_{\text{ATT}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{Y}_i(0))$.

7.1.2.2 Algorithms (matching without replacement)

- **Greedy:** order the treated units (e.g. hardest-to-match first); give each the M closest remaining controls, set them aside, repeat.
- **Optimal:** build the $N_1 \times N_0$ distance matrix D; choose entries so that each row has exactly M entries and each column at most one, minimizing the total distance. The Hungarian algorithm solves this in polynomial time.

Why order matters. With $M = 1$ and the same data, an arbitrary order lets i_2 take the only control near the isolated i_3 (distance 0.9), forcing i_3 to a control 2.9 away (worst distance 2.9). Processing the hardest unit first, i_3 secures its nearby control (1.2), i_2 takes another at 1.1, and the worst distance drops to 1.2.

7.1.2.3 The curse of dimensionality

In one dimension N uniform points have spacing about $1/N$; in p dimensions about $N^{-1/p}$ per axis, so the nearest-neighbor distance is $O(N^{-1/p})$. If μ_0 is Lipschitz with constant L,

$$|\mu_0(Z_i) - \mu_0(Z_j)| \le L \, \|Z_i - Z_j\| = O(N^{-1/p}),$$

which is the bias from mismatch. The rate slows dramatically with p:

p	1	5	10	20
$N^{-1/p}$ for $N = 10{,}000$	0.0001	0.16	0.40	0.63

This motivates dimension reduction, e.g. propensity-score matching (Abadie and Imbens 2006, Thm. 1).

7.1.2.4 ATT matching: independence and consistency

For NN matching with fixed M and replacement, the ATT imputes $\hat{Y}_i(0)$ for each treated unit from controls; each treated unit appears exactly once as a subject, so the imputed counterfactuals are independent across treated units. Independence plus the law of large numbers (and the bias being negligible when the control reservoir is large) give $\sqrt{N}$-consistency for the ATT. Matching is most natural for ATT estimation (Abadie and Imbens 2006, Thms. 1–4).

7.1.2.5 A toy example

$N_1 = 3$ treated, $N_0 = 6$ controls, $p = 2$, $M = 1$, distance $\|\cdot\|_1$:

	Treated				Control		
Unit	Z_1	Z_2	Y	Unit	Z_1	Z_2	Y
$i = 1$	3	5	8	$j = 1$	1	2	2
$i = 2$	6	2	12	$j = 2$	2	6	4
$i = 3$	7	7	10	$j = 3$	4	1	6
				$j = 4$	3	4	5
				$j = 5$	5	3	7
				$j = 6$	8	6	9

Matches: $i = 1 \to j = 4$ (distance 1), $i = 2 \to j = 5$ (2), $i = 3 \to j = 6$ (2). Hence

$$\hat{\tau}_{\text{ATT}} = \tfrac{1}{3}\big[(8 - 5) + (12 - 7) + (10 - 9)\big] = 3.0.$$

The matches are imperfect ($Z_2 - Z_5 \neq 0$), and the mismatch creates bias, which the next estimator corrects.

7.1.2.6 Bias-corrected matching (Abadie and Imbens 2011)

NN matching uses Y_j to estimate $Y_i(0)$, but Y_j approximates $\mu_0(Z_j)$, not $\mu_0(Z_i)$. Adjust each control outcome by the predicted difference due to mismatch:

$$\hat{\tau}_{\text{bc}} = \frac{1}{N_1} \sum_{i:X_i=1} \Big(Y_i - \frac{1}{M} \sum_{j \in \mathcal{M}_i} \big[Y_j + \hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j) \big] \Big),$$

where $\hat{\mu}_0$ is a regression fit of $\mathbb{E}[Y \mid Z, X = 0]$. Naive matching uses Y_j as $Y_i(0)$, ignoring the slope of μ_0 between Z_j and Z_i; the correction shifts by $\hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j)$. This is the default in the R packages `MatchIt` and `Matching`.

7.1.2.7 Inference after matching

Variance estimator for the ATT (Abadie and Imbens 2006):

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{ATT}}) = \frac{1}{N_1^2}\sum_{i=1}^{N}\big(1 + K_M(i)\big)^2\hat{\sigma}_i^2,$$

where $K_M(i)$ is the number of times unit i is used as a match and $\hat{\sigma}_i^2$ estimates $\mathrm{Var}(Y_i \mid Z_i, X_i)$. Units matched often inflate the variance, analogous to extreme IPW weights. **The bootstrap is invalid** for NN matching: bootstrap validity requires small data changes to cause small estimate changes, but the identity of the matched unit is a discontinuous function of the data (Abadie and Imbens 2008).

7.1.2.8 Kernel matching

All controls within a bandwidth h get positive weight, closer ones more. With a kernel K (e.g. Gaussian $K(u) = e^{-u^2/2}$),

$$w_i(j) = \frac{K\Big(\frac{Z_j - Z_i}{h}\Big)}{\sum_{k:X_k=0} K\Big(\frac{Z_k - Z_i}{h}\Big)}, \qquad \hat{\tau}_{\mathrm{kernel}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \sum_{j:X_j=0} w_i(j)Y_j\Big).$$

For large p kernel matching faces the same curse of dimensionality.

7.1.3 Propensity-score matching

Motivation. Reduce the matching problem to one dimension and understand the caveats.

Dimension reduction. NN bias grows like $O(N^{-1/p})$; with 20 binary covariates there are $2^{20} \approx 10^6$ patterns and direct matching is nearly impossible. The propensity score $e(Z) = P(X = 1 \mid Z)$ reduces p dimensions to one, and $Y(0), Y(1) \perp\!\!\!\perp X \mid Z$ implies $Y(0), Y(1) \perp\!\!\!\perp X \mid e(Z)$.

Distance. $d(Z_i, Z_j) = |\hat{e}(Z_i) - \hat{e}(Z_j)|$, or (preferred) on the logit scale

$$d(Z_i, Z_j) = |\mathrm{logit}\,\hat{e}(Z_i) - \mathrm{logit}\,\hat{e}(Z_j)|, \qquad \mathrm{logit}(p) = \ln\frac{p}{1-p}.$$

Distances that are small on the probability scale are compressed near 0 and 1; the logit stretches them out.

Specification: balance, not fit. The PS is usually estimated by logistic regression, but GBMs, random forests, etc. are also used. The goal is covariate balance, not accurate prediction of treatment. Collinearity is not a concern (coefficients are not interpreted); overfitting to the sample can even help (Appendix). Specify iteratively: check balance on covariates, squares and interactions; if imbalanced, add these terms and re-estimate. With missing covariate values, impute and include missingness indicators (Stuart 2010; Rosenbaum and Rubin 1984).

Common support. Each treated unit needs controls with similar covariates. Units outside the overlap region of the two PS distributions have no good matches and should be trimmed.

Caveats. Matching on the *estimated* PS can be more efficient than on the true PS (Abadie and Imbens 2016): estimation noise adapts to the sample, so matched units are closer in covariate space; this relies on correct specification. King and Nielsen (2019) describe the "PS paradox": a narrower PS caliper improves PS balance but can worsen covariate balance, because units with similar $\hat{e}(Z)$ can have very different Z. Always check multivariate covariate balance, not just PS balance.

7.1.4 Practical matching

Motivation. Describe the iterative matching workflow and its design choices.

Workflow. Choose a distance → perform matching → check balance → if imbalanced, adjust (metric, M, caliper) and repeat; if balanced, estimate and do inference. After matching, regression or weighting on the matched sample can correct residual imbalance.

Design choices.

- Which covariates to include (recall the adjustment criterion; never include post-treatment variables).
- Distance metric: Mahalanobis, propensity score, or hybrid.
- Number of matches M: fixed or varying (caliper-based). Smaller M: smaller bias, larger variance; larger M: the reverse. With a much larger control group, 1-to-many matching is natural.
- With or without replacement.

Mahalanobis distance accounts for correlations and scales: $d(x, y) = \sqrt{(x-y)^\top S^{-1}(x-y)}$ with S the sample covariance (treated group or pooled). With few continuous covariates ($p \lesssim 5$–8) it works well; with many covariates or a complex treatment model the PS gives effective dimension reduction. Common practice: match on the PS with a caliper, then refine with Mahalanobis distance within the caliper.

Caliper matching. With caliper width c, all controls within distance $< c$ of a treated unit are potential matches, so M varies by unit (e.g. $M_1 = 2$, $M_2 = 3$, $M_3 = 1$). A common choice is $c = 0.2 \times \mathrm{SD}(\mathrm{logit}\,\hat{e}(Z))$ (Austin 2011, simulation-based).

Full matching (Rosenbaum 1991; Hansen 2004). Standard matching discards unmatched controls. Full matching places every unit into a matched set with variable ratio (some 1:k, others k:1), minimizing total within-set distance (solved efficiently by software). It uses more controls within the overlap and produces a weighted sample, a bridge between matching and IPW. Often combined with a caliper. In R: `MatchIt` with `method = "full"`.

With versus without replacement. With replacement: computationally easier, not order-dependent, but some units (especially with extreme PS) may be matched many times and dominate (like extreme IPW weights). Without replacement: avoids extreme reuse but is order-dependent (greedy vs. optimal matters). Software: `MatchIt` (Ho et al.), `Matching` (Sekhon).

Checking balance. Standardized mean difference of covariate k:

$$\mathrm{SMD}_k = \frac{1}{s_t} \cdot \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big(Z_{ik} - \frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} Z_{jk} \Big),$$

with s_t the SD of covariate k in the treated group before matching. Compare whole distributions via empirical CDFs and Kolmogorov–Smirnov statistics. **Do not use p-values** for balance: they conflate changes in balance with changes in sample size, and balance is an in-sample property (Ho et al. 2007).

Love plot. A dot plot of SMDs per covariate before and after adjustment, with reference lines at ± 0.1. In the LaLonde (1986) job-training data (propensity score, age, education, married, no degree, race indicators, earnings in 1974 and 1975), unadjusted SMDs range up to about 1.5–2 in absolute value; after matching they fall inside ± 0.1.

7.1.5 Model dependence and empirical evidence

Motivation. Explain why matching reduces sensitivity to modeling assumptions, and what the LaLonde benchmark teaches.

The extrapolation problem. When treated and control groups differ on covariates, regression must extrapolate into sparse regions. A linear and a quadratic specification can fit the control region equally well yet give very different values in the treated region. Model dependence is greatest where data are sparse (Ho, Imai, King and Stuart 2007).

Matching as preprocessing. After matching, the distributions overlap and regression interpolates rather than extrapolates; different specifications give similar estimates. Workflow:

(1) match for balance, (2) regress/weight on the matched sample, (3) obtain estimates robust to model choice. "Matching is the starting point, not the end."

The LaLonde benchmark. The National Supported Work (NSW) program was a randomized job-training experiment (LaLonde 1986), with experimental ATT $\approx$ \$1,794. Replacing the experimental controls with observational ones (CPS/PSID), unadjusted estimates range from $-$\$15,000 to $+$\$2,000, because the observational controls differ dramatically in demographics, prior earnings and employment. It is the canonical testbed for matching methods.

Lessons. Dehejia and Wahba (1999): PS matching on a subsample with pre-treatment earnings recovers the experimental benchmark. Smith and Todd (2005): the result is fragile; different subsamples, matching methods and distance metrics lead to different conclusions. A deeper issue: the DW subsample included pre-treatment earnings, arguably the most important confounder; without it, ignorability itself is doubtful whatever the method. Key lessons: covariate sufficiency comes before method choice; matching can recover experimental benchmarks, but with no guarantees.

Practical takeaways.

- Matching is a nonparametric preprocessing step, not an end in itself; then apply regression or weighting on the matched sample.
- Always check multivariate covariate balance, not just PS balance.
- In high-dimensional settings combine matching with regression.
- Sensitivity analysis for unmeasured confounding is an important complement.

7.1.6 Appendix: ATE matching: why it is harder

For the ATE one must impute $\hat{Y}_i(0)$ for treated and $\hat{Y}_i(1)$ for control units. Each unit serves both as a subject and as a match for others, creating cross-group dependence absent for the ATT or ATC alone. Moreover the bias $O(N^{-1/p})$ dominates the $O(N^{-1/2})$ sampling error when $p > 2$: $\text{Bias}(\hat{\tau}_{\text{ATE}}) = O(N^{-1/p}) \gg O(N^{-1/2})$, so simple matching is not $\sqrt{N}$-consistent for the ATE. Use bias correction (Abadie and Imbens 2011) or IPW/DR instead (Abadie and Imbens 2006).

7.1.7 Appendix: Why the estimated PS can beat the true PS

The true $e(Z)$ is a population quantity; two units can have $e(Z_i) = e(Z_j)$ yet very different covariates. The estimated $\hat{e}(Z) = g(Z^\top \hat{\beta})$ is fit to this sample, so $\hat{e}(Z_i) \approx \hat{e}(Z_j)$ means Z_i, Z_j are close in the directions that distinguish treated from control in this sample: estimation noise acts as implicit covariate refinement. Under correct specification (Abadie and Imbens 2016), $\text{Var}(\hat{\tau}_{\text{ATT}} \mid \hat{e}) \leq \text{Var}(\hat{\tau}_{\text{ATT}} \mid e)$. Analogously, IPW with $\hat{e}$ attains the semiparametric efficiency bound while IPW with e does not (Hirano, Imbens and Ridder 2003). The advantage disappears under misspecification.

7.1.8 Appendix: The PS paradox (King and Nielsen 2019)

Many covariate configurations map to the same PS. Unit A (treated) with high income and low education and unit B (control) with low income and high education may both have $\hat{e} = 0.50$: perfect PS balance, opposite covariates. A tighter PS caliper yields more "PS-similar but covariate-different" pairs, so a better PS match can mean worse covariate balance; the remaining imbalance may require more extrapolation than the raw data, increasing model dependence.

7.1.9 Appendix: PS overfitting and the broader picture

Overfitting the PS model can help: the goal is in-sample balance, and a flexible model captures sample-specific patterns so that similar $\hat{e}$ means closer Z. The limit is overlap: if $\hat{e} \to 0$ or 1 for all units, overlap is destroyed. Three results, one message: the estimated PS beats the true PS (sample adaptation), overfitting helps (the same mechanism), but the PS paradox shows that 1D compression can still lose information. The PS is a dimension-reduction tool, not a substitute for checking balance. Practical resolution: use the PS for dimension reduction (especially for large p), always verify multivariate balance, and consider Mahalanobis matching within a PS caliper.

7.1.10 Appendix: Beyond binary treatment

With multiple treatment levels (doses, programs): compare pairs by standard binary matching, or use the generalized propensity score (GPS) for K levels (Imai and van Dyk 2004). For continuous treatments the GPS is the conditional density of treatment given covariates (Hirano and Imbens 2004). In practice subclassification or weighting on the GPS is more common than matching.

7.1.11 Appendix: Bias-corrected matching versus post-matching regression

In bias-corrected matching, regression is a correction term inside each matched pair; matching is primary and regression fixes the residual $Z_i \neq Z_j$ gap. In post-matching regression, matching is preprocessing (balanced sample) and a regression $Y \sim X + Z$ on the matched data does the estimation. Why not regress on the full sample? $\hat{\mu}_0$ can be learned from all controls, and balance is irrelevant if the model is correct; but if the model is wrong, full-sample regression extrapolates and is model-dependent.

	Model correct	Model wrong
Full-sample regression	optimal (more data)	risky (extrapolation)
Post-matching regression	less efficient (data discarded)	safe (interpolation)

If controls lie in $Z \in [0, 3]$ and treated in $[5, 8]$, full regression must extrapolate $\hat{\mu}_0(6)$: a linear model gives answer A, a quadratic answer B. After matching, controls also lie near $[5, 8]$ and both models give similar answers. This is the core of "matching as preprocessing" (Ho et al. 2007).

7.1.12 Appendix: Stratification

With a single discrete covariate Z with levels k that makes assignment ignorable, $\mathbb{E}[Y(1)] = \sum_k \mathbb{E}(Y \mid Z = k, X = 1)P(Z = k)$. With n_k units in cell k and cell means $\bar{Y}_{k,x}$, estimate $\mathbb{E}[Y(1)]$ by $\sum_k \bar{Y}_{k,1} n_k/n$, hence

$$\hat{\tau} = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n}.$$

For continuous Z, split it into k classes: $\hat{\tau}_k = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) n_k/n$, generally biased for τ. Let $R_k = 1 - \frac{\mathbb{E}(\hat{\tau}_k) - \tau}{\mathbb{E}(\hat{\tau}_1) - \tau}$ be the fraction of bias removed by k-stratum stratification. Cochran (1968):

$R_k \geq 90\%$ for $k \geq 5$ across a broad class of models, so stratifying into five or more blocks removes roughly 90% of the confounding bias.

7.2 Instrumental variables

Motivation. The central question is how a causal effect can be learned when confounders are unmeasured and no adjustment set exists, by exploiting a variable that moves the treatment but affects the outcome only through it.

7.2.1 Motivation

Motivation. See instruments in action and state what makes an instrument good.

7.2.1.1 Instrumental variables in action

Study	Instrument Z	Treatment X	Outcome Y
Angrist and Krueger (1991)	quarter of birth	years of schooling	earnings
Card (1995)	proximity to college	years of schooling	earnings
Acemoglu et al. (2001)	settler mortality	institutional quality	GDP per capita
Miguel et al. (2004)	rainfall shocks	economic growth	civil conflict
Levitt (1996)	overcrowding litigation	prison population	crime rate

Returns to education. Ability U confounds schooling X and earnings Y. Angrist and Krueger: compulsory-schooling laws interact with quarter of birth; those born early in the year can drop out sooner and get less schooling. Card: growing up near a four-year college lowers the cost of attending, so proximity leads to more education and, through it, higher earnings.
Institutions and development. Geography and culture U confound institutions X and GDP Y. Acemoglu, Johnson and Robinson: high settler mortality led to extractive institutions, low mortality to inclusive ones; the institutional differences persist and explain modern income gaps. Miguel et al. use rainfall as an instrument for economic growth to study civil conflict in Africa.

7.2.1.2 Why instruments?

So far effects were identified by back-door adjustment, which requires measuring all confounders. If some are unmeasured, no adjustment set blocks the confounding. Idea: find Z that affects Y only through X ($Z \to X \to Y$, $X \leftarrow U \to Y$, $Z \perp\!\!\!\perp U$). Such Z is an *instrumental variable*.

7.2.1.3 Conditions for a good instrument

IVs are very popular in economics, but good ones are hard to find. A good instrument must satisfy

1. **Relevance:** a strong effect on the treatment (large $\mathrm{Cov}(X, Z)$); otherwise the IV estimate has large variance;
2. **Exclusion restriction:** no direct effect on the outcome;

and (implicitly) no confounding between Z and Y.

7.2.2 IV under linearity

Motivation. Derive the classical IV estimators in the linear constant-effect model.

7.2.2.1 The econometric setup

Consider $Y = \beta_0 + \beta_1 X + \beta_2 W + \varepsilon$ with scalar treatment X and covariates W; the estimand is β_1. X is *endogenous* (correlated with ε, i.e. confounded) and W is *exogenous* (uncorrelated with ε). Because ε depends on X, OLS of β_1 is biased. An instrument is a K-vector Z with $(Z, W) \perp \varepsilon$ (uncorrelatedness: $\mathrm{Cov}[\varepsilon, Z] = \mathrm{Cov}[\varepsilon, W] = 0$). If $K > 1$ the model is over-identified; if $K = 1$, just-identified.

7.2.2.2 Just-identified, no covariates

IV estimator. The ratio of covariances:

$$\hat{\beta}_{1,\mathrm{iv}} = \frac{\widehat{\mathrm{Cov}}(Y_i, Z_i)}{\widehat{\mathrm{Cov}}(X_i, Z_i)} = \frac{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})(Z_i - \bar{Z})}.$$

With a binary instrument this is the *Wald estimator*

$$\hat{\beta}_{1,\mathrm{iv}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0},$$

where $\bar{Y}_z, \bar{X}_z$ are sample means within group $Z = z \in \{0, 1\}$ (since $\widehat{\mathrm{Cov}}(Y, Z) = \hat{p}(1-\hat{p})(\bar{Y}_1 - \bar{Y}_0)$ with $\hat{p}$ the fraction with $Z = 1$, and likewise for X).

Indirect least squares (ILS). Fit two reduced-form regressions

$$Y = \pi_{10} + \pi_{11} Z + \varepsilon_1, \qquad X = \pi_{20} + \pi_{21} Z + \varepsilon_2;$$

the ratio of OLS slopes gives $\hat{\beta}_{1,\mathrm{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21} = \hat{\beta}_{1,\mathrm{iv}}$, since each slope is $\widehat{\mathrm{Cov}}(\cdot, Z)/\widehat{\mathrm{Var}}(Z)$.

Two-stage least squares (2SLS). Stage 1: regress X on Z, $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z$. Stage 2: regress $Y = \beta_0 + \beta_1 \hat{X} + \varepsilon$ by OLS to get $\hat{\beta}_{1,\mathrm{2sls}}$. Intuition: the instrument recovers the "unconfounded portion" of X. Since $\widehat{\mathrm{Cov}}(Y, \hat{X}) = \hat{\pi}_{21}\widehat{\mathrm{Cov}}(Y, Z)$ and $\widehat{\mathrm{Var}}(\hat{X}) = \hat{\pi}_{21}^2 \widehat{\mathrm{Var}}(Z)$, $\hat{\beta}_{1,\mathrm{2sls}} = \widehat{\mathrm{Cov}}(Y, Z)/(\hat{\pi}_{21}\widehat{\mathrm{Var}}(Z)) = \hat{\beta}_{1,\mathrm{iv}}$. Hence $\hat{\beta}_{1,\mathrm{iv}} = \hat{\beta}_{1,\mathrm{ils}} = \hat{\beta}_{1,\mathrm{2sls}}$.

7.2.2.3 Just-identified, with covariates

ILS: $Y = \pi_{10} + \pi_{11} Z + \pi_{12} W + \varepsilon_1$, $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $\hat{\beta}_{1,\mathrm{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21}$. 2SLS: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z + \hat{\pi}_{22} W$, then $Y = \beta_0 + \beta_1 \hat{X} + \beta_2 W + \varepsilon$; $\hat{\beta}_{1,\mathrm{2sls}}$ is the OLS coefficient of $\hat{X}$.

IV derivation. Given $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$ and $Y = \beta_0 + \beta_1 X + \underbrace{\beta_2 W + \varepsilon}_{\eta}$, assume in addition $\mathrm{Cov}[W, Z] = 0$. Then $\mathrm{Cov}[X, Z] \neq 0$ but $\mathrm{Cov}[\eta, Z] = 0$, so

$$\mathrm{Cov}[Y, Z] = \mathrm{Cov}[\beta_0 + \beta_1 X + \eta, Z] = \beta_1 \,\mathrm{Cov}[X, Z],$$

so

$$\beta_1 = \frac{\mathrm{Cov}[Y, Z]}{\mathrm{Cov}[X, Z]} = \frac{\mathrm{Cov}[Y, Z]/\,\mathrm{Var}[Z]}{\mathrm{Cov}[X, Z]/\,\mathrm{Var}[Z]} = \frac{\text{reduced form}}{\text{first stage}}.$$

2SLS derivation. Substituting X into the outcome model,

$$\begin{aligned} Y &= \beta_1(\pi_{22} W + \pi_{21} Z + \pi_{20} + \varepsilon_2) + \beta_2 W + \beta_0 + \varepsilon \\ &= \beta_1 \pi_{21} Z + (\beta_2 + \beta_1 \pi_{22}) W + \underbrace{(\beta_1(\pi_{20} + \varepsilon_2) + \beta_0 + \varepsilon)}_{\xi}, \end{aligned}$$

hence

$$Y = \beta_1(\pi_{22} W + \pi_{21} Z) + \beta_2 W + \xi = \beta_1 \underbrace{\mathbb{E}[X \mid W, Z]}_{\text{population analogue of } \hat{X}} + \beta_2 W + \xi,$$

where π_{20} is absorbed into ξ, so $\mathbb{E}[X \mid W, Z] - \pi_{20} = \pi_{21} Z + \pi_{22} W$ (intercept-free). Regressing Y on $\mathbb{E}[X \mid W, Z]$ and W recovers β_1; the sample analogue replaces it by the first-stage fitted value $\hat{X}$.

7.2.2.4 From linear to nonparametric IV

The linear analysis assumes a constant effect β_1 for everyone, and then IV = ILS = 2SLS. If effects are heterogeneous ($Y_i(1) - Y_i(0)$ varies), what does the IV estimand identify? A key setting is the randomized experiment with non-compliance, in which random assignment Z instruments treatment received X; Angrist, Imbens and Rubin (1996) give a potential-outcome framework.

7.2.3 Randomized experiments with non-compliance

Motivation. Identify the effect for compliers under random assignment, monotonicity and exclusion.

7.2.3.1 Setting

A drug is tested in an RCT: Z = random assignment, X = treatment actually received, Y = health outcome. Some patients assigned to treatment refuse the drug; some controls obtain it elsewhere. Comparing $X = 1$ with $X = 0$ is confounded (compliance is self-selected). Use Z as an instrument: it is randomized (independent of potential outcomes) and affects Y only through X (exclusion).

Potential outcomes (Angrist, Imbens and Rubin 1996, JASA). Outcomes are indexed by assignment: $Y(z)$, $z = 0, 1$. Treatment received is a post-assignment variable with potential values $X(z)$. Observed data: Z, $X = X(Z)$, $Y = Y(Z)$. Units are divided into latent *compliance types* $S = (X(0), X(1))$:

	$X(0) = 0$	$X(0) = 1$
$X(1) = 0$	never-taker ($X = 0$ always)	defier (X opposes Z)
$X(1) = 1$	complier (X follows Z)	always-taker ($X = 1$ always)

Only compliers change treatment when Z changes. S is latent, and each observed (Z, X) cell is a mixture: $(0,0)$: {C, NT}; $(0,1)$: {AT, D}; $(1,0)$: {NT, D}; $(1,1)$: {C, AT}. Additional assumptions are needed to identify type-specific effects. Goal: CACE $= \mathbb{E}[Y(1) - Y(0) \mid \text{complier}] = \text{ITT}_y / \text{ITT}_x$.

7.2.3.2 Intent-to-treat effects

S does not depend on Z; it is a baseline characteristic. Define type-specific ITT effects $\text{ITT}_s = \mathbb{E}[Y(1) - Y(0) \mid S = s]$, $s \in \{c, n, a, d\}$. With π_s the proportion of type s, the global ITT $\text{ITT}_y := \mathbb{E}[Y(1) - Y(0)]$ is

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \text{ITT}_n + \pi_a \text{ITT}_a + \pi_d \text{ITT}_d.$$

7.2.3.3 Identification assumptions

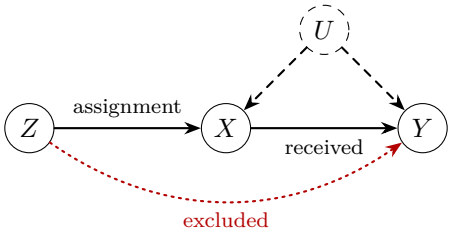


Figure. *Randomized encouragement with non-compliance. Z is randomized (A1), affects Y only through X (exclusion, A2: the dotted arrow is absent), and no one does the opposite of the assignment (monotonicity, A3). Unobserved U confounds treatment received and outcome.*

A1 (random assignment) $\{Y(0), Y(1), X(0), X(1)\} \perp\!\!\!\perp Z$.

A2 (strong monotonicity) (1) no defiers: $X(1) \geq X(0)$ for all units;

(2) compliers exist: $\mathbb{E}[X \mid Z = 1] \neq \mathbb{E}[X \mid Z = 0]$. Monotonicity often holds by design; with one-sided compliance $X(0) = 0$.

A3 (exclusion restriction for non-compliers) $Y(0) = Y(1)$ for all units with $S \in \{n, a\}$. For non-compliers X is the same under both assignments, yet Z could still affect Y directly (e.g. a placebo effect); A3 rules out such direct $Z \to Y$ paths, the same "no direct effect" assumption as in linear IV, phrased in potential-outcome language.

7.2.3.4 Identification

A2 gives $\pi_d = 0$; A3 gives $\text{ITT}_n = \text{ITT}_a = 0$. Hence

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \cdot 0 + \pi_a \cdot 0 + 0 \cdot \text{ITT}_d = \pi_c \text{ITT}_c, \qquad \text{ITT}_c = \text{ITT}_y / \pi_c.$$

With $0 < \pi_c < 1$, $|\text{ITT}_y| < |\text{ITT}_c|$: the global ITT is a conservative (attenuated) estimate.

Compliance-type proportions. Under A1 and A2, when $Z = 0$ only always-takers have $X = 1$, and when $Z = 1$ only never-takers have $X = 0$:

$$\pi_a = P(X = 1 \mid Z = 0) = \mathbb{E}[X \mid Z = 0], \qquad \pi_n = P(X = 0 \mid Z = 1) = 1 - \mathbb{E}[X \mid Z = 1],$$

$$\pi_c = 1 - \pi_a - \pi_n = \mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0].$$

Observed cells under monotonicity. $(Z, X) = (0, 0)$: compliers (with $X(0) = 0$) and never-takers; $(0, 1)$: always-takers only; $(1, 0)$: never-takers only; $(1, 1)$: compliers and always-takers. Two cells are pure and directly identify non-complier means; two are mixed, but the complier means can be solved for using the known proportions:

$$\mathbb{E}[Y \mid X = 0, Z = 0] = \tfrac{\pi_c}{\pi_c + \pi_n} \mathbb{E}[Y(0) \mid \text{complier}] + \tfrac{\pi_n}{\pi_c + \pi_n} \mathbb{E}[Y(0) \mid \text{never-taker}],$$

$$\mathbb{E}[Y \mid X = 0, Z = 1] = \mathbb{E}[Y(0) \mid \text{never-taker}], \qquad \mathbb{E}[Y \mid X = 1, Z = 0] = \mathbb{E}[Y(1) \mid \text{always-taker}],$$

$$\mathbb{E}[Y \mid X = 1, Z = 1] = \tfrac{\pi_c}{\pi_c + \pi_a} \mathbb{E}[Y(1) \mid \text{complier}] + \tfrac{\pi_a}{\pi_c + \pi_a} \mathbb{E}[Y(1) \mid \text{always-taker}].$$

(Here $Y(0)$, $Y(1)$ denote the outcomes under the treatment actually received, which for compliers coincide with the assignment-indexed ones.)

ITT on treatment. $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = \pi_c$ by randomization: under monotonicity only compliers switch, so the fraction who switch is exactly π_c.

Theorem 7.1 (CACE / LATE; Imbens and Angrist 1994, Angrist et al. 1996)

Under A1–A3,

$$\text{CACE} \equiv \mathbb{E}[Y(1) - Y(0) \mid S = c] = \frac{\text{ITT}_y}{\text{ITT}_x} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}.$$

Proof [Supplement]. By randomization (A1), $\text{ITT}_y = \mathbb{E}[Y(1, X(1)) - Y(0, X(0))]$. Split by compliance type $S \in \{c, a, n, d\}$. For always-takers and never-takers $X(1) = X(0)$, so by exclusion (A2) $Y(1, X(1)) = Y(0, X(0))$ and they contribute zero; by monotonicity (A3) there are no defiers. Hence $\text{ITT}_y = P(S = c)\, \mathbb{E}[Y(1) - Y(0) \mid S = c]$. Similarly $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = P(S = c)$, since $X(1) - X(0) = 1$ for compliers and 0 otherwise. Dividing (with $P(S = c) > 0$) gives $\text{CACE} = \text{ITT}_y / \text{ITT}_x$, and randomization identifies each ITT by the difference of observed means. ■

The CACE (local average treatment effect, LATE) is a ratio of two ITT effects; it equals the IV (Wald) estimand when random assignment is viewed as the instrument.

7.2.3.5 A worked example

$N = 1{,}000$ subjects, 500 assigned $Z = 1$ and 500 $Z = 0$.

1. **Proportions.** In the $Z = 0$ group 100 take the drug: $\hat{\pi}_a = 100/500 = 0.20$. In the $Z = 1$ group 150 refuse: $\hat{\pi}_n = 150/500 = 0.30$. So $\hat{\pi}_c = 1 - 0.20 - 0.30 = 0.50$.
2. **ITT effects.** $\bar{Y}_{Z=1} = 7.0$, $\bar{Y}_{Z=0} = 6.0$, so $\text{ITT}_y = 1.0$; $\bar{X}_{Z=1} = 0.70$, $\bar{X}_{Z=0} = 0.20$, so $\text{ITT}_x = 0.50 = \hat{\pi}_c$.
3. **CACE** $= \text{ITT}_y/\text{ITT}_x = 1.0/0.50 = 2.0$.

The ITT (1.0) is attenuated by non-compliance, and the naive comparison of $X = 1$ with $X = 0$ would be confounded.

7.2.3.6 CACE estimator

For two-sided non-compliance,

$$\widehat{\text{CACE}} = \frac{\widehat{\text{ITT}}_y}{\hat{\pi}_c} = \frac{\sum_i Y_i Z_i / \sum_i Z_i - \sum_i Y_i (1 - Z_i) / \sum_i (1 - Z_i)}{1 - \sum_i X_i (1 - Z_i) / \sum_i (1 - Z_i) - \sum_i (1 - X_i) Z_i / \sum_i Z_i},$$

with denominator $\widehat{\text{ITT}}_x = \hat{\pi}_c = 1 - \hat{\pi}_a - \hat{\pi}_n$. Standard errors via asymptotics or the bootstrap. Without A1–A3, nonparametric bounds can be obtained (Chapter 5) but are often too wide to be informative.

7.2.3.7 The "fourth condition": homogeneity

Under homogeneity the IV estimand identifies the ATE for the whole population. Three progressively weaker versions, any one sufficient:

(a) a constant effect for all units, $Y(1) - Y(0) = \beta_1$;

(b) equal average effects across Z, $\mathbb{E}[Y(1) - Y(0) \mid Z = 0] = \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$;

(c) no effect modification by U, $\mathbb{E}[Y(1) - Y(0) \mid U]$ constant.

(a) implies (b) implies (c).

(a) is rarely plausible; (b), (c) are weaker but still strong. Most applied work relies on monotonicity and the LATE interpretation, which needs only "no defiers" but identifies effects for compliers only (Hernán and Robins 2020, §§16.3–16.4).

7.2.3.8 Summary

- **Linear IV:** $\beta_1 = \text{Cov}(Y, Z)/\text{Cov}(X, Z)$; IV = ILS = 2SLS under just-identification; assumes a constant effect.
- **Potential-outcome IV / LATE:** under A1–A3, CACE $= \text{ITT}_y/\text{ITT}_x$, the average effect for compliers; heterogeneous effects allowed, at the price of a local estimand.
- Under constant effects, CACE = ATE $= \beta_1$.

7.2.4 Appendix: Candidate instruments in observational studies

Type	Example	Key concern
Genetic variant (Mendelian randomization)	ALDH2 polymorphism, alcohol metabolism	pleiotropy (direct effect on Y?)
Provider preference	physician's prescribing tendency	unmeasured patient factors drive referral
Geographic/temporal access	distance to facility, policy change date	correlated with SES, urban/rural

Only relevance is empirically testable; exclusion and no confounding require subject-matter knowledge. Always ask whether Z affects Y through any path other than X (*What If*, Fine Point 16.1).

7.2.5 Appendix: IV versus other causal methods

IV estimation	IPW / g-methods
Does not require conditional exchangeability $Y(x) \perp\!\!\!\perp X \mid W$; works with unmeasured confounding.	Requires conditional exchangeability (all confounders measured).
Sensitive to violations of exclusion/no confounding, with bias amplification.	Minor violations give minor bias (no amplification).
Identifies only a local effect (compliers) under monotonicity.	Identifies the population ATE; extends to time-varying treatments.

Prefer IV when unmeasured confounding is likely, a strong instrument exists, treatment is dichotomous, and either homogeneity or monotonicity is plausible (*What If*, §16.6).

7.2.6 Appendix: Practical issues with classic IV studies

Quarter of birth (Angrist and Krueger 1991): weak instrument (QOB explains $< 0.1\%$ of schooling variation); many-instruments bias (180 instruments push 2SLS toward OLS; Bound et al. 1995 obtained similar results with random data); exclusion (birth season may directly affect outcomes via health and school readiness); exogeneity (birth timing correlates with family background; Buckles and Hungerman 2013). Despite the critiques, influential in establishing the IV/LATE framework.
Proximity to college (Card 1995): exclusion (proximity correlates with urban/rural status, SES and school quality); monotonicity (living near a college may not uniformly increase schooling); LATE interpretation (the effect applies to compliers, those marginal with respect to proximity, likely lower-income, so LATE $\neq$ ATE) (Carneiro et al. 2011).

7.2.7 Appendix: Weak instruments

Three problems. (1) Wide confidence intervals: a weak Z–X association gives large variance of $\hat{\beta}_{1,\text{iv}}$.

(2) Bias amplification: $\hat{\beta}_1 = \text{ITT}_y/\text{ITT}_x$, so a small bias in the numerator from a minor exclusion violation is divided by a small denominator and becomes large.

(3) Finite-sample bias: even with a valid instrument, 2SLS is biased toward OLS in finite samples. Rule of thumb: first-stage $F < 10$ (the F-test of $H_0 : \alpha = 0$ in $X = \alpha_0 + \alpha Z + \epsilon$) indicates a weak instrument (Staiger and Stock 1997; *What If*, §16.5).

Example (*What If*, §16.2): effect of smoking cessation X on weight gain Y with cigarette price in the state of birth as instrument Z. $P(X = 1 \mid Z = 1) - P(X = 1 \mid Z = 0) = 25.8\% - 19.5\% = 6.3\%$

(weak), first-stage $F = 0.8 \ll 10$; IV estimate 2.4 kg with 95% CI $[-36.5, 41.3]$: uninformative. A weak instrument with even slight exclusion violations can produce arbitrarily large bias.

What to do. Report the first-stage F and compare it with the threshold for the number of instruments K (Stock and Yogo 2005). Use confidence sets valid under weak instruments: the Anderson–Rubin (AR) set inverts a test that bypasses the first stage, does not rely on instrument strength, and with a single instrument is optimal, so there is no reason not to report it. With multiple instruments prefer LIML (limited-information maximum likelihood) over 2SLS: it is less biased with many or weak instruments (Anderson and Rubin 1949; Andrews, Stock and Sun 2019; Montiel Olea and Pflueger 2013).

What not to do. Screening on $F > 10$ and reporting only passing specifications (pretesting worsens size distortion); trying many instrument sets and picking the highest F (instrument shopping, like p-hacking); bootstrapping IV inference (the bootstrap cannot handle the non-standard distribution under weak instruments); trusting conventional intervals $\hat{\beta} \pm 1.96\,\mathrm{SE}$ when F is small (normality fails). Weak instruments are a research-design problem, not just an estimation problem; the best solution is a stronger instrument.

7.2.8 Appendix: Many instruments and robust IV

With K instruments each gives $\hat{\beta}_k = \mathrm{Cov}(Y, Z_k)/\mathrm{Cov}(X, Z_k)$. If all are valid they agree on β_1; if some are invalid (e.g. pleiotropy) the estimates diverge, and 2SLS, which mixes them all, is contaminated. Robust alternatives: the weighted median (Bowden et al. 2016), consistent if more than 50% of instruments are valid; the mode-based estimator (Hartwig et al. 2017), consistent if a plurality agree; MR-Egger (Bowden et al. 2015), which allows directional pleiotropy through an intercept that corrects the bias. These are widely used in Mendelian randomization, where hundreds of genetic variants serve as instruments and some violate the exclusion restriction.

Chapter 8. Causal Data Science

Motivation. The central question is how to answer a causal query about a target population when the available data sets were collected under different experimental conditions, in different populations, with non-random sampling, with missing entries, or measuring different variables.

8.1 Motivation and big picture

Motivation. Explain why real data rarely match the inferential target and how the mismatches are organized.

Examples. *Genetics:* experiments found a substance carcinogenic in rats; is it carcinogenic in humans? *Advertisement:* ML experts estimated the effect of advertisement strategies on sales of product A; what can be borrowed to advertise product B? *Robotics:* a planetary rover trained in the California desert to dig rocks; how can it use what it learned on Earth to minimize costly exploration on Mars?

"Not all data are created equal." Data are abundant but almost invariably collected under conditions that differ from the target:

(1) different experimental conditions, (2) different populations, (3) non-random sampling, (4) non-random treatment assignment, (5) unmeasured relevant variables. The collected data are messy and rarely match the inferential target. The good news: these dimensions have been formalized, with conditions and algorithms.

Heterogeneous data sets. For a target $Q = P(y \mid \text{do}(x))$, data sets may differ along four dimensions:

		Dataset 1	Dataset 2	Dataset n
d_1	population	Los Angeles	New York	Seoul
d_2	obs./exp.	experimental (randomized Z_1)	observational	experimental (randomized Z_2)
d_3	sampling	selection on age	selection on SES	–
d_4	measured	$\{X_1, Z_1, W, M, Y_1\}$	$\{X_1, X_2, Z_1, N, Y_2\}$	$\{X_2, Z_1, W, L, M, Y_1\}$

Goal. Data science = making the most of multiple, big and heterogeneous data sets: a principled way of performing inferences common in AI, ML and the empirical sciences, with formal machinery (language, conditions, algorithms) for causal inference and *fusion* across heterogeneous studies; imagine an automated scientist.

Classic identifiability takes (1) a query $P_x(y) = P(y \mid \text{do}(x))$, (2) a causal diagram, (3) data $P(\mathbf{V})$, and an inference engine returns either "no" (with evidence) or a formula $P_x(y) = f_{\mathcal{G}}(P(\mathbf{V}))$. In data fusion the target may be the US while data come from NY, LA, Boston, SF, Texas, Arkansas, Utah, Wyoming, each with its own diagram $(\mathcal{G}_{US}, \mathcal{G}_{NYC}, \dots)$ and distribution $(P_{NYC} + P_{LA} + \cdots)$. The sources differ in many ways:

(a) US: census data available (the target population);

(b) NY: survey data resembling the target;

(c) LA: survey data, younger subjects;

(d) Boston: age not recorded, mostly successful lawyers;

(e) SF: high post-treatment blood pressure;

(f) Texas: mostly Spanish subjects, high attrition;

(g) Arkansas: randomized trial on college students;

(h) Utah: RCT with paid volunteers, mainly unemployed;

(i) Wyoming: natural experiment on young athletes.

Tasks. Writing each data collection as a tuple (d_1, d_2, d_3, d_4) (population, obs./exp., sampling, measurement):

1. Descriptive statistics: $(d_1, \text{Samples}(\text{Obs}), d_3, d_4) \to (d_1, \text{Distribution}(\text{Obs}), d_3, d_4)$.
2. Experimental statistics: $(d_1, \text{Samples}(\text{do}(x)), d_3, d_4) \to (d_1, \text{Distribution}(\text{do}(x)), d_3, d_4)$.
3. Causal inference from observational studies: $(d_1, \text{Obs}, d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$. "Causal inference" in 1974–2011 was mainly about confounding bias (Rubin, Robins, Dawid, Pearl, ...).
4. Generalized identification, i.e. using experiments on other variables: $(d_1, \text{do}(Z), d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$.
5. Sampling selection bias: $(d_1, d_2, \text{Select}(\text{Age}), d_4) \to (d_1, d_2, \{\}, d_4)$.
6. Transportability (external validity): $(\text{Bonobos}, d_2, d_3, d_4) \to (\text{Humans}, d_2, d_3, d_4)$.

Each was addressed in isolation in the literature, usually under specific parametric conditions; in practice they appear together, which is what is called "causal data science".

Four dimensions. (1) Experimental conditions: generalized identification.

(2) Environmental conditions: transportability.

(3) Sampling conditions: recovering from selection bias.

(4) Responding conditions: recovering from missingness. Plus measurement (partial observation).

8.2 Experimental conditions: general identifiability

Motivation. Identify an effect from observational data combined with experiments on other variables.

8.2.1 The challenge

Diet Z, cholesterol X, heart attack Y, with $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$. Measured: observational $P(x, y, z)$ and experimental $P(x, y \mid \text{do}(z))$. Needed:

$$Q = P(y \mid \text{do}(x)) = \frac{P(x, y \mid \text{do}(z))}{P(x \mid \text{do}(z))} = P(y \mid \text{do}(z), x).$$

(Rule 3 adds $\text{do}(z)$ because Z affects Y only through X once its incoming edges are cut; Rule 2 exchanges $\text{do}(x)$ for x because in $\mathcal{G}_{\overline{Z}\underline{X}}$ there is no back-door path from X to Y.) Q is not identifiable from $P(x, y, z)$ alone (the bow $X \leftrightarrow Z \leftrightarrow Y$ makes X confounded with its child), so the experiment is essential.

8.2.2 Subtleties of z-ID

Three graphs with the same query (rows: $\mathcal{G}$ with $P(x, y, z)$; $\mathcal{G}_{\overline{Z}}$ with $P(x, y \mid \text{do}(z))$; $\mathcal{G}_{\overline{X}}$ for the query):

- $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$: **z-ID** (as above).
- $Z \to X \to Y$, $X \leftrightarrow Y$, $Z \leftrightarrow Y$: **not z-ID**; $X \leftrightarrow Y$ persists in $\mathcal{G}_{\overline{Z}}$, so the experiment on Z still leaves X confounded with Y. (Linear IV would help, nonparametric IV does not.)

- $Z \leftrightarrow X$, $Z \to Y$, $Z \leftrightarrow Y$, $X \to Y$: **not z-ID**; Z does not cause X, so intervening on Z gives no leverage on X.

8.2.3 z-ID reduced to calculus

Graph: $Z \to X \to W \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $Z \leftrightarrow Y$, $X \leftrightarrow Y$, with $P(\mathbf{v})$ and $P(\mathbf{v} \mid \mathrm{do}(z))$ available.

$$P(y \mid \mathrm{do}(x)) = \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid \mathrm{do}(x)) = \sum_w P(y \mid \mathrm{do}(x, w)) P(w \mid \mathrm{do}(x))$$
$$= \sum_w \underbrace{P(y \mid \mathrm{do}(w))}_{Q[Y]} \underbrace{P(w \mid \mathrm{do}(x))}_{Q[W]}.$$

Neither factor is identifiable from $P(\mathbf{V})$, but both are from $P(\mathbf{v} \mid \mathrm{do}(z))$:

$$Q[Y] = P(y \mid \mathrm{do}(w)) = P(y \mid \mathrm{do}(w, z)) = \sum_x P(y \mid \mathrm{do}(w, z), x) P(x \mid \mathrm{do}(w, z))$$
$$= \sum_x P(y \mid \mathrm{do}(w, z), x) P(x \mid \mathrm{do}(z)) = \sum_x P(y \mid \mathrm{do}(z), w, x) P(x \mid \mathrm{do}(z)),$$
$$Q[W] = P(w \mid \mathrm{do}(x)) = P(w \mid \mathrm{do}(x, z)) = P(w \mid \mathrm{do}(z), x).$$

Within the submodel $\mathcal{M}_z$, $Q[Y]$ is a front-door/back-door type computation and $Q[W]$ an unconfounded effect.

8.2.4 Example: drug–drug interactions

Y: cardiovascular disease, B: blood pressure, X_1: antihypertensive drug, X_2: anti-diabetic drug. Base graph (a): $X_1 \to B \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow B$, $X_1 \leftrightarrow X_2$, $X_2 \leftrightarrow Y$. Goal: the effect of prescribing *both* drugs on cardiovascular risk, from individual-drug experiments, $\mathbb{Z} = \{\{X_1\}, \{X_2\}\}$:

$$P_{x_1,x_2}(y) = \sum_b P_{x_1,x_2}(y, b) = \sum_b P_{x_1}(b) P_{x_2,b}(y) = \sum_b P_{x_1}(b) P_{x_2}(y \mid b).$$

$P_{x_1}(b)$ comes from the X_1 experiment; $P_{x_2}(y \mid b)$ from the X_2 experiment (in $\mathcal{M}_{x_2}$, B is unconfounded with Y). Adding $B \leftrightarrow X_2$ (b) keeps identifiability; adding $B \leftrightarrow Y$ (c) or $X_1 \leftrightarrow Y$ (d) destroys it.

8.2.5 Algorithm and summary

A two-phase algorithm using probability axioms and do-calculus:

(1) the query is modified and decomposed into factors;

(2) each factor is identified from one of the available distributions $\mathbb{P} = \{P_{\mathbf{Z}_1}, \ldots, P_{\mathbf{Z}_m}\}$. It fails only if some factor cannot be identified from any available distribution.

The identifiability of any expression $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$ can be decided for any graph; if identifiable from a combination of observational and experimental studies, the estimand is derived in polynomial time; do-calculus is complete. References: Pearl (1993) back-door; Pearl (1994) do-calculus; Tian and Pearl (2002) c-components and algorithmic ID; Shpitser and Pearl, Huang and Valtorta (2006) completeness of ID; Bareinboim and Pearl (2012) completeness of z-ID; Lee, Correa and Bareinboim (2019) g-ID; Correa and Bareinboim (2019, 2020) stochastic interventions (σ-calculus).

8.3 Environmental conditions: transportability

Motivation. Transfer causal knowledge from source populations to a different target population.

8.3.1 The problem

Can the effect of $\mathbf{X}$ on $\mathbf{Y}$ in a target environment Π^* be computed from observational and experimental findings in different populations (e.g. applying US education policies to South Korea)? "Environment" can mean population, domain, etc. Other sciences call this external validity: "to what populations, settings, treatment variables and measurement variables can this effect be generalized?" (Shadish, Cook and Campbell 2002); extrapolation across studies requires "some understanding of the reasons for the differences" (Cox 1958); an experiment has external validity if the outcome distribution of the treatment group equals that of an actual program (Manski 2007). (Internal validity asks whether a study reliably establishes cause and effect in its own population.)

From lab to real world. At one extreme (H_0) all mechanisms are the same in source and target ($f_z = f_z^*$, $f_x = f_x^*$, ...): trivially transportable. At the other (H_a) everything differs: nothing transports. Real problems lie on the spectrum.

8.3.2 What can an experiment in LA tell us about NYC?

Z (age) $\to X$, $Z \to Y$, $X \to Y$; age distributions differ between LA and NYC, all mechanisms equal otherwise. LA: experimental study, $P(x, y, z)$ and $P(y \mid \mathrm{do}(x), z)$; NYC: observational, $P^*(x, y, z)$. Needed:

$$Q = P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)\, P^*(z)\,:$$

the z-specific effects (LA) reweighted by the NYC age distribution.

Transport formulas depend on the causal story. (a) Z is age ($Z \to X$, $Z \to Y$, differing in distribution): $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z)$.

(b) Z is language skill, a proxy affected by the population difference but not causally relevant: $P^*(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x))$.

(c) Z is a bio-marker on the causal path $X \to Z \to Y$: $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z \mid x)$. ($P^*$: target, P: source.)

8.3.3 Selection diagrams

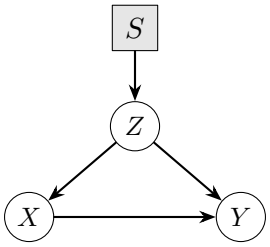


Figure. *A selection diagram for the LA/NYC example: the selection node S points into Z (age), marking the only mechanism that differs between the source and target populations. Since S affects Y only through Z, $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z)$.*

Selection diagrams encode disparities and commonalities among domains. Overlay the source diagram $\mathcal{G}$ and target diagram $\mathcal{G}^*$ and add a *selection node* S (drawn as a small square) pointing to each variable whose mechanism (or exogenous distribution) may differ. Example: $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to W \to Y$; $\mathcal{G}^*$ lacks $Z \to X$. With $f_Z = f_Z^*$, $f_X \neq f_X^*$, $f_W = f_W^*$, $f_Y \neq f_Y^*$, the selection diagram has $S \to X$ and $S \to Y$. (If $Z \to X$ is absent in $\mathcal{G}$ but present in $\mathcal{G}^*$, the union graph is used.) Queries in the target are written $P^*(\cdot) = P(\cdot \mid S = s^*)$.

Theorem 8.1 (Transportability; Pearl and Bareinboim 2011)

Q is transportable from Π to Π^* iff there is a do-calculus reduction of $Q(\Pi^*)$ to an estimand that is a function of the observed distributions.

Proof [Supplement]. *If:* the selection diagram encodes both domains as one causal model with selection nodes S, and $Q(\Pi^*) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), S = s^*)$. Each do-calculus step is valid in this augmented model, so a reduction to an expression in the observable distributions (with S appearing only in factors estimable in Π^*) expresses $Q(\Pi^*)$ through available data. *Only if:* completeness of do-calculus for transportability is proved in Bareinboim and Pearl (2012) by constructing, for non-transportable queries, two pairs of domains that agree on all available distributions but not on $Q(\Pi^*)$; not reproduced here. ■

Example 8.1

Selection diagram with $S \to Z$, $Z \to X$, $Z \to W \to Y$, $X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $X \leftrightarrow Y$:

$$\begin{aligned} Q = P^*(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), S, w) P(w \mid \mathrm{do}(x), S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w) P^*(w). \end{aligned}$$

Step 3 uses $(Y \perp_d S \mid X, W)_{\mathcal{G}_{\overline{X}}}$ (Rule 1); step 4 uses that X is not an ancestor of W (Rule 3). The first factor comes from the source experiment, the second from target observations.

Algorithm. Input: an annotated (selection) diagram. Output: whether Q is transportable, and if so a transport formula specifying which measurements to take in the source and which in the target, e.g.

$$P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) \sum_w P^*(z \mid w) \sum_t P(w \mid \mathrm{do}(x), t) P^*(t).$$

Principle: any c-factor not identifiable from $P^*(\mathbf{V})$ must be identified from source experiments, which is possible only if no selection node points into the variables of that c-factor.

8.3.4 General transportability

Can we predict the effect of X on Y in Π^* using experiments over sets $\{\mathbf{Z}^1, \ldots, \mathbf{Z}^n, \mathbf{Z}^*\}$ from domains $\Pi = \{\Pi^1, \ldots, \Pi^n, \Pi^*\}$? Sometimes; the theory characterizes when and how.

Example 8.2

Transport $Q = P^*(y \mid \mathrm{do}(x))$ from Π^a (LA; experiments on X) and Π^b (NYC; experiments on Z). Both diagrams: $X \to Z \to Y$, $X \leftrightarrow Z$, $X \leftrightarrow Y$; in Π^a selection nodes point to X and Y; in Π^b to X and Z. Separately, Q is provably not transportable from either. Combined (writing S_{xzy} for the selection nodes),

$$\begin{aligned} Q = P(y \mid \mathrm{do}(x), S_{xzy}) &= \sum_z P(y \mid \mathrm{do}(x), z, S_{xzy}) P(z \mid \mathrm{do}(x), S_{xzy}) && \text{def., axioms} \\ &= \sum_z P(y \mid \mathrm{do}(x), \mathrm{do}(z), S_{xzy}) P(z \mid \mathrm{do}(x), S_{xzy}) && \text{Rule 2} \\ &= \sum_z P(y \mid \mathrm{do}(z), S_{xzy}) P(z \mid \mathrm{do}(x), S_{xzy}) && \text{Rule 3} \end{aligned}$$

$$= \sum_z P(y \mid \text{do}(z), S_y) P(z \mid \text{do}(x), S_z) \qquad \text{Rule 1}$$

$$= \sum_z \underbrace{P^{(b)}(y \mid \text{do}(z))}_{\text{NYC}} \underbrace{P^{(a)}(z \mid \text{do}(x))}_{\text{LA}} \qquad \text{def.}$$

($P(y \mid \text{do}(z))$ is needed where S does not point to Y, i.e. NYC; $P(z \mid \text{do}(x))$ where S does not point to Z, i.e. LA.) (Selection diagrams are defined for a pair of domains but generalize to multiple sources.)

8.3.5 Is the gold standard golden?

Before randomization: $Z \to X$, $Z \to Y$, $X \to Y$ with confounding and a selection node into Z. After randomization the edges into X disappear, but $S \to Z \to Y$ remains: the trial population differs from the target in Z. Even a perfect RCT requires a transportability exercise; transportability theory is unavoidable.

Summary. Nonparametric transportability can be decided when the problem is encoded in selection diagrams; transport formulas are derived in polynomial time; the calculus and the algorithm are complete. These results address long-standing problems of external validity, meta-analysis and heterogeneity. References: Pearl and Bareinboim (2011) semantics; Bareinboim and Pearl (2012) completeness; (2013a,b) multiple environments and limited experiments; (2014) completeness for those; Lee, Correa and Bareinboim (2020) subset experiments.

8.4 Sampling conditions: recovering from selection bias

Motivation. Recover population quantities from samples whose inclusion depends on the variables themselves.

8.4.1 The problem

Selection bias, caused by preferential inclusion (exclusion) of samples, is a major obstacle to valid causal and statistical inference. Augment the diagram with a selection indicator S ($S = 1$: in the sample). In Pearl's seven-variable model ($Z_1 \to W_1 \to X$, $Z_1 \to Z_3$, $Z_2 \to Z_3$, $Z_2 \to W_2 \to Y$, $Z_3 \to X$, $Z_3 \to Y$, $X \to W_3 \to Y$) an isolated S means random sampling; with $W_1 \to S$ and $W_2 \to S$, sampling depends on covariates. Under what conditions can we estimate, e.g., $P(y \mid x)$ from $P(\mathbf{v} \mid S = 1)$? The inference engine now receives selection-biased data.

8.4.2 The origin of selection bias

$X = U_x$, $Y = U_y$, $S = \neg(X \oplus Y)$, $P(u_x, u_y) = 1/4$. In the selected sample, $P(X = 1, Y = 1 \mid S = 1) = P(X = 0, Y = 0 \mid S = 1) = 1/2$, a strong correlation:

$$P(Y = 1 \mid X = 1, S = 1) - P(Y = 1 \mid X = 0, S = 1) = 1,$$

whereas in the population $P(X = x, Y = y) = 1/4$ and $P(Y = 1 \mid X = 1) - P(Y = 1 \mid X = 0) = 0$. Conditioning on a common effect ($X \to S \leftarrow Y$) creates dependence.

Confounding versus selection bias. Both are an unblockable flow of information between treatment and outcome (spurious correlation): confounding through a common cause $X \leftarrow Z \to Y$; selection through conditioning on a common effect $X \to S \leftarrow Y$.

Transportability versus selection. Nature-made versus man-made: in transportability S marks difference-producing factors across populations; in selection, S is the selection indicator and its parents are sample-producing factors.

8.4.3 Variants

In statistics:

(i) without external data, is the odds ratio $OR(y,x) = \frac{P(y|x)P(y'|x')}{P(y'|x)P(y|x')}$ recoverable from $P(\mathbf{v} \mid S=1)$? (ii) without external data, is $P(y \mid x)$ recoverable? (iii) with external data $P(\mathbf{t})$ for some $\mathbf{T} \subseteq \mathbf{V}$, is $P(y \mid x)$ recoverable? In causal inference: is $P(y \mid \text{do}(x))$ recoverable from $P(\mathbf{v} \mid S=1)$ and $P(\mathbf{t})$? With instruments, are bounds on $P(y \mid \text{do}(x))$ recoverable?

8.4.4 Recoverability without external data

Theorem 8.2 (Bareinboim, Tian and Pearl 2014)

$Q = P(y \mid x)$ is recoverable from selection-biased data iff $(S \perp\!\!\!\perp Y \mid X)$.

Proof [Supplement]. *If:* $S \perp\!\!\!\perp Y \mid X$ gives $P(y \mid x) = P(y \mid x, S=1)$, which is estimable from the selected sample. *Only if:* if $S \not\perp\!\!\!\perp Y \mid X$ in $\mathcal{G}$, Bareinboim, Tian and Pearl (2014) construct two models compatible with $\mathcal{G}$ that agree on $P(x, y \mid S=1)$ but differ in $P(y \mid x)$; the construction is not reproduced here. ■

If $Y \to S$, not recoverable; if only $X \to S$, recoverable since $P(y \mid x) = P(y \mid x, S=1)$.

8.4.5 Recoverability with external data

In the seven-variable model with $W_1 \to S \leftarrow W_2$, $P(y \mid x)$ is not recoverable by the previous theorem, but what if unbiased $P(W_1, W_2)$ is available?

Theorem 8.3 (Sufficiency)

$P(y \mid x)$ is recoverable if there is a set C with $(Y \perp\!\!\!\perp S \mid C, X)$ in $\mathcal{G}$ and $P(C, X)$ estimable; then

$$P(y \mid x) = \sum_c P(y \mid x, c, S=1) P(c \mid x).$$

Proof [Supplement]. By the law of total probability, $P(y \mid x) = \sum_c P(y \mid x, c) P(c \mid x)$. Since $(Y \perp\!\!\!\perp S \mid C, X)$, $P(y \mid x, c) = P(y \mid x, c, S=1)$, which is estimable from the selected sample, and $P(c \mid x)$ is available by assumption. Substituting gives the formula. ■

$C = \{W_1, W_2\}$: yes (the parents of S separate it from everything). $C = \{W_1, Z_1, Z_2\}$: no ($S \leftarrow W_2 \to Y$ is open). $C = \{W_2, Z_3\}$: yes.

Confounding and selection together. Examples with C given externally:

(1) $W_1 \to X$, $W_1 \to W_2 \to Y$, $X \to Y$, $W_2 \to S$: $C = \{W_2\}$ recovers $P(y \mid \text{do}(x))$; $C = \{W_1\}$ does not.

(2) $X \to Y$, $X \to W \leftarrow Y$, $W \to S$: $C = \{\}$ and $C = \{W\}$ fail, $C = \{W, X\}$ works.

Theorem 8.4 (Generalized adjustment; Correa, Tian and Bareinboim 2018)

$(\mathbf{Z}, \mathbf{Z}^T)$ with $\mathbf{Z}^T \subseteq \mathbf{Z}$ (the part measured without bias) is admissible for generalized adjustment relative to $(\mathbf{X}, \mathbf{Y})$ iff

1. $\mathbf{Z}$ does not disturb any proper causal path;
2. $\mathbf{Z} \cup \{S\}$ blocks all non-causal paths (cut the first edge of every proper causal path; then $\mathbf{Z} \cup \{S\}$ must separate $\mathbf{X}$ and $\mathbf{Y}$);
3. $\mathbf{Z}^T$ d-separates $\mathbf{Y}$ from S in the graph where proper causal paths have been severed.

Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} \underbrace{P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1) P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)}_{\text{biased data}} \underbrace{P(\mathbf{z}^T)}_{\text{unbiased data}} .$$

Proof [Supplement]. *Sufficiency.* By condition 3, $\mathbf{Y} \perp\!\!\!\perp S \mid \mathbf{X}, \mathbf{Z}$ in the relevant graph, so $P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1)$. Conditions 1–2 make $\mathbf{Z}$ an adjustment set, so $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$. Finally $P(\mathbf{z}) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) P(\mathbf{z}^T)$, where $P(\mathbf{z}^T)$ is measured without bias and the remaining condition makes $P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)$. *Necessity* is proved in Correa, Tian and Bareinboim (2018) and not reproduced here. ■

Example 8.3

With treatments $\mathbf{X}$, $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ satisfying the three conditions and $\mathbf{Z}^T = \{Z_2, Z_3\}$: $P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{z_1, z_2, z_3} P(y \mid \mathbf{x}, z_1, z_2, z_3, S=1) P(z_1 \mid z_2, z_3, S=1) P(z_2, z_3)$.

Summary. Nonparametric recoverability can be decided given an augmented diagram with the selection mechanism; estimands are derived in polynomial time; results are complete for pure recoverability and sufficient with external information (Correa, Tian and Bareinboim 2019). The back-door criterion generalizes to handle confounding and selection simultaneously. References: Bareinboim, Tian and Pearl (2014) Markovian completeness; Bareinboim and Tian (2015) semi-Markovian; Correa, Tian and Bareinboim (2018) complete back-door; (2019a) ID with selection bias and external data; (2019b) complete adjustment for transportability with selection bias.

8.5 Responding conditions: missing data

Motivation. Use missingness graphs to decide when quantities are recoverable from incomplete data.

(Partly following Mohan and Pearl 2021, "Graphical models for processing missing data", JASA.)

8.5.1 Motivation

Sensors fail, respondents skip questions, patients cannot recall treatments or outcomes. The statistical literature, rooted in Rubin's work, mostly assumes Missing At Random (MAR): "essentially all the literature on multivariate incomplete data assumes MAR". Popular methods (maximum likelihood via EM, Dempster et al. 1977; multiple imputation, Rubin 1978) need MAR for consistency. But it is almost impossible to decide whether MAR holds, and the literature beyond MAR is limited. Graphical models, long used for missing data in Bayesian-network learning, encode the *reasons* for missingness.

Why graphs? *Transparency:* reasons for missingness are causal, not statistical; MCAR/MAR independences are invisible in the data but visible in the graph. *Recoverability:* decide from the model whether any method yields a consistent estimate of a parameter, and how. *Testability:*

decide whether the model's assumptions are compatible with the (corrupted) data. The graph implies conditional independences.

8.5.2 A motivating example

A school study measures Age A, Gender G, Obesity O (graph $G \to O \leftarrow A$). A and G are fully observed from school records; O is missing for students who do not reveal their weight, recorded as m. A proxy O^* is determined by O and the missingness mechanism R_O:

$$O^* = f(R_O, O) = \begin{cases} O & R_O = 0, \\ m & R_O = 1. \end{cases}$$

R_O masks ($R_O = 1$) or reveals ($R_O = 0$) obesity. Reasons for missingness: random (lost questionnaires); caused by a fully observed variable (rebellious teenagers, $A \to R_O$); caused by a partially observed variable (embarrassed obese students, $O \to R_O$).

8.5.3 m-graphs

Nodes are partitioned into $\mathbf{U} \cup \mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{V}^* \cup \mathbf{R}$: latent variables $\mathbf{U}$; variables observed in every record $\mathbf{V}_o$; variables missing in at least one record $\mathbf{V}_m$; proxies $\mathbf{V}^*$ (V_i^* is actually observed for $V_i \in \mathbf{V}_m$); missingness mechanisms $\mathbf{R}$. No variable in $\mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{U}$ is a child of an R variable unless stated otherwise. The *observed-data distribution* is $P(\mathbf{V}_o, \mathbf{V}^*, \mathbf{R})$; the *underlying distribution* (had there been no missingness) is $P(\mathbf{V}_o, \mathbf{V}_m, \mathbf{R})$.

8.5.4 Three categories

- **MCAR:** $\mathbf{V}_m, \mathbf{V}_o, \mathbf{U} \perp\!\!\!\perp \mathbf{R}$; no edges between $\mathbf{R}$ and $\mathbf{V}_o \cup \mathbf{V}_m$. Example: $O, A, G \perp\!\!\!\perp R_O$.
- **MAR:** $\mathbf{V}_m, \mathbf{U} \perp\!\!\!\perp \mathbf{R} \mid \mathbf{V}_o$;

  (i) no edge between any R and a partially observed variable, (ii) no bidirected edge between an R and a fully observed variable. Example: $A \to R_O$, $O \perp\!\!\!\perp R_O \mid A, G$.
- **MNAR:** neither. Example: $O \to R_O$, $O \not\perp\!\!\!\perp R_O \mid A, G$.

Recovering under MCAR. Since R_O is independent of the variables, $P(G, O, A) = P(G, O, A \mid R_O = 0)$, and $R_O = 0$ implies $O^* = O$, so $P(G, O, A) = P(G, O^*, A \mid R_O = 0)$: list-wise deletion (complete-case analysis), which usually wastes data.

Recovering under MAR. Factorize $P(G, O, A) = P(G, O \mid A)P(A)$. A d-separates (G, O) from R_O, so $= P(G, O \mid A, R_O = 0)P(A) = P(G, O^* \mid A, R_O = 0)P(A)$. Alternatively $P(G, O, A) = P(O^* \mid A, G, R_O = 0)P(A, G)$.

MCAR implies MAR, so MAR methods apply to MCAR. Even though list-wise deletion is unbiased under MCAR, compare the efficiency of $P(G, O^*, A \mid R_O = 0)$ (MCAR estimator) with $P(O^* \mid A, G, R_O = 0)P(A, G)$ (MAR estimator, using all records for $P(A, G)$).

Non-recoverability under MNAR. The edge $O \to R_O$ prevents transforming $P(G, O, A)$ into an expression in observables; in this example the joint distribution is provably non-recoverable.

A recoverable MNAR example. $Z_1 \to Z_2$, $Z_1 \to X$, $Z_1 \to Y$, $Z_2 \to X$, $X \to Y$ (and $Z_2 \to Y$), with Z_1 pointing into R_{Z_2}, R_X, R_Y, and $Z_2 \to R_{Z_2}$ etc. $P(Z_1)$ is recoverable since $R_{Z_1} \perp\!\!\!\perp Z_1$: $P(Z_1) = P(Z_1 \mid R_{Z_1} = 0) = P(Z_1^* \mid R_{Z_1} = 0)$, but *not* from complete cases, since $Z_1 \not\perp\!\!\!\perp R_X$, so $P(Z_1) \neq P(Z_1 \mid \mathbf{R} = 0)$. $P(Y \mid X, Z_1, Z_2)$ is recoverable from complete cases: $P(Y \mid X, Z_1, Z_2) = P(Y \mid X, Z_1, Z_2, \mathbf{R} = 0) = P(Y^* \mid X^*, Z_1^*, Z_2^*, \mathbf{R} = 0)$. (Proxies are often omitted from m-graphs for clarity.)

8.5.5 General results

Theorem 8.5 (Sequential factorization)

Given an m-graph and $P(\mathbf{V}^*, \mathbf{V}_o, \mathbf{R})$, a target Q is recoverable if it can be decomposed into an ordered (chain-rule) factorization, or a sum of such, in which every factor $Q_i = P(Y_i \mid X_i)$ satisfies $Y_i \perp\!\!\!\perp (R_{y_i}, R_{x_i}) \mid X_i$. Then each $Q_i = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0)$.

Proof [Supplement]. Write $Q = \prod_i P(Y_i \mid X_i)$ (or a sum of such products). For each factor, $Y_i \perp\!\!\!\perp (R_{y_i}, R_{x_i}) \mid X_i$ gives $P(Y_i \mid X_i) = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0)$. When the indicators are 0 the proxies equal the true values ($V^* = V$ if $R_V = 0$), so the right-hand side is $P(Y_i^* \mid X_i^*, R_{Y_i} = 0, R_{X_i} = 0)$, a functional of the observed distribution. Hence each factor, and Q, is recoverable. ■

Theorem 8.6 (R-factorization: recoverability of $P(\mathbf{V})$)

Given an m-graph with no edges between R variables, $P(\mathbf{V})$ is recoverable iff there is no $X \in \mathbf{V}_m$ such that (1) X and R_x are neighbors, or (2) X and R_x are connected by a path whose intermediate nodes are all colliders in $\mathbf{V}_m \cup \mathbf{V}_o$. When recoverable,

$$P(\mathbf{v}) = \frac{P(\mathbf{R} = 0, \mathbf{v})}{\prod_i P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)},$$

where $\mathrm{Mb}^o_{r_i} \subseteq \mathbf{V}_o$ and $\mathrm{Mb}^m_{r_i} \subseteq \mathbf{V}_m$ form the Markov blanket of R_i.

Proof [Supplement]. *Formula, given the conditions.* By the chain rule, $P(\mathbf{v}) = P(\mathbf{v}, \mathbf{R} = 0)/P(\mathbf{R} = 0 \mid \mathbf{v})$. With no edges among the R variables, $P(\mathbf{R} = 0 \mid \mathbf{v}) = \prod_i P(R_i = 0 \mid \mathbf{v}, \mathbf{R}_{-i} = 0)$ factorizes over the Markov blankets of the R_i, and each factor equals $P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)$. The two graphical conditions guarantee that the Markov blanket of R_i does not contain the variable whose missingness R_i indicates, so each factor involves only variables observed when the relevant indicators are 0; the numerator $P(\mathbf{v}, \mathbf{R} = 0)$ is observed directly. *Necessity* (non-recoverability when X and R_x are adjacent or joined by a collider path) is proved by Mohan and Pearl (2014, 2021) and not reproduced here. ■

8.5.6 Recovering a causal effect

Graph: $W \to Z \to Y$, $W \leftrightarrow Y$ and $R_Y \leftrightarrow W$, with Y partially observed ($Y \to Y^* \leftarrow R_Y$):

$$\begin{aligned} P(y \mid \mathrm{do}(z)) &= P(y \mid \mathrm{do}(z), r_y) = P(y^* \mid \mathrm{do}(z), r_y) \\ &= \sum_w P(y^* \mid w, \mathrm{do}(z), r_y) P(w \mid \mathrm{do}(z), r_y) = \sum_w P(y^* \mid w, z, r_y) P(w \mid r_y), \end{aligned}$$

with r_y denoting $R_Y = 0$.

Conclusion. Practising statisticians should model the process causing missing data far more often; to do so they need models of that process, which have received little attention in the statistical literature.

8.6 Measurement: general identifiability with partial observation

Motivation. Combine experiments that each measure only some variables.

General identifiability uses experiments $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V})\}_i$. But one cannot always measure all of $\mathbf{V}$: social scientists measure only variables relevant and accessible to their study; clinical trials on

different diseases monitor different variables (oximeter for COVID vs. carbon-monoxide poisoning). *General identifiability with partial observation* (GID-PO) uses $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V}_i)\}_i$ with $\mathbf{V}_i \subseteq \mathbf{V}$.

Example 8.4 (Exercise and stroke)

Age $A \to$ exercise X, $A \to$ BMI B, $X \to B$, $B \to$ blood pressure C, $C \to$ stroke Y, $X \leftrightarrow Y$. Task: the effect of X on Y, given an experimental study on exercise measuring age and blood pressure, $P_X(A, C)$, and an observational study measuring BMI, blood pressure and stroke, $P(B, C, Y)$:

$$
\begin{aligned}
P_x(y) &= \sum_{a,b,c} P_x(a,b,c,y) = \sum_{a,b,c} P(a)P_{a,x}(b)P_b(c)P_c(y) && \text{factorization} \\
&= \sum_{a,c} P(a)\Big[\sum_{b} P_{a,x}(b)P_b(c)\Big]P_c(y) = \sum_{a,c} P(a)P_{a,x}(c)P_c(y) && \text{proposition} \\
&= \sum_{a,c} P_x(a)P_x(c \mid a)\sum_{b} P(y \mid b,c)P(b) && \text{GID of each factor.}
\end{aligned}
$$

The bracketed factors merge into $P(a)P_{a,x}(c)$, identified from $P_X(A, C)$ (since A is not affected by X, $P(a) = P_x(a)$), while $P_c(y) = \sum_b P(y \mid b, c)P(b)$ is identified from $P(B, C, Y)$ by back-door adjustment for B (which blocks $C \leftarrow B \leftarrow X \leftrightarrow Y$). The pieces fit like a jigsaw puzzle.

8.7 Conclusions

- Principled use of abundant data requires representing the underlying causal mechanisms, typically in a coarse and parsimonious way.
- There are necessary and sufficient conditions and algorithms for the most common biases in data collection, solving long-standing challenges: confounding bias, selection bias and generalizability of experiments (sufficient conditions for missing data and partial measurement).
- This gives a principled framework for causal data science: pooling observational and experimental information and knowledge spread across multiple heterogeneous domains.

Chapter 9. Regression Discontinuity, Difference-in-Differences and Synthetic Control

9.1 Regression discontinuity

Motivation. The central question is how a known cutoff rule for treatment assignment identifies a causal effect at the cutoff, how that effect is estimated from data near the boundary, and what the bandwidth choice does to bias, variance and inference.

9.1.1 Where RDD fits

Motivation. Place RDD among empirical designs.

Types of empirical studies, from more to less researcher control:

Experiment (RCT)	Natural experiment	Quasi-experiment	Observational study
researcher randomizes treatment	as-if random assignment by nature/policy (lottery, IV)	known non-random rule exploited (RD, DiD, SCM)	confounders must be assumed away (matching, IPW)

Method	Design type	Key source of variation
Back-door / matching / IPW	observational	adjusted confounders
IV	natural / quasi-experiment	exogenous instrument
RDD	quasi-experiment	cutoff rule
DiD	quasi-experiment	temporal variation across groups
SCM	quasi-experiment	weighted donor pool

IV straddles the boundary: the instrument may come from a natural experiment (lottery) or a quasi-experimental rule (policy threshold). From here on the focus is on quasi-experimental designs that exploit known institutional rules. So far: back-door adjustment (block non-causal paths) and IV (external instrument under unobserved confounding). Today: treatment determined by whether a pre-treatment variable crosses a known threshold.

Notation. $Z_i \in \mathbb{R}$: forcing (running) variable (not an instrument as in Chapter 7); c: cutoff; $X_i = \mathbb{1}\{Z_i \geq c\}$: treatment in the sharp design; $Y_i(1), Y_i(0)$: potential outcomes; τ_{SRD}: sharp-RD effect at the cutoff; $W_i = \mathbb{1}\{Z_i \geq c\}$: the instrument in the fuzzy design (playing the role of Z_i in the IV lecture).

9.1.2 Concept and identification

Motivation. Define the sharp RD effect and show that continuity identifies it.

9.1.2.1 Examples

Incumbency advantage (Lee 2008; Lee et al. 2004). Does winning an election raise the chance of winning the next? Running variable: Democratic vote-share margin at election t; cutoff: 50%; outcome: Democratic vote share at $t+1$ (or the ADA score of the representative). Candidates who barely win and barely lose are comparable in observables and unobservables; near the cutoff the election outcome is as good as random. Barely winning causes a 6–8 percentage-point increase in

next-election vote share: a substantial incumbency advantage. Often regarded as the gold-standard RDD application.

Scholarships and test scores (Thistlethwaite and Campbell 1960, the first RDD): eligibility depends solely on whether an exam score exceeds a threshold; students just above and below are arguably comparable in ability and attitudes. **Financial aid and family income**: eligibility depends on family income below a threshold. In both, the cutoff is known and pre-determined and units near it are plausibly comparable.

9.1.2.2 The design

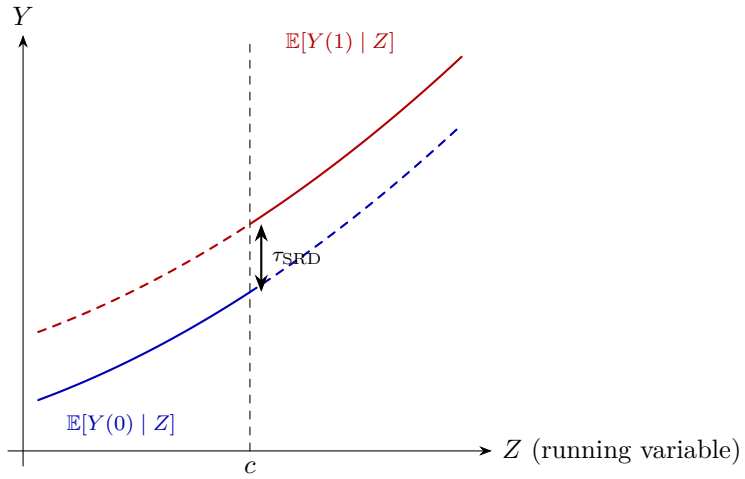


Figure. *Sharp RD: units with $Z \geq c$ are treated. We observe $\mathbb{E}[Y(0) \mid Z]$ only to the left of c and $\mathbb{E}[Y(1) \mid Z]$ only to the right (solid). If both are continuous at c, the jump of the observed regression function at c is the effect at the cutoff.*

In an RDD, treatment status changes discontinuously in a pre-treatment forcing variable, while unobserved confounders vary smoothly around the cutoff. Knowledge of the assignment mechanism identifies a *local* causal effect at the cutoff, by comparing units with similar forcing values but different treatment. The discontinuity is often created by an artificial, pre-fixed threshold.

Sharp RD. Z_i continuous (discrete running variables need extra care; Cattaneo et al. 2020) and $X_i = \mathbb{1}\{Z_i \geq c\}$, a deterministic function of the forcing variable (scores above 1500 get the scholarship). Positivity is violated by construction:

$$P[X_i = 1 \mid Z_i = c - \varepsilon] = 0, \qquad P[X_i = 1 \mid Z_i = c + \varepsilon] = 1,$$

so back-door adjustment and IPW cannot deliver the ATE/ATT; continuity-based extrapolation at the cutoff is used instead.

Fuzzy RD. Crossing the threshold changes the *probability* of treatment discontinuously, but not from 0 to 1 (Section 9.1.5).

9.1.2.3 Estimand and identification

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c] = \mathbb{E}[Y_i(1) \mid Z_i = c] - \mathbb{E}[Y_i(0) \mid Z_i = c].$$

Since Z_i is continuous, no unit has $Z_i = c$ exactly; identification relies on local extrapolation, which needs an assumption.

Continuity assumption. The CEFs $\mu_1(z) = \mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mu_0(z) = \mathbb{E}[Y_i(0) \mid Z_i = z]$ are continuous in z. Then

$$\mathbb{E}[Y_i(0) \mid Z_i = c] = \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid Z_i = z] \qquad \text{(continuity)}$$

$$= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid X_i = 0, Z_i = z] \qquad \text{(SRD: } z < c \Rightarrow X_i = 0\text{)}$$
$$= \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z] \qquad \text{(consistency)},$$

and likewise $\mathbb{E}[Y_i(1) \mid Z_i = c] = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Hence

$$\boxed{\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z].}$$

The estimation problem is to estimate two regression functions at a boundary point: very hard without parametric assumptions; nonparametric regression is consistent but converges slowly.

What can go wrong? Sorting around the threshold violates smoothness: students retake an exam to pass the aid threshold; richer students retake more often, so those just above differ systematically from those just below.

9.1.3 Validity and diagnostics

Motivation. Check the design's assumptions graphically and with falsification tests.

Most RD analyses show some of three figure types:

1. **Outcomes by running variable:** is there a jump at the cutoff? "An RDD is hard to sell if the effect is not visually obvious in the outcome plot" (Imbens and Lemieux 2008).
2. **Covariates by running variable:** covariates should be smooth across c; a discontinuity in covariate means suggests discontinuities in potential-outcome means, and the RD may be invalid.
3. **Density of the running variable:** is there evidence of sorting? Estimate the density separately on each side; a discontinuity suggests sorting.

Manipulation and the McCrary test. RDD assumes units cannot precisely manipulate Z_i to land on a preferred side. If they can, smoothness of potential outcomes is likely violated. McCrary (2008) tests for a discontinuity of the density $f_Z(z)$ at c; a significant jump suggests manipulation, in which case units just above and below are no longer comparable.

Bunching. With administrative data, individuals or firms are often seen to locate at policy thresholds: firms reporting earnings just below a tax or regulation threshold, workers choosing hours just below the full-time cutoff (World Bank blog "Ready, set, bunch"). Example: the distribution of reported wealth in Colombia before (2009) and after (2010) a wealth-tax reform shows bunching just below the new threshold (Londoño-Vélez).

Falsification (placebo) tests. If the RDD is valid, the effect appears only at the true cutoff.

- *Placebo cutoffs:* re-estimate at fake cutoffs away from c; with smooth CEFs no jump should appear. In a simulation with an effect only at $c = 0$, estimates at placebo cutoffs between -2 and 2 are near zero with CIs covering zero, while the true cutoff shows a clear effect.
- *Placebo outcomes:* use pre-treatment covariates as outcomes; treatment cannot affect them, so a jump signals sorting.

9.1.4 Estimation

Motivation. Estimate boundary limits with local polynomial regression and choose the bandwidth.

9.1.4.1 General strategy

The goal is limits such as $\lim_{z\downarrow c}\mathbb{E}[Y_i \mid Z_i = z]$. Two differences from standard nonparametric regression: the target is a single point, and it is a *boundary* point where data exist on one side only, so boundary bias shrinks slowly. The naive difference in means uses data too far from the boundary.

Upper and lower limits: $\mu_+(c) = \lim_{z\downarrow c}\mathbb{E}[Y_i \mid Z_i = z]$, $\mu_-(c) = \lim_{z\uparrow c}\mathbb{E}[Y_i \mid Z_i = z]$, and $\tau_{\text{SRD}} = \mu_+(c) - \mu_-(c)$.

Kernel (local average) regression. With a uniform kernel and bandwidth h,

$$\hat{\mu}_-(c) = \frac{\sum_{i=1}^n Y_i \mathbb{1}\{c-h \le Z_i < c\}}{\sum_{i=1}^n \mathbb{1}\{c-h \le Z_i < c\}},$$

the mean among units at most h below the threshold (similarly above). Equivalently, regress on units within h of c:

$$(\hat{\alpha}, \hat{\tau}_{\text{SRD}}) = \arg\min_{\alpha,\tau} \sum_{i:Z_i\in[c-h,c+h]} (Y_i - \alpha - \tau X_i)^2,$$

where, with binary X_i, the OLS coefficient is the difference in means. This assumes locally constant potential outcomes on each side. h governs the bias–variance trade-off: large h, high bias and low variance; small h, low bias and high variance. Bias shrinks slowly in h, so finite-sample bias may be large. Kernels: uniform $K(u) = \frac{1}{2}\mathbb{1}\{|u| < 1\}$, triangular $K(u) = (1 - |u|)\mathbb{1}\{|u| < 1\}$.

9.1.4.2 Local linear regression

With a wider bandwidth the CEF may not be flat, so allow a slope. On $Z_i \in [c-h, c)$,

$$(\hat{\alpha}_-, \hat{\beta}_-) = \arg\min_{\alpha,\beta} \sum_{i:Z_i\in[c-h,c)} \big(Y_i - \alpha - \beta(Z_i - c)\big)^2,$$

and the same on $[c, c+h]$. Then

$$\hat{\tau}_{\text{SRD}} = \hat{\mu}_+(c) - \hat{\mu}_-(c) = \hat{\alpha}_+ + \hat{\beta}_+(c-c) - \hat{\alpha}_- - \hat{\beta}_-(c-c) = \hat{\alpha}_+ - \hat{\alpha}_-.$$

Single regression. Equivalently,

$$\arg\min_{(\alpha,\beta,\tau,\gamma)} \sum_{i:Z_i\in[c-h,c+h]} \{Y_i - \alpha - \beta(Z_i - c) - \tau X_i - \gamma(Z_i - c)X_i\}^2,$$

with $\hat{\tau}_{\text{SRD}} = \hat{\tau}$. For $X_i = 0$ the fit is $\alpha + \beta(Z_i - c)$; for $X_i = 1$ it is $(\alpha + \tau) + (\beta + \gamma)(Z_i - c)$: one regression on top of the other.

Kernel weighting. Weight points near c more heavily:

$$\arg\min_{(\alpha,\beta,\tau,\gamma)} \sum_{i=1}^n K\left(\frac{Z_i - c}{h}\right) \{Y_i - \alpha - \beta(Z_i - c) - \tau X_i - \gamma(Z_i - c)X_i\}^2,$$

typically with the triangular kernel.

9.1.4.3 Bias–variance trade-off and the MSE-optimal bandwidth

Let $\mu(z) = \mathbb{E}[Y \mid Z = z]$ with one-sided second derivatives $\mu_\pm^{(2)}(c)$, density $f(c)$ of Z at c, and $\sigma_\pm^2(c) = \text{Var}(Y \mid Z = c^\pm)$. For local linear regression with bandwidth h,

$$\text{Bias} \approx C_1 h^2 \underbrace{\big(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\big)}_{\text{curvature difference}}, \qquad \text{Variance} \approx \frac{C_2}{Nh} \cdot \underbrace{\frac{\sigma_+^2(c) + \sigma_-^2(c)}{f(c)}}_{\text{noise / density at } c}.$$

As h grows the variance falls like $1/h$ but the bias grows like h^2; as h shrinks the bias falls but the variance explodes. Minimizing MSE = Bias^2 + Variance (Imbens and Kalyanaraman 2012):

$$h_{\text{MSE}} = C_K \left(\frac{\sigma_+^2(c)/f_+(c) + \sigma_-^2(c)/f_-(c)}{\left(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\right)^2} \right)^{1/5} N^{-1/5}.$$

(Setting the derivative of $C_1^2 h^4 D^2 + C_2 V/(Nh)$ to zero gives $h^5 \propto V/(ND^2)$.) The optimal bandwidth shrinks with N, but slowly. All quantities (curvatures, variances, density) are unknown and estimated, curvature by local quadratic regression.

9.1.4.4 Why the MSE-optimal bandwidth may fail for inference

h_{MSE} minimizes point-estimation error only. Keeping variance low, it picks a relatively wide window ($\propto N^{-1/5}$), so the bias is not negligible relative to the standard error: a nominal 95% CI may cover only about 80% of the time (undercoverage). Remedies: *robust bias correction* (RBC; Calonico, Cattaneo and Titiunik 2014) re-centers the interval by subtracting an estimate of the leading bias term (and accounts for its variability); the *CE-optimal bandwidth* (Calonico, Cattaneo and Farrell 2020) minimizes coverage error, the gap between nominal and actual coverage, with $h_{\text{RBC}} \propto N^{-1/4}$ for local linear fits. Rule of thumb $\hat{h}_{\text{RBC}} \approx (0.7\text{–}0.8) \times \hat{h}_{\text{MSE}}$: valid p-values need a tighter window than the point estimate.

Bandwidth	Rate	Bias	Variance	CI length	Coverage
MSE-optimal	$N^{-1/5}$	high	low	short	fails
Undersmoothing	$N^{-1/3}$	very low	high	too long	valid
CE-optimal	$N^{-1/4}$	moderate	moderate	efficient	optimal

CE-optimal with RBC is the default in modern software (`rdrobust`).

Bandwidth sensitivity. A credible RDD shows stable estimates across a range of bandwidths. In a simulation with true $\tau = 3$, small h gives low bias and high variance (wide CIs, few observations in the window), large h gives high bias and low variance.

9.1.4.5 Covariates and software

Covariates are not needed for identification (continuity at the cutoff is the identifying restriction, unlike back-door adjustment), but they can improve precision:

$$Y_i = \alpha + \beta(Z_i - c) + \tau X_i + \gamma(Z_i - c)X_i + \delta^\top V_i + \varepsilon_i,$$

with V_i pre-treatment covariates (Calonico et al. 2019). Covariates also serve as diagnostics: a jump in V_i at c is evidence against validity.

`rdrobust` (Calonico, Cattaneo, Titiunik; R/Stata) implements MSE- and CE-optimal bandwidths (default CE-optimal with RBC), robust bias-corrected CIs, covariate adjustment, clustering and fuzzy RD. In R: `library(rdrobust); rd <- rdrobust(Y, Z, c = 1500); summary(rd)`.

9.1.5 Fuzzy RD

Motivation. Treat imperfect compliance with the cutoff as a local instrumental-variable problem.

Concept. In the fuzzy design crossing the cutoff only changes the probability of treatment:

$$\lim_{z \downarrow c} P[X_i = 1 \mid Z_i = z] \neq \lim_{z \uparrow c} P[X_i = 1 \mid Z_i = z],$$

but the jump is not from 0 to 1. Reasons: exemptions and discretionary enforcement, imperfect compliance, administrative errors. Examples: a test-score cutoff determines summer-school eligibility, but not all eligible students attend; age-based eligibility with early or late applicants. Many real RD settings are fuzzy.

Fuzzy RD as IV: the Wald estimator. Take $W_i = \mathbb{1}\{Z_i \geq c\}$ as instrument for the treatment received X_i; near c, W_i is as good as randomly assigned. Under continuity,

$$\tau_{\text{FRD}} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z]} = \frac{\text{ITT effect on } Y}{\text{first-stage effect on } X},$$

the Wald estimator, i.e. IV/LATE applied locally at the cutoff.

Assumptions. (1) Continuity of $\mathbb{E}[Y_i(x) \mid Z_i = z]$ and $\mathbb{E}[X_i(w) \mid Z_i = z]$ at c;

(2) first stage: $\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z] \neq 0$;

(3) monotonicity: crossing c makes no one less likely to be treated (no defiers). These mirror relevance, exclusion and monotonicity; exclusion is built in, since W_i affects Y_i only through X_i for compliers near c.

Interpretation. $\tau_{\text{FRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}, Z_i = c]$, a LATE with *double locality*: it applies only to compliers (units whose treatment is actually changed by crossing the cutoff) and only at the cutoff. External validity is limited, internal validity strong; many policy questions naturally produce fuzzy RD settings.

9.1.6 Case study: remedial education in Chicago

Motivation. See a fuzzy RD in action (Jacob and Lefgren 2004, REStat).

Background. In 1996 Chicago Public Schools ended "social promotion": students had to meet minimum scores on the Iowa Test of Basic Skills (ITBS) reading and math tests to advance (e.g. third graders needed 2.8 grade equivalents in both). Students below the cutoff had to attend summer school and could be retained. *Question:* what is the causal effect of summer school and grade retention on achievement?

Design. Treatment: summer school and retention. Fuzzy, since exemptions and discretionary enforcement mean some students above the cutoff are retained and some below are not; but the probability of treatment jumps sharply at the cutoff, identifying the LATE there. Data: CPS administrative records 1993–1999, about 350,000 third and sixth graders; outcomes: changes in ITBS reading and math (in logits) one and two years later.

IV estimation. With the instrument $W_i = \mathbb{1}\{\text{Score}_i \geq \text{cutoff}\}$ (above/below the cutoff; here falling below triggers treatment, so the first-stage jump is negative),

$$\begin{aligned} \text{First stage:} \quad & \text{Treat}_i = \alpha_0 + \alpha_1 W_i + \alpha_2 \text{Score}_i + \alpha_3 (W_i \times \text{Score}_i) + \nu_i, \\ \text{Second stage:} \quad & Y_i = \beta_0 + \tau \widehat{\text{Treat}}_i + \beta_1 \text{Score}_i + \beta_2 (W_i \times \text{Score}_i) + \varepsilon_i, \end{aligned}$$

where τ is the LATE near the cutoff. Relevance: the cutoff strongly predicts treatment; exogeneity: near the threshold, being just above or below is effectively random. (Modern RD software uses local linear regression with $W_i \times \text{Score}_i$ interactions to allow different slopes on each side.)

Results. Net effect (summer school + retention): for third graders roughly 0.11–0.13 logits in the first year (about 20% of a year's typical learning), persisting after two years with some fade-out; for sixth graders no meaningful reading effect and a small, insignificant positive math effect. Retention alone: short-term reading and math gains in grade 3; negative reading and negligible math effects in grade 6. Summer school alone: positive for third graders, especially low performers. Visually, the probability of summer school/retention drops sharply at the promotion cutoff while next-year achievement also drops there, indicating a net positive effect for third graders.

Strengths and limitations. Strengths: a natural experiment via a policy cutoff; fuzzy RD with IV enables causal inference; heterogeneity by grade, gender and SES. Limitations: LATE only (students near the cutoff, compliers); no evidence on long-term (3+ year) outcomes; manual bandwidth choice without MSE- or CE-optimal selection. The study illustrates both the power of fuzzy RD and the importance of modern bandwidth methods.

9.1.7 Summary

- RDD exploits a known cutoff rule for treatment assignment.
- Identification: continuity of the potential-outcome CEFs at the cutoff, $\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$.
- Diagnostics: outcome plot, covariate balance, McCrary density test, placebo tests.
- Estimation: local linear regression with kernel weights.
- Bandwidth: MSE-optimal ($N^{-1/5}$) for point estimation, CE-optimal ($N^{-1/4}$) for inference; rule of thumb, shrink h_{MSE} by 20–30%.
- Fuzzy RD = IV at the cutoff, identifying the LATE for compliers at c.
- Case study: positive effects of remedial education for Chicago third graders, little effect for sixth graders.

9.1.8 Appendix: Kink RD

Sharp kink RD looks for discontinuities in the first derivative (slope) of $\mathbb{E}[Y_i \mid Z_i = z]$ at the threshold rather than in its level (Card, Lee, Pei and Weber 2015). Estimation is similar, but local quadratic regression is preferable.

9.1.9 Appendix: Assumptions for the MSE-optimal bandwidth

(Imbens and Kalyanaraman 2012.) (1) $\{Y_i, Z_i\}$ i.i.d.;

(2) the density $f(z)$ is continuous with $f(c) > 0$;

(3) $\mu(z)$ is three times differentiable near c;

(4) K is symmetric, bounded, with compact support;

(5) $\sigma^2(z)$ is continuous near c with $\sigma^2(c) > 0$.

9.1.10 Appendix: Regularization of the bandwidth

The optimal bandwidth has the squared curvature difference in its denominator; if the estimated curvature is near zero, h explodes. IK (2012) add a penalty $\hat{r}$ (the variance of the curvature estimate) to the denominator:

$$\hat{h}_{\text{reg}} \propto \left(\frac{\text{sum of variances}}{f(c)\,[(\text{curvature})^2 + \text{regularization}]} \right)^{1/5},$$

ensuring a finite bandwidth even for flat data. Implementation: estimate $\hat{\mu}_{\pm}^{(2)}(c)$ by local quadratic regressions and set

$$\hat{r}_+ = \frac{2160\,\hat{\sigma}_+^2(c)}{N_{h,+} h^4}, \qquad \hat{r}_- = \frac{2160\,\hat{\sigma}_-^2(c)}{N_{h,-} h^4},$$

so the penalty grows with less data (N_h) or more noise (σ^2).

9.1.11 Appendix: Why local linear? Polynomial order

RD uses local polynomial regression of order p. Higher p reduces bias but increases variance (more parameters from limited data near c); higher p has faster theoretical rates, but the variance cost dominates in finite samples. Division of labour: local linear ($p = 1$) for the main estimate (good boundary bias–variance balance); local quadratic ($p = 2$) for bias estimation in the MSE-optimal bandwidth (IK 2012) and in robust bias correction (CCT 2014). "Estimate with p, correct bias with $p + 1$" is the `rdrobust` default.

9.1.12 Appendix: Global polynomials: a cautionary tale

An older practice fitted global high-order polynomials (cubic, quartic, ...) over the whole range of the running variable. Gelman and Imbens (2019) show this is problematic: estimates are sensitive to observations far from the cutoff, high-order polynomials oscillate wildly (Runge-type behavior), and confidence intervals have poor coverage. Local polynomials fitted within h of c do not suffer from these problems. Modern best practice is local linear estimation with a data-driven bandwidth. (Further reading: *The Effect*; *The Mixtape*.)

9.2 Difference-in-differences

Motivation. The central question is how repeated measurements on treated and untreated groups identify the effect of treatment on the treated when treatment is not randomized, by assuming that the untreated trend is a valid proxy for the treated group's counterfactual trend.

9.2.1 Motivation

Motivation. Introduce the two sources of variation in panel data.

So far good controls came from randomization, similar covariates (matching/weighting), an exogenous instrument (IV), or proximity to a discontinuity (RD). With repeated measurements on the same units there are two sources of variation: within-unit variation over time (before/after) and cross-sectional variation in treatment. A before–after comparison uses the treated group's own past as its counterfactual; difference-in-differences uses the control group's change as the counterfactual change of the treated group (parallel paths).

9.2.1.1 Case study: minimum wages (Card and Krueger 1994)

Does raising the minimum wage affect employment? Classical theory suggests negative effects, but minimum wages are hard to randomize. In 1992 New Jersey raised its minimum wage from \$4.25 to \$5.05 while neighboring Pennsylvania stayed at \$4.25. Employment is observed in both states before and after the increase, focusing on fast-food restaurants in eastern PA and NJ, which have similar prices, wages and products and are most likely to be affected. In the picture, average employment in NJ (treated) and PA (control) is plotted before and after; the dashed counterfactual line for NJ is parallel to PA's, and the ATT is the gap between NJ's actual and counterfactual post-period levels.

9.2.2 The DiD design

Motivation. Formalize the two-group, two-period design and identify the ATT.

9.2.2.1 Setup

Two groups, two periods. Pre-period ($t = 0$): neither group is treated. Post-period ($t = 1$): one group is treated. $G_i = 1$ for units treated at $t = 1$, $G_i = 0$ for never-treated units. Treatment status: $X_{i0} = 0$ for all; $X_{i1} = G_i$.

	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$X_{i0} = 0$	$X_{i1} = 1$
Control group ($G_i = 0$)	$X_{i0} = 0$	$X_{i1} = 0$

Potential outcomes. $Y_{it}(x)$ is the potential outcome under treatment x at time t; the individual effect is $Y_{it}(1) - Y_{it}(0)$. Consistency: $Y_{it} = X_{it}Y_{it}(1) + (1 - X_{it})Y_{it}(0)$. Hence $Y_{i0}(0) = Y_{i0}$ for all units; $Y_{i1} = Y_{i1}(1)$ if $G_i = 1$; $Y_{i1} = Y_{i1}(0)$ if $G_i = 0$.

9.2.2.2 The identification problem

$$\text{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] = \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{(a)} - \underbrace{\mathbb{E}[Y_{i1}(0) \mid G_i = 1]}_{(b)}.$$

(a) is estimable;

(b) is counterfactual: what would the treated group's average have been had it remained untreated?

9.2.2.3 Three control strategies

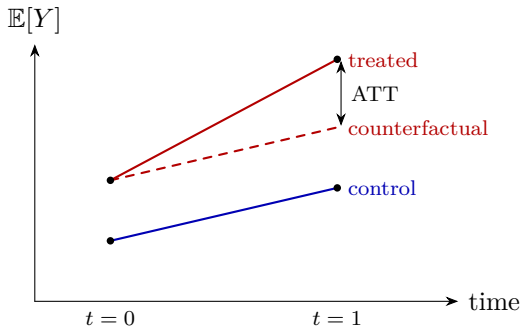


Figure. *Difference-in-differences. Parallel trends assumes that without treatment the treated group would have changed by the same amount as the control group (dashed). The ATT is the treated group's change minus the control group's change.*

1. **Before–after:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i0} \mid G_i = 1]$ (no change over time absent treatment).
2. **Cross-sectional comparison:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1} \mid G_i = 0]$ (no group differences).
3. **Difference-in-differences:** assume parallel trends; then

$$\text{ATT} = \big(\mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i0} \mid G = 1]\big) - \big(\mathbb{E}[Y_{i1} \mid G = 0] - \mathbb{E}[Y_{i0} \mid G = 0]\big).$$

9.2.2.4 Parallel trends

Definition (Parallel trends)

$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$.

The control group's trend proxies how the treated group would have changed without treatment. Why is this weaker than other assumptions? It accounts for time-invariant unobserved differences between groups and allows for common trends over time. But it is *not invariant to nonlinear transformations*: parallel trends in Y_{it} imply non-parallel trends in $\log Y_{it}$, and vice versa.

Theorem 9.1 (Identification)

Under consistency and parallel trends,

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0]).$$

Proof [Supplement]. By parallel trends, $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0}(0) \mid G = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G = 0]$. Consistency gives $Y_{i0}(0) = Y_{i0}$ for everyone and $Y_{i1}(0) = Y_{i1}$ for controls, so $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0} \mid G = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G = 0]$. Substitute into $\text{ATT} = \mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i1}(0) \mid G = 1]$. ■

Threats: non-parallel trends, i.e. unmeasured time-varying confounding. *Ashenfelter's dip:* earnings often fall temporarily just before entry into a training program. The dip makes the treated group's pre-period outcome artificially low, inflating its before–after difference, so the effect is overestimated. *Falsification test:* check parallel trends in pre-treatment periods. This is necessary but not sufficient: passing does not guarantee that parallel trends hold post-treatment.

9.2.2.5 Estimation

With panel data,

$$\widehat{\text{ATT}} = \underbrace{\frac{1}{N_1}\sum_{i=1}^{N} G_i\{Y_{i1} - Y_{i0}\}}_{\text{average trend, treated}} - \underbrace{\frac{1}{N_0}\sum_{i=1}^{N}(1 - G_i)\{Y_{i1} - Y_{i0}\}}_{\text{average trend, control}},$$

with standard errors from the usual difference-in-means estimator. Equivalent regression: with $\Delta Y_i = Y_{i1} - Y_{i0}$, $\Delta Y_i = \alpha + \tau G_i + \varepsilon_i$ gives $\hat{\tau}_{\text{OLS}} = \widehat{\text{ATT}}$ (OLS on a binary regressor equals the difference in group means), convenient for adding pre-treatment covariates. DiD can also be used with repeated cross-sections.

Card and Krueger with numbers (average full-time-equivalent employment per restaurant):

	Before (Feb 1992)	After (Nov 1992)	Difference
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD estimate			+2.76

DiD $= (21.03 - 20.44) - (21.17 - 23.33) = 0.59 - (-2.16) = 2.76$: no evidence of employment loss from the minimum-wage increase.

9.2.2.6 DiD and two-way fixed effects (2×2)

The simplified TWFE model for two groups and two periods is

$$Y_{it} = \mu + \gamma G_i + \delta t + \tau X_{it} + \varepsilon_{it},$$

with fixed effects capturing all time-invariant group differences (γG_i) and unit-invariant time effects (δt). Since $X_{it} = G_i \cdot t$, the four cell means are μ, $\mu + \gamma$, $\mu + \delta$, $\mu + \gamma + \delta + \tau$, and the coefficient τ equals the DiD estimate.

9.2.3 Conditional and nonlinear DiD

Motivation. Weaken parallel trends with covariates, and replace mean trends by rank preservation.

9.2.3.1 Conditional parallel trends

Unconditional parallel trends may fail if groups differ in observed covariates X_i. *Conditional parallel trends*: for all x,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1, X_i = x] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0, X_i = x].$$

This is weaker: trends need only be parallel within covariate strata (states with different demographics may trend differently, but states with similar demographics should evolve similarly). Estimators:

- **Outcome regression:** include X_i in the (TWFE) model.
- **IPW-DiD:** reweight control units by $\hat{e}(X_i)/(1-\hat{e}(X_i))$ to match the treated group's covariate distribution.
- **Doubly robust DiD:** combine both; consistent if either the outcome model or the propensity model is correct (Sant'Anna and Zhao 2020; Callaway and Sant'Anna 2021), with the same logic as DR estimation in Chapter 6.

9.2.3.2 Nonlinear DiD (changes-in-changes)

Standard DiD's parallel trends in levels are not preserved under log, $\sqrt{\cdot}$, etc. Key insight: assume parallel trends in *ranks* (quantiles) instead of levels: absent treatment, a unit at the q-th quantile of its group in period 0 remains at the q-th quantile in period 1. Its value changes, but its rank is preserved. This rank-preservation assumption is invariant to monotone transformations (the q-th quantile of Y maps to the q-th quantile of $\log Y$).

Formalization (Athey and Imbens 2006). Let $F_{gt}(y) = P(Y_{it}(0) \le y \mid G_i = g)$. Observed: F_{00}, F_{01}, F_{10} (untreated outcomes), and the treated post-period CDF $F_{11}^{(1)}$ of $Y_{i1}(1)$. Goal: the counterfactual F_{11}. Quantile treatment effect:

$$\tau(q) = \left(F_{11}^{(1)}\right)^{-1}(q) - F_{11}^{-1}(q).$$

Define the period-0-to-1 value shift at each rank: $h_0(y) = F_{01}^{-1}(F_{00}(y))$ for controls and $h_1(y) = F_{11}^{-1}(F_{10}(y))$ for the treated. *Identification assumption:* $h_0 = h_1$, i.e. the value shift at each rank is the same in both groups. Then $F_{11}^{-1}(F_{10}(y)) = F_{01}^{-1}(F_{00}(y))$, and solving,

$$F_{11}(y) = F_{10}(F_{00}^{-1}(F_{01}(y))).$$

(Take $y' = F_{01}^{-1}(F_{00}(y''))$; the assumption says $F_{11}(y') = F_{10}(y'')$ with $y'' = F_{00}^{-1}(F_{01}(y'))$.)

9.2.3.3 Key takeaways

- Parallel trends are the core of DiD identification.
- DiD handles time-invariant unobserved confounding but is vulnerable to time-varying confounders (Ashenfelter's dip).
- Conditional parallel trends with covariates lead to IPW-DiD and doubly robust DiD.
- Pre-trend tests are useful but do not validate parallel trends; sensitivity analysis is preferred.
- Linear DiD is not invariant to nonlinear transformations; nonlinear DiD needs distributional assumptions.
- TWFE gives the same result in the 2×2 case but needs care with staggered timing.

9.2.4 Appendix: Pre-testing bias

Common practice: run a pre-trend test and proceed with DiD only if it is insignificant. Problem: conditioning on passing biases the DiD estimate. Tests have low power against small pre-trends and often fail to reject; "passing" selects samples in which pre-trends happen to look flat. The same chance variation that hid the pre-trend persists into the post-period and distorts the estimate. Result: under-rejection of the pre-trend test leads to over-rejection of "no treatment effect". A passed pre-trend test gives comfort, not validation (Roth 2022).

9.2.5 Appendix: Sensitivity analysis for parallel trends

Parallel trends are untestable; how robust are the estimates? Allow bounded violations: let M bound the deviation

$$\left|\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]\right| \leq M.$$

For each M construct a confidence set for the ATT; as M grows the set widens. Report the *breakdown value*, the smallest M at which the set first includes 0 (the finding is no longer robust). One can also extrapolate from pre-treatment trend differences to choose plausible bounds (Rambachan and Roth 2023).

9.2.6 Appendix: Staggered treatment timing

Units often enter treatment at different times. Standard TWFE with staggered timing can be biased. The Goodman-Bacon decomposition shows TWFE is a weighted average of all possible 2×2 DiD comparisons: early-treated vs. never-treated (good), late-treated vs. never-treated (good), and late-treated vs. early-treated (dangerous: the early-treated serve as controls). If effects change over time, early-treated units are not valid controls for the late-treated, since their outcomes already reflect treatment; using them attenuates or even reverses the estimate.

Modern solutions. Callaway and Sant'Anna (2021): compute group-time average treatment effects using only never-treated or not-yet-treated units as controls. Event-study plots: plot effects by time relative to treatment onset, $t - t^*$, allowing visual inspection of pre-trends and dynamic effects. (See the lecture on staggered DiD.)

9.3 Staggered difference-in-differences

Motivation. The central question is how to estimate treatment effects with DiD when units adopt treatment at different times and effects may vary across units and over time, a setting in which the usual two-way fixed-effects regression can be badly misleading.

9.3.1 Staggered adoption

Motivation. Describe the setting and why it complicates DiD.

The canonical DiD has two periods and two groups. Much applied work has more periods, with units becoming treated at different times: e.g. group A treated early, group B later, group C later still, and a never-treated group. Plots of staggered DiD without and with treatment reversal show outcome paths of groups A, B, C over 15 periods: after adoption the treated paths jump and may keep changing (dynamic effects) relative to their counterfactual (triangles); with treatment reversal, effects switch on and off.

Staggered DiD generalizes DiD to settings where treatment timing varies, common when policies are rolled out gradually across regions: a *staggered adoption design* (SAD). Units not yet treated can serve as controls for those already treated, so staggered adoption blurs the distinctions treatment/control and pre/post. Once treated, a unit remains treated (no reversal).

9.3.1.1 TWFE as is is not sufficient

The most common approach is the TWFE regression $Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}$. It works when (i) effects are not heterogeneous, or (ii) there are only two periods: then, under parallel trends and no anticipation, τ equals the ATT. But effects are very likely heterogeneous, varying across units or over time. With staggered timing, already-treated units exhibit treatment dynamics and act as controls; these dynamics leak into τ, making it very hard to interpret causally. The fix: use only "good" comparisons between groups instead of all possible comparisons.

9.3.1.2 Terminology

- **Static effects:** constant across post-treatment periods (e.g. a persistent 5% increase).
- **Dynamic effects:** change with time since adoption (grow, decay, fluctuate).
- **Homogeneous effects:** identical across units.
- **Heterogeneous effects:** vary across units (group- or unit-level differences) but are constant over time for each unit.
- **Carryover effects:** a past treatment keeps influencing outcomes after it is withdrawn; a specific form of dynamic effect.

9.3.1.3 Valid and invalid comparisons

At time t, a treated unit can be compared with never-treated units, not-yet-treated (later-treated) units, or already-treated (earlier-treated) units.

- **Valid:** treated vs. never-treated; treated vs. not-yet-treated.
- **Invalid:** treated vs. already-treated; later-treated vs. earlier-treated using the earlier group's post-treatment periods.

Bias arises when effects vary over time (dynamic) or across units (heterogeneous). (The Goodman-Bacon decomposition in Section 9.2 shows TWFE averages all 2×2 comparisons, including the invalid ones.)

Modern alternatives to TWFE modify the set of effective comparison units: Callaway and Sant'Anna (2021), group-time ATT estimators; Sun and Abraham (2021), event-study designs with proper controls; stacked DiD, separate DiDs per cohort to avoid contamination. They differ in which observations serve as comparisons and how covariates enter.

9.3.2 Callaway–Sant'Anna group-time ATTs

Motivation. Define, identify and aggregate effects by adoption cohort and calendar time.

9.3.2.1 Notation

G_i: the period in which unit i becomes treated (groups are defined by adoption time). $Y_{it}(0)$: untreated potential outcome in period t. $Y_{it}(g)$: potential outcome in period t if first treated in period g. Y_{it}: observed outcome. Never-treated units have $Y_{it} = Y_{it}(0)$ for all t; for others

$$Y_{it} = \mathbb{1}\{G_i > t\}Y_{it}(0) + \mathbb{1}\{G_i \leq t\}Y_{it}(G_i) :$$

untreated outcomes before participation and treated outcomes (possibly depending on adoption time) afterwards. This notation embeds a no-anticipation assumption, which can be relaxed. C_i: indicator of the never-treated (control) group. D_{it}: indicator that i has been treated by t. X_i: pre-treatment covariates.

Staggered adoption assumption: for $t = 1, \ldots, T-1$, $D_{it} = 1 \Rightarrow D_{i,t+1} = 1$: once treated, units remain treated (they do not "forget" their treatment). Effects may be heterogeneous across time, groups, treatment sequences, etc.

9.3.2.2 Parallel trends

Based on never-treated units: for all $g = 2, \ldots, T$ and $t = 2, \ldots, T$ with $t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid C = 1].$$

It uses the never-treated for all eventually-treated groups, presuming (i) a large enough never-treated group exists and (ii) these units are similar enough.

Based on not-yet-treated units: for all g and s, t with $s \geq t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D_s = 0, G \neq g].$$

Units not yet treated by time $s \geq t$ can serve as comparisons when computing effects for the group first treated at g. This generally uses more data, but restricts some pre-treatment trends across groups (Marcus and Sant'Anna 2021): no free lunch. (Why is the restriction placed on s relative to t? Because $s \geq t$ guarantees, by staggered adoption, that comparison units are untreated in both periods $t-1$ and t.)

9.3.2.3 Group-time ATT and identification

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g],$$

the average effect for the group first treated in g, at time t. With three periods, $\mathrm{ATT}(g = 2, t = 3)$ is the effect in period 3 for units first treated in period 2.

Under either parallel-trends assumption, group-time ATTs are identified. With never-treated controls, for $t \geq g$,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1];$$

with not-yet-treated controls, for $t \geq g$,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0, G \neq g].$$

Derivation (never-treated case): $\mathbb{E}[Y_t(g) \mid G = g] = \mathbb{E}[Y_t \mid G = g]$ by the observation rule. Telescoping parallel trends from g to t, $\mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid C = 1] = \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$; and $Y_{g-1}(0) = Y_{g-1}$ for group g (no anticipation). Combine. Each $\mathrm{ATT}(g, t)$ is a clean 2×2 DiD between group g and a valid comparison group, with base period $g - 1$.

9.3.2.4 Conditional parallel trends

Parallel trends are often more plausible conditional on pre-treatment covariates. *Never-treated:* for $t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, C = 1]$. *Not-yet-treated:* for $s \geq t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, D_s = 0, G \neq g]$. These allow covariate-specific trends, important when covariate distributions differ across groups (estimated by outcome regression, IPW or doubly robust DiD, as in Section 9.2).

9.3.2.5 Aggregation

By group. The average effect of participation for group g:

$$\theta_S(g) = \frac{1}{T - g + 1} \sum_{t=2}^{T} \mathbb{1}\{g \leq t\} \, \mathrm{ATT}(g, t),$$

highlighting heterogeneity with respect to adoption time. Aggregating further,

$$\theta_S^O := \sum_{g=2}^{T} \theta_S(g) \, P(G = g)$$

is the overall effect of participating across all groups that ever participate: close to a multi-period analogue of the two-period ATT, and the recommended single summary parameter.

By exposure length (event study). To highlight dynamics,

$$\theta_D(e) := \sum_{g=2}^{T} \mathbb{1}\{g + e \leq T\} \, \mathrm{ATT}(g, g + e) \, P(G = g \mid g + e \leq T),$$

the average effect for units exposed to treatment for exactly e periods.

9.3.2.6 Summary

- Staggered DiD uses group-time ATTs $\mathrm{ATT}(g, t)$: the effect in period t for units first treated in period g.
- It allows flexible clean controls: not-yet-treated or never-treated units.
- It accommodates pre-treatment (static) covariates via conditional parallel trends.
- Anticipation can be handled by shifting the base period before any behavioral adjustment.

References: Athey and Imbens (2022); Baker, Larcker and Wang (2022); Callaway and Sant'Anna (2021); Chiu, Lan, Liu and Xu (2023); Goodman-Bacon (2021); Wing et al. (2024).

9.4 Synthetic control

Motivation. The central question is how to estimate the effect of an intervention on a *single* treated unit (a state or country) when no single untreated unit is a credible comparison, by building a weighted combination of untreated units that tracks the treated unit before the intervention.

9.4.1 From DiD to synthetic controls

Motivation. Motivate a data-driven comparison unit when there is only one treated unit.

DiD works well with two groups and parallel trends. But with a single treated unit (a country adopting a policy), which control should be used? Any single control may not be comparable. The **synthetic control method** (SCM) constructs an optimal weighted combination of control units to approximate the counterfactual.

Abadie and Gardeazabal (2003) used a DiD-like approach for "quantitative case studies": an intervention in one country/state at one point in time; one treated group and many controls; compare time series of the treated unit with a control. Since there are many possible control choices, use a *convex combination* of the controls, a synthetic control, with weights minimizing pre-treatment differences between the treated unit and its synthetic counterpart.

9.4.1.1 Case study: California Proposition 99

In 1988 California passed Proposition 99, a \$0.25 per pack cigarette tax increase (the largest state tax increase at that time). Goal: the effect on per-capita cigarette sales. No single state is a good comparison. SCM builds a "synthetic California" from 38 control states; key donors are Utah (33.4%), Nevada (23.4%), Montana (19.9%) and Colorado (16.4%). Weights are chosen to match pre-1988 cigarette sales, income, beer consumption, retail price and the share of population aged 15–24. Result: by 2000 California's per-capita sales were about 26 packs lower than synthetic California's (Abadie, Diamond and Hainmueller 2010). In the figure, actual and synthetic California track each other closely from 1970 to 1988 and diverge afterwards. Without synthetic controls, California compared with the average of the other states differs already before 1988; the synthetic control reproduces California's pre-1988 smoking and predictor values, which, like matching, forces the researcher to demonstrate the affinity between the treated unit and its counterpart and guards against "extreme counterfactuals" far outside the convex hull of the data.

9.4.2 The estimator

Motivation. Define the setting, estimand and weights.

9.4.2.1 Setup

Time period	1	2	$\cdots$	T_0	T_0+1	$\cdots$	T
Treated unit ($i=1$)	0	0	0	0	1	1	1
Control group ($i=2,\ldots,J+1$)	0	0	0	0	0	0	0

All units are untreated for T_0 periods; unit 1 is treated from T_0+1 to T. Potential outcomes $Y_{it}(x)$, $x \in \{0,1\}$; observed $Y_{it} = Y_{it}(X_{it})$. Estimand: the effect on the treated unit over time,

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \qquad t > T_0.$$

9.4.2.2 Missing counterfactuals and synthetic weights

By consistency, for $t > T_0$, $\tau_{1t} = Y_{1t} - Y_{1t}(0)$, so the missing $Y_{1t}(0)$ must be imputed. Choose weights $(w_2, \ldots, w_{J+1})$ with

$$w_j \geq 0, \qquad \sum_j w_j = 1 \quad \text{(convex)},$$

minimizing, over pre-treatment periods $t \le T_0$,

$$\left\| Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right\| \quad \text{and} \quad \left\| Z_1 - \sum_{j=2}^{J+1} w_j Z_j \right\|,$$

where Z_i are pre-treatment predictors. The objectives are combined via a weighted norm with predictor-importance weights V (Appendix 9.4.6); a penalty on the dispersion of the weights can be added. The hope is that, for $t > T_0$, $\sum_{j=2}^{J+1} w_j Y_{jt} \approx Y_{1t}(0)$, giving $\hat{\tau}_{1t} = Y_{1t} - \sum_j w_j Y_{jt}$.

9.4.2.3 Model-based justifications

Model 1: interactive factor model.

$$Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it},$$

with θ_t time-varying coefficients on covariates, α_i a unit fixed effect, δ_t a common time shock, λ_t a $1 \times F$ vector of common factors (unobserved macro shocks) and μ_i an $F \times 1$ vector of factor loadings (each unit's sensitivity to those shocks). The term $\lambda_t \mu_i$ allows time-varying confounding in a structured way; DiD is the special case without it.

Model 2: autoregressive model (no fixed effects):

$$Y_{i,t+1}(0) = \alpha_t Y_{it}(0) + \beta_{t+1}^\top W_{i,t+1} + u_{i,t+1}, \qquad W_{i,t+1} = \gamma_t Y_{it}(0) + \Pi_t W_{it} + v_{i,t+1},$$

with time-varying covariates W_{it}, AR coefficient α_t (the dynamic source replacing the factor structure), coefficients β_{t+1}, feedback γ_t from outcome to covariates, and autoregressive matrix Π_t.

Properties. Suppose perfect balancing weights exist: $\sum_j w_j Y_{jt} = Y_{1t}$ for $t \le T_0$ and $\sum_j w_j Z_j = Z_1$. Let $\hat{Y}_{1t}(0) = \sum_j w_j Y_{jt}$ for $t > T_0$.

- Under Model 1, $\hat{Y}_{1t}(0) \to Y_{1t}(0)$ as $T_0 \to \infty$: consistent, improving as the pre-intervention period grows (matching many pre-period outcomes forces $\sum_j w_j \mu_j \approx \mu_1$).
- Under Model 2, $\mathbb{E}[\hat{Y}_{1t}(0)] = \mathbb{E}[Y_{1t}(0)]$ (given perfect fit): unbiased even with a single pre-treatment period.

9.4.2.4 Convexity: no extrapolation

SCM requires $w_j \ge 0$ and $\sum_j w_j = 1$, so the synthetic control is a convex combination of donors. Example: California's pre-treatment smoking is 120 packs per capita, while all donors lie between 60 and 110; convex weights can reach at most 110, so the imperfect fit is visible and transparent. OLS, in contrast, may assign negative weights (e.g. $w_j = -0.3$); the weights still sum to one, but the synthetic outcome can lie far outside the observed range, and this extrapolation is hidden. SCM trades flexibility for transparency: one can always check whether the treated unit lies inside the convex hull of the donors.

Sparsity: geometric intuition. SC weights are sparse: most $w_j = 0$. Geometrically, the treated unit is approximated by a point on the boundary of the convex hull of donors, so only a few "vertex" donors get positive weight, which aids interpretation. All weight decisions are made without post-treatment data, protecting against specification searching.

9.4.2.5 Practical concerns

- To reduce interpolation bias and overfitting, restrict the donor pool to units similar to the treated unit.

- Donors must not be affected by the treatment (no interference or spillovers).
- "Although we define our synthetic controls as convex combinations of unexposed units, negative weights or weights larger than one can be used at the cost of allowing extrapolation." (Relaxing the no-extrapolation property.)
- Convex weights limit flexibility: the treated unit's pre-treatment outcomes may not be matched perfectly, especially if it lies outside the convex hull; the trade-off is deliberate, for interpretability and robustness.
- With $N = 1$ treated unit, inference uses placebo tests.

9.4.3 Inference and bias

Motivation. Assess significance by placebo permutation, and understand bias from imperfect fit.

9.4.3.1 Placebo tests

With one treated unit, sampling-based inference is unavailable. Iteratively apply SCM to each control state as if it were treated: for each of the 38 control states, build a synthetic control and estimate a "placebo effect". If SCM works well, most placebo effects are near zero. This is a permutation/randomization-inference argument: under the null of no effect, California's gap should look like a typical draw from the distribution of placebo gaps; an unusually large gap is evidence of a real effect. The "p-value" is the fraction of placebo gaps at least as large as California's. In Abadie et al. (2010) only 1 of 34 states had a larger gap, $p \approx 0.03$.

In the placebo plot, gray lines show gaps $Y_{jt} - \hat{Y}_{jt}^{\mathrm{SC}}$ for control states and a red line California's gap: most placebo gaps are small and centered at zero, while California's diverges sharply after 1988. Refinement: exclude states with poor pre-treatment fit (high pre-treatment RMSPE), whose placebo gaps are noisy and uninformative.

9.4.3.2 Why imperfect pre-treatment fit causes bias

Under Model 1, if $\sum_j w_j Y_{jt} = Y_{1t}$ for all $t \le T_0$, the factor loadings are approximately matched, $\sum_j w_j \mu_j \approx \mu_1$. If the match is imperfect for some $t \le T_0$, then $\sum_j w_j \mu_j \neq \mu_1$, and post-treatment the common factors create a systematic wedge:

$$\sum_j w_j Y_{jt}(0) - Y_{1t}(0) \approx \lambda_t \Big(\sum_j w_j \mu_j - \mu_1 \Big) \neq 0,$$

the other terms ($Z_i^\top \theta_t$, α_i, δ_t) cancelling by predictor balancing and ε_{it} averaging out. Intuition: the synthetic control does not capture unit 1's response pattern to macro shocks, so predictions drift after treatment. This is especially problematic when the treated unit lies outside the convex hull of donors.

9.4.3.3 Bias correction: augmented SCM

Use an outcome model to correct residual mismatch (Ben-Michael, Feller and Rothstein 2021). Model $Y_{it}(0) = m_{it} + \varepsilon_{it}$; fit the regression on control units only ($j = 2, \dots, J+1$) and let $\hat{m}_{it}$ denote predictions, including for $i = 1$. The augmented estimator is

$$\hat{Y}_{1t}^{\mathrm{aug}}(0) = \underbrace{\sum_{j=2}^{J+1} w_j Y_{jt}}_{\text{SC estimate}} + \underbrace{\Big(\hat{m}_{1t} - \sum_{j=2}^{J+1} w_j \hat{m}_{jt} \Big)}_{\text{bias correction}} = \underbrace{\hat{m}_{1t}}_{\text{outcome model}} + \underbrace{\sum_{j=2}^{J+1} w_j (Y_{jt} - \hat{m}_{jt})}_{\text{SC-weighted residuals}}.$$

Covariates are easy to add. It is doubly robust: consistent if either the SC weights or the outcome model is approximately correct (analogous to AIPW and doubly robust DiD).

9.4.4 Discussion

Motivation. Weigh strengths and limitations and position SCM among alternatives.

Strengths	Limitations
Transparency: weights are explicit; one can inspect which donors contribute and how much.	Needs a good donor pool: similar units with no spillovers.
Data-driven: no need to pick a single comparison unit by hand.	Designed for a single treated unit; extensions needed for several.
No parallel-trends assumption (uses a factor model instead).	Sensitive to donor-pool and predictor choice.
Pre-treatment fit is a built-in diagnostic of credibility.	Formal inference is limited (permutation-based, not sampling-based); convexity can prevent good fit when the treated unit is an outlier.

(Abadie 2021.)

When to use SCM: one or very few treated units with multiple controls; a rich pre-treatment period; parallel trends implausible but a factor structure reasonable. Consider DiD with multiple treated and control units and plausible parallel trends; consider synthetic DiD to combine both, especially in larger panels. Open research areas: formal inference beyond permutation tests (conformal inference), sensitivity analysis for SCM specifications, experimental design with synthetic controls, external validity.

Extensions. Augmented SCM (bias correction, double robustness); synthetic difference-in-differences (Arkhangelsky et al. 2021: unit weights as in SCM, time weights and unit fixed effects as in DiD, valid large-panel inference); penalized SCM (Abadie and L'Hour 2021: relax strict convexity with L_1/L_2 penalties, allowing regularized partial extrapolation); conformal inference for SCM (Chernozhukov, Wüthrich and Zhu 2021: distribution-free prediction intervals via permutation of residuals); matrix completion (Athey et al. 2021: the counterfactual as a missing-data problem with a low-rank approximation).

DiD versus synthetic control.

	Difference-in-differences	Synthetic control
Setup	two groups, two (or more) periods	one treated unit, many donors
Key assumption	parallel trends	pre-treatment fit (factor model)
Estimand	ATT (group level)	τ_{1t} (unit level)
Strength	simple, transparent	handles one treated unit well
Weakness	sensitive to nonlinear transforms	requires a good donor pool
Extensions	staggered DiD, nonlinear DiD	augmented SCM, matrix completion

9.4.5 Appendix: More treated units

Two estimation approaches generalize to any number of treated units. *Interactive fixed effects:* with $Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it}$ (Model 1), estimate the IFE directly instead of weighting, by iterating:

(1) treat the IFE terms as fixed and fit the parametric part on untreated observations to get $\hat{\theta}$;

(2) treat the covariate coefficients as fixed and estimate the IFE terms by factor analysis (principal components of residuals);

(3) repeat until convergence. *Matrix completion* (Athey et al. 2021): treat the matrix of control potential outcomes $\mathbf{Y}(0)$ as partially missing and estimate a low-rank matrix $\mathbf{L}$ approximating its observed entries, with regularization (as in recommendation systems).

9.4.6 Appendix: Predictor importance weights V

The unit weights minimize a V-weighted predictor mismatch:

$$w(V) = \underset{w\geq 0,\ \sum_j w_j = 1}{\arg\min} \sum_{m=1}^{k} v_m \Big(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm} \Big)^2,$$

where $v_m \geq 0$ says how much predictor m (a covariate or a pre-period outcome) matters. Simple choices: equal weights $v_m = 1$; variance scaling $v_m = 1/\operatorname{Var}(X_m)$ (common scale); regression-based $v_m \propto |\hat{\beta}_m|$ from an outcome regression on donors. Abadie's standard, automatic choice is nested optimization: $V^* = \arg\min_V \mathrm{MSPE}_{\mathrm{pre}}(w(V))$, the outer loop choosing V to minimize pre-treatment outcome MSPE and the inner loop solving for $w(V)$.

Caveats. (1) *Outcome-lag collapse* (Kaul, Klößner, Pfeifer and Schieler 2022): including all pre-period outcome lags drives $v_m \to 0$ for the other predictors, so matching collapses to outcomes only and covariates are ignored; use only a few lags (e.g. every five years).

(2) *Sensitivity:* V depends on the predictor list and lag choice, and different specifications give different donor weights. Practice: choose predictors theoretically related to the outcome; check robustness across several V methods (equal, variance, nested) and report whether the effect is stable; report donor weights explicitly so readers see which donors drive the result.

9.5 Synthetic difference-in-differences

Motivation. The central question is how to combine the strengths of difference-in-differences (invariance to additive unit-level shifts) and synthetic control (re-weighting units to match pre-treatment trends) into one estimator for panel data.

9.5.1 DiD, SC and SDiD

Motivation. Recall DiD and SCM and preview what SDiD changes.

Picture (California Proposition 99, cigarette consumption 1970–2000). *DiD* compares California with the unweighted average of control states, using all pre-periods equally; the level difference is allowed, but the trends must be parallel, and visibly they are not. *SC* re-weights control states so that the synthetic control matches California's level and trend before 1988, using no time weights. *SDiD* re-weights control states so that their trend (not necessarily level) is parallel to California's, and re-weights pre-treatment years, concentrating weight on the few years just before treatment (the shaded time weights).

Recap. DiD estimates effects by comparing outcome changes over time between treated and control groups, under parallel trends (a common trend absent treatment) while allowing time-invariant level differences. SCM addresses comparative case studies with a single treated unit, building a matched synthetic control from many donors. Both are for panel data and similar in spirit, but differ:

(1) SCM adds unit weights;

(2) SCM has time fixed effects but no unit fixed effects and no overall intercept.

SDiD (Arkhangelsky et al. 2021). Like SCM, it re-weights and matches pre-treatment trends, reducing reliance on parallel trends. Like DiD, it is invariant to additive unit-level shifts.

9.5.2 Formulations

Motivation. Write all three estimators as (weighted) TWFE regressions.

With unit fixed effects α_i, time fixed effects β_t, an intercept μ and treatment indicator X_{it}, the effect τ is estimated by

$$\hat{\tau}^{\text{DiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2, \tag{9.1}$$

$$\hat{\tau}^{\text{SCM}} = \arg\min_{\tau,\mu,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i, \tag{9.2}$$

$$\hat{\tau}^{\text{SDiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i\hat{\lambda}_t. \tag{9.3}$$

The unit fixed effects give invariance to additive unit-level shifts; the weights $\hat{\omega}_i$ (and $\hat{\lambda}_t$) match pre-treatment trends.

9.5.2.1 Setting and algorithm

A balanced panel with N units and T periods, outcomes Y_{it}, binary exposure $X_{it} \in \{0,1\}$. The first N_{co} units are never exposed; the last $N_{tr} = N - N_{co}$ units are exposed after T_{pre} ($T_{post} = T - T_{pre}$). Like SCM, SDiD finds unit weights $\hat{\omega}$ aligning pre-exposure trends of unexposed units with those of exposed units; it also finds time weights $\hat{\lambda}$ balancing pre-exposure periods with post-exposure ones.

1. Compute unit weights $\hat{\omega}$.
2. Compute time weights $\hat{\lambda}$.
3. Compute $(\hat{\tau}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i,t}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\hat{\omega}_i\hat{\lambda}_t$.

9.5.2.2 Unit weights

$$(\hat{\omega}_0, \hat{\omega}) = \underset{\omega_0\in\mathbb{R},\ \omega\in\Omega}{\arg\min} \left\|\bar{y}_{pre,tr} - (\omega_0 + \omega_{co}^{\top}Y_{pre,co})\right\|_2^2 + \zeta^2 T_{pre}\|\omega_{co}\|_2^2,$$

where $\bar{y}_{pre,tr} = \frac{1}{N_{tr}}\sum_{i=N_{co}+1}^{N} Y_{it}$ is the average pre-period outcome path of the treated, $\omega_{co}^{\top}Y_{pre,co} = \sum_{i=1}^{N_{co}}\omega_i Y_{it}$ is the synthetic control path, and Ω is the simplex (non-negative weights summing to one). Differences from SCM:

1. SDiD allows an intercept ω_0: the synthetic control need not have the same level as the treated units, only a *parallel trend*.
2. The L_2 penalty spreads non-zero weights across control units, instead of the few contributors typical of SCM, forcing more units to be used.

Regularization (optional). $\zeta = (N_{tr}T_{post})^{1/4}\,\hat{\sigma}(\Delta_{it})$, where $\Delta_{it} = Y_{i(t+1)} - Y_{it}$ is the first difference (temporal fluctuation) of control outcomes in the pre-period,

$$\bar{\Delta} = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}\Delta_{it}, \qquad \hat{\sigma}^2(\Delta_{it}) = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}(\Delta_{it} - \bar{\Delta})^2,$$

the average fluctuation and the variance of fluctuations.

9.5.2.3 Time weights

$\hat{\lambda}$ minimizes the difference between pre- and post-treatment periods for the controls:

$$(\hat{\lambda}_0, \hat{\lambda}) = \underset{\lambda_0 \in \mathbb{R},\ \lambda \in \Lambda}{\arg\min} \left\| \bar{y}_{post,co} - (\lambda_0 + \lambda_{pre}^{\top} Y_{pre,co}) \right\|_2^2,$$

where $\bar{y}_{post,co} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it}$ is a $1 \times N_{co}$ row vector of post-period averages per control unit, $Y_{pre,co}$ is the $T_{pre} \times N_{co}$ matrix of pre-period control outcomes, λ_{pre} a $1 \times T_{pre}$ vector (one entry per pre-period, in the simplex Λ), and λ_0 an intercept allowing the post-period to lie above or below all pre-periods, as in many applications with a clear trend.

9.5.3 Comparisons and takeaways

Motivation. Explain what the weights and fixed effects buy.

SDiD versus DiD. DiD solves the same TWFE problem without unit or time weights. The weights make the regression "local": it emphasizes units similar on average to the treated in their past, and periods similar on average to the treated periods. Using only similar units and periods makes the estimator more robust, and the weights can improve precision by implicitly removing systematic, predictable parts of the outcome.

SDiD versus SCM. SCM omits the unit fixed effects and the time weights: $\hat{\tau}^{\text{SCM}} = \arg\min_{\mu,\beta,\tau} \sum_{i,t} (Y_{it} - \mu - \beta_t - X_{it}\tau)^2 \hat{\omega}_i^{\text{SCM}}$. Time weights can remove bias and improve precision by eliminating periods very different from the post-treatment periods. Unit fixed effects make the model more flexible (strengthening robustness) and often explain much of the outcome variation.

Key takeaways.

- Like SCM, SDiD works with multiple periods when pre-treatment trends are not parallel.
- Unlike SCM, its unit weights ω build a control that is only *parallel* to the treated group, not matched in level.
- Like DiD, it uses time (β) and unit (α) fixed effects, which explain outcome variance and reduce the estimator's variance.
- Its unit weights use an L_2 penalty (more spread out across controls) and allow an intercept ω_0.
- It introduces time weights, which neither DiD nor SCM has.

Chapter 10. Causal Discovery

10.1 Causal discovery

Motivation. The central question is how much of the causal diagram can be recovered from data alone, under which assumptions, and with which algorithms.

10.1.1 The structure-learning problem

Motivation. State what causal discovery takes as input and returns as output, and the assumptions needed.

Earlier lectures assumed the diagram $\mathcal{G}$ induced by an unknown SCM was given. Causal discovery goes the other way: from $P(\mathbf{V})$ (or samples) to $\mathcal{G}$. A cartoon: does the rooster's crow cause the sun to rise? A table of binary observations of (rooster, sun) shows them almost always equal, but the data alone do not say which way the arrow goes.

Structure learning. *Input:* a data set assumed sampled i.i.d. from $P(\mathbf{V})$. *Output:* a set of graphs consistent with what we know about the data (hopefully including $\mathcal{G}$). It is an unsupervised learning problem. Extensions: multiple data sets, observational and interventional data, unknown interventions, ...

10.1.1.1 DAG models

We focus on DAG models: a causal structure on $\mathbf{V}$ is a DAG with one node per variable, each directed edge representing a direct functional relationship. DAG models correspond to Markovian models, i.e. *causal sufficiency*: no unmeasured confounder of the measured variables. Three equivalent definitions, illustrated on $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$:

- **Factorization:** $P(A,B,C,D,E) = P(E \mid D)P(D \mid B,C)P(C \mid A)P(B \mid A)P(A)$.
- **Local Markov property:** $(C \perp\!\!\!\perp B \mid A)$, $(D \perp\!\!\!\perp A \mid B,C)$, $(E \perp\!\!\!\perp A,B,C \mid D)$.
- **Global Markov property:** if $\mathbf{A}$ is d-separated from $\mathbf{B}$ by $\mathbf{C}$ in $\mathcal{G}$, then $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

10.1.1.2 Faithfulness

The global Markov property is one-directional: every d-separation gives a CI, equivalently every dependence gives a d-connection. A complete graph is compatible with every $P(\mathbf{V})$, so we need minimality, an assumption that reverses the implication.

Definition (Faithfulness)

For all $\mathbf{A},\mathbf{B},\mathbf{C}$: $\mathbf{A}$ is d-separated from $\mathbf{B}$ given $\mathbf{C}$ in $\mathcal{G}$ *iff* $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

Faithfulness is an extra property that a $P(\mathbf{V})$ factorizing according to $\mathcal{G}$ may satisfy. Most such distributions are faithful (unfaithful ones have measure zero). However: in finite samples unfaithful and nearly unfaithful distributions are hard to tell apart, and there are many more of the latter; nature does not pick distributions at random and may evolve unfaithful configurations (homeostasis, equilibria). In practice one should justify why faithfulness is sensible for the problem.

Example 10.1 (Violation by cancellation)

$X = \epsilon_X$, $Z = X + \epsilon_Z$, $Y = X - Z + \epsilon_Y$ (edges $X \xrightarrow{1} Z$, $Z \xrightarrow{-1} Y$, $X \xrightarrow{1} Y$). X and Y are not d-separated by $\emptyset$, yet $Y = X - (X + \epsilon_Z) + \epsilon_Y = -\epsilon_Z + \epsilon_Y$, so $X \perp\!\!\!\perp Y$: the two paths cancel exactly.

10.1.2 Constraint-based structure learning

Motivation. Find the CIs that hold, rule out inconsistent graphs, and return what is left.

10.1.2.1 What constraints a DAG encodes

Example: $X \to W \to Y$, $X \to Z \leftarrow Y$. Of the candidate statements

$$(1)\ X \perp\!\!\!\perp Y \mid W, \quad (2)\ X \perp\!\!\!\perp Y, \quad (3)\ X \perp\!\!\!\perp Y \mid W, Z, \quad (4)\ X \perp\!\!\!\perp Y \mid Z,$$

only (1) is forced by the graph; (2)–(4) are dependences ($X \to W \to Y$ open; the collider Z opened). The data must exhibit the graph's independences for the graph to be valid.

With faithfulness, d-separation in $\mathcal{G} \Leftrightarrow$ CI in P, so CIs refine the possible DAGs. Without any CI information the best guess is the *complete undirected* graph (every pair adjacent, no orientation).

Adjacency. If for $A, B \in \mathbf{V}$ some $\mathbf{S} \subseteq \mathbf{V} \setminus \{A, B\}$ gives $A \perp\!\!\!\perp B \mid \mathbf{S}$, then A and B are not adjacent. (Under faithfulness, adjacent variables are dependent given every set.) In the example the only removed edge is X–Y (separated by $\{W\}$); the result, called the *skeleton*, is $X - W - Y$, $X - Z - Y$.

Orientation. For non-adjacent A, B there is a separator $\mathbf{S}$. If $A - C - B$ (an *unshielded triple*) and $C \notin \mathbf{S}$, orient $A \to C \leftarrow B$, an *unshielded collider* (v-structure); otherwise C is an unshielded non-collider (e.g. $A \to C \to B$). In the example the separator $\{W\}$ of X, Y does not include Z, so $X \to Z \leftarrow Y$; this rules out $X \to Z \to Y$, $X \leftarrow Z \leftarrow Y$ and $X \leftarrow Z \to Y$. Can anything else be oriented? $W \in \mathbf{S}$ makes $X - W - Y$ a non-collider, but its direction is undetermined.

10.1.2.2 Markov equivalence

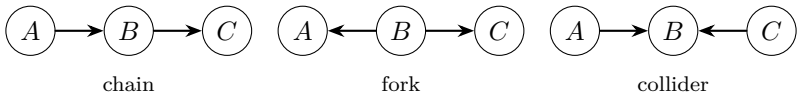


Figure. *The chain and the fork imply the same independence, $A \perp\!\!\!\perp C \mid B$, and are Markov equivalent; the collider implies $A \perp\!\!\!\perp C$ but $A \not\perp\!\!\!\perp C \mid B$. This is why observational data can orient unshielded colliders but not the other edges.*

The three DAGs $X \to W \to Y$, $X \leftarrow W \leftarrow Y$, $X \leftarrow W \to Y$ (each with $X \to Z \leftarrow Y$) entail the same independences: they are *observationally* (Markov) *equivalent*; the local Markov property gives the same CIs. A *Markov equivalence class* (MEC) is a set of Markov-equivalent DAGs.

Theorem 10.1 (Verma and Pearl 1990)

Two DAGs represent the same statistical model iff they have the same skeleton and the same unshielded colliders.

Proof [Supplement]. *If.* Two DAGs with the same skeleton and unshielded colliders have the same d-separations: a path is open given $\mathbf{S}$ depending only on which inner nodes are colliders and their relation to $\mathbf{S}$; for unshielded triples the collider status agrees by assumption, and for a

shielded triple one can replace the path segment by the shielding edge, so a shortest open path has no shielded colliders whose status differs. Same d-separations means the same Markov (statistical) model. *Only if.* Different skeletons: an edge $A - B$ in one DAG and not the other; in the second A and B are separated by some set (e.g. the parents of the later one), but never in the first. Same skeleton but an unshielded collider $A \to C \leftarrow B$ in one and not the other: in the first, A and B are separated by a set excluding C; in the second, every separating set must contain C (the path $A - C - B$ is open otherwise). In both cases the d-separations differ, and so do the models. ■

All DAGs in a class are represented by one mixed graph (directed and undirected edges), the *pattern* or CPDAG (completed partially directed acyclic graph). For three variables (Eberhardt 2017), the 25 DAGs fall into 11 classes; e.g. the chain, reversed chain and fork on X, Y, Z form one class, while the collider $X \to Y \leftarrow Z$ is alone in its class.

10.1.2.3 The PC algorithm

Basic idea:

(1) non-adjacent vertices have a separator;

(2) for an unshielded collider $X \to Z \leftarrow Y$, the separator must not contain Z.

Algorithm 3 PC algorithm (Spirtes and Glymour)

start with the complete undirected graph $\mathcal{G}$ on n vertices
for $i = 0, \ldots, n-2$ **do** ▷ skeleton
  for every adjacent pair A, B in $\mathcal{G}$ **do**
    if $\exists \mathbf{S}$ with $|\mathbf{S}| = i$, $\mathbf{S} \subseteq \mathrm{adj}_{\mathcal{G}}(A) \cup \mathrm{adj}_{\mathcal{G}}(B) \setminus \{A, B\}$, and $A \perp\!\!\!\perp B \mid \mathbf{S}$ **then**
      remove $A - B$; sepset$[\{A, B\}] \leftarrow \mathbf{S}$
for every non-adjacent A, B with $A - C - B$ **do** ▷ v-structures
  if $C \notin$ sepset$[\{A, B\}]$ **then** orient $A \to C \leftarrow B$
orient as many remaining edges as possible without creating new v-structures or directed cycles (Meek's rules)

Meek's rules.

1. Orient $b - c$ as $b \to c$ if there is $a \to b$ with a, c non-adjacent (else a new v-structure $a \to b \leftarrow c$ would appear).
2. Orient $a - b$ as $a \to b$ if there is a chain $a \to c \to b$ (else a cycle).
3. Orient $a - b$ as $a \to b$ if there are two chains $a - c \to b$ and $a - d \to b$ with c, d non-adjacent (if $b \to a$, Rule 2 would force $c \to a$ and $d \to a$, a new v-structure at a).
4. Orient $a - b$ as $a \to b$ if there are two chains $a - c \to d$ and $c \to d \to b$ with c, b non-adjacent and a, d adjacent.

Example 10.2 (Walk-through)

True (unknown) DAG: $A \to B$, $A \to C$, $B \to D$, $C \to D$. CIs revealed by the tests: $B \perp\!\!\!\perp C \mid A$ (fork blocked by A, collider D closed) and $A \perp\!\!\!\perp D \mid \{B, C\}$ (both paths blocked).

- $|\mathbf{S}| = 0$: all pairs dependent; nothing removed.
- $|\mathbf{S}| = 1$: $B \perp\!\!\!\perp C \mid A$: remove $B - C$, sepset$[\{B, C\}] = \{A\}$; other tests dependent.
- $|\mathbf{S}| = 2$: $A \perp\!\!\!\perp D \mid \{B, C\}$: remove $A - D$, sepset$[\{A, D\}] = \{B, C\}$. Skeleton: $A - B$, $A - C$, $B - D$, $C - D$.
- V-structures: $B - D - C$ with $D \notin \{A\}$, so $B \to D \leftarrow C$; $B - A - C$ with $A \in \{A\}$,

not a collider; $A - B - D$, $A - C - D$ with $B, C \in \{B, C\}$, not colliders.

- Meek's rules: Rule 1 applied to $A - B$ (using $B \to D$ with A, D non-adjacent) and to $A - C$, obtaining $A \to B$, $A \to C$, $B \to D \leftarrow C$, which matches the true DAG.

A caution on the last step: Rule 1 requires an arrow *into* B from a vertex non-adjacent to A, whereas $B \to D$ points out of B. Strictly, no Meek rule fires here, and the CPDAG is $A - B$, $A - C$, $B \to D \leftarrow C$. Its class contains $A \to B, A \to C$; $B \to A \to C$; $C \to A \to B$ (each with $B \to D \leftarrow C$), while $B \to A \leftarrow C$ is excluded because it would add a v-structure. The true DAG is one member of this class.

Exercise. Apply PC to $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$ (the earlier five-variable DAG). The v-structure $B \to D \leftarrow C$ is found, and Rule 1 orients $D \to E$ (since $B \to D$ with B, E non-adjacent).

10.1.2.4 Extensions

Conservative PC (Ramsey, Spirtes and Zhang 2006). Split faithfulness into *adjacency faithfulness* (adjacent $\Rightarrow$ no separator) and *orientation faithfulness*: if $X \to Z \leftarrow Y$ is an unshielded collider, X, Y are dependent given every subset of $\mathbf{V} \setminus \{X, Y\}$ containing Z; otherwise every separator contains Z. Algorithmically, check all subsets of X's and of Y's potential parents: if Z is in no separating set, orient a v-structure; if in all, a non-collider; otherwise mark the triple "unfaithful" (shown underlined in the output). This makes the algorithm robust to violations of orientation faithfulness at modest extra cost.

Order-independent PC (Colombo and Maathuis 2014). PC removes an edge as soon as a separator is found; a false-negative CI test early on changes later adjacency sets, so the output depends on the variable ordering (over 25 random orderings the learned skeletons vary). Remedies: average over random orderings, or remove edges only after finishing all tests with the same conditioning-set size; conservative orientation is too restrictive, so a majority rule over separating sets is introduced.

Fast Causal Inference (FCI) relaxes causal sufficiency (allows latent confounders). Example: true graph $X \to Y$, $W \to Z$, and a latent U with $U \to Y$, $U \to Z$. After CI-based edge removal the skeleton is $X - Y - Z - W$; FCI outputs $X \circ\!\!\to Y \leftrightarrow Z \leftarrow\!\!\circ W$, indicating an unmeasured confounder of Y and Z. PAG edge marks:

$X_1 \quad X_2$ (no edge)	X_1 and X_2 are not adjacent
$X_1 \circ\!\!\to X_2$	X_2 is not an ancestor of X_1
$X_1 \circ\!\!-\!\!\circ X_2$	no set d-separates X_1 and X_2
$X_1 \to X_2$	X_1 is a cause of X_2
$X_1 \leftrightarrow X_2$	there is a latent common cause of X_1 and X_2

10.1.2.5 Summary

CI constraints yield partial information about the structure. With a CI oracle (all tests correct), PC is complete (it recovers the full CPDAG). CI testing in practice uses partial correlation (linear Gaussian), Fisher's exact or χ^2 tests (categorical), kernel-based or model-based tests (Appendix 10.1.8). Relaxing sufficiency and/or faithfulness is active research.

10.1.3 Score-based approaches

Motivation. Search for the graph that best fits the data with a complexity penalty.

10.1.3.1 Why scores?

Constraint-based methods rely on CI tests, which can be unreliable in finite samples; a single wrong test can propagate errors through skeleton and orientation. Score-based methods define a function measuring how well a graph fits the data (and how simple it is) and search over graphs. The paradigms are complementary; hybrid methods (e.g. MMHC) combine them.

Why it is possible. Relative to the true structure (e.g. $E \to A \leftarrow B$, $A \to S$), *adding* an arc increases the number of parameters and encodes wrong causal assumptions; *missing* an arc cannot be compensated by fitting parameters accurately and also misses causal structure.

Intuition. True graph $A \to B \leftarrow C$, proposed $A \to B \to C$. In truth $A \perp\!\!\!\perp C$ but $A \not\perp\!\!\!\perp C \mid B$; the proposed graph says $A \not\perp\!\!\!\perp C$ and $A \perp\!\!\!\perp C \mid B$. Data will likely show $A \perp\!\!\!\perp C$ and $A \not\perp\!\!\!\perp C \mid B$, so the proposal fits poorly. Fit is assessed by the (log-)likelihood.

10.1.3.2 Log-likelihood for DAG models

With $P(\mathbf{V}) = \prod_i P(V_i \mid Pa_{\mathcal{G}}(V_i))$ and models $P_{\theta_i}(V_i \mid Pa_{\mathcal{G}}(V_i))$, the log-likelihood of data $D = \{\mathbf{v}^{(j)}\}_{j=1}^{n}$ is

$$\ell(\theta; D) = \sum_{j=1}^{n} \sum_{i=1}^{k} \log P_{\theta_i}(v_i^{(j)} \mid \mathbf{pa}_{\mathcal{G}}^{(j)}(V_i)),$$

a *decomposable* likelihood: each θ_i is fitted separately on the part of the data involving $P(V_i \mid Pa_{\mathcal{G}}(V_i))$.

For the true $A \to B \leftarrow C$ and proposed $B \to A$, $C \to B$:

$$\ell_{\text{left}} = \textstyle\sum_j \left[\log P_A(a_j) + \log P_B(b_j \mid a_j, c_j) + \log P_C(c_j)\right],$$
$$\ell_{\text{right}} = \textstyle\sum_j \left[\log P_A(a_j \mid b_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)\right].$$

Maximized, the true graph has the higher likelihood: with saturated local models, $\ell_{\text{left}}/n - \ell_{\text{right}}/n = I(A; C \mid B) > 0$ (Appendix 10.1.7).

The problem. Compare the true graph with $B \to A$, $C \to A$, $C \to B$ (complete): $\ell_{\text{right}} = \sum_j [\log P_A(a_j \mid b_j, c_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)]$ fits *better*, because the left model $\{P : A \perp\!\!\!\perp C\}$ is a submodel of the right (all of $P(A, B, C)$). With likelihood alone the best model is always a complete graph: overfitting. Too many parameters fit well but miss the sparse structure; reward fit, but punish parameters.

10.1.3.3 BIC

$$\text{BIC} = -2\,\ell(\hat{\theta}; D) + m \log n,$$

with m the number of parameters; lower is better. For a linear Gaussian network $X_i = \beta_{0i} + \sum_{X_k \in Pa(X_i)} \beta_{ki} X_k + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$, node i contributes $|Pa(X_i)| + 2$ parameters (slopes, intercept, variance), so $m = \sum_i (|Pa(X_i)| + 2) = |E| + 2|V|$. BIC is a regularization over graph structures.

10.1.3.4 Search and GES

Naively listing all DAGs and scoring each is impossible (super-exponentially many DAGs, possibly intractable likelihoods). Local search moves between states (DAGs) by local moves able to reach any DAG, ideally finding the best state in a polynomial sequence of moves. **Greedy Equivalence Search** (GES; Chickering 2002) defines states and moves on equivalence classes:

1. Start with the empty graph.

2. *Addition phase:* repeatedly move to the best class obtained by adding a single edge, until no improvement. (Adding an edge to a class can lead to different classes depending on the representative: adding $B - C$ to $\{A \to B \quad C,\ A \leftarrow B \quad C\}$ leads either to $\{A \to B \leftarrow C\}$ or to the class of chains/forks on A, B, C.)
3. *Removal phase:* repeatedly move to the best class obtained by removing a single edge, until no improvement.
4. Return the current class.

Theorem 10.2 (Chickering 2002)

Under faithfulness and as $n \to \infty$, GES identifies the true equivalence class.

Proof [Supplement]. Under faithfulness and with a consistent score such as BIC, two facts hold as $n \to \infty$:

(a) adding an edge that creates a dependence present in P increases the score, and removing an edge whose dependence is absent from P increases the score (local consistency);

(b) the score is equivalent (equal for Markov-equivalent DAGs). The forward phase of GES adds edges while the score improves, and by (a) and Meek's conjecture (proved by Chickering) it stops at an equivalence class whose DAGs contain the true independence model as an I-map. The backward phase then deletes edges while the score improves, and by (a) it can stop only at a minimal I-map, which under faithfulness is the true equivalence class. The detailed argument is Chickering (2002), Theorem 4. ■

Operators (on CPDAGs). Insert$(u, v, \mathbf{T})$: add $u \to v$ for non-adjacent u, v, with $\mathbf{T} \subseteq \mathrm{Ne}(v) \backslash \mathrm{Adj}(u)$ (undirected neighbors of v not adjacent to u); orient every $t \in \mathbf{T}$ as $t \to v$ (possibly creating unshielded colliders); convert to a CPDAG. Delete$(u, v, \mathbf{H})$: remove the edge between u and v, with $\mathbf{H} \subseteq \mathrm{Ne}(v) \cap \mathrm{Adj}(u)$; orient $h \to v$ and $h \to u$ for each $h \in \mathbf{H}$; convert to a CPDAG. Each operator yields a valid CPDAG, so the search stays in class space; at each step GES picks the operator (with its $u, v, \mathbf{T}$ or $\mathbf{H}$) with the best score improvement.

Example 10.3

True DAG $A \to B \leftarrow C$ (a v-structure, a unique class). Forward: start empty; Insert$(A, B, \emptyset)$ gives $A - B$ (undirected, as $A \to B$ and $A \leftarrow B$ are equivalent); Insert$(C, B, \{A\})$ (valid since $A \in \mathrm{Ne}(B) \setminus \mathrm{Adj}(C)$) gives $A \to B \leftarrow C$; no further insertion helps. Backward: no deletion helps. Output $A \to B \leftarrow C$.

10.1.3.5 Continuous optimization: NOTEARS and DAGMA

NOTEARS (Zheng et al. 2018) learns a linear SEM (noise not necessarily Gaussian) by black-box optimization. Example: $X = \epsilon_X$, $Z = aX + \epsilon_Z$, $Y = bX + cZ + \epsilon_Y$, i.e.

$$\begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{pmatrix} \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} + \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix} \iff \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b + ac & c & 1 \end{pmatrix} \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix}.$$

With data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and coefficients $W \in \mathbb{R}^{d \times d}$,

$$\min_{W \in \mathbb{R}^{d \times d}} \frac{1}{2n} \|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda \|W\|_1 \quad \text{subject to} \quad h(W) = \mathrm{tr}\left(e^{W \circ W}\right) - d = 0.$$

Why this characterizes acyclicity: for a binary adjacency matrix B, B^k counts walks of length k ($B^0 = I$), $e^B = \sum_k B^k / k!$, and $\mathrm{tr}(e^B)$ counts closed walks weighted by $1/k!$, including the d trivial

ones; hence $\operatorname{tr}(e^B) = d$ iff B is a DAG. $\operatorname{tr}(e^{W \circ W})$ counts weighted cycles (including 0-length ones) and is smooth, so the problem is solved by an augmented Lagrangian (NOTEARS: Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning).

NOTEARS-MLP (Zheng et al. 2020) learns general mechanisms f: $\min_f L(f)$ subject to $\mathcal{G}(f) \in$ DAGs, with $L(f) = \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, f_j(\mathbf{X}))$, $\mathbb{E}[X_j \mid X_{\mathrm{pa}(j)}] = g_j(f_j(X))$, $\mathbb{E}f_j(X) = 0$, where $k \to j$ is an edge iff f_j depends on X_k. A surrogate for W: f_j is independent of X_k iff $\|\partial_k f_j\|_{L^2} = 0$, so set $[W(f)]_{kj} = \|\partial_k f_j\|_{L^2}$ and solve $\min_{f_j \in H^1(\mathbb{R}^d)} L(f)$ s.t. $h(W(f)) = 0$ (H^1: Sobolev space of square-integrable functions with square-integrable derivatives). With MLPs $\mathrm{MLP}(u; A^{(1)}, \ldots, A^{(h)}) = \sigma(A^{(h)} \cdots \sigma(A^{(2)}\sigma(A^{(1)}u)))$, the output is independent of u_k if the k-th column of $A^{(1)}$ is zero, so the problem becomes

$$\min_{\theta} \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, \mathrm{MLP}(\mathbf{X}; \theta_j)) + \lambda \|A_j^{(1)}\|_{1,1} \quad \text{s.t.} \quad h(W(\theta)) = 0,$$

with

$$[W(\theta)]_{kj} = \|k\text{-th column of } A_j^{(1)}\|_2.$$

DAGMA (Bello, Aragam and Ravikumar 2022). The trace-exponential penalty is smooth but numerically demanding (matrix exponentials) and its optimization can be difficult for larger graphs. DAGMA uses a log-determinant penalty

$$h_{\mathrm{DAGMA}}(W) = -\log\det(sI - W \circ W) + d\log s, \qquad s > 0 \text{ (e.g. } s = 1).$$

Acyclic graphs get zero penalty, graphs with cycles positive penalty, which grows quickly near highly cyclic regions and gives a useful signal before the graph is acyclic. It keeps the NOTEARS setup, optimizing fit, sparsity and acyclicity together, $\min_\theta Q(f_\theta; \mathbf{X}) + \beta_1\|\theta\|_1 + h_{\mathrm{DAGMA}}(W(\theta))$, for linear or nonlinear SEMs, and reports faster optimization with competitive accuracy. It is not a new causal assumption, only another optimization device for enforcing DAGness.

10.1.3.6 Summaries

Score-based. BIC and GES guarantees are large-sample; with finite data correctness is not guaranteed. Local search is intractable if the score cannot be evaluated in polynomial time. If the local models $P(V_i \mid Pa_{\mathcal{G}}(V_i))$ are misspecified, the graph is likely wrong. No hidden variables are assumed.

Constraint- vs. score-based. Constraint-based (PC, FCI): CI tests prune edges, orientation via sepsets and Meek's rules; output a CPDAG (PAG for FCI); sensitive to CI errors. Score-based (GES, NOTEARS, DAGMA): a score is optimized over graph space; tends to do better with small samples, requires correct model specification. Both assume faithfulness and (except FCI) causal sufficiency; hybrids (MMHC) combine both.

10.1.4 Parametric approaches

Motivation. Use functional-form assumptions to orient edges that are indistinguishable from CIs alone.

Under what conditions can we orient $X \to Y$ versus $Y \to X$? With two variables there are no CIs to exploit; asymmetries in the joint distribution must be used.

Additive noise model (ANM). $y = f_y(x) + u_y$ with $x \perp\!\!\!\perp u_y$. If the distribution is known to come from an ANM, compare

$$\text{H1: } y = f_y(x) + u_y,\ x \perp\!\!\!\perp u_y\ (x \to y) \qquad \text{vs.} \qquad \text{H2: } x = f_x(y) + u_x,\ y \perp\!\!\!\perp u_x\ (y \to x).$$

Two main cases:

(1) f non-linear;

(2) f linear but u_y non-Gaussian.

10.1.4.1 Non-linear functions

For non-linear f (generically), an ANM holds in only one direction: in the wrong direction the residual's spread depends on the regressor. Procedure, given (x_i, y_i):

1. Fit both directions with any ML method: $\hat{y} = \hat{f}_y(x)$ and $\hat{x} = \hat{f}_x(y)$.
2. Compute residuals (guesses of u): $\hat{u}_x = x - \hat{f}_x(y)$, $\hat{u}_y = y - \hat{f}_y(x)$.
3. Test residual independence (e.g. HSIC): $\hat{u}_y \perp\!\!\!\perp x$ suggests $X \to Y$; $\hat{u}_x \perp\!\!\!\perp y$ suggests $Y \to X$. If neither holds, the ANM assumption is suspect.

10.1.4.2 Linear functions with non-Gaussian noise: LiNGAM

Cramér's decomposition theorem: if the sum of two independent real random variables is Gaussian, both summands are Gaussian; by induction the same holds for any finite sum. This complements the CLT: sums approach Gaussianity but cannot be exactly Gaussian unless every summand is.

LiNGAM (linear non-Gaussian acyclic model). For $y = ax + u_y$ with non-Gaussian errors,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \qquad x = u_x, \quad y = au_x + u_y.$$

In the backward regression of x on y, the residual is a mixture of u_x and u_y that is uncorrelated with, but (by Darmois–Skitovich/Cramér-type results) not independent of, y; in the forward direction the residual u_y is independent of x.

The trick: ICA. Suppose independent non-Gaussian sources $s_1, \ldots, s_n$ are observed only through a mixture $\mathbf{x} = A\mathbf{s}$. How can A and $\mathbf{s}$ be recovered? By the CLT intuition, $x = as_1 + bs_2$ looks more Gaussian than s_1 or s_2; ICA finds directions maximizing non-Gaussianity. Measures:

- *Kurtosis* (fourth cumulant) $\kappa(y) = \frac{\mathbb{E}[(y-\mu)^4]}{\mathrm{Var}(y)^2} - 3$: zero for Gaussians, positive for super-Gaussian (heavy-tailed, e.g. Laplace), negative for sub-Gaussian (light-tailed, e.g. uniform); simple but outlier-sensitive.
- *Negentropy* $J(y) = H(y_{\text{Gauss}}) - H(y) \geq 0$, with y_{Gauss} Gaussian with the same mean and variance; $J(y) = 0$ iff y is Gaussian (the Gaussian maximizes entropy for a given variance).

ICA takes $\mathbf{x} = A\mathbf{s}$ and returns an unmixing matrix W_{ICA} with $\hat{\mathbf{s}} = W_{\text{ICA}}\mathbf{x}$, up to permutation and scaling (including sign): among all $\mathbf{w}^\top\mathbf{x}$, the maximally non-Gaussian ones are the sources (Hyvärinen and Oja 2000). In the two-variable case, run ICA on (x, y) and look for the zero in the mixing matrix.

LiNGAM (Shimizu et al. 2006). For ordered variables $x_i = \sum_{j<i} b_{ij}x_j + e_i$, i.e. $\mathbf{x} = B\mathbf{x} + \mathbf{e}$, $(I - B)\mathbf{x} = \mathbf{e}$, $\mathbf{x} = (I - B)^{-1}\mathbf{e}$. Let $W = I - B$. ICA gives $W_{\text{ICA}} = PDW$ (P a permutation, D diagonal: the permutation, scaling and sign indeterminacies), but the LiNGAM structure resolves them:

1. $\mathbf{e} = (I - B)\mathbf{x} = W\mathbf{x}$, with W a row-permuted, rescaled version of $I - B$.
2. Run ICA: $W_{\text{ICA}} = \text{ICA}(\mathbf{X})$.

3. Permute the rows of W_{ICA} so that the diagonal has no zeros (the unique such permutation, since $I - B$ has a non-zero diagonal and is permutable to lower-triangular), then rescale rows so the diagonal is one: $W = \text{normalize}(\text{rowpermute}(W_{\text{ICA}}))$.
4. $\hat{B} = I - W$; a causal order is the one making $\hat{B}$ lower-triangular.

Summary. Given an ANM $y = f(x) + u_Y$ with f non-linear or U_Y non-Gaussian, the direction of the arrow is identifiable, assuming no latent confounding ($X \perp\!\!\!\perp U_Y$).

10.1.4.3 Score matching (NoGAM)

(Montagna et al. 2023.) Consider an additive non-linear model $X_i = f_i(\text{PA}_i) + N_i$ with arbitrary noise. The *score function* is the gradient of the log-density, $s(X) = \nabla_X \log p(X)$, $s_i(X) = \partial_{X_i} \log p(X)$ (unrelated to BIC/GES scores): it describes how the log-density changes when one variable is moved. Score matching estimates $s(X)$ from samples without estimating $p(X)$, and the score identifies *leaf* nodes; recursively removing leaves gives a topological order, after which extra edges are pruned.

Why leaves are special. Writing $\log p(X) = \sum_k \log p_k(N_k)$ with $N_k = X_k - f_k(\text{PA}_k)$,

$$s_l(X) = \partial_{N_l} \log p_l(N_l) \quad \text{for a leaf } X_l,$$

$$s_i(X) = \partial_{N_i} \log p_i(N_i) - \sum_{k \in \text{CH}_i} \partial_{N_k} \log p_k(N_k)\, \partial_{X_i} f_k \quad \text{otherwise.}$$

For a leaf, regressing X_l on the rest gives residual $R_l = X_l - q(X_{-l}) = N_l$ (in population), so $s_l(X)$ is a function of R_l alone; for non-leaves the children terms remain and R_i alone generally cannot predict $s_i(X)$. Population statement (under ANM identifiability conditions):

$$X_i \text{ is a leaf} \iff \min_h \mathbb{E}\big[\{h(R_i) - s_i(X)\}^2\big] = 0.$$

Algorithm (NoGAM). (1) Estimate $\hat{s}(X) \approx \nabla_X \log p(X)$.

(2) For each remaining node, regress X_i on X_{-i} and compute residuals $\hat{R}_i$.

(3) Predict $\hat{s}_i(X)$ from $\hat{R}_i$ and record the MSE.

(4) Declare the node with smallest MSE a leaf, remove it, and repeat.

(5) Reverse the removal order to get a topological order, then prune extra edges. This recovers an order without assuming Gaussian noise.

10.1.5 Appendix: GES operators in detail

Insert$(u, v, \mathbf{T})$ adds $u \to v$ (with $u \notin \text{Adj}(v)$), orients $t - v$ as $t \to v$ for every $t \in \mathbf{T}$ ($t \notin \text{Adj}(u)$), and completes the PDAG to a CPDAG. Key set: $\mathbf{T} \subseteq \text{Ne}(v) \setminus \text{Adj}(u)$, and $(\text{Ne}(v) \cap \text{Adj}(u)) \cup \mathbf{T}$ must be a clique; full validity also requires every semi-directed path from v to u to pass through this blocking set. In the v-structure example, after $A - B$ the correct move is $\text{Insert}(C, B, \{A\})$.

Delete$(u, v, \mathbf{H})$ removes the edge between u and v ($u - v$ or $u \to v$), orients $h - v$ as $h \to v$ and $h - u$ as $h \to u$ for every $h \in \mathbf{H}$ (h points into both endpoints), and completes to a CPDAG. Key set: $\mathbf{H} \subseteq \text{Ne}(v) \cap \text{Adj}(u)$, and $(\text{Ne}(v) \cap \text{Adj}(u)) \setminus \mathbf{H}$ must be a clique. These are Chickering's (2002) operators; XGES reparametrises the same score-relevant sets (Nazaret et al. 2025, "Extremely greedy equivalence search").

10.1.6 Appendix: Post-nonlinear causal model (optional)

PNL extends the ANM with an invertible transformation of the effect: for $X_1 \to X_2$, $X_2 = g_2\{f_1(X_1) + E_2\}$, $E_2 \perp\!\!\!\perp X_1$, g_2 invertible. Then $g_2^{-1}(X_2) = f_1(X_1) + E_2$, i.e. $E_2 = g_2^{-1}(X_2) - f_1(X_1)$.

Find ψ_2, ϕ_1 such that $E_2 = \psi_2(X_2) - \phi_1(X_1)$ is independent of X_1; if this works in one direction but not the other, orient the edge. Identifiability holds generically, with special functional/noise cases still ambiguous (Zhang and Hyvärinen 2009).

10.1.7 Appendix: Scores in detail

Why the left likelihood is larger. For left $A \to B \leftarrow C$ and right $B \to A$, $C \to B$, with saturated local models the maximized population log-likelihoods are negative conditional entropies:

$$\ell_{\text{left}}/n = -H(A) - H(C) - H(B \mid A, C), \qquad \ell_{\text{right}}/n = -H(C) - H(B \mid C) - H(A \mid B).$$

If the truth is the collider, $A \perp\!\!\!\perp C$ and $H(A) + H(C) + H(B \mid A, C) = H(A, B, C)$, so

$$\ell_{\text{left}}/n - \ell_{\text{right}}/n = H(A \mid B) + H(B \mid C) - H(A) - H(B \mid A, C) = I(A; C \mid B).$$

Conditioning A on B helps the right graph, but dropping A from B's parents hurts more, exactly when $A \not\perp\!\!\!\perp C \mid B$.

Bayesian Dirichlet scores (categorical data). The BD score (Cooper and Herskovits) computes the probability of a network given data, under assumptions. Drawbacks: the user must specify α_{ijk} for every variable–parent configuration, and equivalent structures need not get equal scores. BDe is the score-equivalent variant obtained via an equivalent sample size α; BDeu sets $\alpha_{ijk} = \alpha/(r_i q_i)$, spreading α uniformly over local CPT cells (not a uniform prior over graphs). With i indexing variables, j parent configurations (q_i of them) and k values (r_i of them), and the Dirichlet as conjugate prior of the multinomial,

$$\text{BD}(\mathcal{G}, D; \alpha) = \prod_{i=1}^{N} \text{BD}(X_i \mid Pa_{X_i}; \alpha_i) = \prod_{i=1}^{N} \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + n_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + n_{ijk})}{\Gamma(\alpha_{ijk})},$$

α_i being the imaginary (equivalent) sample size. $\alpha_{ijk} = 1$ gives the K2 score; $\alpha_{ijk} = 1/2$ the BD score with Jeffreys' prior; $\alpha_{ijk} = \alpha/(r_i q_i)$ the BDeu score (with $\alpha_i = \alpha$ for all X_i).

BIC (maximization form). Dividing by -2,

$$\text{BIC}(\mathcal{G}, D) = \sum_{i=1}^{N} \Big[\ell_i(\hat{\theta}_i) - \frac{\log n}{2} q_i (r_i - 1) \Big], \qquad \ell_i(\hat{\theta}_i) = \sum_{m=1}^{n} \log \hat{P}_i(x_i^{(m)} \mid \pi_i^{(m)}),$$

which is score-equivalent. BIC is the large-sample Laplace approximation of log BDeu; they agree up to $O(1)$ as $n \to \infty$.

10.1.8 Appendix: CI tests for causal discovery

χ^2 test (categorical). Tests $P(X, Y \mid Z) = P(X \mid Z) P(Y \mid Z)$ for all z. Example (2008 General Social Survey, "Where do you get most of your information about current news events?"; credit Yibi Huang):

Age	TV	Internet	Newspapers	Other	Total
18–29	109	92	25	36	262
30–49	272	157	88	63	580
50+	345	59	165	63	632
Total	726	308	278	162	1474

(Did the way people get news change with age?) For each level of Z, expected counts under CI are $E(x, y \mid z) = N(x, z) N(y, z)/N(z)$, and

$$\chi^2(Z = z) = \sum_{x,y} \frac{(O(x, y \mid z) - E(x, y \mid z))^2}{E(x, y \mid z)}, \qquad \chi^2_{\text{total}} = \sum_z \chi^2(Z = z),$$

with $df = (|X|-1)(|Y|-1)|Z|$; the p-value compares χ^2_{total} with the χ^2_{df} distribution.

Partial correlation (linear).

$$r_{XY\cdot Z} = \frac{r_{XY} - r_{XZ}r_{YZ}}{\sqrt{1-r_{XZ}^2}\sqrt{1-r_{YZ}^2}}$$

for a single conditioning variable (for multivariate Z use recursion or regression residuals). If it remains high, X and Y are correlated beyond Z; if it drops, Z explains most of their correlation.

HSIC (non-linear marginal independence). Tests $H_0 : X \perp\!\!\!\perp Y$ without assuming linearity. With kernels k, l, Gram matrices K, L and $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, $\widehat{\text{HSIC}}(X,Y) = \frac{1}{n^2}\operatorname{tr}(KHLH)$. With characteristic kernels, population HSIC is zero iff $X \perp\!\!\!\perp Y$. Null calibration by permutation, gamma approximation or asymptotics (Gretton et al. 2005). Useful when marginal dependence may be non-linear and non-Gaussian; for conditional independence use KCI.

Kernel conditional independence test (KCI). Targets $H_0 : X \perp\!\!\!\perp Y \mid Z$ via RKHS conditional covariances: represent X, Y, Z by kernels K_X, K_Y, K_Z; regress out Z in the RKHS, e.g. $R_Z = \epsilon(K_Z + \epsilon I)^{-1}$, $\tilde{K}_X = R_Z K_X R_Z$, $\tilde{K}_Y = R_Z K_Y R_Z$; test whether the residualised dependence $T_{\text{KCI}} = \frac{1}{n}\operatorname{tr}(\tilde{K}_X \tilde{K}_Y)$ is zero. It replaces Fisher-z or χ^2 tests in PC-style discovery when non-linear CI matters, at higher computational cost and with sensitivity to kernel bandwidth and regularization (Zhang, Peters, Janzing and Schölkopf 2011).

10.2 Local causal discovery

Motivation. The central question is what happens to structure learning when the reasoning is restricted to local objects (Markov blankets, a target's neighborhood, order prefixes) instead of the whole graph.

10.2.1 The map

Motivation. Organize discovery algorithms by signal and reach.

The previous lecture gave the global workhorses: constraint-based (PC, FCI), using CI tests over the whole graph, and score-based (GES, NOTEARS, DAGMA), optimizing one global score. This lecture is not a new family but a direction: *what happens when we go local?* Two knobs organize everything:

- **Signal** (vertical): test independences or optimize a score?
- **Reach** (horizontal): reason over the whole graph or a local object?

	global	local
CI tests	PC, FCI	GS, HITON
CI → score		MMHC
score	GES, DAGMA	BOSS

The previous lecture filled the global column; this one fills the local column.

10.2.1.1 One primitive: grow–shrink

A recurring two-step move, reused on different local objects:

- **Grow:** add while the candidate still helps. CI version: add Y while $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; score version: add Y while the score improves.

- **Shrink:** remove what became redundant. CI version: drop Y if $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; score version: drop Y if the score improves.

Algorithm	Local object	Signal
GS	Markov blanket	CI
HITON	one target's parents and children	CI
MMHC	skeleton, then DAG	CI + score
BOSS	order prefix	score

10.2.2 Going local on CI

Motivation. Learn Markov blankets and target neighborhoods by CI tests.

10.2.2.1 The Markov blanket

For a node X,

$$\mathrm{MB}(X) = \big(\mathrm{pa}(X) \cup \mathrm{ch}(X) \cup \mathrm{pa}(\mathrm{ch}(X))\big) \setminus \{X\},$$

its parents, children and spouses (co-parents of children). Conditioning on $\mathrm{MB}(X)$ makes every other variable irrelevant for X. It is small and local, yet enough to drive skeleton and orientation (Margaritis and Thrun 1999; Margaritis 2003).

10.2.2.2 GS: grow–shrink the blanket

Grow (permissive): add any Y with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; some additions are real, others artefacts of an incomplete $\mathbf{S}$. **Shrink** (clean up): drop Y once a better separator appears, $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$. Example: in the chain $X \to C \to G$, $X \not\perp\!\!\!\perp G$, so grow may add G; once C is in $\mathbf{S}$, $X \perp\!\!\!\perp G \mid C$ and shrink removes G. Then test which blanket members are true neighbors and orient unshielded colliders as in PC.

Algorithm 4 GS-MB(X)

1: $\mathbf{S} \leftarrow \emptyset$
2: **repeat** find $Y \in \mathbf{V} \setminus (\{X\} \cup \mathbf{S})$ with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \cup \{Y\}$
3: **until** no such Y exists
4: **repeat** find $Y \in \mathbf{S}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \setminus \{Y\}$
5: **until** no such Y exists
6: **return S**

Correctness sketch. Under faithfulness, a parent or child Y of X stays dependent on X given any set not containing X, Y, so grow includes it. If Y is a parent of a child of X, conditioning on that child creates dependence, so grow eventually includes Y. Thus after grow $\mathrm{MB}(X) \subseteq \mathbf{S}$. If $W \notin \mathrm{MB}(X)$ but $W \in \mathbf{S}$, the true blanket (itself a separating set) separates X from W; since $\mathrm{MB}(X) \subseteq \mathbf{S} \setminus \{W\}$, shrink removes W via $X \perp\!\!\!\perp W \mid \mathbf{S} \setminus \{W\}$.

Structure induction. (1) Compute $\mathrm{MB}(X)$ for every X.

(2) For each $Y \in \mathrm{MB}(X)$ decide whether it is a direct neighbor: no edge if $\exists \mathbf{S} \subseteq \mathbf{R}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S}$, where $\mathbf{R}$ is the smaller of $\mathrm{MB}(X) \setminus \{Y\}$ and $\mathrm{MB}(Y) \setminus \{X\}$.

(3) Orient unshielded colliders using local tests.

(4) Repair cycles and propagate forced directions.

Randomized GS. Plain GS becomes expensive with large blankets. Randomized GS samples conditioning sets from the smaller blanket instead of enumerating all subsets and combines multiple tests as evidence for edge existence and direction: an anytime-style algorithm (more tests, more stable decisions) with controlled runtime but weaker exactness guarantees.

10.2.2.3 HITON: target-focused, not whole-graph

GS computes a blanket for every node to assemble the full network. HITON wants only the local neighborhood of one target T (e.g. feature selection for a disease variable). Its workhorse HITON-PC$(\cdot)$ returns a node's *parents and children* ("PC" here is not the PC algorithm) by interleaved grow-while-shrink with small conditioning sets, which is sample-efficient. "Focus" means running this subroutine on T and a bounded local ring around it, not on all n variables: local, not literally single-node. Two pieces: HITON-PC (one node's neighbors) and HITON-MB (assembling T's blanket) (Aliferis et al. 2003; 2010).

HITON-MB: two rings, then prune. Example: $P \to T \to C \leftarrow Z$, $C \to G$.

1. Ring 1: PC$(T) \leftarrow$ HITON-PC(T), T's parents and children (P, C).
2. Ring 2 (PC-of-PC): run HITON-PC(Y) for each $Y \in$ PC(T). A spouse Z is not adjacent to T, so it only surfaces here, as a neighbor of a child.
3. Prune: drop ring-2 candidates still separable from T (e.g. the grandchild G, separated by C). Ring 2 keeps the spouse Z (no set separates it from T once the collider C is conditioned on) and prunes G.

This calls HITON-PC on several nodes, but only on T's local ring. The PC step finds direct

Algorithm 5 HITON-MB(T)

PC$(T) \leftarrow$ HITON-PC(T); $\mathbf{C} \leftarrow$ PC(T)
for each $Y \in$ PC(T) **do** $\mathbf{C} \leftarrow \mathbf{C} \cup$ HITON-PC(Y)
for each candidate $X \in \mathbf{C} \setminus$ PC(T) **do**
    if $\exists \mathbf{S} \subseteq \mathbf{C} \setminus \{X\}$ with $T \perp\!\!\!\perp X \mid \mathbf{S}$ **then** remove X from $\mathbf{C}$
return C

neighbors of the target; the PC-of-PC search brings in co-parents; CI tests remove false positives.

Halfway: the CI column is filled (GS: a blanket per node; HITON: one target). The score-local corner remains: MMHC and BOSS.

10.2.3 Bringing in the score

Motivation. Combine local CI skeletons with scores, and search over orders.

10.2.3.1 MMHC: a hybrid bridge

Start at CI-local, then switch to score (Tsamardinos, Brown and Aliferis 2006). Run local parents-and-children discovery for every node with MMPC, the max-min sibling of HITON-PC (same grow/shrink, different admission rule); the union is a candidate skeleton. A score-based hill-climb then picks a DAG but may add edges only inside that skeleton:

$$\text{local PC per node (MMPC)} \;\to\; \text{union} = \text{skeleton} \;\to\; \text{score hill-climb} \;\to\; \text{one DAG}.$$

It combines a sound skeleton restriction with heuristic score-based orientation: the first time a score drives the graph.

10.2.3.2 BOSS: best order score search

Score-local (Andrews et al. 2023). The search state is a permutation π, not a graph. Each v's candidate parents are its *prefix*, the variables before it:

$$\mathrm{pre}_\pi(v) = \{w : \pi.\mathrm{index}(w) < \pi.\mathrm{index}(v)\};$$

e.g. for $\pi = (A, C, B, D, E)$, $\mathrm{pre}_\pi(B) = \{A, C\}$.

Scoring an order. The projection $\mathcal{G}_\pi$ picks each v's parents from its prefix by greedy grow/shrink, approximating

$$\mathrm{pa}_\pi(v) \approx \underset{\mathbf{W} \subseteq \mathrm{pre}_\pi(v)}{\arg\max}\ \mathrm{score}_v(\mathbf{W}), \qquad \mathrm{score}_v(\mathbf{W}) = \mathrm{BIC}(X_v \mid X_{\mathbf{W}}),$$

a penalized log-likelihood (higher = better fit with fewer parameters). The order score sums the local terms, $\mathrm{score}(\pi) = \sum_v \mathrm{score}_v(\mathrm{pa}_\pi(v))$. Grow–shrink trees (GSTs) cache repeated prefix scores; this is the key to BOSS's scaling. It relies on the local decomposition $\mathrm{BIC}(\mathcal{G}, D) = \sum_{v \in \mathbf{V}} \mathrm{BIC}(X_v \mid X_{\mathrm{pa}_\mathcal{G}(v)})$: changing v's parents affects only one local term, which makes projection and best-move scoring efficient.

Algorithm 6 project$(\mathbf{X}, \pi)$ and best-move$(\mathcal{T}, \pi, v)$

$E \leftarrow \emptyset$
for each v in π **do**
  $\mathbf{Z} \leftarrow \mathrm{pre}_\pi(v)$; $\mathbf{W} \leftarrow \mathrm{grow}(\mathbf{X}, v, \mathbf{Z})$; $\mathbf{W} \leftarrow \mathrm{shrink}(\mathbf{X}, v, \mathbf{W})$
  add $w \to v$ to E for each $w \in \mathbf{W}$
return $\mathcal{G}_\pi = (\mathbf{V}, E)$

best $\leftarrow \mathcal{T}.\mathrm{score}(\pi)$
for $i = 1, \ldots, |\pi|$ **do**
  $\pi' \leftarrow \pi.\mathrm{move}(v, i)$
  if $\mathcal{T}.\mathrm{score}(\pi') >$ *best* **then** *best* $\leftarrow \mathcal{T}.\mathrm{score}(\pi')$; $\pi \leftarrow \pi'$
return π

Example 10.4 (Best-move)

Truth: the collider $A \to C \leftarrow B$. Move one variable to every position, re-project, and keep the move only if $\mathrm{score}(\mathcal{G}_{\pi'}) > \mathrm{score}(\mathcal{G}_\pi)$. With $\pi = (A, C, B)$, C precedes B, so C is a candidate parent of B; conditioning on the collider C couples A and B, so the projection includes an extra (spurious) A–B dependence: 3 edges. Moving C last, $\pi' = (A, B, C)$, gives the true collider with 2 edges. The independences represented are those of the true model with fewer parameters, so $\mathrm{score}(\mathcal{G}_{\pi'}) > \mathrm{score}(\mathcal{G}_\pi)$ and π' is kept.

Benchmarks. BOSS is designed for large, relatively dense models. In reported simulations it matches GRaSP (an earlier order-based search) in accuracy while scaling far better in runtime, compares favorably with fast GES, PC, DAGMA and LiNGAM, and has been demonstrated on resting-state fMRI with hundreds of cortical parcels (dense graphs, hundreds to 1000 variables).

10.2.4 Zoom out

Motivation. Choose among the methods and remember what none of them solve.

The filled map. Going local keeps the signal but shrinks the object; MMHC bridges CI and score; GSTs make the score-local corner scale.

What to use when.

- Sparse graph, reliable CI tests: GS (full graph) or HITON (one target).
- High-dimensional variable/feature selection: HITON's target blanket.
- Large, dense, score-friendly data: BOSS (or MMHC as a simpler hybrid).
- Hidden confounders: none of these; use FCI-type methods.
- Only one effect is needed, not the whole graph: effect-specific local discovery (Entner et al.; LDECC).

What none of these solve / limitations checklist.

- Causal sufficiency: GS, HITON, MMHC and BOSS all assume no hidden confounders; none replaces FCI.
- Faithfulness: separations must mirror independences (or score consistency must hold).
- Finite samples: weak effects and large conditioning sets hurt every method.
- Specification and tuning: the score family (BOSS needs an appropriate local likelihood) and the penalty (e.g. BIC) set the precision–recall balance between sparse and dense recovery.

Locality changes the cost of search, not the assumptions.

Comparison.

	GS	HITON	MMHC	GES	BOSS
Main signal	CI tests	CI tests	CI + score	BIC-like score	BIC-like score
Search object	blankets	target blanket	skeleton + DAG	CPDAGs	orders
Locality	Markov blankets	PC/co-parents	local skeleton	edge operators	prefixes
Output target	MEC graph	variables for T	one DAG	CPDAG	CPDAG
Main risk	CI error	target scope	heuristic orient.	many score calls	model spec.

Summary. One primitive, grow–shrink, on four local objects: GS (CI on Markov blankets), HITON (CI on one target's neighborhood), MMHC (a CI skeleton restricting a score hill-climb), BOSS (score on order prefixes, with GSTs caching repeated work). The same signals as before, made local: cheaper search, same assumptions.

Chapter 11. Mediation, SWIGs and G-Methods

11.1 Direct and indirect effects

Motivation. The central question is how to decompose a total causal effect into the part transmitted directly and the part transmitted through mediators, in nonlinear models where "holding the mediator fixed" and "subtracting" no longer give a well-defined indirect effect.

11.1.1 Preliminaries: potential responses

Motivation. Fix counterfactual notation, including nested counterfactuals.

In an SCM, $Y(u)$ is the value of Y for unit $U = u$ (the potential response). $Y_x(u)$ is the value of Y for the same unit under $\text{do}(X = x)$, so $P(Y_x = y) = P(y \mid \text{do}(x))$. $Y_{xz}(u)$ is the value under the joint intervention $\text{do}(X = x, Z = z)$. $Z_{x^*}(u)$ is the mediator value that would naturally occur under the reference intervention $\text{do}(X = x^*)$. The nested counterfactual

$$Y_{x,Z_{x^*}}(u): \quad \text{set } X = x, \text{ but set } Z \text{ to the value it would have taken under } X = x^*.$$

11.1.2 Introduction

Motivation. Motivate direct and indirect effects and the prescriptive/descriptive distinction.

For $X \to Z \to Y$ with $X \to Y$:

- **Total effect:** the change in the distribution of Y when X is set to x by external intervention, $P(Y_x = y) = P(y \mid \text{do}(x))$.
- **Direct effect:** measured by holding all intermediate variables constant.
- **Indirect effect:** in a linear system, indirect = total − direct; in a nonlinear system it cannot be measured by holding some variables constant.

Informally, a direct effect quantifies an influence not mediated by other variables; more formally, the sensitivity of Y to changes in X while all other factors in the analysis are held fixed. Graphically, holding those factors fixed eliminates all causal paths from X to Y except the direct link $X \to Y$. Indirect effects cannot be defined this way: no choice of variables to hold constant makes the measured effect circumvent the direct pathway. Hence the definition of indirect effects remained incomplete.

11.1.2.1 Examples

Birth-control pill and thrombosis. The pill may increase thrombosis directly but reduce it indirectly by lowering pregnancy rates (pregnancy itself increases thrombosis). The direct effect is biologically stable and invariant to social factors (e.g. marital status), more transportable and more useful in scientific explanation and policy analysis than the total effect.

Race/sex discrimination in hiring. Courts care about the direct effect of race or gender on hiring, not the indirect effect via qualifications. The legal test: would the employer have made the same decision had the applicant's race or gender been different, all else being equal?

Smoking, hypertension and heart disease (Hernán and Robins 2020, Ch. 23). A randomized trial finds that quitting smoking ($A = 0$) reduces myocardial infarction Y. Does the benefit operate through reduced hypertension M (measured at 6 months)? Direct effect: $A \to Y$ not through

M (oxidative stress, platelet effects); indirect effect: $A \to M \to Y$. Policy question: if a drug eliminated smoking-induced hypertension, how much of the MI benefit would remain?

Drug, aspirin and headache. A drug has a side effect, headache; patients with headaches tend to take aspirin, which may affect the disease directly or modify the impact of the drug. The total effect reflects existing aspirin usage patterns; the direct effect isolates the drug's impact regardless of aspirin intake. A direct effect such as

$$P(Y_{xz} = y) - P(Y_{x^*z} = y),$$

with z a specified aspirin level, informs whether aspirin should be encouraged or discouraged during treatment. In linear systems direct effects are given by path coefficients and do not depend on the level at which intermediates are held; in nonlinear systems that level generally modifies the effect and must represent the target policy. This leads to two conceptualisations.

11.1.2.2 Prescriptive versus descriptive

- **Prescriptive (controlled direct effect):** would an untreated patient improve if treated, while aspirin intake is held at a predetermined level $Z = z$? It does not rely on individual natural behavior: $P(Y_{xz} = y) - P(Y_{x^*z} = y)$.
- **Descriptive (natural direct effect):** would the untreated patient improve if treated, while aspirin intake is held at the level the patient currently consumes under no treatment? It requires knowledge of individual natural behavior.

Consider a patient who takes aspirin iff treated, and for whom treatment works only with aspirin. Then the natural direct effect is zero, whereas the controlled direct effect (at $z =$ aspirin) is non-zero. For population averages: the prescriptive formulation is *pragmatic* (predict the difference in recovery when a prescribed aspirin dose is given to everyone), while the descriptive formulation is *attributional* (is an observed improvement due to the treatment itself, as opposed to preferential aspirin use among the treated?).

Why the average natural direct effect? Suppose the manufacturer considers eliminating the headache side effect. Would the drug retain its effectiveness in the population? The CDE does not answer this, since it refers to one aspirin level imposed uniformly. In the target population aspirin intake varies with factors other than drug-induced headache, which may also modulate effectiveness; hence the average natural direct effect is needed.

Descriptive indirect effects. There is no prescriptive definition of indirect effects, but the descriptive one transfers: to assess a patient's natural indirect effect, withhold treatment and ask whether the patient would recover if given as much aspirin as under treatment. (The prescriptive formulation cannot be transported, since the direct effect cannot be blocked by holding variables constant.) Their operational meaning concerns non-standard policies: if asking applicants about gender is made illegal (and no other indication of gender is available), any residual sex preference in hiring is attributable to the indirect effect of sex via qualifications. Its magnitude, predicted from data collected before the policy, is an average indirect effect.

11.1.3 Mathematisation

Motivation. Define controlled and natural direct effects and identify the latter.

Let X be the control variable, Y the response, Z the set of all intermediate variables between X and Y, and $Y_x(u)$ the value of Y for unit u under $\text{do}(X = x)$.

Definition (Controlled direct effect)

$\text{CDE}_z(x, x^*; Y, u) = Y_{xz}(u) - Y_{x^*z}(u)$, and $\text{CDE}_z(x, x^*; Y) = \mathbb{E}[Y_{xz} - Y_{x^*z}]$, the expectation taken over u.

(The difference is the most common measure; other contrasts, e.g. ratios, can be used.)

Definition (Natural direct effect)

$\text{NDE}(x, x^*; Y, u) = Y_{x,Z_{x^*}(u)}(u) - Y_{x^*}(u)$, and $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$.

11.1.3.1 The cross-world problem

$Y_{x,Z_{x^*}}$ refers to two intervention worlds at once: world 1 with $X = x$ (to evaluate Y) and world 2 with $X = x^*$ (to determine Z_{x^*}). No experiment, not even joint randomization of X and Z, observes both worlds for the same individual. NDE and NIE are therefore empirically unverifiable in principle, and identification requires an untestable *cross-world independence* $Y_{xz} \perp\!\!\!\perp Z_{x^*}$ (Hernán and Robins 2020, §23.1; Robins and Richardson 2010).

11.1.3.2 Experimental identification

Formally, $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$ is not in general reducible to expressions of the form $P(Y_x = y)$ or $P(Y_{xz} = y)$. (Pearl's paper is from 2001; general nested-counterfactual identification results came later: Shpitser and Pearl 2008; Correa, Lee and Bareinboim 2021.)

Theorem 11.1 (Pearl 2001)

If there is a set W of covariates, non-descendants of X or Z, with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$, the average NDE is experimentally identifiable:

$$\text{NDE}(x, x^*; Y) = \sum_{w,z} [\mathbb{E}(Y_{xz} \mid w) - \mathbb{E}(Y_{x^*z} \mid w)] P(Z_{x^*} = z \mid w) P(w).$$

Proof [Supplement].

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid Z_{x^*} = z, w) P(Z_{x^*} = z \mid w) P(w) = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid w) P(Z_{x^*} = z \mid w) P(w),$$
$$\mathbb{E}[Y_{x^*}] = \mathbb{E}[Y_{x^*,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{x^*z} \mid w) P(Z_{x^*} = z \mid w) P(w),$$

using (in the first line) $Y_{xz} = Y_{x,Z_{x^*}}$ on the event $Z_{x^*} = z$ and the conditional independence, and (in the second) the composition axiom $Y_{x^*} = Y_{x^*,Z_{x^*}}$. Subtract. ■

Graphical criterion. Example (a): $X \to Z \to Y$, $X \to Y$, and W a common cause of Z and Y. In (b), the subgraph $\mathcal{G}_{\underline{XZ}}$ obtained by deleting all arrows emanating from X and Z, W d-separates Y from Z: $(Y \perp_d Z \mid W)_{\mathcal{G}_{\underline{XZ}}}$, which ensures the condition.

11.1.3.3 Markovian models

The condition $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ holds with $W = \emptyset$ when Y_{xz} is independent of all variables in the model. In Markovian models:

- $P(Y_{xz} = y) = P(y \mid x, z)$, where $X \cup Z$ is the parent set of Y;

- $P(Z_{x^*} = z) = \sum_s P(z \mid x^*, s)P(s)$;
- $P(Y_{x,Z_{x^*}} = y) = \sum_s \sum_z P(y \mid x, z)P(z \mid x^*, s)P(s)$,

where S are the parents of Z other than X (or any set satisfying the back-door criterion). Hence the NDE is identifiable from non-experimental data:

$$\text{NDE}(x, x^*; Y) = \sum_s \sum_z [\mathbb{E}(Y \mid x, z) - \mathbb{E}(Y \mid x^*, z)]P(z \mid x^*, s)P(s).$$

11.1.3.4 The mediation formula

In the simplest case ($X \to Z \to Y$, $X \to Y$, no confounding of Z–Y),

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]\, P(Z = z \mid X = x^*) :$$

evaluate Y under treatment x with the mediator distribution of treatment x^*. Smoking example (A treatment, M mediator): what would MI risk be if patients continued smoking ($a = 1$) but had the blood pressure of quitters ($a = 0$)?

$$\mathbb{E}[Y_{1,M_0}] = \sum_m \mathbb{E}[Y \mid A = 1, M = m]\, P(M = m \mid A = 0).$$

11.1.4 Natural indirect effects

Motivation. Define the NIE, relate it to the total effect, and identify it.

Definition (Natural indirect effect)

$\text{NIE}(x, x^*; Y, u) = Y_{x^*, Z_x(u)}(u) - Y_{x^*}(u)$, and $\text{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*})$: keep the treatment at the reference level but let the mediator respond as under treatment.

Decomposition.

$$\begin{aligned} \text{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}), \\ \text{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}), \\ -\text{NIE}(x^*, x; Y) &= -\mathbb{E}(Y_{x, Z_{x^*}}) + \mathbb{E}(Y_x). \end{aligned}$$

Adding the last two lines gives

$$\text{TE}(x, x^*; Y) = \text{NDE}(x, x^*; Y) - \text{NIE}(x^*, x; Y) :$$

the total effect is the direct effect of the transition $x^* \to x$ minus the indirect effect of the reverse transition. Strange as it looks, in the linear case it reduces to the standard additive relation $\text{TE}(x, x^*; Y) = \text{NDE}(x, x^*; Y) + \text{NIE}(x, x^*; Y)$ (Appendix 11.1.6).

Theorem 11.2 (Identification of the NIE)

If there is a set W of non-descendants of X or Z with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ for all x and z, then

$$\text{NIE}(x, x^*; Y) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)[P(Z_x = z \mid w) - P(Z_{x^*} = z \mid w)]P(w).$$

In non-experimental studies the NIE is identifiable when $\mathbb{E}(Y_{x^* z} \mid w)$, $P(Z_x = z \mid w)$ and $P(Z_{x^*} = z \mid w)$ are. In a simple Markovian model,

$$\text{NIE}(x, x^*; Y) = \sum_z \mathbb{E}(Y \mid x^*, z)[P(z \mid x) - P(z \mid x^*)].$$

Proof [Supplement]. By definition $\mathrm{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*}) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*, Z_{x^*}})$ (composition). For the first term, condition on W and on $Z_x = z$:

$$\mathbb{E}(Y_{x^*, Z_x}) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid Z_x = z, w) P(Z_x = z \mid w) P(w) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w) P(Z_x = z \mid w) P(w),$$

using $Y_{x^* z} \perp\!\!\!\perp Z_x \mid W$. The same computation with x^* in place of x gives the second term, and subtracting yields the formula. In a Markovian model with $W = \emptyset$, $\mathbb{E}(Y_{x^* z}) = \mathbb{E}(Y \mid x^*, z)$ and $P(Z_x = z) = P(z \mid x)$, which gives the last display. ■

11.1.5 Path-specific effects

Motivation. Generalize direct/indirect effects to effects along any chosen set of paths.

With several mediators ($X \to W \to Z \to Y$, $X \to Z$, $X \to Y$, ...) one may want the effect transmitted along a specific subset of paths. In the counterfactual graph, the inputs of each mechanism are switched between their values under x and under x^* according to whether the edge belongs to the selected paths.

Definition (Path-specific effect; Pearl 2001)

Let $\mathcal{G}$ be the causal graph of $\mathcal{M}$ and g an edge-subgraph of $\mathcal{G}$ containing the paths selected for analysis. The g-specific effect of x on Y (relative to the reference x^*) is the total effect of x on Y in a modified model $\mathcal{M}_g$ built as follows. Partition each parent set as $Pa_i = \{Pa_i(g), Pa_i(\bar{g})\}$, where $Pa_i(g)$ are the parents linked to X_i in g and $Pa_i(\bar{g})$ the rest. Replace each $f_i(\mathbf{pa}_i, u)$ by

$$f_i(\mathbf{pa}_i, u; g) = f_i(\mathbf{pa}_i(g), \mathbf{pa}_i^*(\bar{g}), u),$$

where $\mathbf{pa}_i^*(\bar{g}) = Pa_i(\bar{g})_{x^*}$ are the values those variables would attain (in $\mathcal{M}$ and u) under $X = x^*$. Then $SE_g(x, x^*; Y, u)_{\mathcal{M}} = TE(x, x^*; Y, u)_{\mathcal{M}_g}$.

With $g = \{X \to Y\}$ this gives the NDE; with g the mediated paths, the corresponding indirect effect. Such effects lead to nested counterfactuals such as $Y_{W_x, Z_x, Z_{x^*}}$-type expressions, represented in counterfactual graphs.

Conclusions. Pearl (2001) defines path-specific effects by *path switching* rather than variable fixing, extending the interpretation and estimation of direct and indirect effects to nonlinear models. Under certain conditions these effects are identifiable even when Z is affected by X; the conditions always hold in Markovian models. The definition also gives an operational interpretation of indirect effects.

11.1.6 Appendix: The linear additive relation

Take $Z = \alpha X + U_Z$, $Y = \beta X + \gamma Z + U_Y$, so $Z_x = \alpha x + U_Z$ and $Z_{x^*} = \alpha x^* + U_Z$. Then

$$\begin{aligned}
\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}) = (\beta + \alpha\gamma)(x - x^*), \\
\mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}) = \beta(x - x^*), \\
\mathrm{NIE}(x, x^*; Y) &= \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*}) = \alpha\gamma(x - x^*),
\end{aligned}$$

so TE = NDE + NIE. (Also $\mathrm{NIE}(x^*, x; Y) = -\alpha\gamma(x - x^*)$, consistent with $\mathrm{TE} = \mathrm{NDE}(x, x^*) - \mathrm{NIE}(x^*, x)$.) This uses the linear, no-interaction form; nonlinear models do not generally preserve the same-direction sum.

11.1.7 Appendix: Counterfactual axioms

Composition. For singleton Y, W and any set X: $W_x(u) = w \Rightarrow Y_{xw}(u) = Y_x(u)$. Forcing a variable to the value it would have had anyway has no effect on other variables.
Consistency. $X(u) = x \Rightarrow Y(u) = Y_x(u)$.
Effectiveness. For all X, W: $X_{xw}(u) = x$.
Reversibility. For Y, W and any X: $(Y_{xw}(u) = y) \wedge (W_{xy}(u) = w) \Rightarrow Y_x(u) = y$.

11.2 Single-world intervention graphs

Motivation. The central question is how potential outcomes can be placed on a causal graph, so that counterfactual independences can be read off by d-separation, without the extra untestable assumptions of a fully specified structural-equation model.

11.2.1 Potential outcomes and graphs

Motivation. Recall the two languages that SWIGs unify.

11.2.1.1 Potential outcomes

For binary treatment X and response Y: $Y(x = 0)$ (also $Y(x_0)$) is the value of Y that would be observed for a unit assigned $X = 0$; $Y(x = 1)$ (or $Y(x_1)$) likewise for $X = 1$. This presumes the outcomes are well defined: one version of treatment, and no interference between units (SUTVA). "Potential outcome" and "counterfactual" are used synonymously.

Response types for binary Y:

$Y(x_0)$	$Y(x_1)$	Name
0	0	never recover
0	1	helped
1	0	hurt
1	1	always recover

Relating counterfactuals and do-notation. $P(Y(x) = y) \equiv P(Y = y \mid \mathrm{do}(X = x))$, but counterfactual notation is more general. Example: the distribution of outcomes that would have arisen among those who took treatment $(X = 1)$ had they, counter to fact, not received it: $P(Y(x = 0) = y \mid X = 1)$. If treatment is randomized, $X \perp\!\!\!\perp Y(x = 0)$ and this equals $P(Y(x = 0) = y)$; in an observational study they may differ.

11.2.1.2 Graphs

A DAG is associated with the factorization $P(\mathbf{V}) = \prod_{X \in \mathbf{V}} P(X \mid \mathrm{pa}(X))$; e.g. for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$, $P(A, B, C, D, E) = P(A)P(B \mid A)P(C \mid A)P(D \mid B, C)P(E \mid D)$. d-separation reads independences such as $C \perp\!\!\!\perp B \mid A$, $D \perp\!\!\!\perp A \mid B, C$, $E \perp\!\!\!\perp A, B, C \mid D$.

In the graphical approach, edges represent direct causal relations, and by convention confounders (e.g. an unobserved H with $X \leftarrow H \to Y$) are included. It originates in Sewall Wright's path diagrams of the 1920s. For $X \to Y$ without confounding, $P(x, y) = P(x)P(y \mid x)$ and the causal model asserts

$$P(Y(x) = y) = P(Y = y \mid \mathrm{do}(X = x)) = P(Y = y \mid X = x),$$

where $P(y \mid \text{do}(x))$ is the distribution under an experiment fixing X to x. How does this relate to the counterfactual approach? No confounding corresponds to $X \perp\!\!\!\perp Y(x_0)$ and $X \perp\!\!\!\perp Y(x_1)$; confounding to $X \not\perp\!\!\!\perp Y(x_0)$ or $X \not\perp\!\!\!\perp Y(x_1)$. The elephant in the room: $Y(x_0)$ and $Y(x_1)$ do not appear on these graphs.

11.2.2 Node splitting

Motivation. Put counterfactual variables on the graph.

Setting X to 0. Start from $P(X = \tilde{x}, Y = \tilde{y}) = P(X = \tilde{x})P(Y = \tilde{y} \mid X = \tilde{x})$ for $X \to Y$. Split X into a random node X and a fixed node $x = 0$; the outcome becomes $Y(x = 0)$:



Now $X \perp\!\!\!\perp Y(x = 0)$ can be read off, and a new factorization is associated:

$$P(X = \tilde{x}, Y(x = 0) = \tilde{y}) = P(X = \tilde{x})\, P(Y(x = 0) = \tilde{y}), \qquad P(Y(x = 0) = \tilde{y}) = P(Y = \tilde{y} \mid X = 0).$$

The last equation links a term of the original factorization to the new one: the *modularity assumption*.

Marginals represented by SWIGs are identified. The SWIG $\mathcal{G}(x_0)$ represents $P(X, Y(x_0))$ and $\mathcal{G}(x_1)$ represents $P(X, Y(x_1))$; under no confounding both are identified from $P(X, Y)$. In contrast $P(X, Y(x_0), Y(x_1))$ is not identified. $Y(x = 0)$ and $Y(x = 1)$ never appear on the same graph: although $X \perp\!\!\!\perp Y(x = 0)$ and $X \perp\!\!\!\perp Y(x = 1)$, we do not assume $X \perp\!\!\!\perp \{Y(x = 0), Y(x = 1)\}$. A single graph containing both would have been impossible; hence *single-world* intervention graphs.

Templates. Both graphs are represented by a template $\mathcal{G}(x)$: $X \mid x \to Y(x)$. Formally the template is a graph-valued function (not a graph): it takes a value x and returns a SWIG $\mathcal{G}(x)$; each instantiation represents a different margin.

Intuition (Robins, VanderWeele and Richardson 2007). How could we learn whether someone would choose treatment $(X = 1)$ and, at the same time, what happens to that person without treatment $(Y(x = 0))$? Imagine that whenever a patient is observed to swallow the drug $(X = 1)$, we instantly administer a safe emetic that makes them regurgitate the pill before any drug enters the bloodstream. With no side effects of the emetic, the recorded outcome is $Y(x = 0)$, observed jointly with the natural choice X.

11.2.3 SWIG construction and properties

Motivation. Construct SWIGs in general and state what they imply.

Given $\mathcal{G}$ and a set $\mathbf{A} = \{A_1, \ldots, A_k\}$ to intervene on, form $\mathcal{G}(\mathbf{a})$ in two steps:

1. **Node splitting:** split every $A \in \mathbf{A}$ into a random node A and a fixed node a. The random half inherits all edges directed *into* A; the fixed half inherits all edges directed *out of* A.
2. **Relabelling:** every descendant Y of a fixed node is relabelled $Y(\mathbf{a}_Y)$, indexed by the fixed nodes that are its ancestors (e.g. $B(a, \ldots)$, $C(a, \ldots)$, $D(a, \ldots)$); non-descendants keep their names.

Factorization and modularity. The original graph $\mathcal{G}$ carries the observed $P(\mathbf{V})$; the SWIG $\mathcal{G}(\tilde{\mathbf{a}})$ carries the counterfactual distribution $P(\mathbf{V}(\tilde{\mathbf{a}}))$.

- *Factorization:* $P(\mathbf{V}(\tilde{\mathbf{a}}))$ factorizes with respect to $\mathcal{G}(\tilde{\mathbf{a}})$, ignoring fixed nodes.

- *Modularity:* the conditional density of $Y(\tilde{\mathbf{a}}_Y)$ in $\mathcal{G}(\tilde{\mathbf{a}})$ equals that of Y in $\mathcal{G}$ after substituting $\tilde{a}_i$ for any $A_i \in \mathbf{A}$ that is a parent of Y.

Consequence: if $P(\mathbf{V})$ is observed, $P(\mathbf{V}(\tilde{\mathbf{a}}))$ is identified.

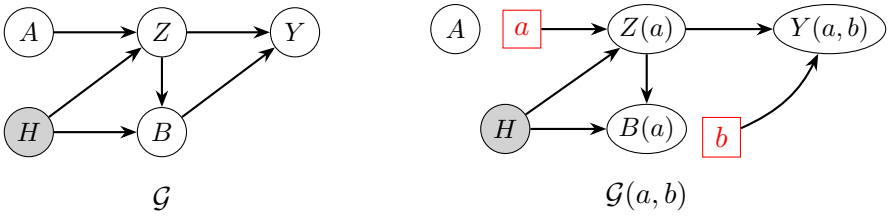
d-separation on SWIGs. If subsets $\mathbf{B}(\tilde{\mathbf{a}})$ and $\mathbf{C}(\tilde{\mathbf{a}})$ of random nodes are d-separated by $\mathbf{D}(\tilde{\mathbf{a}})$ together with the fixed nodes $\tilde{\mathbf{a}}$ in $\mathcal{G}(\tilde{\mathbf{a}})$, then

$$\mathbf{B}(\tilde{\mathbf{a}}) \perp\!\!\!\perp \mathbf{C}(\tilde{\mathbf{a}}) \mid \mathbf{D}(\tilde{\mathbf{a}}) \quad \text{in } P(\mathbf{V}(\tilde{\mathbf{a}})).$$

11.2.4 A harder inferential problem

Motivation. Use a SWIG to settle a counterfactual independence query.

Graph $\mathcal{G}$: $A \to Z$, $Z \to Y$, $Z \to B$, $B \to Y$, and an unobserved H with $H \to Z$, $H \to B$. Query: does $\mathcal{G}$ imply $Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A$?



Pearl (2009, Ex. 11.3.3) claims the DAG does not imply

$$Y(a,b) \perp\!\!\!\perp B \mid Z, A = a. \tag{3}$$

The SWIG shows that (3) does hold. In $\mathcal{G}(a,b)$, every path from $B(a)$ to $Y(a,b)$ passes through $Z(a)$ as a non-collider ($B(a) \leftarrow Z(a) \to Y(a,b)$, $B(a) \leftarrow H \to Z(a) \to Y(a,b)$), and $B(a)$ has no edge to $Y(a,b)$ because the edge $B \to Y$ now leaves the fixed node b. Hence

$$Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A \quad \Longrightarrow \quad Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A = a,$$

which is equivalent to (3) by consistency ($B(a) = B$ and $Z(a) = Z$ when $A = a$). (Pearl had inferred that a claim of Robins was false because, if true, (3) would hold.)

11.2.5 Adjustment for confounding

Motivation. Prove the back-door formula in the SWIG language.

Graph: $L \to X$, $L \to Y$, $X \to Y$. In the SWIG $\mathcal{G}(x)$: $L \to X$, $L \to Y(x)$, $x \to Y(x)$, so $X \perp\!\!\!\perp Y(x) \mid L$. Then

$$\begin{aligned} P[Y(\tilde{x}) = y] &= \sum_l P[Y(\tilde{x}) = y \mid L = l] P(L = l) \\ &= \sum_l P[Y(\tilde{x}) = y \mid L = l, X = \tilde{x}] P(L = l) && (X \perp\!\!\!\perp Y(\tilde{x}) \mid L) \\ &= \sum_l P[Y = y \mid L = l, X = \tilde{x}] P(L = l) && \text{(consistency)}. \end{aligned}$$

Connection to do-calculus. Factorization and modularity suffice to imply all identification results of Pearl's (1995) do-calculus (see also Spirtes et al. 1993): $P(Y = y \mid \text{do}(A = a))$ is identified iff $P(Y(a) = y)$ is identified.

11.2.6 Conclusion

Eliminating a false trichotomy. Previously the only unification of counterfactuals and graphs was Pearl's non-parametric structural equation models with independent errors, which left three options: use graphs without counterfactuals (Dawid); use counterfactuals without graphs (many statisticians); or use both but accept many additional assumptions that are not experimentally testable even in principle and not needed for most identification results. SWIGs allow graphs and counterfactuals together without these extra assumptions.

Summary and extensions. SWIGs unify graphs and counterfactuals via node splitting, working by linking the factorizations of the two graphs. The new graph represents a counterfactual distribution identified from the distribution of the original DAG, giving a common language for counterfactual and graphical researchers. (Not covered: combining information on the absence of individual- and population-level direct effects; models in which interventions on only some variables are well defined.)

References: Pearl (2009); Richardson and Robins (2013), CSSS Technical Report 128; Robins (1986); Robins, VanderWeele and Richardson (2007); Shpitser and Pearl (2008); Spirtes, Glymour and Scheines (1993).

11.3 G-methods for time-varying treatments

Motivation. The central question is how to estimate the effect of a sequence of treatment decisions when covariates measured along the way are both affected by earlier treatment and drive later treatment, a situation in which ordinary regression, stratification and matching all fail.

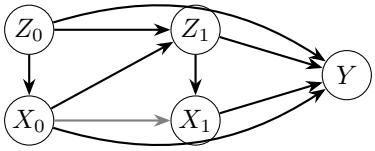
11.3.1 Time-varying treatments

Motivation. Set up sequential decisions and explain why feedback breaks standard adjustment.

Recall the single time point. Identification by back-door adjustment, $\mathbb{E}[Y(x)] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]P(Z = z)$, and estimation by IPW, $\mathbb{E}[Y(x)] = \mathbb{E}[\mathbb{1}(X = x)Y/P(X = x \mid Z)]$. What if treatment is applied repeatedly?

Many interventions involve repeated decisions: medication updated at each visit, policies applied in stages, algorithms adapting to the current state. Later decisions are made after earlier information has been observed, so the data-generating process is history-dependent: action $\to$ state $\to$ action $\to \cdots \to$ outcome.

Causal diagrams. Treatments X_k with history $\bar{X}_k = (X_0, \ldots, X_k)$; covariates (states) Z_k with history $\bar{Z}_k = (Z_0, \ldots, Z_k)$; outcome Y at the end of follow-up. For two periods: $Z_0 \to X_0$, $Z_0 \to Z_1$, $X_0 \to Z_1$, $Z_1 \to X_1$, $X_0 \to X_1$, and all of them $\to Y$.



Treatment–confounder feedback occurs when (1) past treatment X_{k-1} affects a future confounder Z_k, and (2) Z_k affects future treatment X_k. Example (HIV): antiretroviral therapy (ART, X) $\to$ CD4 count or viral load (Z, health biomarkers) $\to$ next ART decision $\to \cdots$. Previous treatment may also guide later decisions directly ($X_{k-1} \to X_k$). This feedback loop makes standard methods fail.

Treatment strategies. A strategy specifies treatment at each time. *Static*: the same for everyone regardless of covariates, e.g. "always treat" $\bar{x} = (1, \ldots, 1)$, "never treat" $\bar{x} = (0, \ldots, 0)$. *Dynamic*: treatment at time k depends on observed history, $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, e.g. "initiate ART only when CD4 drops below 500". Causal estimands contrast mean counterfactual outcomes, $\mathbb{E}[Y(\bar{x})] - \mathbb{E}[Y(\bar{x}')]$.

11.3.1.1 Identification assumptions

Sequentially randomized experiments: X_k is randomly assigned with probability depending only on $(\bar{X}_{k-1}, \bar{Z}_k)$. The observational analogue: no unmeasured common causes of X_k and Y given $(\bar{X}_{k-1}, \bar{Z}_k)$. An unmeasured U may affect Y and Z_1 as long as it has no arrow into X_0 or X_1; then adjusting for the measured past suffices.

- **Sequential exchangeability:** $Y(\bar{x}) \perp\!\!\!\perp X_k \mid \bar{X}_{k-1}, \bar{Z}_k$ for $k = 0, \ldots, K$ (treatment as-if randomized given the observed past).
- **Consistency:** $Y(\bar{x}) = Y$ when $\bar{X} = \bar{x}$.
- **Positivity:** $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k) > 0$.

11.3.1.2 The fundamental dilemma

Should we adjust for Z_1? Adjusting removes confounding of $X_1 \to Y$ (via Z_1) but over-adjusts by blocking $X_0 \to Z_1 \to Y$ (and, with $U \to Z_1$, $U \to Y$, opens the collider $X_0 \to Z_1 \leftarrow U \to Y$). Not adjusting preserves X_0's effect through Z_1 but leaves X_1 confounded. Traditional methods (regression, stratification, matching) cannot do both under feedback.

G-methods (Robins' generalized methods) are consistent under time-varying confounding:

(1) the g-formula (outcome modeling);

(2) IP-weighted marginal structural models (treatment modeling);

(3) g-estimation of structural nested models (effect modeling). All rely on the same assumptions. Key idea: handle Z_1 without blocking the causal path through it.

What you already know. Apply the truncated factorization (Chapter 2) to (Z_0, X_0, Z_1, X_1, Y): from

$$P(z_0) \cdot P(x_0 \mid z_0) \cdot P(z_1 \mid x_0, z_0) \cdot P(x_1 \mid z_1, x_0, z_0) \cdot P(y \mid x_1, z_1, x_0, z_0)$$

remove the treatment factors ($\mathrm{do}(x_0)$, $\mathrm{do}(x_1)$) and keep the rest:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_0, z_1} \mathbb{E}[Y \mid x_0, z_0, x_1, z_1]\, P(z_1 \mid x_0, z_0)\, P(z_0).$$

This is the *g-formula*, the truncated factorization in temporal order; sequential exchangeability ensures each remaining factor is estimable. What is new: why feedback breaks standard estimation, and estimation strategies (sequential weight products, MSMs, SNMs).

11.3.2 The g-formula

Motivation. Standardize over the time-varying confounder instead of conditioning on it.

Idea (two time points, no baseline confounder, so X_0 is marginally exchangeable). Instead of conditioning on Z_1, standardize over its distribution *given past treatment*:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_1} \mathbb{E}[Y \mid X_0 = x_0, Z_1 = z_1, X_1 = x_1]\, P(Z_1 = z_1 \mid X_0 = x_0),$$

weighting by $P(z_1 \mid x_0)$, not by the marginal of Z_1.

Derivation. The observed density factorizes temporally (writing f and P interchangeably):

$$P(y, x_1, z_1, x_0) = P(y \mid x_1, z_1, x_0) \underbrace{P(x_1 \mid z_1, x_0)}_{\text{remove}} P(z_1 \mid x_0) \underbrace{P(x_0)}_{\text{remove}}.$$

Under $\mathrm{do}(X_0 = x_0, X_1 = x_1)$ the assignment terms are replaced by the intervention, giving outcome model $\times$ confounder model as above. For $K+1$ time points,

$$\mathbb{E}[Y(\bar{x})] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{X} = \bar{x}, \bar{Z} = \bar{z}] \prod_{k=0}^{K} P(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}),$$

with $\bar{x}_{-1}, \bar{z}_{-1}$ empty so that $P(z_0 \mid \bar{x}_{-1}, \bar{z}_{-1}) = P(z_0)$.

11.3.2.1 Iterated conditional expectations (ICE)

The sum over $\bar{z}$ needs covariate densities $P(z_k \mid \text{past})$, hard to model for high-dimensional Z_k. ICE replaces the sum by nested regressions computed backward in time: each step is an outcome regression averaging out one confounder, the data doing the averaging, so no density model is needed:

$$Y \xrightarrow{\text{regress}} \hat{Q}_K \to \cdots \to \hat{Q}_0 \xrightarrow{\text{average}} \hat{\mathbb{E}}[Y(\bar{x})].$$

Two time points. (1) $\hat{Q}_1(z_1) = \hat{\mathbb{E}}[Y \mid x_0, z_1, x_1]$, an outcome regression at (x_0, x_1);

(2) $\hat{\mathbb{E}}[Y(x_0, x_1)] = \hat{\mathbb{E}}[\hat{Q}_1(Z_1) \mid X_0 = x_0]$, the average over the $X_0 = x_0$ group. The second step performs the g-formula's averaging through the empirical distribution of Z_1 in the selected treatment-history group.

Three time points as variable elimination. Eliminate Z_2, then Z_1:

$$Q_2(z_1, z_2) = \mathbb{E}[Y \mid x_0, z_1, x_1, z_2, x_2], \qquad Q_1(z_1) = \mathbb{E}[Q_2(Z_1, Z_2) \mid x_0, z_1, x_1],$$
$$\mathbb{E}[Y(x_0, x_1, x_2)] = \mathbb{E}[Q_1(Z_1) \mid X_0 = x_0].$$

For Q_1, create the pseudo-outcome $\tilde{Y}_i^{(2)} = \hat{Q}_2(Z_{1i}, Z_{2i})$ with treatments set to (x_0, x_1, x_2), and regress it on Z_1 among rows with $X_0 = x_0$, $X_1 = x_1$; the data distribution averages over Z_2.

General recursion. Starting from $\hat{Q}_{K+1} = Y$, for $k = K, \ldots, 0$:

$$\hat{Q}_k = \hat{\mathbb{E}}[\hat{Q}_{k+1} \mid \bar{X}_k = \bar{x}_k, \bar{Z}_k], \qquad \hat{\mathbb{E}}[Y(\bar{x})] = \hat{\mathbb{E}}[\hat{Q}_0(x_0, Z_0)].$$

Every step is an outcome regression; ICE never models $P(z_k \mid \text{past})$, which is why it scales to many time points and high-dimensional Z_k.

11.3.3 IP weighting and marginal structural models

Motivation. Re-weight to a pseudo-population in which treatment is unconfounded.

Idea. Instead of modeling the outcome, weight each individual inversely to the probability of their observed treatment given confounders; this "breaks" the arrows $Z_k \to X_k$ (and U, which never points into treatment, stays harmless).

One time point: why $1/P(X \mid Z)$. Under exchangeability $Y(x) \perp\!\!\!\perp X \mid Z$, consistency and positivity,

$$\mathbb{E}\Big[\frac{\mathbb{1}(X = x)Y}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[\mathbb{1}(X = x)Y \mid Z]}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = \mathbb{E}_Z\big[\mathbb{E}[Y(x) \mid Z]\big] = \mathbb{E}[Y(x)].$$

Dividing by $P(x \mid Z)$ up-weights under-represented (X, Z) combinations, building a pseudo-population with $X \perp\!\!\!\perp Z$: the arrow $Z \to X$ is gone and the crude association becomes causal.

Two time points (no baseline confounder), for a fixed regime (x_0, x_1):

$$W^X(x_0, x_1) = \frac{\mathbb{1}(X_0 = x_0)}{P(x_0)} \cdot \frac{\mathbb{1}(X_1 = x_1)}{P(x_1 \mid x_0, Z_1)},$$

$$\begin{aligned}\mathbb{E}[W^X(x_0, x_1)Y] &= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] \frac{P(x_0, z_1, x_1)}{P(x_0)P(x_1 \mid x_0, z_1)} \\ &= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] P(z_1 \mid x_0) = \mathbb{E}[Y(x_0, x_1)].\end{aligned}$$

The first factor undoes the observed assignment of X_0, the second that of X_1 given history (X_0, Z_1). With baseline Z_0, replace $P(X_0 = x_0)$ by $P(X_0 = x_0 \mid Z_0)$.

The weight product. Confounding $Z_k \to X_k$ enters at every step, so divide by $P(X_k \mid \text{past})$ at every k:

$$W^X = \prod_{k=0}^{K} \frac{1}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)} \qquad \text{(non-stabilized)}.$$

Stabilized weights. The mean of W^X blows up: for binary treatments,

$$\mathbb{E}[W^X] = \sum_{\bar{x}, \bar{z}} \frac{P(\bar{x}, \bar{z})}{\prod_k P(x_k \mid \text{past})} = \sum_{\bar{x}, \bar{z}} \prod_k P(z_k \mid \text{past}) = \sum_{\bar{x}} 1 = 2^{K+1},$$

so the pseudo-population is 2^{K+1} times larger, and rare treatment cells give extreme weights and high variance. Fix: put $P(X_k \mid \bar{X}_{k-1})$ (treatment given past treatment only) in the numerator,

$$SW^X = \prod_{k=0}^{K} \frac{P(X_k \mid \bar{X}_{k-1})}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

The numerator uses treatment history only, so it does not reintroduce $Z \to X$ confounding (the estimator remains valid), and $\mathbb{E}[SW^X] = 1$: the pseudo-population has the data's size, with weights centered at 1 and much smaller variance.

Proof (one time point). With $SW = P(X)/P(X \mid Z)$,

$$\begin{aligned}\mathbb{E}\Big[\frac{P(x)}{P(x \mid Z)} \mathbb{1}(X = x) Y\Big] &= P(x)\, \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z] P(x \mid Z)}{P(x \mid Z)}\Big] \\ &= P(x)\, \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = P(x)\, \mathbb{E}[Y(x)].\end{aligned}$$

With $Y \equiv 1$, $\mathbb{E}[SW\, \mathbb{1}(X = x)] = P(x)$, so $\mathbb{E}[SW] = \sum_x P(x) = 1$ and $\mathbb{E}[SW\, \mathbb{1}(X = x)Y]/\mathbb{E}[SW\, \mathbb{1}(X = x)] = \mathbb{E}[Y(x)]$. The extra factor $P(x)$ merely rescales the $X = x$ arm to its marginal size, so the pseudo-population is not doubled; it cancels in the ratio (and in an MSM), leaving the causal target unchanged and only shrinking variance.

Proof (general). Let $r_k(x_k \mid \bar{x}_{k-1}) \equiv P(X_k = x_k \mid \bar{X}_{k-1} = \bar{x}_{k-1})$. The observed law factorizes as $p(\bar{z}, \bar{x}, y) = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}) p(x_k \mid \bar{x}_{k-1}, \bar{z}_k) \cdot p(y \mid \bar{x}, \bar{z})$. For $SW^X = \prod_k r_k(X_k \mid \bar{X}_{k-1})/p(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$, the weighted population has density

$$p_{SW}(\bar{z}, \bar{x}, y) = p(\bar{z}, \bar{x}, y)\, SW^X = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r_k(x_k \mid \bar{x}_{k-1}) \cdot p(y \mid \bar{x}, \bar{z}),$$

a proper density (so $\mathbb{E}[SW^X] = 1$) in which treatment depends only on past treatment: the $Z_k \to X_k$ arrows are removed. For a static regime $\bar{x}$ let $r(\bar{x}) = \prod_k r_k(x_k \mid \bar{x}_{k-1})$. Then

$$\mathbb{E}\big[SW^X \mathbb{1}(\bar{X} = \bar{x}) Y\big] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{x}, \bar{z}] \prod_k p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r(\bar{x}) = r(\bar{x})\, \mathbb{E}[Y(\bar{x})]$$

by the g-formula, and $\mathbb{E}[SW^X \not\Vdash (\bar{X} = \bar{x})] = r(\bar{x})$, so $\mathbb{E}[SW^X \not\Vdash (\bar{X} = \bar{x})Y]/\mathbb{E}[SW^X \not\Vdash (\bar{X} = \bar{x})] = \mathbb{E}[Y(\bar{x})]$. Stabilization changes the marginal treatment distribution in the pseudo-population, not the causal target.

11.3.3.1 Marginal structural models

An MSM specifies the relation between treatment strategy and counterfactual mean, e.g.

$$\mathbb{E}[Y(x_0, x_1)] = \beta_0 + \beta_1 x_0 + \beta_2 x_1 + \beta_3 x_0 x_1.$$

It is a model for potential outcomes, not for observed data, estimated by weighted regression in the IP-weighted pseudo-population. More complex MSMs: saturated (one parameter per strategy); dose–response $\mathbb{E}[Y(\bar{x})] = \beta_0 + \beta_1 \text{cum}(\bar{x})$ with $\text{cum}(\bar{x}) = \sum_k x_k$; conditional MSMs including baseline covariates.

11.3.4 G-estimation of structural nested models

Motivation. Model and remove the history-specific effect of each decision.

The blip. An MSM models $\mathbb{E}[Y(\bar{x})]$ marginally (averaged over history). A structural nested mean model (SNMM) targets the local, history-specific effect of each decision, the *blip*: a contrast of potential outcomes with the future held at the reference (0):

$$\gamma_1(x_0, z_1) = \mathbb{E}[Y(x_0, 1) - Y(x_0, 0) \mid X_0 = x_0, Z_1 = z_1] \text{ (time-1 blip, given history)},$$

$$\gamma_0(x_0) = \mathbb{E}[Y(x_0, 0) - Y(0, 0)] \text{ (time-0 blip)}.$$

Blips accumulate: in a three-decision example from "never" (000) to "always" (111), $\mathbb{E}[Y(\bar{x})]$ climbs $000 \to 100 \to 110 \to 111$ by steps $+\gamma_0, +\gamma_1, +\gamma_2$; the MSM is the whole climb, the SNMM the history-specific steps.

G-estimation: find the blip by removing it. If the late blip were known, subtracting it would rebuild each patient's outcome had they not taken the late treatment: $Y_{\text{removed}} = Y - \psi_1 X_1$ (a candidate blip). G-estimation solves for the ψ_1 that makes Y_{removed} balanced across X_1 given history: an estimating equation, not an outcome regression. Two handles, one target: IPW models the treatment (re-weighting to break confounding); SNMM models the effect (the blip) and removes it. Both recover $\mathbb{E}[Y(\bar{x})]$.

In practice: one time point. With an additive, history-homogeneous blip $\gamma(z; \psi) = \psi x$, blipping down gives $H(\psi) = Y - \psi X$, the candidate $Y(0)$. At the true ψ, consistency and the blip make $H(\psi) = Y(0)$, and exchangeability gives $Y(0) \perp\!\!\!\perp X \mid Z$, so $H(\psi)$ must be uncorrelated with X given Z. With a fitted propensity $\hat{e}(Z) = \hat{P}(X = 1 \mid Z)$, solve

$$U(\psi) = \sum_i (X_i - \hat{e}(Z_i)) \underbrace{(Y_i - \psi X_i)}_{H_i(\psi)} = 0 \quad \Longrightarrow \quad \hat{\psi} = \frac{\sum_i (X_i - \hat{e}_i) Y_i}{\sum_i (X_i - \hat{e}_i) X_i},$$

an IV-type ratio in which the propensity residual $X - \hat{e}$ plays the role of an instrument. Recipe:

(1) fit $\hat{e}(Z)$ (e.g. logistic);

(2) form residuals $X_i - \hat{e}_i$;

(3) plug in. It is consistent if $\hat{e}$ is correct; adding an outcome term $\mathbb{E}[H(\psi) \mid Z]$ makes it doubly robust. With effect modification, e.g. $\gamma(z; \psi) = (\psi_0 + \psi_1 z)x$, H uses the z-specific blip and the instrument becomes the vector $(1, z)(X - \hat{e})$.

Two time points. Order $X_0 \to Z_1 \to X_1 \to Y$ with history-homogeneous blips $\gamma_1 = \psi_1 x_1$, $\gamma_0 = \psi_0 x_0$ (history enters only through the propensities). Fit the latest blip first, then work backward.

1. Remove the time-1 blip: $H_1(\psi_1) = Y - \psi_1 X_1$ is the candidate $Y(x_0, 0)$, and exchangeability gives $Y(x_0, 0) \perp\!\!\!\perp X_1 \mid (X_0, Z_1)$. With $\hat{e}_1 = \hat{P}(X_1 = 1 \mid X_0, Z_1)$, solving $\sum_i (X_{1i} - \hat{e}_{1i}) H_{1i}(\psi_1) = 0$ gives
$$\hat{\psi}_1 = \frac{\sum_i (X_{1i} - \hat{e}_{1i}) Y_i}{\sum_i (X_{1i} - \hat{e}_{1i}) X_{1i}}.$$

2. Remove the time-0 blip from the blipped-down outcome: $H_0 = H_1(\hat{\psi}_1) - \psi_0 X_0 = Y - \hat{\psi}_1 X_1 - \psi_0 X_0$, the candidate $Y(0, 0)$. With $\hat{e}_0 = \hat{P}(X_0 = 1 \mid Z_0)$,
$$\hat{\psi}_0 = \frac{\sum_i (X_{0i} - \hat{e}_{0i}) H_{1i}(\hat{\psi}_1)}{\sum_i (X_{0i} - \hat{e}_{0i}) X_{0i}}.$$

The same one-step equation is applied backward with history-indexed propensity residuals $X_k - \hat{e}_k$. Telescoping, $\mathbb{E}[Y(1,1)] - \mathbb{E}[Y(0,0)] = \hat{\psi}_0 + \hat{\psi}_1$. Each step becomes doubly robust by adding an outcome term $\mathbb{E}[H_k \mid \text{history}]$.

11.3.5 Doubly robust estimation

Motivation. Combine outcome and treatment models for a second chance at consistency.

The g-formula needs a correct outcome model $\mathbb{E}[Y \mid \bar{X}, \bar{Z}]$; IP weighting needs correct treatment models $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$; misspecification biases either. Doubly robust (augmented IPW, AIPW) estimators are consistent if either model is correct.

Time-fixed review. With outcome model $\hat{m}(x, z) = \hat{\mathbb{E}}[Y \mid X = x, Z = z]$ and propensity $\hat{\pi}(z) = \hat{P}(X = 1 \mid Z = z)$,
$$\hat{\tau}_{DR} = \frac{1}{n} \sum_{i=1}^{n} \left[\hat{m}(1, Z_i) - \hat{m}(0, Z_i) + \frac{X_i \{Y_i - \hat{m}(1, Z_i)\}}{\hat{\pi}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{m}(0, Z_i)\}}{1 - \hat{\pi}(Z_i)} \right].$$

If $\hat{m}$ is correct, the IPW residual terms have mean zero; if $\hat{\pi}$ is correct, the residuals correct the bias of $\hat{m}$.

Time-varying extension. Fit treatment models $\hat{\pi}_k$ at each time and outcome models backward from K to 0 (ICE-style); the estimator augments IPW with outcome predictions at each step and is consistent if, at each time point, either the treatment or the outcome model is correct. The full sequential AIPW involves iterated regressions and cumulative weights (Bang and Robins 2005). Modern variants: TMLE, sequential regression with cross-fitting.

11.3.6 Practical considerations

Motivation. Dynamic strategies and positivity.

Dynamic strategies. All three g-methods generalize to $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$ with $g_k(\cdot) \in \{0, 1\}$. G-formula: simulate forward, drawing Z_k and setting $X_k = g_k(\cdot)$. IPW: weight individuals who happened to follow g,
$$W^g = \prod_{k=0}^{K} \frac{\mathbb{1}(X_k = g_k(\bar{X}_{k-1}, Z_k))}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

G-estimation: SNMMs naturally accommodate dynamic regimes via effect modification by Z_k.

Positivity violations. With K binary treatment times there are 2^K treatment histories $(2, 4, 8, \ldots)$, so cells become sparse (near-positivity violations), IP weights extreme, and estimates unstable. Remedies: stabilized weights, weight truncation, parametric models.

11.3.7 Comparison of g-methods

Motivation. Summarize what each method models, targets and risks.

	G-formula	IPW / MSM	G-estimation / SNM
Key model	outcome + confounder dist.	treatment assignment	treatment + blip function
Target	$\mathbb{E}[Y(\bar{x})]$ (marginal)	$\mathbb{E}[Y(\bar{x})]$ (marginal)	conditional blip effects
Effect modification by Z_k	outcome-model interactions	not directly	directly modeled
Extreme weights?	no	yes	no
G-null paradox?	yes (parametric)	no	no
DR version?	yes	yes (AIPW)	yes

All three are consistent under correct specification and the same identifiability conditions. (The g-null paradox: with parametric models for both outcome and time-varying confounders, the parametric g-formula can reject the null of no effect even when it is true, because the models cannot all be correct under the null.)

Estimand, model, estimator.

Method	Estimand (what)	Model (what must be right)	Estimator (how)
g-formula	$\mathbb{E}[Y(\bar{x})]$	outcome (+ confounders)	standardize / ICE
IPW	$\mathbb{E}[Y(\bar{x})]$	treatment	weighting
MSM	$\mathbb{E}[Y(\bar{x})]$, marginal	*marginal form* + treatment	weighted regression
SNMM / g-est.	blip γ_k (history-specific)	*blip form* + treatment	g-estimation (balance)
AIPW / DR	$\mathbb{E}[Y(\bar{x})]$	outcome or treatment	augmented IPW

Two kinds of "model": *structural* (an assumption on the estimand: MSM form, SNMM blip; saturated means no assumption, italic above) and *nuisance* (a working model of the observed data, outcome or propensity). So MSM/SNMM carry an extra structural assumption that the g-formula and IPW do not. Mind the level: MSM and SNMM are models placed on an estimand; IPW, g-estimation and AIPW are estimators (MSMs are fitted by IPW, SNMMs by g-estimation). SNMM is the outlier, targeting a different estimand, the history-specific blip, which enables time-varying effect modification and optimal dynamic regimes.

The big picture. Treatment–confounder feedback ($X_0 \to Z_1 \to X_1 \to \cdots$) defeats standard methods. The g-formula standardizes over time-varying Z; IPW/MSM re-weights to remove $Z \to X$ arrows; g-estimation/SNM subtracts blip effects backward in time; doubly robust AIPW combines outcome and treatment models.

Key takeaways.

1. Time-varying treatments require extending the causal framework to multiple time points and treatment strategies.
2. When a time-varying confounder is affected by prior treatment, standard regression, stratification and matching all fail.
3. The g-methods handle this under sequential exchangeability, consistency and positivity.
4. Doubly robust methods add protection against misspecification.
5. In practice choose by problem structure, available models, and sensitivity to positivity violations.